

APPENDIX TO THE ANNEX XV RESTRICTION REPORT

PROPOSAL FOR A RESTRICTION

SUBSTANCE NAME(S): Octocrilene

IUPAC NAME(S): 2-ethylhexyl 2-cyano-3,3-diphenylacrylate, 2-Ethylhexyl 2-cyano-3,3-diphenylprop-2-enoate, 2-Ethylhexyl 2-cyano-3,3-diphenyl-2-propenoate, 2-cyano-3,3-diphenyl-acrylic acid, 2-ethyl-hexyl ester, 2-Ethylhexyl 2-cyano-3,3-diphenylacrylate, (RS)-2-Cyan-3,3-diphenylacrylsäure-2-ethylhexylester, Propen-2-säure, 2-cyano-, 3,3-diphenyl-, 2-ethylhexylester, Octocrilene, Octocrilene

EC NUMBER(S): 228-250-8

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CONTENT

Appendix A: Manufacture and uses.....	11
A.1. Manufacture, import and export.....	11
A.2. Uses	11
A.2.1. Uses of octocrilene in cosmetic products	11
A.2.1.1. Presence of octocrilene in cosmetics	12
A.2.1.2. Survey of prices and composition of sun protection products in five European countries	15
A.3. Consumer preferences regarding sun protection products	17
A.4. Uses advised against by the registrants	17
Appendix B: Information on hazard and risk	18
B.1. Identity of the substance(s) and physical and chemical properties.....	18
B.2. Manufacture and uses (summary)	18
B.3. Classification and labelling	18
B.3.1. Classification and labelling in Annex VI of Regulation (EC) No 1272/2008 (CLP Regulation)	18
B.3.2. Classification and labelling in classification and labelling inventory/ Industry's self classification(s) and labelling.....	18
B.4. Environmental fate properties	18
B.4.1. Degradation.....	18
B.4.1.1. Abiotic degradation.....	18
B.4.1.2. Biodegradation.....	19
B.4.1.3. Summary of degradation.....	30
B.4.2. Environmental distribution.....	30
B.4.2.1. Adsorption/desorption.....	30
B.4.2.2. Volatilisation	31
B.4.2.3. Distribution modelling	31
B.4.3. Bioaccumulation	31
B.4.3.1. Aquatic bioaccumulation.....	31
B.4.3.2. Terrestrial bioaccumulation.....	33
B.4.4. Secondary poisoning.....	33
B.5. Human health hazard assessment	33

B.6. Human health hazard assessment of physicochemical properties	33
B.6.1. Explosivity	33
B.6.2. Flammability	33
B.6.3. Oxidising potential	33
B.7. Environmental hazard assessment	34
B.7.1. Aquatic compartment (including sediments)	34
B.7.1.1. Fish	34
B.7.1.2. Aquatic invertebrates	41
B.7.1.3. Algae and aquatic plants	49
B.7.1.4. Conclusion for surface water	51
B.7.1.5. Sediment organisms	51
B.7.1.6. Other aquatic organisms	57
B.7.2. Terrestrial compartment	61
B.7.3. Atmospheric compartment	66
B.7.4. Microbiological activity in sewage treatment systems	66
B.7.5. Non compartment specific effects relevant for the food chain (secondary poisoning)	67
B.7.6. Hazard information - conclusion	68
B.8. Environmental exposure assessment and associated risk	69
B.8.1. Physicochemical properties, environmental fate and distribution properties	69
B.8.2. Overview on exposure scenarios and coverage of substance life cycle	70
B.8.3. Tonnage approach - Environmental exposure assessment and associated risk	72
B.8.3.1. Tonnage approach - Local exposure assessment	72
B.8.3.2. Tonnage approach - Regional exposure assessment	77
B.8.3.3. Tonnage approach - Environmental risk assessment	79
B.8.4. Refined approach for the dispersive use of cosmetics end-products - Environmental exposure assessment and associated risk	79
B.8.4.1. Refined tonnage approach	79
B.8.4.1. Consumer approach	82
B.8.5. Environmental monitoring data	90
B.8.5.1. Wastewater	90
B.8.5.1. Aquatic ecosystems	107
B.8.5.1. Sediment	126
B.8.5.2. Terrestrial compartment	133

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

B.8.5.3. Biota.....	133
Appendix C: Justification for action on a Union-wide basis.....	137
Appendix D: Baseline.....	137
Appendix E: Impact Assessment.....	137
E.1. Risk Management Options	137
E.1.1. Discarded restriction options.....	137
E.1.1.1. Definition of an updated and lower concentration limit for octocrilene in cosmetic products.....	137
E.1.1.2. Elimination of octocrilene by treatment of sewage plant effluents	138
E.1.2. Other Union-wide risk management options than restriction	139
E.2. Alternatives	139
E.2.1. Description of the use and function of the restricted substance(s)	139
E.2.2. Identification of potential alternative substances and techniques fulfilling the function	141
E.2.2.1. Alternatives authorised in cosmetic regulation.....	141
E.2.2.2. Specific information on substances not used anymore as UV filters despite being authorised in annex VI of the cosmetic regulation.....	152
E.2.2.3. Other potential alternatives to be used as UV filters.....	153
E.2.3. Risk reduction, technical and economic feasibility, and availability of alternatives	156
E.3. Restriction scenario(s)	158
E.4. Economic impacts	158
E.4.1. Description of the cosmetic industry	158
E.4.1.1. Cosmetic Value Chain	158
E.4.1.2. Cosmetic market	159
E.4.1.3. Sun care market.....	160
E.4.1.4. Market for adult sun protection products and market for baby and child specific sun care products.....	161
E.4.2. Number of expected reformulations	163
E.4.3. Sensitivity analyses	166
E.4.3.1. Impact of alternative assumptions for reformulation cost per formulation and useful life of a formulation on reformulation costs	166
E.4.3.2. Impact of alternative assumptions on the reformulation rate.....	167
E.4.3.3. Impact of the length of the transitional period on reformulation costs	167

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

E.4.3.4. Impact of voluntary reformulations on reformulation costs	168
E.4.3.5. Costs of the restriction based on CosmEthics data	170
E.4.3.6. Total costs of the restriction under the worst-case scenario	172
E.4.4. Economic valuation of environmental impacts of the restriction options based on a stated preference	173
E.4.4.1. Relevance of discrete choice experiment and contingent valuation for the benefit assessment of the proposed restriction.....	173
E.4.4.2. Survey development	174
E.4.4.3. Pre-testing of the survey	179
E.4.4.4. Geographical scope and sampling	180
E.4.4.5. Model estimation	180
E.4.4.6. Descriptive statistics – Results and discussion	185
E.4.4.7. Willingness to pay estimates	187
E.4.4.8. Policy implications	188
E.4.4.9. Additional information	189
E.4.5. Benefits assessment of the proposed restriction based on DCE results	229
E.4.5.1. Main assessment	229
E.4.5.2. Sensitivity analysis – Processing of statistically insignificant WTP coefficients in the benefits assessment of the proposed restriction	231
E.5. Practicality and monitorability	232
Appendix F: Assumptions, uncertainties and sensitivities	232
Appendix G: Stakeholder information	233
References	236

TABLES

Table 1: Proportion of cosmetics products containing octocrilene.	12
Table 2: Proportion of cosmetics products containing octocrilene.	13
Table 3: Selected online retailers for sunscreen protection in 5 European countries.	16
Table 4: Share of primary sunscreens containing octocrilene in five EEA countries.	16
Table 5: Leading reasons to stop using a sun protection product.	17
Table 6: Screening tests for biodegradation in water.....	19
Table 7: Simulation tests for biodegradation in water and sediments (AR: Applied Radioactivity).....	19
Table 8: Measured concentration of octocrilene (% AR) in the whole freshwater system and the NER concentration for each sampling time and replicate (n=2).	23
Table 9: Degradation of parent and formation of its metabolites in the total Calwich Abbey Lake system under aerobic conditions after application of [¹⁴ C]Octocrilene (mean values of 2 replicates, values in % AR).	24
Table 10: Summary of the different DT50 for octocrilene and its degradation products in the whole system estimated with different kinetics models (Cake 3.7).	25
Table 11: Measured concentration of octocrilene (% AR) in the whole estuarine water system and the NER concentration for each sampling time and replicate (n=2).	27
Table 12: Degradation of parent and formation of its metabolites in the total system of Dyfi Estuary under aerobic conditions after application of [¹⁴ C]Octocrilene (mean values of 2 replicates, values in % AR).	28
Table 13: Summary of the different DT ₅₀ for octocrilene and its degradation products in the whole system estimated with different kinetics models (Cake 3.7).	29
Table 14: Studies on adsorption/desorption.	30
Table 15: Studies on volatilisation.	31
Table 16: Studies on aquatic bioaccumulation.	31
Table 17: Short-term effects on fish.	34
Table 18: Long-term effects of etocrilene on fish.	38
Table 19: Physico-chemical properties, environmental e-fate and behaviour of octocrilene and etocrilene (data from the registration dossier ⁹).	39
Table 20: Other studies available on fish.	40
Table 21: Short-term effects on aquatic invertebrates.....	41
Table 22: Long-term effects on aquatic invertebrates.....	46
Table 23: Effects on algae and aquatic plants.....	49
Table 24: Effects on sediment-dwelling organisms.	51
Table 25: Effects on other aquatic organisms.	57

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Table 26: Effects on soil macro-organisms.....	61
Table 27: Effects on microorganisms.....	66
Table 28: Physicochemical properties- Environmental fate and distribution properties -used for exposure estimation.....	69
Table 29: Dissipation rate of octocrilene in WWTP.....	70
Table 30: Emission scenario (ES) parameter values for each use and by registrant.	72
Table 31: total regional releases.....	78
Table 32: regional releases for uses of cosmetic products (industrial and wide disperse sources)	79
Table 33: Parameters used to estimate local PEC values after release to a municipal WWTP.	79
Table 34: Quantity of sunscreen lotion applied per day by a consumer ($Q_{\text{sunscreen_appl}}$) – applicable to swimming and shower scenarios	82
Table 35: Number of swimmers (N_{swimmer}) in function of the water body dimension ($V_{\text{waterbody}}$) – applicable to swimming scenario only	84
Table 36: Fraction of swimmers using a sunscreen product ($F_{\text{swim_sunscreen}}$) – applicable to swimming scenario only	85
Table 37: Fraction of sunscreen products containing octocrilene ($F_{\text{sunscreen_OC}}$) – applicable to swimming scenario only	86
Table 38: Fraction of octocrilene released after swimming ($F_{\text{after_swimming}}$) – applicable to swimming scenario only	87
Table 39: Number of emission days (T_{emission}) – applicable to swimming scenario only	88
Table 40: All remaining parameters	88
Table 41: results of wastewater monitoring studies in Europe	90
Table 42: results of sludge monitoring studies in Europe	102
Table 43: results of lake monitoring studies in Europe.....	107
Table 44: results of river monitoring studies in Europe	108
Table 45: results of marine water in Europe.....	115
Table 46: results of Sediment in freshwater environment in Europe	126
Table 47: results of Sediment in marine water environment in Europe	129
Table 48: Occurrence of Octocrilene in the aquatic biota in European countries	133
Table 49: Authorised UV filters with/without UV absorption and light stabilisation properties.	143
Table 50 (summary of table 23 of the main report): Authorised cosmetic UV filters under ongoing REACH process.....	146
Table 51: Physico-chemical properties of best suitable alternatives to octocrilene according to stakeholders consultation.	152

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Table 52: Potential alternatives to octocrilene and current market price.	156
Table 53: Herfindahl-Hirschmann Index of the relevant sun care markets under review. ..	163
Table 54: Number of expected reformulations per product category based on YUKA data	164
Table 55: Reformulation costs for major reformulation of sun protection products (cost per formulation in €2024).	166
Table 56: Reformulation costs for reformulation of 'other' cosmetic products (cost per formulation in €2024).	166
Table 57: Reformulation costs based on YUKA data for the three cost scenarios and two scenarios regarding the useful life of a formulation (2024 PV, 3% discount rate, 2 years transitional period).	167
Table 58: Reformulation cost in million euros based on YUKA data for two scenarios regarding the reformulation rate (2024 PV, 3% discount rate, mid-point cost).....	167
Table 59: Impact of the length of the transitional period on reformulation cost.	168
Table 60: Number of reformulations per year under the three alternative baseline scenarios.	169
Table 61: Total cost of the proposed restriction for the three alternative baseline scenarios in million euros (2024 PV, 3% discount rate).	170
Table 62: Reformulation costs based on CosmEthics data for three cost scenarios and two scenarios regarding the useful life of a formulation (2024 PV, 3% discount rate, 2 years transitional period).	170
Table 63: Values considered regarding the formulation cost and the increase in formulation cost in the minimum, mid-point and maximum cost scenarios.	171
Table 64: Raw material costs based on CosmEthics data for three cost scenarios and yearly extra cost (in million euros).	172
Table 65: Total costs of the restriction in million euros under the worst-case scenario (20 years PV 2024 at 3%)	172
Table 66: Attributes and attribute levels included in the survey	175
Table 67: Comparison of goodness of fit of linear and non-linear models.	183
Table 68: Mean WTP by country from individual WTP (Values are expressed in Euro PPP per household and per year for one level of improvement)	187
Table 69: Mean WTP by Country (Values are presented in Euro PPP per household and per year)	188
Table 70: characteristics of the surveyed samples.	210
Table 71: Characteristics of national populations.	211
Table 72: Survey perception by country (in %).	212
Table 73: Estimation results - Bulgaria.	212
Table 74: Estimation results – Germany.	213

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Table 75: Estimation results – Spain.	213
Table 76: Estimation results – France.	214
Table 77: Estimation results – Hungary.	215
Table 78: Estimation results – Sweden.	215
Table 79: Mean and Median WTP for Bulgaria from non-linear individual WTP.	216
Table 80: Mean and Median WTP for Germany from non-linear individual WTP.	216
Table 81: Mean and Median WTP for Hungary from individual WTP.	217
Table 82: Mean and Median WTP for Spain from non-linear individual WTP.	217
Table 83: Mean and Median WTP for France from non-linear individual WTP.	217
Table 84: Mean and Median WTP for Sweden from non-linear individual WTP.	218
Table 85: Determinants of WTP for coastal water improvement.	218
Table 86: Determinants of WTP for freshwater improvement.	220
Table 87: WTP estimates for a low improvement in coastal and freshwater ecosystems (Values are presented in Euro PPP per household and per year).	220
Table 88: WTP estimates for a medium improvement in coastal and freshwater ecosystems (Values are presented in Euro PPP per household and per year).	220
Table 89: WTP estimates for a high improvement in coastal and freshwater ecosystems (Values are presented in Euro PPP per household and per year).	221
Table 90: Determinants of WTP – Pooled model.	222
Table 91: Determinants of WTP – Bulgaria.	223
Table 92: Determinants of WTP – France.	224
Table 93: Determinants of WTP – Germany.	225
Table 94: Determinants of WTP – Hungary.	226
Table 95: Determinants of WTP – Spain.	227
Table 96: Determinants of WTP – Sweden.	228
Table 97: Mean willingness to pay to reduce the risks associated with the use of octocrilene in cosmetic products for freshwater and coastal water ecosystems – Results of the discrete choice experiment.	229
Table 98: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE (simple average - 6 countries, NS included).	230
Table 99: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE (simple average - 6 countries – NS as zero).	232
Table 100: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE Simple average – (NS excluded : average over 5 countries for coastal water and 4 for freshwater).	232
Table 101: Guide for stakeholders hearing.	234

FIGURES

Figure 1: Proportion of cosmetics products containing octocrilene per product category. (Source: YUKA, 2024).....	15
Figure 2: Pattern of degradation of [¹⁴ C] octocrilene in Calwich Abbey Lake system (% AR: Percentage of applied radioactivity).	21
Figure 3: Degradation kinetics of octocrilene in Calwich Abbey Lake system. (A) SFO model (χ^2 error: 7.5% ; t-test=1.09E-05) ; (B) Hockey-Stick (HS) model (χ^2 error: 3.17% ; t-test=4.9E-01) (CAKE 3.7)	22
Figure 4: Pattern of degradation of [¹⁴ C]Octocrilene (Parent) in the Dyfi Estuary system (% AR: Percentage of applied radioactivity).....	26
Figure 5: Degradation kinetics of octocrilene in the Dyfi Estuary system. (A) SFO model (χ^2 error: 10% ; t-test=8.2E-04) ; (B) Hockey-Stick (HS) model (χ^2 error: 8.6% ; t-test=0.5) (CAKE 3.7).....	27
Figure 6: Mapping of the octocrilene's life cycle with the declared tonnages.....	71
Figure 7: Conceptual release pathway of octocrilene to the environment through the use of cosmetic products (releases in tonnes per year determined through a tonnage approach).	81
Figure 8: Illustration of potential adverse effect on DNA from UV radiation (from <i>Tang et al., 2024</i>).....	139
Figure 9: Illustration of potential adverse effect on Human Health from UV radiation (from <i>Tang et al., 2024</i>).	140
Figure 10: Illustration of nanocarriers used in sunscreen formulations (from Santos et al., 2022).	154
Figure 11: Advantages and disadvantages of using nanocarriers in cosmetics formulation (adapted from Chavda et al., 2023).....	155
Figure 12: Cosmetic Value Chain.....	159
Figure 13: Share of retail channels in the distribution of sun care products in Western Europe.....	161
Figure 14: Evolution of markets' size over time (in litres) over 2009-2023.	162
Figure 15: Impact of the length of the transitional period on reformulation cost.	168
Figure 16: Comparison of reformulation costs calculated using YUKA vs CosmEthics data for three cost scenarios (2024 NPV, 3% discount rate, 2 years transitional periods and partial reformulation).....	171
Figure 17: Example of a choice card	179
Figure 18: Share of improvement achieved under the three benefits assessment scenarios (under RO1).	231

Appendix A: Manufacture and uses

A.1. Manufacture, import and export

Octocrilene is an organic molecule characterized by an aromatic structure and is manufactured through a condensation reaction between cyanoacetic acid ethylhexyl ester with benzophenone imine (Knoevenagel reaction) in a closed vessel to form crude octocrilene. First distillation fraction is recycled and used in the next reaction batch. The second fraction consists of benzophenone and octocrilene and is recycled into the next distillation step. The main fraction of the distillation is dissolved in organic solvent and washed. Finally, unreacted starting materials and by-products are removed by a final distillation step. The purified octocrilene is then isolated as bottom product and collected in a closed vessel.

A.2. Uses

There are two main uses for octocrilene: one as UV filter like in cosmetic products and the other as UV stabilisers like in plastisol for the production of plastic films.

A plastisol is a colloidal dispersion of small polymer particles, usually polyvinyl chloride (PVC), in a liquid plasticizer. When heated to around 180 °C (356 °F), the plastic particles absorb the plasticizer, causing them to swell and fuse together forming a viscous gel. Once this is cooled to below 60 °C (140 °F) it becomes a flexible, permanently plasticized solid product. This process is called 'curing'. Octocrilene is a liquid UV stabilizer and the technical function within a plastisol is the absorption of UV light. The recommended concentration of octocrilene within plastisol is up to 30% in masterbatches and up to 0.8% in plastic articles. Photo-oxidation (and sometimes, oxidative photodegradation) is the degradation of a polymer surface due to the combined action of UV light and oxygen. It is the most significant factor in the weathering of plastic applications derived from plastisols. Photo-oxidation causes the polymer chains to break, resulting in the material becoming increasingly brittle. This leads to optical defects and mechanical failure (CfE#1752).

A.2.1. Uses of octocrilene in cosmetic products

The light emitted by the sun, called solar light, correspond to electromagnetic radiation. Ultraviolet (UV) radiation are part of the electromagnetic radiations emitted by the sun and is subdivided into UV-C (100–290 nm), UV-B (290–320 nm), UV-A1 (320–340 nm) and UV-A2 (340–400 nm). UV radiations are able to generate potential benefit for human health but also deleterious effects for the human health (see annex E.2.). To avoid these deleterious effects, it is necessary to protect the body from the solar radiations.

There are several possibilities to protect the skin from the UV radiation, and as clearly stated by the Center for Disease and Prevention (CDC), sun safety needs to be practice¹. These practices advices are derivated from the International Agency for Research on Cancer (IARC) that clearly state that *“Protection of the skin from solar damage ideally involves a number of actions which include wearing tightly woven protective clothing that adequately covers the arms, trunk and legs, a hat that provides adequate shade to the whole of the head, seeking shade whenever possible, avoiding outdoor activities during*

¹[https://www.cdc.gov/skin-cancer/prevention/index.html#:~:text=Wear%20clothing%20that%20covers%20your,SPF\)%20of%2015%20or%20higher](https://www.cdc.gov/skin-cancer/prevention/index.html#:~:text=Wear%20clothing%20that%20covers%20your,SPF)%20of%2015%20or%20higher)

periods of peak insolation and use of sunscreens.” Another solution, as evoked by the IARC, is to use sunscreens, which are cosmetic products.

Cosmetic regulation defines a cosmetic products as ‘*any substance or mixture intended to be placed in contact with the external parts of the human body (epidermis, hair system, nails, lips and external genital organs) or with the teeth and the mucous membranes of the oral cavity with a view exclusively or mainly to cleaning them, perfuming them, changing their appearance, protecting them, keeping them in good condition or correcting body odours*’. Specifically, a sunscreen is a skin product that is defined in this regulation as a “cosmetic product which is intended to be applied on the skin”; and a ‘UV-filters’ means substances which are exclusively or mainly intended to protect the skin against certain UV radiation by absorbing, reflecting or scattering UV radiation.

A sunscreen is a multicomponent product that contains at least one UV filter, but frequently a mixture of them, and additional substances like surfactants, preservatives, and stabilizers. Sunscreens are categorised into two main types - physical or mineral and chemical - based on the type of filters present in their formulations. The term “physical filter” can be misleading because mineral filters are chemicals. Mineral filters in sunscreens are zinc oxide and/or titanium dioxide as active protective components. Chemical sunscreens typically include a blend of UV filters to covert to the maximum extent the full spectra of UV radiation. Upon application, these chemical UV filters are absorbed into the skin. When the skin is subsequently exposed to UV radiation, these chemical active components absorb UV radiation and transform it into thermal energy (heat). However, this process may generate free radicals as a by-product, which can be detrimental on skin (Ambruso *et al.*, 2023). To avoid and encounter the potential adverse effects of these by-products, sunscreens formulations are often complemented with antioxidants such as vitamins C and E.

UV filters are characterized primarily by the value of their absorption peak, lambda max (λ_{max}), where the substance absorbs the most of the wavelength of light and the value of the molar absorptivity coefficient (ξ) at their λ_{max} . These parameters are directly influenced by the UV filter’s chemical structure and determine the range of UV protection.

A.2.1.1. Presence of octocrilene in cosmetics

Table 1: Proportion of cosmetics products containing octocrilene.

Cosmetic	Product category	Proportion of octocrilene
Sunscreen	Tanning	36.74%
Self-tanner	Tanning	1.66%
After sun	Tanning	0.82%
Lip balm with sunscreen	Tanning	30.20%
Moisturisers/Face cream	Facial care	3.91%
Anti-age cream	Facial care	6.19%
Lip balm	Facial care	3.33%
Serums and treatments	Facial care	1.61%
Toners and mists	Facial care	0.13%
Eye moisturiser	Facial care	0.49%
Masks	Facial care	0.14%
Cleansers	Facial care	0.14%
Lotion	Baby and Children's Products	0.74%

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Other baby products	Baby and Children's Products	0.27%
Nail polish	Hands and Nails	2.71%
Creams and lotions	Hands and Nails	0.93%
Nail polish remover	Hands and Nails	1.18%
Foundation/BB Cream	Make up	2.62%
Tinted lip balm	Make up	1.66%
Pressed powder	Make up	0.19%
Lipstick	Make up	0.30%
Concealer	Make up	0.19%
Lip gloss	Make up	0.20%
Body lotion/Balm/Cream/Gel	Bath and Body Products	0.49%
Body oil	Bath and Body Products	0.36%
Other	Bath and Body Products	0.08%
Shower gel	Bath and Body Products	0.05%
Serum/oil	Hair care	0.90%
Treatments	Hair care	0.13%
Masks	Hair care	0.44%
Styling cream	Hair care	0.18%
Hair spray	Hair care	0.20%
Hair color	Hair care	0.10%
Shampoo	Hair care	0.09%
Conditioner	Hair care	0.19%
Facial care	Male grooming	0.29%
After shave	Male grooming	0.21%
Perfume/Parfum/Eau de Parfum	Fragrances	1.29%
Others	Others	0.11%
Grand total		2.31%

(source: YUKA, 2024)

Table 2: Proportion of cosmetics products containing octocrilene.

Cosmetic	Product category	Proportion of octocrilene
Sunscreen	Tanning	62.14%
Sunscreen	Baby and Children's Products	48.02%
Sunscreen	Facial care	40.19%
Lip balm	Male grooming	33.33%
Facial moisturizers	Male grooming	9.55%
Anti-age cream	Facial care	8.33%
After sun moisturiser	Facial care	6.52%
Self-tanner	Tanning	6.40%
Moisturisers/Face cream	Facial care	6.36%
Nail polish	Hands and Nails	5.83%
After sun gel	Facial care	4.17%

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Cosmetic	Product category	Proportion of octocrilene
Lip balm	Facial care	4.06%
Tinted lip balm	Make up	3.32%
Foundation/BB Cream	Make up	3.23%
Other nail or cuticle products	Hands and Nails	2.67%
Self tanner face	Facial care	2.59%
Facial care	Male grooming	2.58%
Serum/oil	Hair care	2.34%
Creams and lotions	Male grooming	2.13%
Scalp Care	Hair care	1.69%
Creams and lotions	Hands and Nails	1.45%
After sun	Tanning	1.15%
Other baby products	Baby and Children's Products	1.12%
Lotion	Baby and Children's Products	1.04%
Serums and treatments	Facial care	0.91%
Eye moisturiser	Facial care	0.86%
Massage oil	Bath and Body Products	0.79%
Treatments	Hair care	0.75%
Nail polish remover	Hands and Nails	0.73%
Body oil	Bath and Body Products	0.69%
Body lotion/Balm/Cream/Gel	Bath and Body Products	0.66%
Toners and mists	Facial care	0.64%
Styling cream	Hair care	0.62%
Highlighter	Make up	0.50%
Masks	Hair care	0.48%
Perfume/Parfum/Eau de Parfum	Fragrances	0.44%
Pressed powder	Make up	0.29%
After shave	Male grooming	0.26%
Concealer	Make up	0.24%
Dry shampoo	Hair care	0.19%
Hair gel	Hair care	0.19%
Hair spray	Hair care	0.17%
Hand wash	Hands and Nails	0.16%
Lip gloss	Make up	0.14%
Blush/Bronzer/Contour	Make up	0.13%
Lipstick	Make up	0.11%
Masks	Facial care	0.10%
Exfoliators/Body scrub	Bath and Body Products	0.08%

Cosmetic	Product category	Proportion of octocrilene
Hair color	Hair care	0.07%
Cleansers	Facial care	0.02%
Grand Total		3.87%

(source: CosmEthics, 2023)

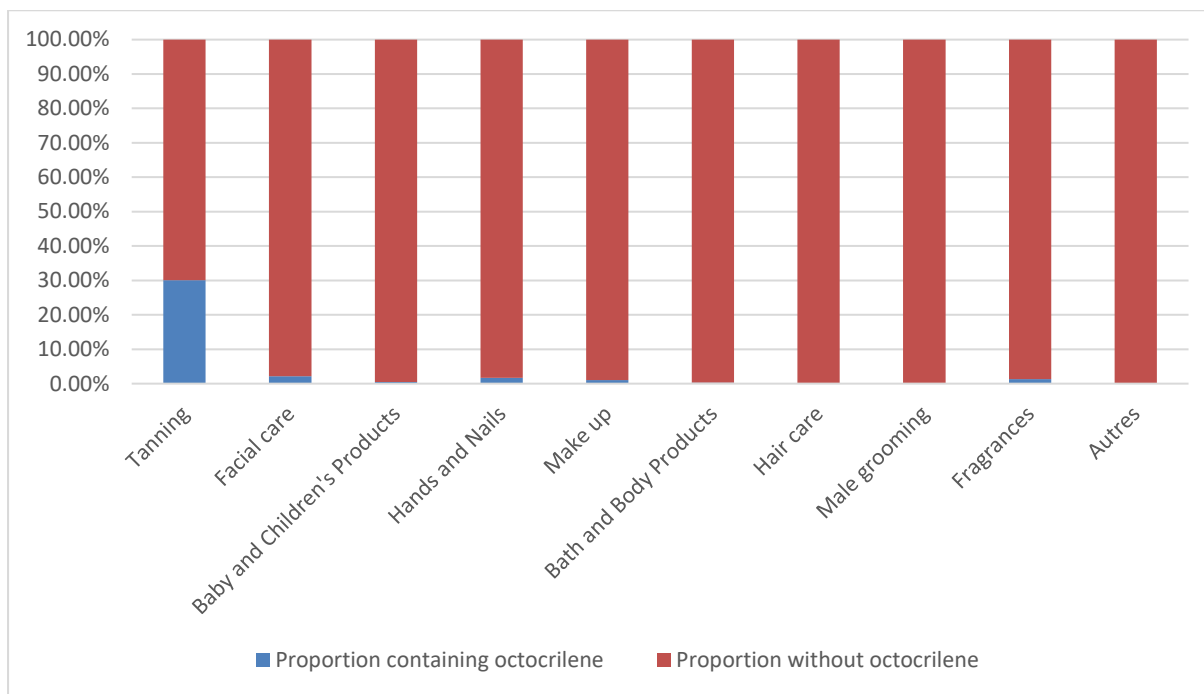


Figure 1: Proportion of cosmetics products containing octocrilene per product category. (Source: YUKA, 2024)

A.2.1.2. Survey of prices and composition of sun protection products in five European countries

A.2.1.2.1. Data collection method

This data collection process aimed at estimating **the share of cosmetic products that contain octocrilene on the European Economic Area (EEA) market** and cross-referencing the latter with the information derived from the CosmEthics and YUKA database. Due to time constraints, the data collection focused on primary sunscreens and took place online for five EEA countries only. This focus on primary sunscreens is relevant given the relative volumes of octocrilene used to formulate these products. Indeed, according to YUKA data primary sunscreens represent 55% of cosmetics products (in number of formulations) containing octocrilene (see Annex E.4.2). Besides, according to Cosmetics Europe in 2019 more than 80% of octocrilene (by weight) was used to formulate primary sunscreens (Cosmetics Europe, personal communication).

EEA countries considered for this data collection were selected while aiming to capture the heterogeneity of both preferences and use of sun protection within the EEA. In an attempt to do so, the Dossier Submitter selected five countries representing distinct geographical zones of the EEA:

- Finland (Northern Europe)
- France (Western Europe)

- Germany (Central Europe)
- Ireland (Western Europe)
- Poland (Eastern Europe)

Due to time constraints, it was not possible to collect data from a Southern European country.

The Dossier Submitter collected the data from the websites listed in Table 3 below. The latter were selected to reflect actual purchasing behaviour as closely as possible. According to an Emergen Research report, “the global sun protection is segmented into supermarket and hypermarket, online, pharmacy and drugstore, specialty store and others”. Besides, according to Euromonitor², inhabitants of Western Europe consume mainly in non-grocery stores such as health and beauty specialist (pharmacy and parapharmacy) and in supermarkets/hypermarkets. Based on this information, the Dossier Submitter choose to consider the following sales channels: pharmacies/para-pharmacies, supermarkets/hypermarkets and online stores. For each ‘physical’ distribution channel, data were collected on an online equivalent. For each distribution channel, the brand/website with the largest market share and/or the highest number of visits was selected (depending on available data).

Table 3: Selected online retailers for sunscreen protection in 5 European countries.

	Pharmacy	Supermarket/hypermarket	Online store
Finland	yliopistonapteekki.fi	www.s-kaupat.fi	lyko.com/fi
France	pharma-gdd.com	e.leclerc	www.nocibe.fr
Germany	shop-apotheke.com	edeka24.de	dm.de
Ireland	inishpharmacy.com	dunnesstoresgrocery.com	boots.ie
Poland	doz.pl	rossmann.pl	Hairstore.pl

A.2.1.2.2. Results

Table 4: Share of primary sunscreens containing octocrilene in five EEA countries.

	Finland	Poland	Ireland	Germany	France
Tanning	17.3%	44.3%	50.6%	24.4%	35.2%
Baby and children’s product	0%	16.7%	30.3%	13%	10.6%
Facial care	29.5%	18.9%	29.9%	19.9%	24.1%
Total	20.4%	27.4%	42.5%	20.7%	29.1%

² Euromonitor database (Passport) is a database that contains information on the consumption of different categories of sun care products in terms of value and volume between 2009 and 2028 in different countries of the EEA. It should be noted, first, that it does not provide any information on the other sun care products available (but not purchased) to consumers during their shopping trips. Thus, the choice set of each consumer is not available. Second, the data accessed and used in this section do not cover the whole EEA and cover some countries outside the EEA (Switzerland, Turkey and UK).

A.3. Consumer preferences regarding sun protection products

According to industrial stakeholders, consumers attach great importance to the sensory performance of sun protection products (i.e., transparent, non-greasy sunscreens that blend well), and poor performance in these respects can lead consumers to stop using the product. This information is supported by the international literature examining consumer preferences for sunscreen attributes showing that the sensory aspects are highlighted as decisive factors when purchasing a product (*Xu et al., 2016; Tong et al., 2020; Vergilio and Rocha Filho, 2020; Harlington et al., 2022*). However, these studies are based on non-European samples. Regarding consumer preferences in the EEA, the Dossier Submitter was able to identify national surveys supporting the importance of the sensoriality of sun protection products (see Table 5): being difficult to apply, leaving an unpleasant feeling on the skin or traces on the body are designated as leading reasons to stop using a sun protection product (*Statista, 2024*).

However, the presence of natural ingredients, the absence of chemical filters and the sustainability of the products is also valued by consumers (albeit heterogeneously across countries). The latter are designated as the most important criterion when buying a sun protection product in Italy for 36%, 22% and 16% of respondents respectively; in France for 6%, 3% and 4% of respondents respectively and in Spain 31%, 18% and 14% of respondents respectively (*Statista, 2024*).

Table 5: Leading reasons to stop using a sun protection product.

	Italy	France	Spain
Difficult to apply	1 (38%)	1 (37%)	2 (30%)
Leaves traces on clothes/swimsuit	2 (31%)	7 (27%)	6 (24%)
Unpleasant scent	3 (30%)	2 (31%)	5 (24%)
Skin reactions (rashes, itchiness, allergy, etc.)	4 (30%)	6 (28%)	4 (27%)
Unpleasant feeling on the skin	5 (27%)	3 (31%)	1 (40%)
Leaves traces on the body	6 (27%)	4 (29%)	8 (14%)
Value for money	7 (25%)	5 (28%)	3 (29%)
Packaging makes it difficult to dispense the product	8 (20%)	/	7 (20%)
Non-environmentally friendly	/	8 (20%)	/
Results obtained from national YouGov survey conducted in May 2022 (Spain – 728 respondents, Italy – 621 respondents) and June 2022 (France – 541 respondents). No information on representativeness was provided.			

(Source: Statista 2024)

A.4. Uses advised against by the registrants

There are no uses advised against in the registrations of the substance.

Appendix B: Information on hazard and risk

B.1. Identity of the substance(s) and physical and chemical properties

See Annex XV report.

B.2. Manufacture and uses (summary)

See Appendix A.

B.3. Classification and labelling

B.3.1. Classification and labelling in Annex VI of Regulation (EC) No 1272/2008 (CLP Regulation)

There is no harmonised classification for octocrilene under CLP.

B.3.2. Classification and labelling in classification and labelling inventory/ Industry's self classification(s) and labelling

See Annex XV report.

B.4. Environmental fate properties

B.4.1. Degradation

B.4.1.1. Abiotic degradation

B.4.1.1.1. Hydrolysis

According to the results from Hydrowin v2.00, the substance has an estimated half-life at pH 7 and 25°C of about 206 years. Therefore, the substance is considered as slowly hydrolysing in water.

B.4.1.1.2. Phototransformation/photolysis

- Phototransformation in air

According to the AOPWIN v1.92 model, octocrilene is considered to have a half-life of 6.19 hours in the atmosphere (within the following conditions: $1.5E+06$ OH/ cm^3). It is therefore not considered to be persistent in air based on this estimated rapid photodegradation potential.

- Phototransformation in water

No relevant information available.

- Phototransformation in soil

No relevant information available.

B.4.1.2. Biodegradation

B.4.1.2.1. Biodegradation in water

B.4.1.2.1.1. Screening tests

The studies on biodegradation in water (screening tests) are summarised in the Table 6.

Table 6: Screening tests for biodegradation in water

Method	Results	Remarks	Reference
Test type: ready biodegradability activated sludge, non-adapted according to EU Method C.4-D (Determination of the "Ready" Biodegradability – Manometric Respirometry Test)	no biodegradation % Degradation of test substance: 0 after 28d (O ₂ consumption)	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene	Unpublished study report (1995a)
Test type: ready biodegradability activated sludge, domestic, non-adapted according to OECD Guideline 301 F (Ready Biodegradability: Manometric Respirometry Test)	Poorly biodegradable % Degradation of test substance: 0 - 10 after 28d (O ₂ consumption)	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene	Unpublished study report (1991a)

The ready biodegradability of octocrilene was tested in two Manometric Respirometry tests according to EU Method C.4-D and OECD TG 301F. Both tests used non-adapted activated sludge as inoculum. At termination of the test after 28 days of inoculation, the detected degradation rate was in the range of 0 – 10 % (*Unpublished study report (1995a)*; *Unpublished study report (1991a)*). Thus, octocrilene is not readily biodegradable according to OECD criteria. The substance is considered as poorly biodegradable.

B.4.1.2.1.2. Simulation tests (water and sediments)

Two new studies were provided and investigated the rate and route of degradation of [¹⁴C]Octocrilene in water-sediment systems under aerobic conditions:

- Octocrilene - Rate and Route of Degradation of [¹⁴C]Octocrilene in An Aerobic Freshwater Sediment System (*Unpublished study report (2024a)*);
- Octocrilene - Rate and Route of Degradation of [¹⁴C]Octocrilene in An Aerobic Marine Water Sediment System (*Unpublished study report (2024b)*) ; Note that in this study, the water and sediment were sampled in an estuarine rather than in a marine location.

The methods and the results are summarized in Table 7.

Table 7: Simulation tests for biodegradation in water and sediments (AR: Applied Radioactivity).

Method	Results	Remarks	Reference
Test type: simulation testing on degradation in aquatic sediment	Total mean recovery: 96.7% AR Recovery at day 100:	2 (reliable with	Unpublished study report

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
system Test system: natural water/sediment freshwater (Fw) (aerobic) Site: Calwich Abbey Lake (UK) Silt loam sediment (fine texture): Clay 5% + silt 74% (+sand 21%); C_{org}=4,2%. Observations: 0, 7, 14, 30, 64 and 100 days Temperature: 12.3 °C, dark Solvent: acetonitrile (<1%) Replicate: two per treatments according to OECD Guideline 308 (Aerobic and Anaerobic Transformation in Aquatic Sediment Systems)	Water: 37.2% AR Sediment: 50.9% AR NER: 5.4% AR Mineralization: 3.3% AR <u>Degradation time (days, total system):</u> <u>Parent compound:</u> DT₅₀, whole system (Fw) = 1810 d (SFO, mineralization data); <u>Relevant Metabolites (M):</u> M1 [2-ethylhexyl 2-cyano-3,3-diphenyl-propanoate]: max 8.6% AR day 30; DT₅₀ = 14.8 d; M2 [2-cyano-3,3-diphenylpropenoic acid]: max 18.2% AR day 64; Degradation time not defined; M3 [2-cyano-3,3-diphenyl-propanoic acid]: max 31.1% AR at d64; Degradation time not defined; M6: max 5.6% AR at d100; Degradation time not defined	restriction) key study experimental result Test material (EC name): octocrilene [Phenyl-U-¹⁴C]-Octocrylen purity: 99.9%	(2024a)
Test type: simulation testing on degradation in aquatic sediment system Test system: natural water/sediment estuarine water (Ew) (aerobic) Site: Dyfi Estuary, Ceredigion (Wales) Sand sediment (coarse texture): Clay 1% + silt 2% (+sand 97%); C_{org}=0,11% Observations: 0, 7, 14, 30, 64 and 100 days Temperature: 12.3 °C days Solvent: acetonitrile (<1%) Replicate: two per treatments according to OECD Guideline 308 (Aerobic and Anaerobic Transformation in Aquatic Sediment Systems)	Total mean recovery: 96.6% AR Recovery at day 100: Water: 43.8% AR Sediment: 41.8% AR NER: 5.2% AR Mineralization: 3.2% AR <u>Degradation time (days, total system):</u> <u>Parent compound:</u> DT₅₀, whole system (Ew) = 1870 d (SFO mineralization data) <u>Relevant Metabolites (M):</u> M1 [2-ethylhexyl 2-cyano-3,3-diphenyl-propanoate]: max 5.9% AR day 30; DT₅₀ = 16 d. M2 [2-cyano-3,3-diphenylpropenoic acid]: max 18.9% AR day 64 max; Degradation time not defined M3 [2-cyano-3,3-diphenyl-propanoic acid]: max 30.2% AR at d100; Degradation time not defined	2 (reliable with restriction) key study experimental result Test material (EC name): octocrilene [Phenyl-U-¹⁴C]-Octocrylen purity: 99.9%	Unpublished study report (2024b)

After two weeks acclimation period in glass flasks under exactly the same conditions as the test incubation (12.3°C, dark, ventilation), the systems were treated with 54 µg/L ^{14}C -labelled octocrilene solution applied into the water phase. Two samples of treated water and sediment were taken for extraction and analysis immediately after treatment (day 0) and after 7, 14, 30, 64 and 100 days of incubation. Physico-chemical parameters (pH, oxygen concentration and redox potential) and microbial biomass were monitored during incubation in control samples treated in duplicate with acetonitrile or unlabelled test item. The degradation rates were calculated using a step-wise approach according to the FOCUS Guidance (2014) on estimating persistence and degradation kinetics from Environmental Fate Studies. Evaluations were performed with the software tool CAKE (version 3.7).

These two studies were performed under test conditions that deviated with the OECD Testing Guideline 308:

- One sediment type was used in the freshwater-sediment system (fine texture/high organic carbon content) and one another sediment type was used in the estuarine-sediment system (coarse texture/low organic carbon content sediment, while at least two sediment types and their associated waters are required for each of the two water-sediment systems;
- It is not clear if duplicates samples are true, independent replicates or rather replicate analytical results from a single sample;
- High variability between duplicates analysis is noted in both systems.

Both studies are considered reliable with restriction. The deviations are discussed below and the choice of DT_{50} justified accordingly.

B.4.1.2.1.3. Freshwater-sediment system (Unpublished study report (2024a)):

B.4.1.2.1.4. Octocrilene

The distribution of the radioactivity in the freshwater system treated with [^{14}C]Octocrilene is presented Figure 2:

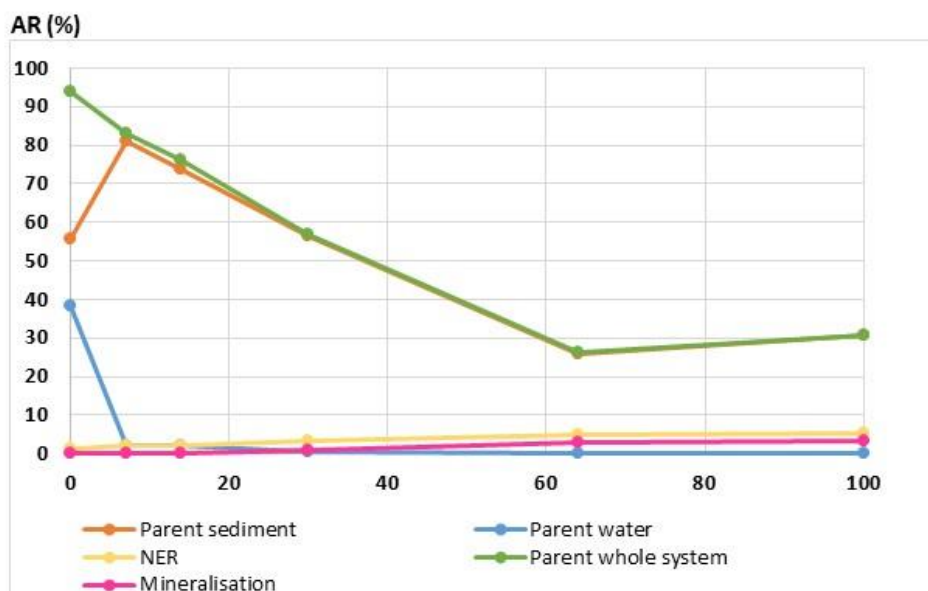


Figure 2: Pattern of degradation of [^{14}C] octocrilene in Calwich Abbey Lake system (% AR: Percentage of applied radioactivity).

The total mean recovery was $96.7 \pm 1.3\%$ of applied radioactivity (AR). Mineralisation to $^{14}\text{CO}_2$ was low throughout the study, with a maximum mean amount of 3.3% AR at the

end of incubation period. The recovered amount of organic volatiles was $\leq 0.1\%$ AR at any sampling interval. For the sediment phase, the amount of non-extractable radioactivity (bound residues, NER) increased from 1.2% AR (day 0) to 5.4% AR (day 100).

In the whole system, the mean amount of [^{14}C]Octocrilene decreased from 94% AR (day 0) to 31% AR (day 100). The dissipation of octocrilene from aqueous to sediment phase was observed during the first 7 days of the incubation ($\text{Log } K_{oc}=4.69$). In the aqueous phase, octocrilene decreased rapidly from 38% AR at day 0 to 2% AR at day 7 and then was undetected until the test end. In the sediment extracts, the radioactivity increased from 56% AR at day 0 to 82% AR at day 7, then continuously decreased up to 31% AR at day 100. For the further parent kinetics evaluation, the whole system has been considered.

The rate of degradation of octocrilene was estimated with Cake 3.7. The **single first order (SFO) kinetics model** is generally the preferred model for estimating DT_{50} according to the FOCUS guidance (2014). This model was used by the registrant resulting in **$\text{DT}_{50}=45.5$ days** (χ^2 error: 7.5%; t-test= $1.09\text{E-}05$). Despite the kinetic of decline clearly shows a fast initial decrease in test substance concentration followed by a slower decline phase (up to 30.9% AR at day 100), other models such as bi-phasic kinetic ones have not been tested by the registrant. This was done here, as presented in the Figure 3. Results showed that the **bi-phasic Hockey-Stick (HS) model** fits the data better than SFO kinetics, resulting in the half-life **$\text{DT}_{50}=7030$ days** (slow phase) with a better χ^2 value (3.17%) than for SFO (7.5%). However, the degradation rate constant approaches zero in the second phase (t-test $p>10\%$) that could be due to a high variability observed between the two replicates at the two last sampling time. Then, the HS estimation of DT_{50} should be interpreted with caution.

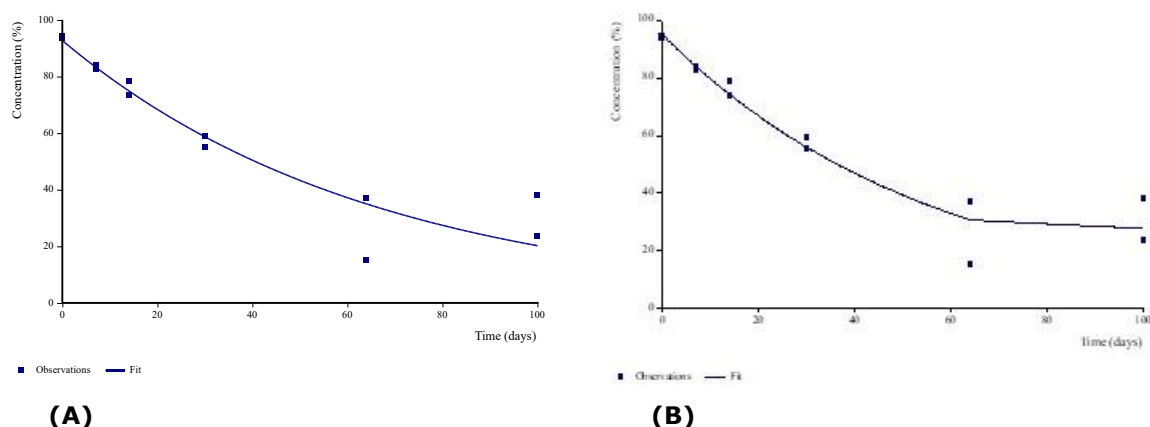


Figure 3: Degradation kinetics of octocrilene in Calwich Abbey Lake system. (A) SFO model (χ^2 error: 7.5% ; t-test= $1.09\text{E-}05$) ; (B) Hockey-Stick (HS) model (χ^2 error: 3.17% ; t-test= $4.9\text{E-}01$) (CAKE 3.7)

Deviations from first-order kinetics cannot be attributed to experimental artefacts. We wondered about the first replicate low value at day 64 (15% AR) as a possible outlier. The Table 8 shows the percentages of the applied radioactivity for each replicate and sampling times in the whole system (water+sediment):

Table 8: Measured concentration of octocrilene (% AR) in the whole freshwater system and the NER concentration for each sampling time and replicate (n=2).

Time (days)	Parent _{whole syst} (NER) % AR
	Freshwater
0	93.6 (1.2)
0	94.2 (1.2)
7	84.0 (2.2)
7	82.5 (2.4)
14	78.5 (2.2)
14	73.6 (2.1)
30	55.2 (3.1)
30	59.0 (3.3)
64	15.3 (6.4)
64	36.9 (3.7)
100	38.1 (3.6)
100	23.7 (7.3)

A deviation from the trend of the values is noted at day 64: in the first replicate the concentration of octocrilene was about half that of the second replicate while it has twice as many NER as the second replicate. We used the Grubbs' test to quantify the outliers. This statistical test confirmed that the 15.3% AR value is furthest from the rest, but it is not a significant outlier ($p > 0.05$). Note that only two points per replicate are available reducing the statistical robustness. No experimental explanation can be provided to explain this deviation at day 64. Consequently, this value cannot be excluded from the data set.

Results from this study also showed a very low mineralization rate of octocrilene in the whole system. Although not proposed by the registrant, the half-life degradation time estimation based on mineralization data has been performed and resulted in **DT_{50,mineralization} = 1810 days³** (SFO model ; χ^2 error: 0.23 % ; t-test = 2.07E-05). Uncertainties related to the persistence of some degradation products have been underlined as explained in the next section. Consequently, the DT₅₀ based on mineralization data rather than that of the registrant was considered as the most conservative half-life for the parent.

B.4.1.2.1.5. Degradation products

Four relevant degradation products (out of nine detected) were observed (Table 9). **M2** and **M3** were detected at amounts $\geq 10\%$ AR with a maximum of 18% AR and 31% AR respectively at day 64. **M1** and **M6** were detected at amounts $> 5\%$ AR with a maximum of 9% AR for M1 day 30 and a constant total amounts of around 5% AR for the last three incubation intervals for M6. Another metabolite M16 was detected with increasing amounts of radioactivity towards the end of the incubation period but remained at irrelevant concentration (aqueous phase only, up to 0.7% AR on day 100). Four additional metabolites (M5, M7, M8 and M14) were detected with maximum total amounts of 0.5% (day 7), 1.5% (day 64), 0.6% (day 30) and 0.3% AR (day 30), respectively. All four

³ Calculation: Parent after mineralization (%AR) = 100% - %AR ¹⁴CO₂

metabolites (M5, M7, M8 and M14) showed a decreasing degradation pattern towards the end of the incubation period or were below the LOD.

Table 9: Degradation of parent and formation of its metabolites in the total Calwich Abbey Lake system under aerobic conditions after application of [¹⁴C]Octocrilene (mean values of 2 replicates, values in % AR).

Time (days)	Parent	M1	M2	M3	M5	M6	M7	M8	M14	M16
0	93.9	0.2	0.2	0.2	0.1	0.2	0.2	0.2	0.0	0.0
7	83.3	4.2	2.4	3.0	0.5	1.9	0.0	0.0	0.0	0.0
14	76.1	6.9	3.5	4.9	0.3	1.2	0.3	0.0	0.0	0.0
30	57.1	8.6	8.7	13.2	0.1	5.1	0.3	0.6	0.3	0.3
64	26.1	7.3	18.2	31.1	0.0	4.6	1.6	0.0	0.0	0.5
100	30.9	4.7	16.1	30.1	0.0	5.6	0.0	0.0	0.2	0.7

The determination of the metabolite degradation rate was possible for **M1 only**, with decreasing concentration towards the end of the test. Then, according to the registrant, M1 degraded in the total system with a **DT₅₀ of 13.7 days** (χ^2 error: 1.9%; t-test=1.56E-06) based on the SFO as the best-fit kinetic model for the parent. Considering the HS bi-phasic model for the parent (see the previous section), a similar DT₅₀ for M1 was estimated to be **14.8 days** (χ^2 error: 2.9%; t-test=1.50E-03).

The radioactivity measured for **M2, M3 and M6** increased at the three last sampling time preventing an estimation of their DT₅₀ (Default value **DT₅₀ > 1000 days**).

B.4.1.2.1.6. Conclusions for the Freshwater system

This OECD TG 308 study presented deficiencies. Only one sediment was used instead of two recommended. It is not clear if duplicates samples are true, independent replicates or rather replicate analytical results from a single sample; A high variability was observed between duplicates at the two last sampling time without justification. One questionable value was noted but without experimental justification and was not considered as a relevant outlier in the statistical test (runned with only two values). Consequently, excluding this value from the dataset to improved the model-fits and determine the most relevant DT₅₀ was not possible. The model considered to best fitted the experimental data for octocrilene degradation was the bi-phasic Hockey-Stick (HS) model (rather than the SFO model) leading to a DT₅₀=7030 days (slow phase), however the t-test >10% (absence of degradation at the test end) lowered the confidence of the estimated parameter and this (HS) DT₅₀ should be interpreted with caution.

The degradation kinetics based on the mineralization of octocrilene in the whole system resulted in an estimated **DT₅₀, mineralization=1810 days**. Regarding the degradation products data, the DT₅₀ determination was possible only for M1. The DT₅₀ of M2, M3 and M6 were undefined caused by their accumulation and slow degradation until the end of the test. Consequently, the DT₅₀ based on octocrilene mineralization in the single tested sediment was considered as the most conservative value covering the potential P/vP concern of its degradation products.

The uncertainties for metabolites degradation was also mentioned in the registrant dossier **"Further information is necessary to conclude on the P or vP properties in the context of the PBT assessment. All as relevant identified metabolites may fulfill**

the screening criteria for P/vP. Therefore, further information may be considered necessary for the assessment of P/vP criteria."

The different discussed DT₅₀ are summarised in the Table 10.

Table 10: Summary of the different DT₅₀ for octocrilene and its degradation products in the whole system estimated with different kinetics models (Cake 3.7).

Kinetic models	Chi ²	DT _{50, whole system} days	Prob. > t
Parent			
SFO (IES, 2024a)	7,5%	45	1,09E-05
Hockey-Stick (HS)	3,2%	7030*	4,90E-01
SFO (Mineralisation)	0,2%	1810	2,07E-05
M1			
SFO (IES, 2024a)	1,9%	13,7	1,56E-06
SFO (HS Parent)	2,9%	14,8	1,50E-03

**k₂DT₅₀ slow phase*

Based on the available data (freshwater environment), we consider the following DT₅₀ for the persistence criteria:

Octocrilene: DT_{50, whole system} (Fw) = 1810 days

M1: DT_{50, whole system} (Fw) = 14.8 days

M2, M3, M6: DT_{50, whole system} (Fw) > 1000 days (Default value)

Octocrilene fullfills the P/vP criteria (DT₅₀> 180 days)) based on the half-life of degradation products.

B.4.1.2.1.7. Estuarine water sediment system (*Unpublished study report (2024b)*):

B.4.1.2.1.8. Octocrilene

The distribution of the radioactivity in the estuarine system treated with [^{14}C]Octocrilene is presented Figure 4:

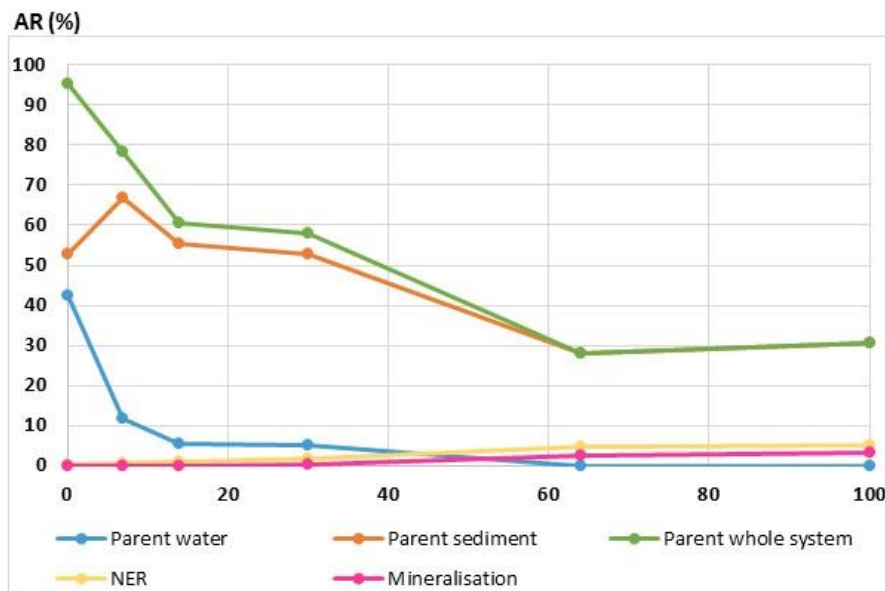


Figure 4: Pattern of degradation of [^{14}C]Octocrilene (Parent) in the Dyfi Estuary system (% AR: Percentage of applied radioactivity).

The total mean recoveries were $96.6 \pm 1.4\%$ applied radioactivity (AR). Mineralisation to $^{14}\text{CO}_2$ was low throughout the study, with a maximum mean amount of 3.2% AR at the end of incubation period. The recovered amount of organic volatiles was low with a maximum of 0.9 % AR at end of study. For the sediment phase, the amount of non-extractable radioactivity (bound residues, NER) increased from 0.3% AR (day 0) to 5.2% AR (day 100).

In the whole system, the mean amount of [^{14}C]Octocrilene decreased from 95% AR (day 0) to 31% AR (day 100). The increase and decrease of extracted radioactivity from the sediment phase can be explained by the dissipation of octocrilene from aqueous to sediment phase ($\text{Log } K_{oc} = 4.69$). For the further parent kinetics evaluation, the whole system has been considered.

The rate of degradation of octocylene was estimated with Cake 3.7. The **single first order (SFO) kinetics model** is generally the preferred model for estimating DT_{50} according to the FOCUS guidance (2014). This model was used by the registrant, resulting in **$\text{DT}_{50} = 47.2$ days**. Despite the kinetic of decline clearly shows a fast initial decrease in test substance concentration followed by a slower decline phase (up to 30.7% AR at day 100), other models such as bi-phasic kinetic ones have not been tested by the registrant. This was done here, as presented in the Figure 5. Results showed that the **bi-phasic Hockey-Stick (HS) model** fits the data better than SFO kinetics. The estimated half-life was **$\text{DT}_{50} > 1000$ days** (slow phase) with a better X^2 value (8.6%) than for SFO (10%). However, there is high variability between replicates throughout the test particularly at day 100, leading to a t-test $p > 10\%$ and to an undefined DT_{50} (> 1000 days, default value).

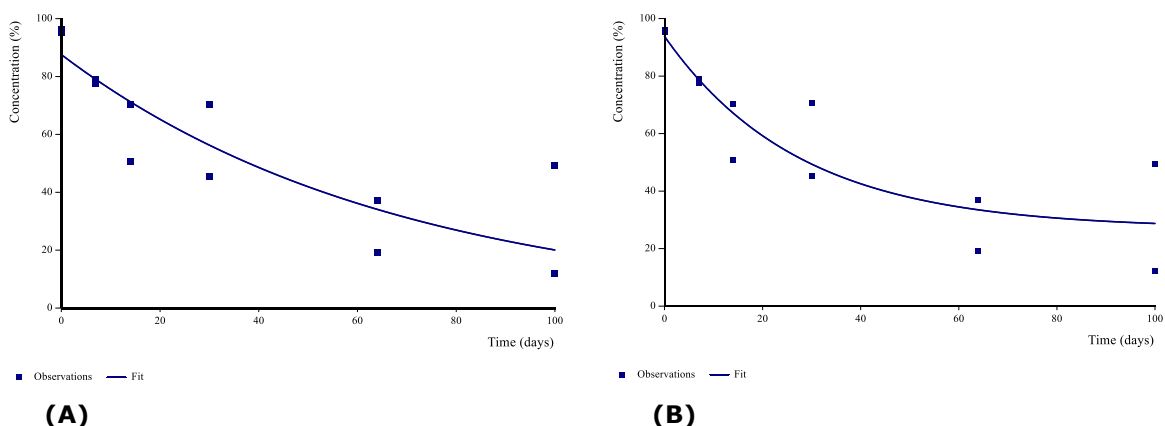


Figure 5: Degradation kinetics of octocrilene in the Dyfi Estuary system. (A) SFO model (χ^2 error: 10% ; t -test=8.2E-04) ; (B) Hockey-Stick (HS) model (χ^2 error: 8.6% ; t -test=0.5) (CAKE 3.7).

As in the previous freshwater study (*Unpublished study report (2024a)*), the potential outliers were quantified with the Grubbs' test in order to improve the model-fits. The Table 11 shows the percentages of the applied radioactivity for each replicate and sampling times in the whole system (estuarine water+sediment):

Table 11: Measured concentration of octocrilene (% AR) in the whole estuarine water system and the NER concentration for each sampling time and replicate (n=2).

Time (days)	Parent _{whole syst} (NER) % AR
	Estuarine water
0	95.0 (0.3)
0	95.8 (0.3)
7	79.0 (0.7)
7	77.6 (0.7)
14	50.8 (1.2)
14	70.4 (1.2)
30	45.3 (2.6)
30	70.5 (1.1)
64	19.2 (6.1)
64	37.1 (3.5)
100	12.1 (6.7)
100	49.3 (3.7)

This statistical test indicated that the 49.3% AR value at day 100 is furthest from the rest without being a significant outlier ($p > 0.05$). Note that only two points per replicate are available reducing the statistical robustness. No clear experimental explanation can be provided to explain this deviation at day 100. We only note a decrease in the microbial biomass in the untreated sediment at the end of the incubation (5.8 mg microbial Carbon/kg dw, corresponding to 0.5% of the TOC). The microbial biomass should constitute more than 1% of the total organic carbon in the OECD TG 307 soil transformation study but no microbial biomass is recommended for the water-sediment study. Consequently, this value cannot be excluded from the data set.

Results from this study also showed a very low mineralization rate of octocrilene in the whole system. Although not proposed by the registrant, the half-life degradation time estimation based on mineralization data has been performed and resulted in the

DT_{50,mineralization}=1870 days⁴ (SFO model ; χ^2 error: 0.26 % ; t-test=2.01E-04). Uncertainties related to the persistence of some degradation products have been underlined as explained in the next section. Consequently, the DT₅₀ based on mineralization data rather than that of the registrant was considered as the most conservative half-life for the parent.

B.4.1.2.1.9. Degradation products

Three relevant degradation products (out of 14 detected) were observed (Table 12). **M2** and **M3** were detected at amounts $\geq 10\%$ AR, with a maximum reached at day 64 (19% AR) and at day 100 (30% AR) respectively. **M1** was detected at amounts $>5\%$ AR at two consecutive sampling time with a maximum of 5.9% AR at day 30. M6 was detected at a maximum of 6.7% AR at day 14 only then was not considered relevant. Both M1 and M6 decreased towards the end of test. Ten additional metabolites (M5, M7, M8, M9, M10, M11, M12, M13, M15 and M16) were detected at $<3\%$ AR with a decreasing degradation pattern towards the end of the incubation period, except for M13 and M16 which increased up to 1.5% AR and 1% AR respectively at day 100.

Table 12: Degradation of parent and formation of its metabolites in the total system of Dyfi Estuary under aerobic conditions after application of [¹⁴C]Octocrilene (mean values of 2 replicates, values in % AR).

Time (days)	Parent	M1	M2	M3	M5	M6	M7	M8	M9	M10	M11	M12	M13	M15	M16
0	95.4	0.1	0.1	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0
7	78.3	1.6	5.3	3.8	0.0	4.4	3.1	0.2	0.1	0.1	0.1	1.1	0.1	0.1	0.1
14	60.6	5.3	5.8	8.9	0.2	6.8	1.7	2.2	0.3	1.6	0.5	1.2	0.8	0.3	1.6
30	57.9	5.9	12.1	13.8	0.7	2.6	1.4	0.2	0.1	0.1	0.1	0.2	1.1	0.1	0.1
64	28.1	5.2	18.9	29.9	0.0	3.1	0.4	0.0	0.1	0.1	0.0	0.0	1.5	0.1	0.1
100	30.7	2.1	17.2	30.2	0.0	1.7	0.7	0.0	0.4	0.3	0.0	0.0	1.5	0.4	0.3

The determination of the metabolite degradation rate was possible for **M1 only**, with decreasing concentration towards the end of the test. Then, according to the registrant, M1 degraded in the total system with a **DT₅₀ of 13.1 days** (χ^2 error: 15.4%; t-test=3.2E-02) based on the SFO as the best-fit kinetic model for the parent. Considering the HS bi-phasic model for the parent (see above), a similar DT₅₀ for M1 was estimated to be **24.7 days** (χ^2 error: 12.8% ; t-test=0.09).

The radioactivity measured for **M2 and M3** increased at the three last sampling time preventing an estimation of their DT₅₀ (Default value **DT₅₀ > 1000 days**).

B.4.1.2.1.10. Conclusions for the estuarine water system

This OECD TG 308 study presented some deficiencies. Only one sediment was used instead of two recommended; It is not clear if duplicates samples are true, independent replicates or rather replicate analytical results from a single sample; A high variability was observed between duplicates from day 14 till the end of the test without any justification. The second replicate concentration value at day 100 was detected as an outlier (Grubb's test)

⁴ Calculation: Parent after mineralization (%AR) = 100% - %AR ¹⁴CO₂

but was not statistically relevant (but runned with only two values), then not excluded from the dataset.

The model considered to best fit the experimental data for octocrilene degradation should be the bi-phasic Hockey-Stick model rather than the SFO model. However, the high variability and the absence of degradation during the slow-phase led to a $DT_{50} > 1000$ days with a t-test $> 10\%$ (absence of degradation at the test end) which lowered the confidence of the estimated parameter. Then, this (HS) DT_{50} should be interpreted with caution.

The degradation kinetics based on the mineralization of octocrilene in the whole system resulted in an estimated **$DT_{50, \text{mineralization}} = 1870$ days**. This DT_{50} could be considered as the more relevant conservative degradation time value for octocrilene. Indeed, regarding the degradation products data, the DT_{50} determination was possible only for M1. The DT_{50} of M2 and M3 were undefined caused by their accumulation and slow degradation until the end of the test. Consequently, the DT_{50} based on octocrilene mineralization was considered as the most relevant covering the potential P/vP concern of its degradation products.

As mentioned by the registrant, ***"Further information is necessary to conclude on the P or vP properties in the context of the PBT assessment. All as relevant identified metabolites may fulfill the screening criteria for P/vP. Therefore, further information may be considered necessary for the assessment of P/vP criteria."***

The different discussed DT_{50} are summarised in the Table 13.

Table 13: Summary of the different DT_{50} for octocrilene and its degradation products in the whole system estimated with different kinetics models (Cake 3.7).

Kinetic models	Chi ²	$DT_{50, \text{whole system}}$ days	Prob. > t
Parent			
SFO (IES, 2024b)	10,01%	47	8,21E-04
HS	8,61%	>1000*	4,90E-01
SFO (Mineralisation)	0,26%	1870	2,01E-04
M1			
SFO (IES, 2024a)	15.4%	13.1	3,20E-02
SFO (HS Parent)	12.8%	24.7	0,09

* k_2DT_{50} slow phase

Based on the available data (estuarine environment), we consider the following DT_{50} as relevant values for the persistence criteria:

Octocrilene: $DT_{50, \text{whole system}} (E_w) = 1870$ days

M1: $DT_{50, \text{whole system}} (E_w) = 24.7$ days

M2, M3: $DT_{50, \text{whole system}} (E_w) > 1000$ days (Default value)

Octocrilene fullfills the P/vP criteria ($DT_{50} > 180$ days) based on the half-life of degradation products.

B.4.1.2.2. Biodegradation in soil

Data waiving: The substance is poorly biodegradable in water. Comparable degradation is expected for biodegradation in soil.

B.4.1.3. Summary of degradation

Abiotic degradation

The substance will slowly hydrolyse in water.

Biotic degradation

According to the two studies performed in compliance with EU method C.4 -D and OECD TG 301F, octocrilene is poorly biodegradable.

According to the two simulation tests in water-sediment systems under aerobic conditions (OECD TG 308), the degradation kinetics of octocrilene and its metabolites in the whole system resulted in the following estimated DT₅₀:

- *Freshwater environment (Fw):*

Octocrilene: DT₅₀, mineralization (Fw)=1810 days;

Metabolite M1: DT₅₀, whole system (Fw)=14.8 days; Metabolites M2, M3 and M6: DT₅₀ >1000 days (Default value).

- *Estuarine environment (Ew):*

Octocrilene: DT₅₀, mineralization (Ew)=1870 days;

Metabolite M1: DT₅₀, whole system (Ew)=24.7 days; Metabolites M2, M3: DT₅₀ >1000 days (Default value).

B.4.2. Environmental distribution

B.4.2.1. Adsorption/desorption

The study on adsorption/desorption is summarised in the Table 14:

Table 14: Studies on adsorption/desorption.

Method	Results	Remarks	Reference
Study type: OECD Guideline 106 (Adsorption - Desorption Using a Batch Equilibrium Method)	Adsorption coefficient at 20°C: Koc: 53 355 L.kg ⁻¹ (Org. C (%): 0.718) Koc: 29 934 L.kg ⁻¹ (Org. C (%): 1.47) Koc: 79 018 L.kg ⁻¹ (Org. C (%): 0.412) Koc: 38 744 L.kg ⁻¹ (Org. C (%): 1.74) Koc: 43 438 L.kg ⁻¹ (Org. C (%): 0.916)	1 (reliable without restriction) key study experimental study Test material (EC name): octocrilene	Unpublished study report (2018a)

Discussion

A Koc value at 20°C of 48 898 L.kg⁻¹ (arithmetic mean, range: 29 934 -79 018) is determined. The substance has a high adsorption potential on organic matrices.

B.4.2.2. Volatilisation

The studies on volatilisation are summarised in the Table 15:

Table 15: Studies on volatilisation.

Method	Results	Remarks	Reference
Calculated using SRC HENRYWIN v3.10 (Bond estimation method)	Henry's Law constant H: 0.00031 Pa m ³ /mol at 25°C	2 (reliable with restrictions) key study estimated by (Q)SAR Test material (EC name): octocrilene	HENRYWIN v3.10

Discussion

From the water surface, the substance will not evaporate into the atmosphere. A Henry's law constant (H) of 0.00031 in Pa.m³.mol⁻¹ at 25°C. The substance has a very low potential to evaporate from the water surface to the air.

B.4.2.3. Distribution modelling

No relevant information available.

B.4.3. Bioaccumulation

B.4.3.1. Aquatic bioaccumulation

Table 16: Studies on aquatic bioaccumulation.

Method	Results	Remarks	Reference
<i>Danio rerio</i> aqueous (freshwater) flow-through Total uptake duration: 28 d Total depuration duration: 16 d Nominal concentration: 0.1 µg.l ⁻¹ and 1 µg.l ⁻¹ OECD Guideline 305 (Bioconcentration: Flow-through Fish Test)	Bioaccumulation factor (kinetic, corrected for growth and lipid fraction): BCF: 858 dimensionless (geometric mean of two values) BCF (at 0.1 µg.l ⁻¹): 830 dimensionless BCF (at 1 µg.l ⁻¹): 887 dimensionless Lipid content: 5.5 % (Lipid content was determined during each sampling occasion during the entire test duration (uptake and depuration))	2 (reliable with restriction) supporting study experimental result Test material (EC name): octocrilene	Unpublished study report (2008)
<i>Oncorhynchus mykiss</i> Feed flow-through	Biomagnification factor: BMF: 0.034 dimensionless	1 (reliable without restriction)	Unpublished study report (2018d)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
Natural water: freshwater Uptake duration: 14 d Depuration duration: 7 d According to OECD Guideline 305: Dietary Exposure Bioaccumulation Fish Test	(whole body w.w.) (Growth-corrected and lipid-normalised)) Lipid content: 3.5 % (start of exposure) (- average from 3 fish) 3.6 % (end of exposure) (- average from 3 fish) 3.9 % (end of depuration period) (- average from 3 fish) Transformation products: Possible metabolites were not identified; however, as the BMF was based on total radioactivity, it has to be mentioned that the measured amount on fish represents both the parent molecule as well as any possible metabolite.	key study experimental result Test material (EC name): octocrilene	

Additional information:

In the first study, the bioaccumulation of octocrilene in *Danio rerio* was investigated using a flow through system set up according to OECD TG 305. This study was conducted at low concentrations of radio-labelled octocrilene (0.1 µg.l⁻¹). Fish were exposed *via* the water phase to the ¹⁴C-radiolabeled test substance for 28 days. After a depuration period of 15 days the test was terminated and a mean growth and lipid corrected BCF of 858 based on the whole body wet weight was determined. This BCF takes into account both parent and any possible metabolite. The BCF value was however calculated from octocrilene measurements in unfiltered water samples whereas concentrations are reported lower in filtered water samples (raw data are not shared in the full study report). A more reliable BCF value was expected to be calculated from octocrilene measurements in filtered samples in order to get rid of likely adsorption/absorption behavior to particles considering that fish are fed during the test. Moreover the real dissolved concentration of octocrilene in water is expected much lower than the available provided concentration. Thus the provided result by this study is considered unsatisfactory because underestimated.

Conclusion on aquatic bioaccumulation

octocrilene is not classified as bioaccumulative and very bioaccumulative according to the REACH PBT/vPvB criteria.

However, it should be noted that data from the scientific literature show the presence of octocrilene in the aquatic biota in almost all European area studied (from Spain, to Norway, in sea, lake, river, in protected area) (see section B.9.6.4.). Octocrilene demonstrated the ability to bioaccumulate in several aquatic species. Field bioaccumulation factors was estimated in marine clam *Ruditapes philippinarum* caged in cadiz bay (Spain) with log BAF value up to 4.89 (*Pintado-Herrera et al., 2020*) as well as in two species of cold-water corals with log BAF calculated at 3.8 and 4.3 (*Tovar-Salvador et al., 2025*). Furthermore, it has to be underlined the tendency of octocrilene transformed into fatty acid conjugates *via* oxidation of the ethylhexyl chain, yielding very lipophilic octocrilene analogues that accumulate in tissues (*Stien et al., 2019; Clergeaud et al., 2022*). The formation of octocrilene analogues suggests that octocrilene concentrations measured in the organisms

may be largely underestimated as those conjugates were not measured in these studies.

Those data suggest that octocrilene could be considered as a 'bioaccumulative' compound even if not classified according to criteria set out in annex XIII on PBT, vPvB.

B.4.3.2. Terrestrial bioaccumulation

No relevant information available.

B.4.4. Secondary poisoning

Not investigated. However, biomagnification of octocrilene was suggested as shown in the published scientific studies in marine species of high trophic level such as tuna (*Pintado-Herrera et al., 2024*) and dolphin (*Gonzalez-Bareiros et al., 2023*).

B.5. Human health hazard assessment

As the restriction proposal covers solely risks arising from the wide dispersive uses of cosmetic products for the environment, this section will not be further developed. See report annex XV for an extended summary on endocrine disruption data in human health, as these data are considered in addition to environmental data for the whole ED assessment of octocrilene (one health concept and conservation of the modalities considered). Full details of human health hazard assessment will be available in the conclusion document of the substance evaluation under REACH when it is completed.

B.6. Human health hazard assessment of physicochemical properties

B.6.1. Explosivity

No relevant information available.

B.6.2. Flammability

No relevant information available.

B.6.3. Oxidising potential

No relevant information available.

B.7. Environmental hazard assessment

B.7.1. Aquatic compartment (including sediments)

B.7.1.1. Fish

B.7.1.1.1. Short-term toxicity to fish

The results are summarised in the following Table 17:

Table 17: Short-term effects on fish.

Method	Results	Remarks	Reference
<i>Leuciscus idus</i> freshwater 96h, static Nominal concentration: 5000 and 10000 mg.l ⁻¹ Measured concentration: unknown Positive control (chloroacetamide LC ₅₀ = 31 mg.l ⁻¹) Photoperiod : 16:8h light:dark Water temperature: 22-25°C Replicate: n= 10 fish per condition	LC50 (96 h) > 10000 mg.l ⁻¹ test mat. (nominal) based on: mortality	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene Purity >98% German standard DIN 38412, part 15 (1982) GLP: no	Unpublished study report (1990)
<i>Brachydanio rerio</i> freshwater 96h, static Nominal concentration: 0.5 mg.l ⁻¹ Measured concentration: unknown Photoperiod : 16:8h light:dark Water temperature: 26°C ± 1°C Replicate: n=10 fish per condition	LC50 (96 h): > 0.5 mg.l ⁻¹ test mat. (nominal) based on: mortality	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene Purity: unknown Equivalent or similar to EU Method C.1 (Acute Toxicity for Fish) GLP: yes	Unpublished study report (1995c)
Zebrafish Embryo-larval (<i>Danio rerio</i>) Spiked artificial sediment Exposure duration: 48 hours Nominal concentration: 10, 100, 1000 mg.kg dw ⁻¹ Measured concentration: unknown Negative and positive controls (3,4-dichloraniline = 3.7 mg.l ⁻¹) Solvent: acetone (<0.01%) Photoperiod : 14h light - 10h dark Water temperature: 26°C ± 1°C Water pH: 7.8 ± 0.5 Dissolved oxygen: >7.8 ± 0.3 mg.l ⁻¹ Replicates: 5 fish embryos per well, 4 wells per concentration	No mortality	2 (reliable with restriction) supporting study Test material (EC name): octocrilene Purity: unknown Based on fish embryo test DIN 38415-6 (DIN 2001), and the methods in Nagel (2002), Braunbeck et al. (2005), Hollert et al. (2003) and Lammer et al. (2009) Not GLP	Kaiser et al., 2012a

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
Embryos Zebrafish (<i>Danio rerio</i>) Exposure duration: 144 hpf Exposure medium: renewal every 24h Mean measured concentrations (% of the nominal concentration): 69 (69%), 293 (29%) and 925 (46%) $\mu\text{g.l}^{-1}$ Photoperiod : 16:8h light:dark Water temperature: 27°C $\pm$ 1°C Replicates: 5 groups with 4 replicates (200 fertilizate/tank)	No effect on survival rate and hatching rate No mortality after 96h and 10 dpf ↓ Hatching success (96h) No effect on cardiac frequency and aerobic metabolism (5dph)	2 (reliable with restriction) Test material: octocrilene Purity: > 97% OECD guideline No. 204 (Fish, prolonged toxicity test)* *Deleted guideline on 2 nd April 2014 Not GLP	Blüthgen et al., 2014
Zebrafish Embryo-larval (<i>Danio rerio</i>) Spiked artificial sediment Exposure duration : 96h with an additional 6 days period in E3 medium (10 days post fertilization) Nominal concentration: 10 mg.kg dw ⁻¹ Measured concentration: unknown Negative (sediment only) and solvent controls Solvent: Dichloromethane Photoperiod: 14:10h light:dark Water temperature: 28°C $\pm$ 1°C Water pH: 7.8 $\pm$ 0.5 Oxygen saturation > 80% Replicates: 3 replicates of 30 eggs embryos per treatment	No mortality after 96h and 10 dpf ↓ Hatching success (96h) No effect on cardiac frequency and aerobic metabolism (5dph)	2 (reliable with restriction) supporting study Test material: octocrilene Purity: unknown Guideline: no Not GLP	Lucas et al., 2022
Embryo-larval Zebrafish (<i>Danio rerio</i>) Freshwater, renewed every 24h Exposure duration: 96 hpf (hours post fertilization) Nominal concentration: 5, 50, 500 $\mu\text{g.l}^{-1}$ Measured concentration: unknown Solvent: ethanol (<0.01%) Photoperiod : 14:10h light:dark Water temperature: 27°C $\pm$ 2°C Water pH: 7.2 $\pm$ 0.06 Dissolved oxygen: 6.6 $\pm$ 0.4 mg.l ⁻¹ Replicates: 50 embryos per concentration, triplicate	Antioxidant enzyme activity (SOD, CAT, GST, LPO): ↑ at 50 and 500 $\mu\text{g.l}^{-1}$ AChE: ↓ at 50 and 500 $\mu\text{g.l}^{-1}$ ↑ at 5 $\mu\text{g.l}^{-1}$ Malformations: tail deformities, yolk sac edema and pericardial edema: ↑ at 50 and 500 $\mu\text{g.l}^{-1}$ Hatching rate and heart beat (72 hpf): ↓ at 50 and 500 $\mu\text{g.l}^{-1}$ Histopathology (elongated yolk sac, pericardial edema, abnormal muscle, swim bladder inflammation, ganglion cell layer, inner and outer nuclear layer, yolk sac): ↑ at 50 and 500 $\mu\text{g.l}^{-1}$	2 (reliable with restriction) Test material: octocrilene Purity: 98% Guideline: similar to OECD 236 Not GLP	Gayathri et al., 2023
Embryo Zebrafish (<i>Danio rerio</i>) freshwater Exposure duration: 96 hpf renewed every 24h	No mortality Hatching rate (72 hpf): ↑ at 1 $\mu\text{g.l}^{-1}$ No malformation	2 (reliable with restriction) Test material: octocrilene	Cahova et al., 2021

Method	Results	Remarks	Reference
Nominal concentration: 0.1, 1, 10, 50, 100, 250 µg.l ⁻¹ Measured concentration: unknown Water and solvent (ethanol <0.01%) controls Photoperiod : 12:12h light:dark Water temperature: 26°C ± 1°C Water pH: 7.8 Replicates: 24 eggs per concentration, test in duplicate		Purity: 98% Guideline: modified Fish Embryo Acute Toxicity Test OECD 236 Not GLP	

Discussion

The toxicity of octocrilene to fish was tested in two guideline studies.

In an *unpublished study report (1990)* study according to German standard DIN 38412, part 15, *Leuciscus idus* was exposed for 96 hours to test substance concentrations of 5000 mg.l⁻¹ and 10000 mg.l⁻¹. At termination, a LC₅₀ of > 10000 mg.l⁻¹ was determined. However, the concentration tested exceeded the water solubility (40 µg.l⁻¹) and undissolved test substance at the bottom of the aquaria and on the water surface was observed. The study protocol did not follow the guidance document on aquatic toxicity testing of poorly water-soluble substances. Furthermore, no analytical measurements were done.

In a second *unpublished study report (1995c)* study conducted according to EU Method C.1, the test fish (*Brachydanio rerio*) were exposed under static conditions to the substance for 96 hours at a max. nominal test concentration of 0.5 mg.l⁻¹ (prepared from a solution at saturated concentration). At test termination a LC₅₀>0.5 mg.l⁻¹ was determined (Bayer AG 1995). Nonetheless, it should be noted that no analytical measurements have been performed throughout this test. Therefore, no information about the exact measured value of the saturated concentration determined in a preliminary experiment is known. Only an approximated value is given. In addition, the concentrations tested exceeded the water solubility.

Results from these two studies should be considered as supportive data.

Several non guideline studies from the scientific literature are available:

In *Kaiser et al., 2012a*, the toxic effects of three common UV filters including octocrilene were studied in a sediment contact test with zebrafish embryos. The test was performed according to a modified fish embryo test (DIN 2001) with sediment contaminated up to 1000 mg.kg_{dw}⁻¹ (nominal) octocrilene. No effect on embryos mortality (coagulation, lack of heartbeat, missing somite development and failure of tail detachment from the yolk sack) was observed. Note that the maintenance of the nominal chemical concentration throughout the test was not monitored, then the bioavailable fraction of octocrilene is not known. Furthermore, the results for octocrilene were not presented (since no effect was observed), then validity criteria can not be checked.

In *Blüthgen et al., 2014*, octocrilene did not affect survival and hatching rate of zebrafish embryos exposed up to 2000 µg.l⁻¹ (nominal) or 925 µg.l⁻¹ (measured). In this study, the authors used glass and stainless steel tanks without organic matter and they showed the high instability of octocrilene in water occurring during storage and due to sorption to

exposure-container material. This underlines the importance to perform the analytical monitoring throughout a toxicity test with octocrilene.

In *Lucas et al., 2022*, hatching success decreased after 96h exposure of zebrafish eggs to sediment contaminated with octocrilene at 10 mg.kg_{dw}⁻¹ (nominal). No mortality was observed even after an additional 6 days period in a freshly prepared E3 medium 10 days. No impact on cardiac frequency and aerobic metabolism was observed in 5 dpf embryos. This study showed that octocrilene induced a significant delayed hatching which could impact fish development under environmental conditions. Note that the viability in the control of 84%-86% was lower than the validity criteria (>90%) and exposed temperature is relatively high (28°C) compared to the maximal of 26+/-1°C recommended for this species in guidelines (for ex., ZET OECD 236). Furthermore, only one concentration was tested and no analytical measures were performed. Then, the results should be used as supportive information.

In *Gayathri et al., 2023*, exposure during 96h of zebrafish embryo/larval to 50 and 500 µg.l⁻¹ (nominal) of octocrilene decreased hatching rate and heartbeat rate (72 hpf), induced various developmental deformities. Octocrilene also inhibited the activity of AchE, may interfere with lipid metabolism, increased the number of apoptotic cells in the heart region and led to histopathological injuries that might be responsible for the observed developmental toxicity in zebrafish embryos/larvae. Some limitations should be noted: solvent and positive controls not available, the mortality in the control at the end of the test not specified, a higher temperature and a photoperiod different from OECD 236 recommendations (26 +/-1°C and 16h-12h respectively), the use of large test chambers (50 ml) with a higher loading rate than in OECD 236. Analytical monitoring throughout the test was not performed although the 24h renewal should have minimised the possible loss of octocrilene highly adsorptive. Those results should be used as supportive data.

In *Cahova et al., 2021*, the embryotoxicity of octocrilene was evaluated in zebrafish using a modified OECD 236. The toxic effects were studied by assessing mortality, hatching and the occurrence of malformations at 24, 48, 72 and 96 hpf. The tested concentrations were 0.1, 1, 10, 50, 100, 250 µg.l⁻¹ (nominal), and the lowest ones corresponded to environmentally relevant concentrations of octocrilene in surface water. No mortality was recorded in controls and in all exposed groups up to 96 hpf. All controlled embryos were hatched at 96 hpf and no significant differences ($p > 0.05$) in the hatching rate were found between the control and experimental groups at 96 hpf, except in 1 µg.l⁻¹ group where a significant earlier hatching was recorded. No explanation was done and this result was not confirmed in an other study for higher exposed concentrations (*Blüthgen et al., 2014*). No malformations were observed in the control and exposed groups. This study did not report any embryotoxicity for zebrafish in the environmentally relevant concentrations of octocrilene.

The acute toxicity of octocrilene to fish was tested in two guideline studies and several publications from the scientific literature. Considering the methodological missing points in those studies, acute fish toxicity endpoints are considered as supportive data.

B.7.1.1.2. Long-term toxicity to fish

No long-term toxicity studies were provided by registrants for the octocrilene.

B.7.1.1.2.1. Read-across with etocrilene

A fish early life-stage toxicity test (OECD TG 210) has been provided but for another

substance, etocrilene (CAS 5232-99-5). The results of this study are summarised in the following Table 18:

Table 18: Long-term effects of etocrilene on fish.

Method	Results	Remarks	Reference
<i>Danio rerio</i> (<5 hours old) Freshwater Exposure: duration: 36 days Flow through Nominal concentrations of etocrilene: 0.09, 0.2, 0.4, 0.9 and 2 mg.l ⁻¹ Measured concentrations of etocrilene: 0.09, 0.19, 0.33, 0.75 and 1.67 mg.l ⁻¹ Temperature: 25°C-26°C O ₂ >60% Replicates: 4 replicates of 25 embryos	NOEC (36d): 0.09 mg.l ⁻¹ test mat. (nominal) based on: mortality - (Juvenile survival/overall survival)	2 (reliable with restriction) Experimental study Test material Etocrilene (CAS 5232-99-5) Purity: 100% OECD Guideline 210 (Fish, Early-Life Stage Toxicity Test) ; guideline EPA OPPTS 850.1400 (Fish Early-life Stage Toxicity Test) GLP: yes	Unpublished study report (2024c)

The study was conducted for a period of 36 days under flow-through conditions in accordance with the OECD TG 210 and the US EPA OCSPP 850.1400 guidelines. The 5 etocrilene tested nominal concentrations were 0.09, 0.2, 0.4, 0.9 and 2 mg.l⁻¹. They were prepared using a saturation column according to OECD TG 23 (2019) in order to achieve a saturated solution without using solvent and considering 2.6 mg.l⁻¹ as the maximum water solubility value (unavailable report, 2021). It is worth noting that the water solubility value reported in all the studies from the registration dossier was 5 µg.l⁻¹. The concentrations were monitored weekly by RP-HPLC with UV detection. The overall mean measured concentrations during the exposure period were within the range ± 20% of the nominal concentrations.

Biological parameters evaluated included the survival of the embryos until hatch, the number of surviving larvae at end of swim-up (day 6) and the survival of young zebrafish until the end of exposure (day 36), the time to hatch and swim-up, signs of toxicity, and the body weight and the total body length of surviving fish at the end of exposure. The study is valid according to the test guideline. Results showed no significant decrease in hatching success and in survival from hatch to the end of swim-up (day 6) and no test substance-related effect on the time to swim-up. However, survival significantly decrease from 0.2 mg.l⁻¹ and a NOEC (mortality) of 0.09 mg.l⁻¹ (nom) was determined.

B.7.1.1.2.2. Relevance of this study in a read-across approach

The test item studied was **Etocrilene (CAS 5232-99-5)**. The registrant did not indicate explicitly whether they are making an adaptation using read-across, and did not provide a comprehensive justification for the use of a read-across approach.

- Physico-chemical and e-fate properties

The analysis of the chemical structure of octocrilene and etocrilene showed that both substances belong to the same chemical family of aromatic ester compounds with UV-absorbing properties. Both contain aromatic and ester moieties, which dominate their chemical reactivity and biological interactions. The key structural difference between

octocrilene and etocrilene lies in the alkyl chain length, octocrilene having a longer chain (six carbons more) compared to etocrilene which may favor octocrilene distribution in lipid compartment and its bioaccumulation potential.

Based on the available (non evaluated) data from the registration dossier of etocrilene⁵ (Table 19), lower lipophilicity (two orders lower for etocrilene, Qsars values) precludes any comparison in tissue distribution and long term effects.

Table 19: Physico-chemical properties, environmental e-fate and behaviour of octocrilene and etocrilene (data from the registration dossier⁹).

	Octocrilene (CAS 6197-30-4)	Etocrilene (CAS 5232-99-5)
Molecular weight	361.47 g.mol ⁻¹	277.31 g.mol ⁻¹
Vapor pressure	4E-07 Pa (20 °C)	1.6E-04 Pa (20°C)
Water solubility	40 µg.l ⁻¹ (20°C)	< 5 µg.l ⁻¹ (LOD, 23°C)
Biodegradation	Poorly biodegradable	Poorly biodegradable
LogKow	6.1	4.01 (Qsars)
Bioaccumulation	830 < BCF < 887 L.kg ⁻¹	BCF < 500 L.kg ⁻¹ (Qsars)
LogKoc	4.69	3.99 (Qsars)

- Aquatic toxicity properties

Information on the aquatic toxicity of etocrilene is limited. In the registration dossier, two acute and two chronic limit tests (fish OECD TG 203, algae OECD TG 201, Daphnia OECD TG 211) performed with 100% saturated solution (max water solubility <5µg.l⁻¹) and without analytical measurements are available. It should be noted that these studies have never been evaluated. No toxic effects were reported. The third chronic available study is the actual fish OECD TG 210 (*unpublished study data, 2024c*). No other data on aquatic toxicity of etocrilene have been found in the literature.

In view of these considerations, it is not possible to conclude on a similarity level of aquatic toxicity between octocrilene and etocrilene.

B.7.1.1.2.3. Conclusion about the read-across approach

Based on the available data from the registration dossier of etocrilene and the aquatic toxicity data available, it is not possible to conclude on a similarity level between octocrilene and etocrilene:

- of the physicochemical, environmental fate and behaviour properties;
- of aquatic toxicity.

In conclusion, **a read-across with etocrilene has not been take into account in the weight of evidence.**

B.7.1.1.2.4. Others studies available

Several studies in which fish were exposed to octocrilene over a long exposure period are available. These studies have been designed to observe the effects potentially related to an endocrine disruption (EAS modalities). They provides some indications of the systemic

[Etocrilene Registration Dossier](#)

toxicity and about parameters considered as 'sensitive to, but not diagnostic of, EATS' in an ED assessment. These studies give relevant indications for ecotoxicological hazard and risk assessment of octocrilene (*i.e.* to determine more sensitive species and to establish the safety factor for the PNEC_{aquatic} calculation). However, they cannot be used to determine the No (Lowest) Observed Effect Concentration (N(L)OEC) necessary for the risk assessment (e.g. no study include 5 concentrations for enables dose-response relationship, concentration may be too low to detect statistically significant apical effects, the response may be not monotonous, ...).

These studies indicate a low systemic toxicity of octocrilene on adult fish. They show effects of octocrilene on the early life stages of fish (embryonic and larval) but without being able to distinguish at this stage any EAS-mediated effect.

Table 20: Other studies available on fish.

Method	Results	Reference
Female adult <i>Gasterosteus aculeatus</i> Freshwater Exposure duration: 21 days Nominal concentrations: 3.5, 8, 20 µg.l ⁻¹ Measured concentrations: 0.70, 1.85 and 4.44 µg.l ⁻¹ (centrifuged samples) 1.41, 3.22 and 8.64 µg.l ⁻¹ (non-centrifuged samples) Replicates: 4 per treatment, 4 fish each	A single mortality was observed in the 20 µg.l and DHT treatment group within the first 48 hours of the study and was not considered to be a direct result of the exposure.	Unpublished study report (2018b)
Male adult Zebrafish (<i>Danio rerio</i>) Freshwater Exposure duration: 16 days Nominal concentration: 100, 1000 and 2000 µg.l ⁻¹ Measured concentration: 22, 209 and 383 µg.l ⁻¹ Replicates: 5 groups with 5 replicats (15 fish/tank)	No effect on survival rate, body length, body weight and gonadosomatic index (GSI)	Blüthgen et al., 2014
Adult zebrafish (<i>Danio rerio</i>) Freshwater Exposure duration: 28d Nominal concentration: 100, 1,000, 3000 µg.l ⁻¹ Measured concentration (0-48h): 28.61, 505.62, 1248.7 µg.l ⁻¹ Replicates: 5 fish embryos per well, 4 wells per concentration	No mortality No abnormal behavior No effect on the condition factor (CF=weight/length) ↑ GSI in female at 1248 µg.l ⁻¹	Zhang et al., 2016
Adult Zebrafish (<i>Danio rerio</i>) Exposure duration: 60d 7 and 700 µg.l ⁻¹ (nominal) Replicate: 3 per treatment with 4 fish each	No mortality of adult zebrafish Observation on F1 After 25 days, no difference in embryo mortality (24h) and hatching rate (72h) After 47 days, embryo mortality 100% at 700 µg.l ⁻¹ , 28% at 7 µg.l ⁻¹ ↓ hatching rate (72h) at 7 µg.l ⁻¹	Li et al., 2018
Japanese medaka (<i>Oryzias latipes</i>), 3-4 months old	Adults F0: No mortality	Yan et al., 2020

Method	Results	Reference
Freshwater Exposure duration: 28d Japanese medaka (<i>Oryzias latipes</i>), embryo Freshwater Exposure duration: 14d Nominal concentration: 5, 50, 500 µg.l⁻¹ Measured concentration: 2.71, 14.29, 232.62 µg.l⁻¹ Replicates: breeding pairs in 18L: n=15 and in 1L: n=5 – three replicates per groups	↑ GSI and HSI in female at 232.6 µg.l⁻¹ F1 embryos exposed to octocrilene (14d): ↑ time to hatching at 14.3 and 232.6 µg.l⁻¹ ↑ morphological abnormality rates at 2.7, 14.3, 232.6 µg.l⁻¹ ↑ accumulative death rates at 2.7, 14.3, 232.6 µg.l⁻¹ ↓ hatchability but not significant F1 larvae 14 dph: ↓ body length at 2.7, 14.3, 232.6 µg.l⁻¹	

No OECD study on the long-term toxicity of octocrilene on fish is provided to derive a chronic endpoint value in the context of an environmental risk assessment. Long-term toxicity studies from literature are considered as supportive data.

B.7.1.2. Aquatic invertebrates

B.7.1.2.1. Short-term toxicity to aquatic invertebrates

The results are summarised in the following Table 21:

Table 21: Short-term effects on aquatic invertebrates.

Method	Results	Remarks	Reference
<i>Daphnia magna</i> Freshwater, 48h, static Nominal concentrations: 2, 4, 8 mg.l⁻¹ in M4 medium Measured concentrations: <2 (<1/2 LOQ 1 mg.l⁻¹), 2, 4 mg.l⁻¹ Replicates: 2 with 10 daphnia per treatment Temperature : 19.6–19.9°C pH 7.7 Photoperiod: unknown Dissolved O2: 9.8 mg.l⁻¹	EC50= 3.6 mg.l⁻¹ (geometric mean EC0/EC100), based on: mobility	2 (reliable with restrictions) supporting study Test material (EC name): octocrilene Purity: unknown According to the test guideline "Acute toxicity to daphnia" (C.2) GLP: yes	Unpublished study report (1995d)
<i>Daphnia magna</i> Freshwater, 48h, static Mean measured concentration (calculated by results of HPLC analysis): 0.023 mg.l⁻¹ in water Positive control:	EC50 (48 h) >0.023 mg.l⁻¹ test mat. (meas. (arithm. mean)) based on: behavior, mobility	2 (reliable with restriction) supporting study Test material (EC name): octocrilene Purity: 98.4 % OECD Guideline 202 (1984), EU C.2	Unpublished study report (2001b)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
potassium dichromate (EC_{50,24h}=2.1 mg.l⁻¹) Replicates: 4 groups of 5 animals per concentration Temperature : 18 - 22 °C ± 1 °C Photoperiod: 16/8 h light/dark Dissolved O2 >60%		(<i>Daphnia</i> sp. Acute Immobilisation Test) GLP: yes	
Daphnia magna Freshwater, 48h, static Nominal concentrations: 12.5, 25, 50, 100 mg.l⁻¹ in M4 medium Measured concentrations: unknown Positive control: potassium dichromate (EC_{50,24h}= 0.75 mg.l⁻¹) Replicates: 4 groups of 5 animals per concentration Temperature : 18 - 22 °C, ± 2 °C Photoperiod: 16/8 h light/dark	EC50 (48 h) >100 mg.l⁻¹ test mat. (nominal) based on: mobility	2 (reliable with restriction) supporting study Test material (EC name): octocrilene Purity: 99.3% OECD Guideline 202 (<i>Daphnia</i> sp. Acute Immobilisation Test) GLP: yes	Unpublished study report (2000)
Daphnia magna Freshwater 24h exposure Fed with <i>C. vulgaris</i> 5x10⁵ cells.ml⁻¹ Nominal concentrations: 25, 50, 100, 200, 400 µg.l⁻¹ Measured concentrations: unknown Solvent: DMSO 0.01 % Photoperiod: dark (avoid algal growth) Replicate: 4 with five neonates per groups	↓ feeding rate at 0.1, 0.2, 0.4 mg.l⁻¹	2 (reliable with restriction) Test material: octocrilene Purity: unknown No Guideline Feeding bioassay according to Barata et al. (2008) Not GLP	Pablos et al., 2015
Daphnia magna Freshwater, 48h exposure Nominal concentration: 0.0625, 1.25, 2.5, 5, 10, 20, 40, 80, 160, 320, 640 mg.l⁻¹ Measured concentration: unknown	EC50 (48h): 3.18 mg.l⁻¹ based on: immobility + mortality	2 (reliable with restriction) Test material: octocrilene Purity> 99% Guideline: based on OECD 202 (<i>Daphnia</i> sp. Acute Immobilisation Test)	Park et al., 2016b

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
Solvent: DMSO (0.0001%) Photoperiod: 16:8h light:dark Water temperature: 23°C ± 0.5°C Water pH: 7.2 to 8.2 Dissolved oxygen saturation: 6.8 to 8 Replicates: 10 daphnids (<15h old) into each of the 3 replicates per test group		Not GLP	
Artemia salina (Brine shrimps) Marine water 48h exposure (Nauplii instar II-III stage) Nominal concentration: 0.02, 0.2, 2, 20, 200, 2000 µg.l⁻¹ Measured concentration: unknown Solvent (DMSO 2.5% v/v) and blank (artificial seawater) controls Photoperiod : 12:12h light:dark Water temperature: 25°C Replicates: 10 nauplii, six replicates	↑ Mortality at 2 mg.l⁻¹ LC50 (48h) = 0.610 mg.l⁻¹	2 (reliable with restriction) Test material: octocrilene Purity: unknown Guideline: no Not GLP	Thorel et al., 2020
Daphnia magna Freshwater 48h exposure (+ 19 days clean water post-exposure) Fed daily Semi-static, renewal every 2 days Nominal concentration: 0.2, 2, 20, and 200 µg.l⁻¹ Measured concentration: unknown (stock solution 100 g.l⁻¹ stable) Solvent (DMSO 1%) and water controls Replicates: 5 neonates (<24 h old) into each of the 6 replicates per test group	No mortality LC50 (48h) = 3.6 mg.l⁻¹ EC50 (48h): 0.03 mg.l⁻¹ Based on: immobilization Behavioural impairment: ↑ phototactic response time at 0.2 mg.l⁻¹ 19 days clean water post-exposure: ↑ mortality (87% day 8) at 0.2 mg.l⁻¹	2 (reliable with restriction) Test material: octocrilene Purity: unknown Based on OECD Guideline 202 (2004) Daphnia sp. Acute Immobilisation Test) Not GLP	Boyd et al., 2020
Daphnia magna Freshwater, 48h exposure	<i>Daphnia magna</i> EC50: 2.57 mg.l⁻¹ Based on immobilization	2 (reliable with restriction) Test material:	De C S de Paula et al., 2022

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
Nominal concentration: 0.5, 0.75, 1, 1.25, 1.75, 2, 2.5, 3.5, 5, 7 mg.l⁻¹ Measured concentration: unknown Solvent (DMSO<0.03%) control Temperature 20 +/- 2°C Obscurity Replicates: 10 daphnids into each of the 3 replicates per test group Artemia salina Seawater, 48h exposure Nominal concentration: 0.75, 1, 1.25, 1.75, 2, 2.5, 3.5, 5 mg.l⁻¹ Measured concentration: unknown Solvent (DMSO<0.03%) control Replicates: 1 artemia per well, 10 wells into each of the 3 replicates per test group	<i>Artemia salina</i> EC50: 2.97 mg.l⁻¹ Based on immobilization	octocrilene Purity> 99.8% Guideline: no Not GLP	
Daphnia magna Freshwater, 48h exposure Nominal concentration: 2, 4, 6, 8, 10 mg/L Measured concentration: unknown Solvent (DMSO 1%) control Replicates: 5 daphnids into each of the 3 replicates per test group	EC50 (48h): 2.66 mg.l⁻¹	2 (reliable with restriction) Test material: octocrilene Purity> 98% Guideline: according to OECD 202 Daphnia sp. Acute (2004) (immobilisation Test) Not GLP	Marcin et Aleksander., 2023

Discussion:

In a guideline study following OECD TG 202 (*Unpublished study report, 2001b*), a preliminary range finding test (up to 100 mg.l⁻¹) was conducted under static conditions without effects observed on the percentage of immobility. In the definitive test, no biologically significant effect neither in the tested limit concentration of 0.023 mg.l⁻¹ nor in control was observed and the EC₅₀>0.023 mg.l⁻¹ was proposed. However, the saturated octocrilene concentration of 0.023 mg.l⁻¹ is a calculated value by extrapolation of measured concentrations instead of a direct measured concentration.

In a second study performed according to OECD TG 202 (*Unpublished study report, 2000*), no effect was reported after 48h exposure to octocrilene in M4 medium up to 100 mg.l⁻¹ (nominal concentration). However, the study protocol did not include analytical

measurement performed throughout the test to determine the measured value of the saturated concentration. In addition, the concentrations tested exceed the water solubility.

In an *Unpublished study report (1995d)* study performed according to the test guideline EU C2, 48h exposure to 2 and 6.5 mg.l⁻¹ (meas.) octocrilene in M4 medium immobilized 65% and 100% daphnia respectively. Note that the substance has been tested above its solubility limit in M4 (54 µg.l⁻¹) and no solvent was used.

In these studies, the absence of analytical measures and the high tested concentrations question the bioavailable fraction of octocrilene in water or M4 medium is not known. Considering the high adsorptive property and lipophilicity of the substance, the results should be considered only as supportive data.

Several studies from the scientific literature are available:

Pablos et al., 2015 studied the *D. magna* feeding inhibition after 24h sublethal exposure to several personal care products among with octocrilene tested at 0.025, 0.05, 0.1, 0.2, 0.4 mg.l⁻¹ (nominal). The absence of inhibition of the feeding rate was shown until 0.05 mg.l⁻¹. These results provided by such assays help to complement information on chronic responses since it is well-known that effects on feeding are linked with reproduction and growth.

In *Park et al., 2016b*, *D. magna* were exposed to octocrilene at 0.0625, 1.25, 2.5, 5, 10, 20, 40, 80, 160, 320, 640 mg.l⁻¹ (nominal) for a period of 48 hours. No analytical measurements were done. The immobilized Daphnia in the control was less than 10% throughout the test and at the test end, immobilisation (mortality + immobility) was recorded giving an EC₅₀ of 3.8 mg.l⁻¹. Immobilized Daphnia by octocrilene sticking to the body surface were observed in the ≥10 mg.l⁻¹ exposed groups.

Thorel et al., 2020 presented preliminary results on the toxicity of octocrilene assessed on marine organisms from two major trophic levels, including heterotrophs (*Artemia salina*) and autotrophs (*Tetraselmis sp.*, see §B.7.1.3.). Among the ten UV filters tested, OC was shown to be the most toxic molecule on *Artemia* reaching 88 % mortality at 2 mg.l⁻¹ and showing the lowest LC₅₀ = 0.61 mg.l⁻¹. Any alterations in these populations may result in chain reaction effects on organisms at higher trophic levels. These results were presented before investigating a large diversity of species representative of phytoplankton. Note that no analytical measurement was performed.

De C S de Paula et al., 2022 evaluated the 48h acute toxicity of OC in *D. magna* and *A. salina* (also in *D. subspicatus* see §B.7.1.3.). The tested nominal concentrations were between 0.5 and 7 mg.l⁻¹ for *D. magna* and 0.75 and 5 mg.l⁻¹ for *A. salina* and the EC₅₀ were 2.57 mg.l⁻¹ and 2.97 mg.l⁻¹ respectively. The concentrations were not measured during the test. *Marcin et Aleksander., 2023* studied the toxicity of octocrilene at 2, 4, 6, 8, 10 mg.l⁻¹ (nominal) on species from different stages of the trophic chain among with the marine (*Artemia franciscana*) and the freshwater (*Daphnia magna*) crustaceans. After 48h exposure, the LC₅₀ was 0.55 mg.l⁻¹ for *A. franciscana* and *D. magna* EC₅₀ was 2.66 mg.l⁻¹. Similar toxic concentration values were reported by *De C S de Paula et al., 2022*, *Thorel et al., 2020* and *Park et al., 2016b*. In both crustacean tests, octocrilene was the most toxic UV filter among the nine investigated.

Boyd et al., 2020 exposed daphnids neonates for 48 h to 0.2, 2, 20, and 200 µg.l⁻¹ (nominal) prior to being raised an additional 19 days in water to an age of 21 days total. No significant mortality was observed during the 48h-acute exposure (LC₅₀=3.6 mg.l⁻¹). An CE₅₀ (48h) closed to environmental concentrations was recorded at 30 µg.l⁻¹ (immobilization). A significant increased of the phototactic response latency with a log-

linear relationship with dose was detected at 200 µg.l⁻¹; however, recovery was observed following 19 days post-exposure in clean water. Time to first brood and the molting behavior were unaffected whereas a delayed mortality up to 87% was observed over the following 7 days post-exposure period in daphnids acutely exposed to 200 µg.l⁻¹. Those results corroborate the high chronic lethality observed (§ B7.1.2.2). This study showed that 48-hours octocrilene exposure reduced survivorship and behaviorally impaired daphnids at environmentally realistic concentration.

The acute toxicity of octocrilene to invertebrates was tested in three guideline studies. Several data are also available in the scientific literature. Considering the methodological missing points in these three GLP guideline studies and in the available literature, acute invertebrates toxicity endpoints are considered as supportive data.

B.7.1.2.2. Long-term toxicity to aquatic invertebrates

The results are summarised in the following Table 22:

Table 22: Long-term effects on aquatic invertebrates.

Method	Results	Remarks	Reference
<i>Daphnia magna</i> Freshwater Exposure duration: 22d semi-static, daily renewal Nominal concentrations: 3.1, 6.3, 12.5, 25 and 50 µg.l ⁻¹ (based on test substance mass not corrected for purity) Measured concentrations: 2.66, 5.25, 9.62, 19.2, 40.6 µg.l ⁻¹ (two mean meas.) Solvent (DMF 0.1 mL/L) and water controls Photoperiod: 16:8h light:dark Water temperature: 21°C ± 1°C Replicates: one neonate (<24h old) <i>D. magna</i> into each of the 10 replicates per test group Stocking density: 0.02 animals/mL	Parent mortality: 100% at 40.6 µg.l ⁻¹ Number of living young: ↓ 16%, 30%, 66% and 100% at 5.25, 9.62, 19.2, 40.6 µg.l ⁻¹ ↑ 15.2% time to first brood at 19.2 µg.l ⁻¹ LOEC = 5.25 µg.l ⁻¹ NOEC = 2.66 µg.l⁻¹ EC10 = 4.58 µg.l ⁻¹ based on reproduction (living offspring per surviving adult)	1 (reliable without restriction) key study experimental study Test material (EC name): octocrilene 14C-Octocrilene Purity: 99.4 % OECD Guideline 211 (2012) (<i>Daphnia magna</i> Reproduction Test) GLP: yes	Unpublished study report (2018c)
<i>Daphnia magna</i> Freshwater 21 days exposure Semi-static, renewal every 3 days Nominal concentrations: 3.12, 6.25, 12.5, 25, 50 µg.l ⁻¹ Measured concentrations: Unknown Solvent (DMSO 0.01%)	NOEC (21d): > 50 µg.l ⁻¹ (nominal based on reproduction) NOEC (21d): > 50 µg.l ⁻¹ (nominal based on growth)	2 (reliable with restriction) Test material: octocrilene Purity: unknown Guideline: no Not GLP	Pablos et al., 2015

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
control Replicates: one neonate (<24h old) <i>D. magna</i> into each of the 10 replicates per test group Stocking density: 0.06 animals/mL			
<i>Daphnia magna</i> Freshwater Exposure duration: 21 days Semi-static, renewal every 2 days Nominal concentration: 0.5, 7.5 µg.l ⁻¹ Measured concentration: unknown (stock solution (100 g/L) stable) Solvent (DMSO 1%) and water controls Replicates: 5 neonates (<24 h old) into each of the 6 replicates per test group Photoperiod: 12:12h light:dark Water temperature: 20°C ± 1°C	Mortality (day 21): 0% at 0.5 µg.l ⁻¹ 100% at 7.5 µg.l ⁻¹ (from day 7) No effect on: Time to first brood Brood size per daphnid Proportion of reproductive adults Number of neonates/individual Number of molts Metabolic rate (respiration rate)	2 (reliable with restriction) Test material: octocrilene Purity: unknown Based on OECD Guideline 211 (2008) (<i>Daphnia magna</i> Reproduction Test) with modifications Not GLP	Boyd et al., 2020
<i>Daphnia magna</i> Two generations (F0 and F1) exposed 21 days Semi-static, renewal every 2 days Nominal concentration: 4.45 µg.l ⁻¹ Measured concentration: unknown Solvent (DMSO <0.03%) and M4 medium controls Photoperiod: 16:8h light:dark Water temperature: 20°C ± 2°C Replicates: one neonate into each of the 12 replicates per test group (reproduction) four neonate into each of the 3 pooled replicates per test group (biochemical biomarkers)	No effect (F0 and F1) Longevity Time for first reproduction Reproduction rate Oxidative stress (CAT and GST activity)	2 (reliable with restriction) Test material: octocrilene Purity: unknown Based on OECD Guideline 211 (2008) (<i>Daphnia magna</i> Reproduction Test) with modifications Not GLP	De C S de Paula et al., 2022

Discussion

In an *unpublished study report (2018c)*, a 21d *Daphnia magna* reproduction GLP study was performed according to OECD TG 211 using ^{14}C test material. Octocrilene was tested at 3.1, 6.3, 12.5, 25 and 50 $\mu\text{g.l}^{-1}$ (nominal). The measured concentrations deviated markedly from the nominal concentrations (56-85%), then the time-weighted mean measured concentrations 2.66, 5.25, 9.62, 19.2, 40.6 $\mu\text{g.l}^{-1}$ (77%–86% of the nominal) were used for the evaluation of the test substance. In this study, the parent mortality, reproduction and sublethal effects were observed daily. At the test end, a clear concentration response on daphnid reproduction was observed: a significant 16% to 100% decreases of the number of living young per surviving adult were observed from 5.25 to 40.6 $\mu\text{g.l}^{-1}$ respectively. At the highest concentration, results showed 100% parent mortality. A significant 15.2% increase of mean days to first brood was shown at 19.2 $\mu\text{g.l}^{-1}$. No significant effect on growth, the number of immobile young and aborted eggs was shown. In the water and solvent controls, the parent mortality was <20% and mean number of living young per parent ≥ 60 , then this test was compliant with the validity criteria. A NOEC of 2.66 $\mu\text{g.l}^{-1}$ (TWM measured concentration) was derived based on reproduction (living offspring per surviving adult).

In *Pablos et al., 2015*, a semi-static reproduction test with juveniles *D. magna* (less than 24h old) was performed according to OECD TG 211 (2008). Daphnids were exposed to 3.12, 6.25, 12.5, 25, 50 $\mu\text{g.l}^{-1}$ (nominal). Analytical determinations of the test compounds were not conducted. This paper did not give any information on the exposure conditions and the results were not detailed. Based on the available informations, the estimated ration level of *C. vulgaris* was 0.067 mg C/daphnids/day which is slightly lower than the guideline recommendations (between 0.1 and 0.2 mg C/Daphnia/day required to be sufficient for achieving the required number of living offspring to meet the test validity criteria). At the test end (21 days), no significant effect on the number of living offspring per parent animal and on the parental size were detected up to the highest tested concentration (50 $\mu\text{g.l}^{-1}$) then, the proposed NOEC was >50 $\mu\text{g.l}^{-1}$.

In *Boyd et al., 2020* study, 21-day chronic exposure to 7.5 $\mu\text{g.l}^{-1}$ (nominal) octocrilene yielded complete mortality within 7 days whereas 0.5 $\mu\text{g.l}^{-1}$ did not affect the mortality rate throughout the 21 days exposure. No effect on physiological and reproductive outputs were observed until the test end. However, this study followed the OECD TG 211 with modifications and exposure conditions were not detailed. The tested concentration was calculated from the dilution of the stock solution that was the only one measured and showed to be stable. It can be noted that the mean neonates per daphnia recorded at the test end in the control is about 43% and did not meet the validity criteria of 60%. Nonetheless, combined with the acute exposure, results showed that octocrilene exerts toxic effects in freshwater invertebrates and could pose a widespread threat to population at environmentally realistic concentrations.

De C S de Paula et al., 2022 investigated the chronic effects of octocrilene in two consecutive generations of *D. magna* exposed 21 days at 4.45 $\mu\text{g.l}^{-1}$ nominal. The longevity, time to produce the first offspring, reproduction rate, and oxidative stress (activities of the catalase and glutathione-S-transferase enzymes) were evaluated at day 21. No significant effect was observed in both generations for all tested parameters. However, information on the study protocol are lacking. The supplied diet based on the amount of organic carbon is not clearly detailed, the mean number of living offspring produced per parent surviving at the end of the test is 47% in F0 and 46% in F1 (reproduction rate) which is lower than the validity criteria of 60%. The tested volume per daphnia is also lower than recommended in OECD TG 211. Furthermore, only one concentration was tested without measure of its stability.

The NOEC (21d) of 2.66 µg.l⁻¹, has been considered as the relevant endpoint for long-term toxicity to aquatic invertebrates.

B.7.1.3. Algae and aquatic plants

The results are summarised in the following Table 23:

Table 23: Effects on algae and aquatic plants.

Method	Results	Remarks	Reference
<i>Desmodesmus subspicatus</i> (algae) freshwater static, 72h exposure Nominal concentrations: 10, 22, 46, 100, 220 mg.l ⁻¹ Reference substance: dichromate de potassium ErC50=0.86 mg.l ⁻¹ Temperature 22.1-22.4°C Replicates: 6 for the control, 3 for each test item concentration	ErC50 (72 h): > 220 mg.l ⁻¹ test mat. (nominal) based on: growth rate NOErC (72 h): 100 mg.l ⁻¹ test mat. (nominal) based on: growth rate	2 (reliable with restriction) supportive study experimental result Test material (EC name): octocrilene Purity >99.5% OECD Guideline 201 (<i>Alga, Growth Inhibition Test</i>) GLP yes	Unpublished study report (2010)
<i>Chlorella vulgaris</i> (green algae) Freshwater Static, 72h exposure Nominal concentrations: 0.19, 0.39, 0.78, 1.56, 3.12, 6.25, 12.5, 25 mg.l ⁻¹ Measured concentrations: unknown Solvent: DMSO 0.01 % Photoperiod: constant light Temperature: 20+/-2°C Replicate: 6 per sample	NOEC (72h): 3.12 mg.l ⁻¹ test mat. (nominal) based on: growth rate IC10 (72h): 0.4 mg.l ⁻¹ test. Mat (nominal) based on: growth inhibition. IC50 (72h): 20.3 mg.l ⁻¹ test. Mat (nominal) based on: growth inhibition.	2 (reliable with restriction) supportive study experimental result Test material (EC name): octocrilene Modified OECD Guideline 201 (<i>Freshwater Alga and Cyanobacteria, Growth Inhibition Test</i>) Not GLP	Pablos et al., 2015
<i>Tetraselmis sp</i> (green algae) Marine water, 7 days exposure Nominal concentrations: 0.01, 0.1, 1 mg.l ⁻¹ Measured concentration: unknown Solvent (DMSO 2.5% v/v) control Temperature: 18°C Photoperiod: 12:12h light:dark Triplicate with initial cell density of 5.10 ⁴ cells/mL	↑ Growth rate at 1 mg.l ⁻¹ (hormesis) ↓ Esterase activity at ≥ 0.1 mg.l ⁻¹	2 (reliable with restriction) supportive study Experimental result Test material (EC name): octocrilene Purity unknown Guideline: no Not GLP	Thorel et al., 2020
<i>Desmodesmus subspicatus</i> Freshwater static, 72h exposure Nominal concentrations: 0.5, 1, 1.5, 2, 2.5, 3 mg.l ⁻¹ Measured concentration: unknown Solvent (DMSO 0.03%) and water	EC50 = 1.54 mg.l ⁻¹ based on growth inhibition	2 (reliable with restriction) supportive study Experimental result Test material (EC name): octocrilene Purity >99.8%	De C S de Paula et al., 2022

Method	Results	Remarks	Reference
control Temperature 23-25°C		Guideline: no Not GLP	
Lemna minor Freshwater, 7 days exposure Nominal concentrations: 2, 4, 6, 8, 10 mg.l ⁻¹ Measured concentration: unknown Solvent (DMSO 1%) control Photoperiod : Water temperature: 20°C ± 0.5°C Replicates: one plant with 2-3 fronds into 10mL of test solution in triplicate for each treatment	IC ₅₀ (7d): 1.95 mg.l ⁻¹ based on growth inhibition (leaf area)	2 (reliable with restriction) Test material: octocrilene Purity >98% according to OECD Guideline 221	Marcin et Aleksander., 2023

Discussion:

An *unpublished study report (2010)* conducted in compliance with OECD TG 201 showed a slight toxicity of the substance to *Desmodesmus subspicatus*. The exposure was terminated after 72 hours and an EC₅₀>220 mg.l⁻¹ based on algal growth rate was determined. The study protocol did not include analytical measurements performed throughout the test to determine the measured value of the saturated concentration. Even if, in according to guideline OECD TG 23 (on aquatic toxicity testing of difficult substances and mixtures), the effect concentration can be expressed based on the nominal concentrations for tests with chemicals that cannot be quantified by analytical methods at the concentrations causing slight effects, the nominal value is considered far from the solubility limit. Consequently this endpoint should be considered as supportive endpoint.

In a test modified from OECD TG 201 for low volumes, *Pablos et al., 2015* showed the low toxicity of octocrilene for the freshwater algae *C. vulgaris* with an IC₅₀ of 3.6 mg.l⁻¹ nominal after 72h exposure. In an other freshwater species *D. subspicatus*, the EC₅₀ was observed in the same order of magnitude at 1.54 mg.l⁻¹ (nominal) (*De C S de Paula et al., 2022*). In these studies, the substance stability was not monitored throughout the test.

Thorel et al., 2020, studied octocrilene toxicity on marine organisms from two major trophic levels including autotrophs (*Tetraselmis sp.*) and heterotrophs (*Artemia salina*; see §B.7.1.2.1.). After 7 days of exposure, a significant decrease of metabolic activity of the marine microalgae *Tetraselmis sp.* at 0.1 mg.l⁻¹ (nominal) and above was observed. Any alterations in these populations may result in chain reaction effects on organisms at higher trophic levels. This study showed preliminary results before investigating a large diversity of species representative of phytoplankton.

The toxicity of octocrilene on freshwater plant was studied by Marcin and Aleksander (2023) with *Lemna minor*. Octocrilene induced a significant decrease in frond number and frond area after 7 days exposition. The calculated IC₅₀ = 1.95 mg.l⁻¹ (nominal).

The available OECD TG 201 study on Algae is not valid to determine an endpoint value (nominal value far from the solubility limit). Nevertheless, it can be concluded that algae are not the most sensitive trophic level.

B.7.1.4. Conclusion for surface water

Invertebrate *Daphnia magna* has been considered that the most sensitive species with the NOEC determined at 2.66 µg.l⁻¹. Even if a fish study from the literature (Yan *et al.*, 2020) showed effects on the early life stage of fish (larval stage) at similar concentration levels (from 2.7 µg.l⁻¹). However, this fish study cannot be used to provide the Lowest Observed Effect Concentration (LOEC) and NOEC which are necessary for a risk assessment as it was not designed to investigate adverse effects and did not follow any relevant OECD test guideline. This fish study is therefore only used as supportive information.

B.7.1.5. Sediment organisms

The results are summarised in the following Table 24:

Table 24: Effects on sediment-dwelling organisms.

Method	Results	Remarks	Reference
<i>Chironomus riparius</i> 2 days old first-instar larvae 28-day static sediment-water systems Nominal concentrations: 62.5, 125, 250, 500 and 1000 mg.kg ⁻¹ dw Measured concentrations: recoveries inside 80 ⁻¹ 20% of spiked amount Solvent control (acetone) Temperature : 19.4 – 20.9°C pH: 7.1 – 8.3 O ₂ : 5.9 – 8.6 mg.l ⁻¹ Light intensity: 505-990 lux Photoperiod :16:8h light:dark Replicate: 8 per treatment	NOEC (28d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: emergence rate EC10 (28d) > 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: emergence rate EC50 (28d) > 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: emergence rate NOEC (28d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: development rate, sex ratio, egg rope production, fertility, emergence (F1); development rate (F1), sex ratio (F1) EC10 (28d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: development rate, sex ratio, egg rope production, fertility, emergence (F1); development rate (F1), sex ratio (F1) EC50 (28d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: development rate, sex ratio, egg rope production, fertility, emergence (F1); development rate (F1), sex ratio (F1)	2 (reliable with restriction) key study experimental result Test material (EC name): octocrilene Purity: 99.8% According to guideline OECD 233 Guideline Sediment-Water Chironomid Life-Cycle Toxicity Test Using Spiked Water or Spiked Sediment	Unpublished study report (2023a)
<i>Myriophyllum spicatum</i> Freshwater-sediment system Smart and Barko medium 14d exposure Static test Tested concentrations: 62.5, 125, 250, 500 and 1000 mg.kg ⁻¹ dw (nominal) Solvent control (acetone)	NOEC (14d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: Total shoot length (growth rate) LOEC (14d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: Total shoot length (growth rate) EC10 (14d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: Total shoot length (growth rate) EC50 (14d) ≥ 1000 mg.kg ⁻¹ sediment dw test mat. (nominal) based on: Total shoot length	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene	Unpublished study report (2023b)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
16 hours light: 8 hours dark Temperature: 20.9°C (mean) [20.2-21.6°C] Soil pH: 7.8⁻¹0 Oxygen concentration (all treatments): 8.6 – 16.3 mg.l⁻¹ Replicates: 4 per condition (3 plants eah control)	(growth rate) NOEC (14d) ≥ 1000 mg.kg⁻¹ sediment dw test mat. (nominal) based on: Fresh weight (growth rate) or Dry weight (growth rate) LOEC (14d) ≥ 1000 mg.kg⁻¹ sediment dw test mat. (nominal) based on: Fresh weight (growth rate) or Dry weight (growth rate) EC10 (14d) ≥ 1000 mg.kg⁻¹ sediment dw test mat. (nominal) based on: Fresh weight (growth rate) or Dry weight (growth rate) EC50 (14d) ≥ 1000 mg.kg⁻¹ sediment dw test mat. (nominal) based on: Fresh weight (growth rate) or Dry weight (growth rate)	Purity: 99.8% According to guideline OECD Guideline 239 (Water-Sediment Myriophyllum Spicatum Toxicity Test)	
<i>Lumbriculus variegatus</i> Sediment dwelling worm Freshwater-sediment system 28 days, Static Artificial spiked sediment Nominal concentrations: 9.7, 21, 47, 103, 227, and 500 mg.kg⁻¹ sediment dw Measured concentrations: recoveries inside 80-120% of spiked amount Water, solvent (acetone) and positive (PCP) controls Temperature: 20.4°C (mean) pH 7.9-8.3 (controls) 7.9-8.4 (treatments) Oxygen concentration (all treatments): 8 – 8.8 mg.l⁻¹ 16:8h light:dark Replicates: 6 (controls) and 4 (treatments) with n=10 synchronised worms each	<u>Reproduction (total number of surviving worms) at day 28:</u> NOEC repro = 47 mg.kg⁻¹ dw (nom) EC10 repro = 62.8 mg.kg⁻¹ dw (nom) <u>Biomass (dry weight):</u> NOEC biomass = 47 mg.kg⁻¹ dw (nom) EC10 biomass = 46.3 mg.kg⁻¹ dw (nom)	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene Purity: 99.8 % According to guideline OECD GD 225 (Sediment-Water Lumbriculus Toxicity Test Using Spiked Sediment) GLP yes	Unpublished study report (2023c)
<i>Chironomus riparius</i> Freshwater-sediment system Artificial spiked sediment 28d exposure, static Nominal concentrations: 0.08 (0.052*) – 0.4 – 2 – 10 and 50 (51.6*) mg.kg⁻¹ dw, *measured concentrations at the test end Replicates: 4 per condition Solvent control (ethyl acetate) According to OECD Guideline 218 (Sediment-Water	NOEC (28 d) > 50 mg.kg⁻¹ dw (nominal) based on: mortality, number of produced clutches, number of eggs per egg mass, mean emergence time, sex ratio	2 (reliable with restrictions) Supporting study Test material (EC name): octocrilene Purity: unknown	Kaiser et al., 2012a

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
<i>Chironomid Toxicity Test Using Spiked Sediment)</i>			
Potamopyrgus antipodarum Freshwater-sediment system 56 days of exposure, static 0.00064 (0.0019*), 0.016 (0.045*), 0.08, 0.4 (0.37*), 2, 10, 50 mg.kg _{dw} ⁻¹ , nominal * measured concentrations at the test end Solvent control (ethyl acetate)	No effect on the embryo numbers (reproduction)		
Lumbriculus variegatus Freshwater-sediment system 28 days of exposure, Static Sediment: artificial spiked sediment Nominal concentrations: 0.08 (0.059*), 0.4, 2, 10, 50 (36.8*) mg.kg _{dw} ⁻¹ *measured concentrations at the test end Replicates: 4 per condition Solvent control (ethyl acetate) OECD Guideline 225 (Sediment-Water Lumbriculus Toxicity Test Using Spiked Sediment)	No effect on the biomass and the number of worms (reproduction)		
Chironomus riparius 4th instar larvae Freshwater system (no sediment) Static, 24h exposure Nominal concentrations: 0.1, 1, 10 mg.l ⁻¹ Measured concentrations: unknown Solvent control (ethanol 0.02%) Replicates: N=30, 3 per condition (n = 90 at survival experiments) n = 30 at expression analysis	No significant mortality NOEC >10 mg.l ⁻¹ (nominal)	2 (reliable with restrictions) Supporting study Test material (EC name): octocrilene Purity> 97% According to toxicity testing guidelines (EPA, 2000; OECD, 2004). Not GLP	Ozaez et al., 2013

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
<i>Chironomus riparius</i> Freshwater system (no sediment) Expression studies: 24h exposure, static Eggs mass and 4th instar larvae 1 mg.l⁻¹ (nominal) N=30 larvae Survival test: 3 days exposure, renewal every 24h 4th instar larvae, N=90 0.1, 1, 10*, 100* mg.l⁻¹ (nominal) *Measured concentrations (24h) = 39% of the starting concentration. Replicates: 3 per condition Solvent control (ethanol)	LC₅₀, larvae 96h = 95.9 mg.l⁻¹ (nominal) ↓ % eclosion at 1 mg.l⁻¹ (nominal) Gene expression in embryos (24h): ↑ Heat Shock Protein (<i>hsp70</i>) at 1 mg.l⁻¹ nominal Gene expression in larvae (24h): No effect	2 (reliable with restrictions) Supporting study Test material (EC name): octocrilene Purity> 97% According to toxicity testing guidelines (EPA, 2000; OECD, 2004). Not GLP	Ozaez et al., 2016
<i>Chironomus riparius</i> 1st instar (<48h post-hatching) Freshwater-sediment system 28d exposure, static Sediment: artificial spiked sediment Nominal concentration: 2.5, 5, 10 mg.kg dw⁻¹ Measured concentration (5 days after spiking): 0.53 – 1.27 and 2.33 mg.kg dw⁻¹ Solvent control (ethanol 96%) Replicates per condition: 6; n=5 larvae	↓ Length at 2.33 mg.kg dw⁻¹ NOEC (10d): 1.27 mg.kg dw⁻¹ (meas. (not specified)) based on length NOEC (28d): >2.33 mg.kg dw⁻¹ (meas. (not specified)) based on development time (days until emergence)	2 (reliable with restriction) Supporting study Test material (EC name): Octocrilene Purity >97% Similar to OECD 218 (Sediment-Water Chironomid Toxicity Test Using Spiked Sediment) Not GLP	Campos et al., 2017
<i>Chironomus riparius</i> 4th instar larvae Freshwater system (no sediment) 24h exposure Nominal concentrations: 0.1, 1, and 10 mg.l⁻¹ Measured concentration: unknown Solvent (ethanol 0.02%) control Exposure in obscurity Replicate: three larvae into	No effect Based on: gene transcription <i>HYOU1</i>, <i>I(2)efl</i> (Stress response); <i>CypP450</i>, <i>GST</i> (detoxification mechanisms); <i>Defensin</i>, <i>PO1</i> (immune system))	2 (reliable with restriction) Supporting study Test material (EC name): Octocrilene Purity >97% No guideline	Muñiz-González et Martínez-Guitarte., 2018

Method	Results	Remarks	Reference
each of the 3 replicates per test group			

Discussion:

Three OECD guideline GLP studies on *Myriophyllum spicatum* (Unpublished study report, 2023b), *Chironomus riparius* (Unpublished study report, 2023a) and *Lumbriculus variegatus* (Unpublished study report, 2023c), are available.

In an unpublished study report (2023b), the toxic effects of 62.5, 125, 250, 500 and 1000 mg.kg_{dw}⁻¹ octocrilene (nominal) on the submerged aquatic plant *Myriophyllum spicatum* were determined in a 14-day static water-sediment growth inhibition test. The measured concentrations of octocrilene in sediment samples ranged between 76 and 88% of the nominal values at the start of the test and between 78 and 97% at the end of the test. Thus, the analytical results confirm the correct dosage of the test item and its stability in the sediment phase. The biological results were related to the nominal concentrations of the test item. This study meet the OECD TG 239 validity criteria. After 14 days of exposure, no toxic effects were found based on the growth parameters assessed including total shoot length, fresh weight and dry weight in any of the test item treatments. In addition, the plants showed no visible deformities or symptoms of chlorosis in any of the test item treatments. The NOEC of the test was determined to be at least the test concentration of nominal 1000 mg.kg_{dw}⁻¹.

In an unpublished study report (2023a), first-instar larvae of *Chironomus riparius* were exposed to the octocrilene spiked into the sediment in a static sediment-water system with nominal test concentrations of 62.5, 125, 250, 500 and 1000 mg.kg_{dw}⁻¹. Except for two values in the sediment systems used for the F1 generation (68% of 250 mg.kg_{dw}⁻¹ at day 0; 79% of 62.5 mg.kg_{dw}⁻¹ at day 28), all determined octocrilene concentrations were ≥80 % of nominal and thus confirmed the correct dosing of the test item. The nominal concentrations were used to derive effects. The total number of adults emerged, development rate and sex ratio of the fully emerged and alive adults (for both 1st and 2nd generations) as well as the number of egg ropes per female and fertility of the egg ropes (1st generation only) were compared with the corresponding parameters in the control. After 28 days exposure of parental generation, none of the endpoints were affected by the exposure to the test item up to and including the highest test concentration of 1000 mg octocrilene.kg_{dw}⁻¹. Neither the production of egg ropes nor fertility was affected by the exposure to the test item up to and including the highest test concentration of 1000 mg.kg_{dw}⁻¹. No effect was also observed after 28 days exposure of F1 generation. Due to the absence of toxic effects, the NOEC of the test item octocrilene for *Chironomus riparius* in the life-cycle toxicity test using spiked sediment was determined to be 1000 mg/kg ds (nominal). This study meet the OECD TG 233 validity criteria.

However, it should be noted that the toxicity was probably underestimated due to octocrilene behaviour ($K_{oc}=48898 \text{ l.kg}^{-1}$). For strongly adsorbing substances, ingestion of contaminated food is a more significant exposure route. In this study, the route of exposure was not modified. The fish food flakes were suspended in test water and was not contaminated. The analytical results showed that the test item was not found in water nor pore water. Then, the organisms were mostly non exposed to the substance.

In an *unpublished study report (2023c)*, the toxic effects of octocrilene was evaluated on the reproduction (total number of surviving worms) and biomass (dry weight) of a sediment-endobenthic dwelling worm, the freshwater oligochaete *Lumbriculus variegatus*. Synchronized worms were exposed during 28 days to octocrilene spiked into the artificial sediment at 9.7, 21, 47, 103, 227, and 500 mg.kg_{dw}^{-1} . Food was mixed into the sediment during spiking the sediment with the test item. The analytical results confirmed the correct dosage of octocrilene and its stability in the sediment phase. No octocrilene was recovered from the water phase, except in one pore water sample with very small amount ($<0.01 \%$ of the total spiked amount). The recoveries of the test substance were inside $80^{-1} 20\%$ of spiked amount and the nominal concentrations were used to derive effects. The mean total number of worms and the dry biomass in the water and solvent controls did not differ statistically significantly from each other. Then all control replicates were pooled. This study fulfilled the OECD TG 225 validity criteria. The TOC content in the sediment at 3.1% (3.6% with the added food) was noted to be higher than the 2% ($\pm 0.5\%$) recommended in the guideline. This deviation was considered not to have influenced the test results. A statistically significant reduction in worm reproduction and biomass was found at day 28 in the three higher treatments 103, 227 and 500 mg.kg_{dw}^{-1} . The mean total number of worms were 19.0, 14.5, and 11.8 respectively (compared to 22.5 for the control) and the mean worm dry weight were 17.1, 15.1, and 12.6 mg respectively (compared to 22.4 mg for the control). The NOEC was determined to be at 47 mg.kg_{dw}^{-1} . The EC_{10} was determined to be at 62.8 and 46.3 mg.kg_{dw}^{-1} for reproduction and dry biomass, respectively.

Additional data are available from the scientific literature:

In *Campos et al., 2017*, octocrilene induced a significant reduction in the size of *C. riparius* larval at the highest tested concentration 2.33 mg.kg_{dw}^{-1} . In spite of not statistically significant, a weight reduction in female (18.63%, $p=0.04$) and to a lesser extent in male (8.47%) imagoes was also observed at 2.33 mg/kg_{dw} . Octocrilene did not significantly affect development time (days until emergence) of female nor male. Note that according to the guidance OECD TG 218, growth is defined as an "increase in dry weight of test organisms during the experiment and expressed as mean dry weight per surviving chironomid". Then, the proposed NOEC is based on length rather than growth as stated by the authors, and no significant effect on growth was observed up to the highest tested concentration. In this study, the concentrations of octocrilene in the sediments were measured only at 5 days after spiking. The measured concentrations at the test end were not specified.

A study, *Kaiser et al., 2012a* aimed to evaluate toxic effects and endocrine disruption of three common UV filters including octocrilene on aquatic organisms, focusing particularly on infaunal and epibenthic invertebrates. Tests were performed according to OECD guidelines with minor changes on the following species:

- *Potamopyrgus antipodarum*, sensitive for endocrine active substances,
- Aquatic oligochaete *Lumbriculus variegatus* extensively used as a standard test organism for sediment toxicity and bioaccumulation tests (OECD TG 225),

- *Chironomus riparius* widely distributed in a broad range of water bodies with an important ecological role (OECD TG 218).

The results showed that octocrilene did not affect any of the test organisms at the tested concentrations up to 50 mg.kg_{dw}⁻¹ (nominal).

Ozaez et al., 2016 investigated the effect of octocrilene on the expression level of genes coding for the heat shock protein as biomarker of stress effects at the cellular level in embryos and larvae of *Chironomus riparius*. An increased of *hsp70* gene expression after 24h exposure of egg mass to 1 mg.l⁻¹ octocrilene was observed whereas no effect was detected at the larval stage. A significant decreased in the percentage of eclosed eggs was also shown at 1 mg.l⁻¹ suggesting a lethal impairment in embryo development. Those results revealed a differential sensitivity dependent on the stage of development in insects. Note that concentrations were measured only for 10 and 100 mg.l⁻¹ after 24h and were at 39% of the starting concentration.

Muniz et al., 2018 analyzed the transcriptional activity of genes related to the endocrine system, stress response, detoxification mechanisms, and the immune system in the fourth instar of *C. riparius* larvae. A 24h exposure to 0.1, 1 and 10 mg.l⁻¹ octocrilene did not affect the transcriptional activity of either gene. Longer exposure should be tested as well as earlier developmental stage as embryo with higher sensitivity (*Ozaez et al., 2016*). The concentration stability was not analysed.

However, in those studies the range of tested concentration is above the solubility limit. Furthermore, only nominal concentrations were reported, and when concentration were measured they were not available at the test end. Without analytic measurement, the bioavailable fraction of the substance remain unknown given octocrilene's low solubility, high lipophilicity and high adsorption capacity. These studies can be use as supportive data.

Experimental data on the toxicity of octocrilene are available for three different sediment dwelling species: *Chironomus riparius*, *Myriophyllum spicatum* and *Lumbriculus variegatus*. Test results show that *L. variegatus* is the most sensitive species in the 28-day sediment tests performed, with the EC₁₀ for biomass as the most sensitive endpoint determined at 46.3 mg.kg_{dw}⁻¹.

B.7.1.6. Other aquatic organisms

The results are summarised in the following Table 25:

Table 25: Effects on other aquatic organisms.

Method	Results	Remarks	Reference
<i>Tetrahymena thermophila</i> 24h exposure Nominal concentration: 0.001, 0.01, 0.1, 1, 5, 10, 15 mg.l ⁻¹ Measured concentration: unknown Solvent (DMSO 0.1%) Temperature: 30°C Triplicate	EC50 >15 mg.l ⁻¹ Based on growth inhibition	2 (reliable with restriction) Supporting study Test material (EC name): Octocrilene Purity >97% No guideline	Gao et al., 2013

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
Marine species Nominal concentration: 5, 20, 150 µg.l⁻¹ Measured concentrations (48h): 0.9, 1.25, 42,38 µg.l⁻¹ Solvent (DMSO 0.1%) FSW (filtered Seawater) controls <i>Isochrysis galbana</i> (algae) 72h exposure Replicates : three per concentration and control <i>Mytilus galloprovincialis</i> (wild mussel) 48h, 20°C 40 eggs/mL Replicates : three per concentration, ten per control (FSW and DMSO) <i>Paracentrotus lividus</i> (Sea urchin) 48h, 20°C 40 eggs/mL	NOEC (72h): 40 µg.l⁻¹ test mat. (nominal) based on: cell density (<i>Isochrysis galbana</i>) NOEC (48h): 20 µg.l⁻¹ test mat. (nominal) based on: development stage (<i>Mytilus galloprovincialis</i>) NOEC (48h): 20 µg.l⁻¹ test mat. (nominal) based on: growth rate (<i>Paracentrotus lividus</i>) EC10 (72h): 103 µg.l⁻¹ test mat. (nominal) based on: cell density (<i>Isochrysis galbana</i>) EC10 (48h): 511 µg.l⁻¹ test mat. (nominal) based on: development stage (<i>Mytilus galloprovincialis</i>) EC10 (48h): 162 µg.l⁻¹ test mat. (nominal) based on: growth rate (<i>Paracentrotus lividus</i>) EC50 (72h): >150 µg.l⁻¹ test mat. (meas. (not specified)) based on: cell density (<i>Isochrysis galbana</i>) EC50 (48h): >650 µg.l⁻¹ test mat. (nominal) based on: development stage (<i>Mytilus galloprovincialis</i>) EC50 (48h): 737 µg.l⁻¹ test mat. (nominal) based on: growth rate (<i>Paracentrotus lividus</i>)	2 (reliable with restrictions) supporting study Test material (EC name): Octocrilene Purity >98% No guideline followed	Giraldo et al., 2017
<i>Pocillopora damicornis</i> (coral) Seawater 7 days of exposure Nominal concentrations: 1000, 300, 50, and 5 µg.l⁻¹ Measured concentrations: unknown Solvent control (DMSO 0.25% v/v) Salinity 36g/L pH 8 Temperature 24°C Photoperiod: 10:14h light:dark Replicate: n=5	<i>P. damicornis</i> significantly impacted at 50 µg.l⁻¹ NOEC = 5 µg.l⁻¹ (nominal) Based on fatty acids metabolism	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene Purity >98% No guideline followed	Stien et al., 2018

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
<i>Seriatopora. caliendrum</i> <i>Pocillopora. Damicornis</i> (coral) 7 days of exposure Tested concentrations: octocrilene: 0.1 – 1 – 10 – 100 – 1000 µg.l⁻¹ (nominal). Maintained concentrations on Day 0 and Day 1. Strong decrease on day 7. Octocrilene in combination with EHMC: 5 - 30 - 100 µg.l⁻¹ (nominal). Maintained concentrations on Day 0. Strong decrease on day 7. Sunscreen water 5%: 3.6% w/w as active ingredient in sunscreen water (with 7% w/w of EHMC). Maintained concentrations on Day 0 and Day 1. Strong decrease on day 7. Solvent control (methanol 0.01%) Blank control Salinity 35.7±0.6 ‰ pH: 7.7±0.2 Temperature: 25.3 ± 0.8 °C Photoperiod: 14h:10h light:dark 200-300 µmol quanta m⁻² s⁻¹ Dissolved oxygen: 6.6 ± 0.5 mg.l⁻¹ Replicate: 3 of three nubbins per concentration of exposure	<i>S. caliendrum</i> Total polyp retraction: LOEC = 1000 µg.l⁻¹ (Single OC) LOEC = 5 µg/L (combined EHMC and OC) LOEC = 5 µg.l⁻¹ (sunscreen water) Mortality, Visual bleaching, Zooxanthellae density: LOEC = 100 µg.l⁻¹ (combined EHMC and OC) LOEC = 30 µg.l⁻¹ (sunscreen water) <i>P. damicornis</i> Total polyp retraction: LOEC = 1000 µg.l⁻¹ (Single OC) LOEC = 100 µg.l⁻¹ (combined EHMC and OC) LOEC = 5 µg.l⁻¹ (sunscreen water) Mortality, Visual bleaching, Zooxanthellae density: LOEC = 100 µg.l⁻¹ (combined EHMC and OC) Mortality, Visual bleaching: LOEC = 30 µg.l⁻¹ (sunscreen water) Zooxanthellae density: LOEC = 100 µg.l⁻¹ (sunscreen water)	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene single octocrilene in combination with EHMC octocrilene in sunscreen water No guideline followed	He et al., 2018b
<i>Stylophora pistillata</i> (coral) 7 days exposure Artificial seawater Renewal every 24h Nominal concentration: 300, 1000 µg.l⁻¹ Measured concentration: unknown Solvent control (DMSO 0.25% v/v) Salinity 36g.l⁻¹ pH 8 Temperature 24°C Photoperiod: 10:14h light:dark Replicate: 5	↑ acylcarnitines and sphingoid at 300 and 1000 µg.l⁻¹	2 (reliable with restriction) Test material: octocrilene Purity unknown No guideline followed Not GLP	Thorel et al., 2022
<i>Aliivibrio fischeri</i> (Marine bacteria)	No effect	2 (reliable with restriction)	Marcin et Aleksander., 2023

Method	Results	Remarks	Reference
5, 15, and 30 min exposure Nominal concentration : 10 mg.l ⁻¹ Measured concentration: unknown	Based on: bioluminescence inhibition	Test material: octocrilene Purity > 98% Guideline: according to ISO 11348-3 standard (2019) (<i>Luminescent bacteria test</i>) Not GLP	

Additional information:

In the scientific literature, the toxicity of octocrilene was studied on the marine bacteria *Aliivibrio fischeri* (Marcin et Aleksander., 2023) and the freshwater protist *Tetrahymena thermophila* (Gao et al., 2013). No decrease in luminescence or growth inhibition were recorded at the tested nominal concentrations (up to 10 and 15 mg.l⁻¹ respectively).

Giraldo et al., 2017 investigated the acute toxicity of octocrilene towards various marine organisms (i.e. *Isochrysis galbana*, *Mytilus galloprovincialis* and *Paracentrotus lividus*). All tests were carried out as non-GLP using non-standardised protocols. The resulting effect concentrations were as follows:

1. *Isochrysis galbana* (cell density): EC₅₀ > 150 µg.l⁻¹, EC₁₀: 103 µg.l⁻¹ ; NOEC: 40 µg.l⁻¹
2. *Mytilus galloprovincialis* (development stage): EC₅₀ > 650 µg.l⁻¹, EC₁₀: 511 µg.l⁻¹ ; NOEC: 20 µg.l⁻¹
3. *Paracentrotus lividus* (growth rate): EC₅₀: 737 µg.l⁻¹, EC₁₀: 162 µg.l⁻¹ ; NOEC 20 µg.l⁻¹.

However toxicity parameters were calculated based on nominal concentrations. The study protocol included analytical measurement performed at the beginning of the exposures and after 48h. The chemical analyses showed that octocrilene has a low stability in seawater, because after 48h the dissolved concentration was below nominal concentrations (83% lower for octocrilene). The measured concentrations at 48h were 0.90, 1.25, 42.38 µg.l⁻¹ for the tested nominal concentration 5, 20, 150 µg.l⁻¹ respectively. Furthermore, the range of concentrations tested was not clearly defined. Indeed, inconsistencies were noted between the higher concentration in the analytical checking table (150 µg.l⁻¹) and the graphical results where responses were recorded at concentration higher than 600 µg.l⁻¹. This is probably the reason why some EC₅₀ values were largely higher than 150 µg.l⁻¹ (nominal).

In Stien et al., 2018, the toxicity of octocrilene in corals was evaluated. Adult *Pocillopora damicornis* coral was treated at concentrations of 5, 50, 300, and 1000 µg.l⁻¹ (nominal). Most polyps were closed at concentrations of 300 µg.l⁻¹ and higher. Octocrilene was transformed into fatty acid conjugates via oxidation of the ethylhexyl chain, yielding very lipophilic octocrilene analogues that accumulate in coral tissues. Second, the differential analysis of coral profiles revealed higher levels of 15 acylcarnitines. Acylcarnitines are common biomarkers of cell toxicity, and, in many models including human pathologies, elevated acylcarnitine concentration has been correlated with mitochondrial dysfunction in which mitochondrial fatty acid β-oxidation and cell bioenergetics are affected. The coral reef builder species *P. damicornis* significantly impacted by octocrilene after only 7 days of exposure to 50 µg.l⁻¹ (nominal) octocrilene. The highest non-toxic octocrilene concentration in our assay was 5 µg.l⁻¹ (nominal).

In *He et al., 2018b*, the toxicity of octocrilene in corals was evaluated. In this study, 7-day exposures were conducted to evaluate the toxicity and bioaccumulation of two organic UV filters, ethylhexylmethoxy-cinnamate (EHMC; octinoxate) and octocrilene (OC) (single- and combined-chemical tests), and diluted sunscreen wash-off water containing both active ingredients to the adult life stage of two hard coral species, *Seriatopora caliendrum* and *Pocillopora damicornis*. In the sunscreen product exposures, 5% sunscreen water (containing EHMC and OC) caused high mortality in *S. caliendrum* (66.7–83.3%) and *P. damicornis* (33.3–50%), and tissue concentrations were up to 10 times greater than in the single-chemical exposures. At similar levels, tissue concentrations seems to be similar between single-chemical exposure and combined exposure (EHMC + OC). These results confirm the bioaccumulation potential of EHMC and OC and show that other ingredients (coformulants) in sunscreen products may increase the bioavailability of active ingredients to corals and exacerbate the toxicity of sunscreen products.

Thorel et al., 2022 measured the effect of octocrilene on another reefbuilding species scleratinian coral *Stylophora pistillata* using the metabolomic profiling approach developed with *P. damicornis* in *Stien et al., 2018*. After 7 days exposure *S. pistillata* accumulated octocrilene derivatives similar to *P. damicornis* (*Stien et al., 2018*). Octocrilene–fatty acid conjugates were found, as well as octocrilene heterosides. The relative concentrations of 8 acylcarnitines increased on average by a factor of 4 at 300 µg.l⁻¹ octocrilene and by a factor of 6 at 1000 µg.l⁻¹ octocrilene. This increase was always statistically significant. As in the case of *P. damicornis*, the large and significant increase in relative acylcarnitine concentrations in *S. pistillata* suggests an abnormal metabolism of fatty acids. Furthermore, the tissue concentrations of three sphingoid bases increased significantly in *S. pistillata*. This phenomenon was indicative of the induction of cellular senescence processes in exposed corals. Similar results were observed by *Stien et al., 2018*, then the proven impact of octocrilene on this second coral species only after 7 days exposure suggests that potential environmental octocrilene pollution could impact many reef-building species, at lower concentration following prolonged exposure.

B.7.2. Terrestrial compartment

The results are summarised in the following Table 26:

Table 26: Effects on soil macro-organisms.

Method	Results	Remarks	Reference
Soil microorganisms in natural soil 28d exposure Tested concentrations: 9.54, 30.5, 97.7, 313, 1000 mg.kg _{dw} ⁻¹ (nominal) Solvent (acetone) and positive (Nitrapyrin 6.7 mg.kg ⁻¹) controls Temperature: 20.0°C (mean) [19.6–20.7°C] Soil pH: 6.69 (mean) Soil humidity: 45 % of maximum water holding capacity Replicates: 3 per condition	NOEC (28d) ≥ 1000 mg.kg _{dw} ⁻¹ (nominal) based on: nitrate formation rate LOEC (28d) ≥ 1000 mg.kg _{dw} ⁻¹ (nominal) based on: nitrate formation rate EC10 (28d) ≥ 1000 mg.kg _{dw} ⁻¹ (nominal) based on: nitrate formation rate EC50 (28d) ≥ 1000 mg.kg _{dw} ⁻¹ (nominal) based on: nitrate formation rate	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene Purity: 99.8% According to guideline OECD Guideline 216 (Soil Microorganisms: Nitrogen Transformation Test)	Unpublished study Report (2023c)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
<i>Eisenia fetida</i> Static, 8 weeks Artificial soil nominal concentrations: 15.6 -31.3 - 62.5 - 125 -250 - 500 mg.kg _{dw} ⁻¹ Replicates: controls: 10 earthworms in each of 8 replicates (n=80); 10 earthworms in each of 4 replicates per treatment (n=40/group)	NOEC (8wk): 125 mg.kg _{dw} ⁻¹ (nominal) based on: reproduction NOEC (8wk): >500 mg.kg _{dw} ⁻¹ (nominal) based on: mortality NOEC (8wk): >500 mg.kg _{dw} ⁻¹ (nominal) based on: biomass	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene Purity according to OECD Guideline 222 (Earthworm Reproduction Test (<i>Eisenia fetida</i> / <i>Eisenia andrei</i>))	Unpublished study Report (2017)
Terrestrial plants Seedling emergence and seedling growth test (semi-field study) 21 days exposure after seedling emergence in the solvent control Natural soil (loamy sand, 0.63% OC) Monocotyledonae <i>Hordeum sativum</i> (Barley) <i>Allium cepa</i> (Onion) Dicotyledonae <i>Vigna radiata</i> (Mung Bean) <i>Fagopyrum esculentum</i> (Buckwheat) <i>Brassica napus</i> (oilseed rape) <i>Solanum lycopersicum</i> (tomato) Nominal concentrations: Trial 1: 12.3, 37, 111, 333, 1000 mg.kg _{dw} ⁻¹ Trial 2 (mung bean): 1.4, 4.1, 12.3, 37, 111, 333, 1000 mg.kg _{dw} ⁻¹ Solvent (acetone only) and standart (no test item) controls Temperature (mean): 22, 22.6°C (trial 1) – 23.7°C (trial 2) Relative humidity (rH): 75.7, 77-6 (trial 1) – 65.9 (trial 2) 16:8h light:dark Replicates: 8 pots per species with 4 seeds each (32 seeds per treatment)	No mortality Emergence rate: NOEC >1000 mg.kg _{dw} ⁻¹ Shoot Length: NOEC = 37 mg.kg _{dw} ⁻¹ Shoot weight: LOEC = 12.3 mg.kg _{dw} ⁻¹ Phytotoxicity (chlorosis): NOEC = 12.3 mg.kg _{dw} ⁻¹	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene Purity: 99.8% according to guideline OECD Guideline 208 (Terrestrial Plants, Growth Test)	Unpublished study report (2024d)
<i>Eisenia fetida</i> Artificial soil, 48h exposure Nominal concentrations:	No mortality Hormesis effect at 0.01 mg.l ⁻¹ ↑ avoidance rate:	2 (reliable with restriction) Test material: octocrilene	Santo et al., 2022

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Method	Results	Remarks	Reference
0.01, 0.1, and 1 mg.l⁻¹ Measured concentrations: unknown (stock solution stable for 7 days) Positive (boric acid 750 mg.kg⁻¹ soil) control Soil humidity: adjusted to 60% maximum water holding capacity using water or test solutions Based on: NBR ISO 17.512-1 (2011)	20% at 0.1 mg.l⁻¹ 80% at 1 mg.l⁻¹ Oxidative stress: ↑ APX activity at 1 mg.l⁻¹ ↓ SOD activity at 0.01, 0.1, 1 mg.l⁻¹	Purity unknown (analytical grade or higher) Not GLP	
Allium cepa Nominal concentrations: 0.01, 0.1, and 1 mg.l⁻¹ Measured concentrations: unknown (stock solution stable for 7 days) <u>Phytotoxicity:</u> 5 days exposure Replicates: n=5 of 20 seeds (germination), n=10 (root measurement) <u>Cyto/genotoxicity:</u> 24h and 48h exposure Positive (MMS 4E-04 mol/L) control Replicates: n=5 roots No guideline followed (except Based on OECD TG 208 for phytotoxicity parameters)	↓ germination ↓ roots elongation ↓ cell division in roots ↑ genotoxic cellular changes (micronucleus, chromosome missegregation in metaphase, chromosome loss in metaphase and anaphase, anaphase bridge) at 100 and 1000 µg.l⁻¹ Oxidative stress: ↑ CAT activity at 0.01 mg.l⁻¹ ↑ GPX activity at 1 mg.l⁻¹ ↑ SOD activity at 1 mg.l⁻¹		
Eisenia fetida Artificial soil <u>Acute test (14 days):</u> Nominal concentration: 500, 1000, 1500 & 2000 mg.kg⁻¹ Solvent (acetone 1.57 ml/kg) and water controls <i>according to OECD Guideline 207 (Earthworm acute toxicity tests)</i> <u>Reproduction test (56 days):</u> Nominal concentration: 2 mg.kg⁻¹ Solvent (acetone 500 µl.kg⁻¹) Photoperiod: 16:8h light:dark for reproduction test Replicate: 10 earthworms in each triplicate <i>according to OECD Guideline 222 (Earthworm Reproduction Test (Eisenia fetida/Eisenia andrei))</i>	<u>Acute test:</u> LC50 (14d) > 2000 mg.kg⁻¹ mortality Body weight ↓ at 1500 and 2000 mg.kg⁻¹ (14d) <u>Survival and growth</u> No effect at 2 mg.kg⁻¹ (14 and 56 days) <u>Reproduction/growth:</u> (cocoon formation, weight, and juvenile production) No effect at 2 mg.kg⁻¹ <u>Oxidative stress and neurotoxicity:</u> (ROS, enzyme activity: SOD, CAT, GST, MDA, AchE, CES)	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene Purity: 97% Not GLP	Gautam et al., 2023

Method	Results	Remarks	Reference
<u>Oxidative stress and neurotoxicity (14 days):</u> Nominal concentration: 2 mg.kg ⁻¹ Replicate: 2 earthworms in each of triplicate (n=6)	↑ CAT activity at 2 mg.kg ⁻¹		

Discussion:

In an *unpublished study report (2023c)*, the toxic effects of octocrilene on the nitrogen transformation activity of soil microorganisms were determined following 28 days exposure to the test item in the soil at 9.54, 30.5, 97.7, 313 and 1000 mg.kg_{dw}⁻¹. This study met the OECD TG 216 validity criteria. After 28 days of exposure, the mean nitrate formation rates for all the test item treatments did not differ statistically significantly from the pooled control. Octocrilene did not induce significant toxicity on the soil microbial activity.

In an *unpublished study report (2017)*, effects of octocrilene on mortality, biomass and the reproductive potential of the earthworm species *Eisenia fetida* were determined according to OECD TG 222 (2016). All validity criteria recommended by the test guidelines were fulfilled. After 28 days of exposure in soil, neither evident earthworm mortalities (<10%) nor pathological symptoms or changes in the behaviour of adult earthworms were observed in the controls or all test item concentrations. There were no statistically significant differences in earthworm body weights in all test item concentrations compared to the control. After a further four weeks, the reproduction rate (average number of juveniles produced) was 142 in the control and ranged from 65 to 143 in the treated groups. The results showed a statistically significant reduction of earthworm reproduction at the two highest tested concentrations 250 and 500 mg.kg_{dw}⁻¹. Then, the NOEC (reproduction) was determined to be 125 mg.kg_{dw}⁻¹.

In *unpublished study report (2024d)*, the effects of octocrilene on seedling emergence and early growth of six terrestrial plants (2 monocotyledonous and 4 dicotyledonous) after pre-emergence application of the test item into a natural soil were tested according to OECD TG 208. The five tested item concentrations were 12.3, 37, 111, 333, 1000 mg.kg_{dw}⁻¹ (Trial 1). For mung beans species, the Trial 1 was repeated with two additional low concentrations 1.4 and 4.1 mg.kg_{dw}⁻¹ (Trial 2). The concentration of the application solution was confirmed by the analysis of the highest concentration of 1000 mg.kg_{dw}⁻¹ subsequently diluted to prepare the lower tested concentrations (recovery of 101% of the nominal concentration for both trials). The test duration was 21 days after 50% seedling emergence in the solvent control for all test species. The plants were evaluated weekly for emergence, visual phytotoxicity (i.e. leaf and stem deformations, chlorosis, necrosis and overall appearance) compared with the controls, and plant mortality for the 21-day test period. The shoot length and shoot weight were determined at day 21. This study fulfilled all validity criteria.

Results showed no effect on emergence and mortality of emerged plants for all species. Statistically significant differences between the solvent control and test item treatments were observed for the following species and endpoints: barley (shoot length), mung bean (phytotoxic symptoms, shoot length, shoot weight Trial 1 and Trial 2), buckwheat (phytotoxic symptoms, shoot length, shoot weight) and tomato (shoot length, shoot weight). Based on visual inspection of the data for all plants species, any concentration-effect relationship for the abovementioned endpoints was flat and effect size small

compared to the biological variability of the test plants. Therefore, ECx values could not reliably be determined by concentration-effect modelling and the NOEC were proposed as relevant measure for evaluation of test item effects observed in this study.

Regarding mung beans, the Registrant excluded the Trial 1 results from the overall data set "*based on the statistical evaluation of the first trial (NOEC and EC₁₀ values could not be determined for all endpoints)*". In Trial 1, a statistically significant reduction (25%) in shoot weight compared to the solvent control was observed at test end for all test item concentrations then, only a LOEC could be determined at 12.3 mg.kg_{dw}⁻¹. In Trial 2, a significant shoot weight reduction (35%) was detected only at 1000 mg.kg_{dw}⁻¹ (LOEC (chlorosis) = 111 mg.kg_{dw}⁻¹). Those two trials on mung beans did not showed reproducible results. Different endpoint were impaired at different concentrations. At 21-days, control mung beans showed discrepancies in their shoots weight and length: 2497 mg and 85.7 mm in Trial 1, and 1351 mg and 97.8 mm in Trial 2. Reasons for such differences were not explained. The soil batch used in Trial 1 and Trial 2 was the same. However, differences between the two trials were noted for the irrigation of the test system (watering day, volume of tap water per tray of 8 pots, 3.25L of fertiliser used in Trial 1 versus 2.6L in Trial 2) and for the light intensity higher in Trial 2 (light adjustments that might be linked to the test period (spring for Trial 1, autumn for Trial 2). The unit used for light intensity was kLux and not µE/m²/s as indicated in the OECD TG 208, making difficult the comparison with the recommendations (conversion factor for artificial light not provided). Temperatures were similar in Trial 1 (22°C) and Trial 2 (23.7°C) and relative humidity slightly higher in trial 1 (77.6 rH) than in trial 2 (65.9 rH). Finally, the high biological variability observed in this study for all plant species could be the result of lack of homogeneity in soil contamination which could not be verified as soil analysis was not performed (concentration analytically confirmed only for the application solution of 1000 mg.kg⁻¹ ds).

Results showed that the lowest significant effect was observed at 12.3 mg.kg⁻¹ d.s. (mung bean shoot weight, Trial 1). Even if slightly above 20% effect, and considering the study limits described above, this LOEC should be considered in the overall dataset as relevant to be used for the NOEC calculation (LOEC/2; ECHA, 2008c). In conclusion, the overall most sensitive endpoint was shoot weight of mung beans. The **NOEC=6.15 mg.kg_{dw}⁻¹**. After conversion to a standard soil defined as a soil with organic carbon content of 2% (Table R16-8, Echa 2016), the **NOEC = 19.52 mg.kg_{dw}⁻¹⁶**. mg.kg⁻¹

Two other supportive studies from the scientific literature are available:

Santo et al., 2022 evaluated the phytotoxicity, cytogenotoxicity, and oxidative stress of octocrilene in *A. cepa* and the acute toxicity and oxidative stress in *E. fetida*. After 48 h exposure, the concentrations 0.1 and 1 mg.l⁻¹ caused lipid peroxidation in both species. In *A. cepa*, an induction of cellular changes in significant numbers was observed at 0.1 and 1 mg.l⁻¹, characterizing them as genotoxic (micronucleus, chromosome missegregation in metaphase, chromosome loss in metaphase and anaphase, and anaphase bridge). These two highest tested concentrations also reduced cell proliferation in root meristems as well as a significant number of prophase to the detriment of the other phases of cell division, characterizing a delayed mitosis induced by octocrilene at these concentrations. Seeds germination was reduced at these concentrations. After 48h exposure of *E. fetida*, there was no death of earthworms in the treatments with the three concentrations of octocrilene. However, the avoidance rate of earthworms from the contaminated soil increased to 20% and 80% at 0.1 and 1 mg.l⁻¹ respectively. Those

⁶ Tested soil with 0.63%OC, NOEC=6.15*(2/0.63) mg.kg⁻¹ ds

results showed that octocrilene caused toxicity on terrestrial organisms.

In *Gautam et al., 2023*, octocrilene treated earthworm showed a reduction in body weight at higher doses, i.e., 1500 mg.kg⁻¹ without causing lethality. The LC₅₀ value was >2000 mg.kg⁻¹. No significant changes in earthworm mortality, biomass, and reproduction in terms of hatching and cocoon production and weight were observed after short and long term exposures at 2 mg.kg⁻¹. At cellular level, exposure to 2 mg.kg⁻¹ octocrilene has negligible effects on oxidative stress and neurotoxicity markers following a 14-day exposure, except for CAT activity which significantly increased.

Experimental data on the toxicity of octocrilene are available for three different terrestrial species: soil microorganisms, terrestrial worm and terrestrial plants. Test results show that terrestrial plants are the most sensitive species with a NOEC of 19.52 mg.kg_{dw}⁻¹ (After conversion to a standard soil defined as a soil with organic carbon content of 2%, (Table R16-8, Echa 2016)).

B.7.3. Atmospheric compartment

Not investigated

B.7.4. Microbiological activity in sewage treatment systems

The results are summarised in the following Table 27:

Table 27: Effects on microorganisms.

Method	Results	Remarks	Reference
activated sludge of a predominantly domestic sewage freshwater static ISO 8192 (Test for Inhibition of Oxygen Consumption by Activated Sludge)	IC ₅₀ (30 min): > 10000 mg.l ⁻¹ test mat. (nominal) based on: respiration rate (Octocrilene water solubility < 0.5 mg.l ⁻¹)	1 (reliable without restriction) key study experimental result Test material (EC name): octocrilene <i>equivalent or similar to OECD Guideline 209 (Activated Sludge, Respiration Inhibition Test)</i>	Unpublished study Report (1995b)
activated sludge, domestic freshwater static OECD Guideline 209 (Activated Sludge, Respiration Inhibition Test)	EC ₅₀ (30 min): > 1000 mg.l ⁻¹ test mat. (nominal) based on: respiration rate	2 (reliable with restrictions) supporting study experimental result Test material (EC name): octocrilene	Unpublished study Report (1991a)

Discussion:

The toxicity of octocrilene to microorganisms was tested in three studies investigating the respiration inhibition by the substance in activated sludge and *Pseudomonas putida* respectively. Two tests on respiration inhibition in activated sludge were performed. In a study following guideline ISO 8192, an $IC_{50} > 10000 \text{ mg.l}^{-1}$ was determined. A guideline study according to OECD TG 209 detected no respiration inhibition up to a substance concentration of 1000 mg.l^{-1} . The results are comparable to a test on *P. putida* conducted in compliance with DIN 38412, part 27 and reporting an $EC_{50} > 10000 \text{ mg.l}^{-1}$. However, a solubiliser was used in this study.

Moreover toxicity control of the manometric respirometry test to determine the “ready” biodegradability revealed that octocrilene is not toxic for bacteria. It should be noted that the initial biomass concentration recommended in the OECD TG 301F guideline is 1.6 g.l^{-1} instead of 800 mg.l^{-1} . This argument justified that this data is reliable with restriction.

The inhibition of the degradation activity of activated sludge is not anticipated when introduced in appropriate low concentrations. An IC_{50} for microorganisms $> 1000 \text{ mg.l}^{-1}$ is considered as the most sensitive endpoint value.

B.7.5. Non compartment specific effects relevant for the food chain (secondary poisoning)

Not investigated. However, biomagnification of octocrilene was suggested as shown in the scientific literature in marine species of high trophic level such as tuna (*Pintado-Herrera et al., 2024*) and dolphin (*Gonzalez-Bareiros et al., 2023*).

B.7.6. Hazard information - conclusion

Assessment Factor (AF) to derive the PNEC_{aquatic}

No OECD TG study on the long-term toxicity of octocrilene on fish is available to derive a chronic endpoint value (NOEC/LOEC) in the context of an environmental risk assessment. Nevertheless, in a Weight of Evidence approach, long term exposure data from several fish tests (from literature) with octocrilene are available to define the assessment factor (AF) for the PNEC value. These studies indicate a low systemic toxicity of octocrilene on adults and effects on the early life stage of fish (larval stage) from $2.7 \mu\text{g.l}^{-1}$, but without following a testing protocol similar to OECD test guidelines for testing chronic or reproductive effects or fish life-cycle test (Yan *et al.*, 2020), without being able to distinguish at this stage any EAS-mediated effect.

An OECD TG 211 test is considered as the only reliable long term aquatic toxicity test for octocrilene. These results show that *Daphnia magna* is the most sensitive species in a 21-day test with a NOEC_{Reproduction} of $2.66 \mu\text{g.l}^{-1}$ (living offspring per surviving adult).

The available OECD TG 201 study on algae is not valid to determine an endpoint value (nominal value far from the solubility limit). However, based on the overall available data including literature studies, it can be concluded that algae is clearly not the most sensitive trophic level and this endpoint can be used to reduce the assessment factor in the determination of a PNEC_{aquatic}.

In a weight of evidence approach, an AF of 10 for freshwater and 100 for marine water were considered. In this case, the PNEC_{aquatic} is determined to be $0.27 \mu\text{g.l}^{-1}$ for freshwater based on the NOEC of $2.66 \mu\text{g.l}^{-1}$ in *Daphnia* and $0.027 \mu\text{g.l}^{-1}$ for marine water. Nevertheless, and given uncertainties about the toxicity of octocrilene on fish at the larval stage, an AF of 50 for freshwater and 500 for marine water could not be proposed as an acceptable surrogate. However, risks are already identified with PNEC based on *Daphnia* and an AF of 10, the question of increasing the AF does not need to be elaborated further.

Assessment Factor (AF) to derive the PNEC_{sediment}.

Regarding sediments, three chronic tests with species representing different living and feeding condition (*Chironomus*, *Lumbriculus*, *Myriophyllum*) are available to derive the PNEC value. It is important to note that the study with *chironomus* was judged reliable with restriction since the toxicity was probably underestimated due to octocrilene behaviour ($K_{oc}=48898 \text{ l.kg}^{-1}$). For strongly adsorbing substances, ingestion of contaminated food is a more significant exposure route. In this study, the route of exposure (spiked sediment) was not modified, and the fish food flakes suspended in test water were not contaminated. The analytical results confirmed the absence of the test item in water and pore water. Then, the organisms were mostly non exposed to the substance. The most sensitive species was demonstrated to be *Lumbriculus variegatus* in the OECD TG 225 spiked-sediment dwelling worm study with an EC₁₀ of $46.3 \text{ mg.kg dw}^{-1}$ nominal.

Chronic data are considered available for two trophic levels leading to an AF of 50 for the freshwater environment and 500 for the marine environment. The PNEC_{sediment} is determined to be 0.926 and $0.0926 \text{ mg.kg dw}^{-1}$ sediment (for freshwater and marine water respectively) based on the EC₁₀ of $46.3 \text{ mg.kg dw}^{-1}$ in *Lumbriculus variegatus*.

Assessment Factor (AF) to derive the PNEC_{soil}.

Regarding soil, long-term toxicity tests are available for three trophic levels (microorganisms, worms, and plants) to derive PNEC value. The most sensitive species was demonstrated to be the mung bean, with a NOEC of 6.15 mg.kg_{dw}⁻¹ nominal for the emergence of seeds.

Chronic data are considered available for three trophic levels leading to an AF of 10. In this case, the PNEC_{soil} is determined to be 1.95 mg.kg_{dw}⁻¹ considering the organic carbon normalisation (at 2%) (Table R16-8, Echa 2016).

B.8. Environmental exposure assessment and associated risk**B.8.1. Physicochemical properties, environmental fate and distribution properties**

The physicochemical properties, the environmental fate and distribution properties used for exposure estimation are summarized in the Table 28.

Table 28: Physicochemical properties- Environmental fate and distribution properties - used for exposure estimation.

Input	Value	Unit	Reference
Physicochemical properties			
Molecular weight	361.47	g.mol ⁻¹ [25°C]	EPI Suite v4.11
Vapour pressure	3.15E-09	mmHg [25°C]	MPBPWIN v1.43
Henry's law constant	3.1E-04	Pa.m ³ .mol ⁻¹ [25°C]	HENRYWIN v3.10
Water solubility	40	µg.l ⁻¹ [20°C]	Unpublished study report (2018b)
Partition coefficient n-octanol/water (Log Kow)	6.1	[log10]	Unpublished study report (2018d)
Environmental fate and distribution properties			
Organic carbon/water partition coefficient (Koc)	48 898	L.kg ⁻¹ [20°C]	Unpublished study report (2018a)
Hydrolysis (DT ₅₀)	206	years [pH 7, 25°C]	HYDROWIN v2.00
Photo-transformation in air (DT ₅₀)	6.19	hours [25°C]	AOPWIN v1.92 model
Biodegradability	Not readily biodegradable	[-]	Unpublished study report (1995a) Unpublished study report (1991a)
Dissipation in aerobic freshwater (DT _{50, dissipation})	1.7*	days [12°C]	Unpublished study report (2024a)

Degradation in aerobic freshwater sediment system (DT ₅₀ , degradation whole system)	45.5*	days [12°C]	Unpublished study report (2024a)
Dissipation in aerobic estuarine water (DT ₅₀ , dissipation)	4.1*	days [12°C]	Unpublished study report (2024b)
Degradation in aerobic estuarine water sediment system (DT ₅₀ , degradation whole system)	47*	days [12°C]	Unpublished study report (2024b)
Total rate constant for removal from agricultural top soil	1.00E+06	days [12°C]	EUSES model 2.2.0

* According to the R.16 guidance, in case of indirect emission *via* the STP, the DT₅₀ values for the aquatic compartment (dissipation from the aqueous phase and degradation in the whole system) are used only for the regional tonnage approach. In fact, the local assessment in case of indirect emission *via* the STP does not take into account the degradation of the substance in the estimation of local aquatic PECs (including sediment) considering a continuous release over a short distance from the emission source.

The multimedia model SimpleTreat, evaluates the distribution and elimination of chemicals by municipal sewage treatment plants (STP). The output results (percentage of the contaminant mass in the influents that ends up in the sludge and effluents, and degradation/volatilization removal efficiency) are summarized in the Table 29.

Table 29: Dissipation rate of octocrilene in WWTP.

Calculated fate and distribution in the STP – SimpleTreat V4	
COMPARTMENT	PERCENTAGE [%]
Air	4.24E-06
Water	18.66
Sludge	81.35
Degraded in STP	0

B.8.2. Overview on exposure scenarios and coverage of substance life cycle

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

The Figure 6 below relates the tonnages declared by each registrant for each stage in the substance's life cycle.

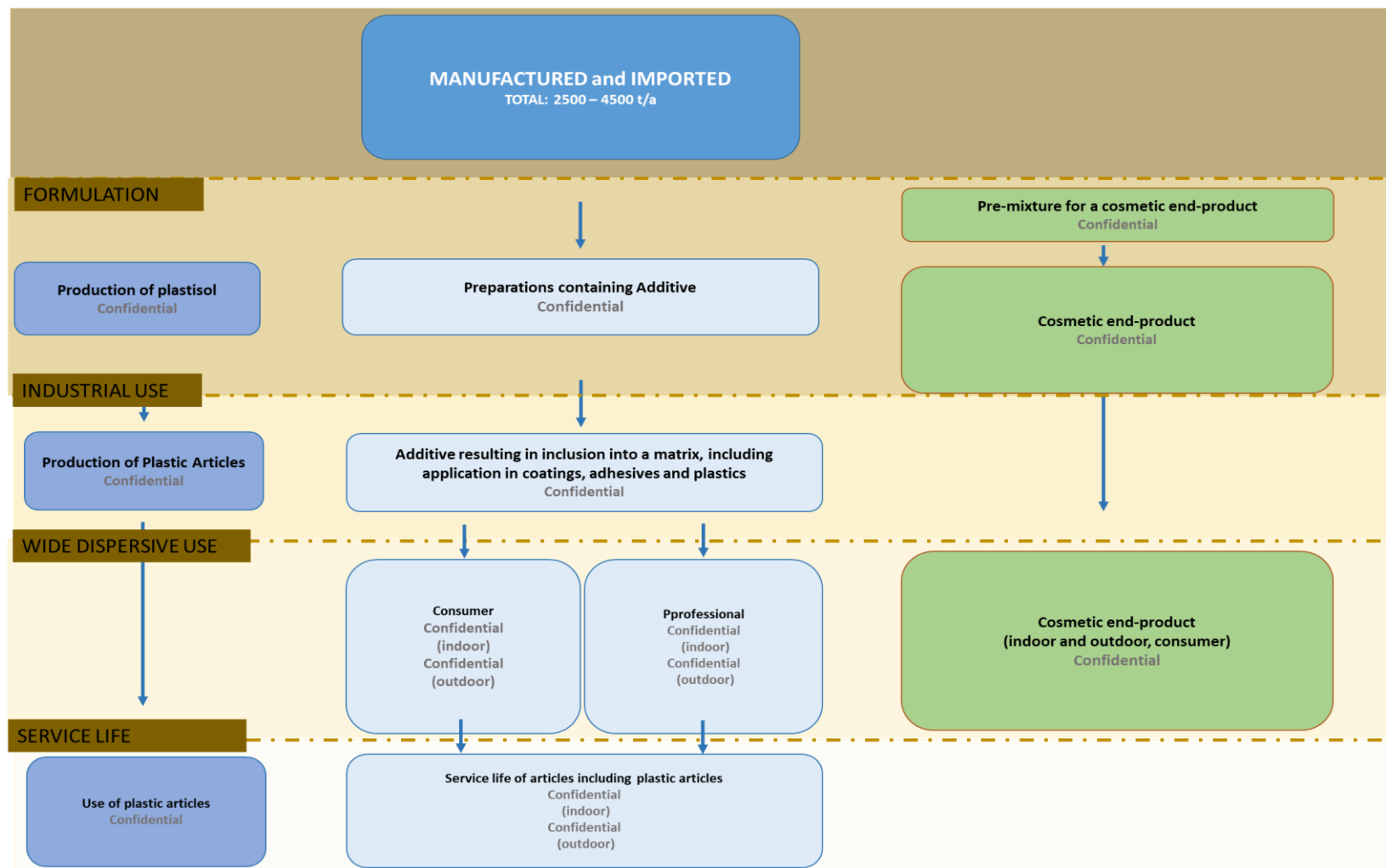


Figure 6: Mapping of the octocrilene's life cycle with the declared tonnages.

B.8.3. Tonnage approach - Environmental exposure assessment and associated risk

B.8.3.1. Tonnage approach - Local exposure assessment

Table 30: Emission scenario (ES) parameter values for each use and by registrant.

MANUFACTURE		
	ES1	ES2
Annual tonnage (t. year ⁻¹)	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential
FORMULATION: production of plastisol		
	ES3	ES4
Annual tonnage (t. year ⁻¹)	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Discharge rate of STP (m³.d⁻¹)	Confidential	Confidential
River flow rate (m³.d⁻¹)	Confidential	Confidential
FORMULATION: preparations containing additives		
	ES5	
Annual tonnage (t. year⁻¹)	Confidential	
Release times per year (days. year⁻¹)	Confidential	
Release fraction to water before wastewater (%)	Confidential	
Release fraction to soil (%)	Confidential	
Release fraction to air (%)	Confidential	
Fraction tonnage to region (%)	Confidential	
Fraction used at main source (%)	Confidential	
Connected to STP municipal (yes/no)	Confidential	
Application of the STP sludge on agricultural soil (yes/no)	Confidential	
Discharge rate of STP (m³.d⁻¹)	Confidential	
River flow rate (m³.d⁻¹)	Confidential	
FORMULATION: pre-mixture for cosmetic end-products		
	ES6	
Annual tonnage (t. year⁻¹)	Confidential	
Release times per year (days. year⁻¹)	Confidential	
Release fraction to water before wastewater (%)	Confidential	
Release fraction to soil (%)	Confidential	
Release fraction to air (%)	Confidential	
Fraction tonnage to region (%)	Confidential	
Fraction used at main source (%)	Confidential	
Connected to STP municipal (yes/no)	Confidential	
Application of the STP sludge on agricultural soil (yes/no)	Confidential	
Discharge rate of STP (m³.d⁻¹)	Confidential	
River flow rate (m³.d⁻¹)	Confidential	

1)					
FORMULATION: cosmetic end-products					
	ES7	ES8	ES9	ES10	ES11
Annual tonnage (t. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential	Confidential	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential	Confidential	Confidential	Confidential
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
INDUSTRIAL USE: production of Plastic Articles					
	ES12		ES13		
Annual tonnage (t. year ⁻¹)	Confidential		Confidential		
Release times per year (days. year ⁻¹)	Confidential		Confidential		
Release fraction to water before wastewater (%)	Confidential		Confidential		
Release fraction to soil (%)	Confidential		Confidential		
Release fraction to air (%)	Confidential		Confidential		
Fraction tonnage to region (%)	Confidential		Confidential		
Fraction used at main source (%)	Confidential		Confidential		
Connected to STP municipal (yes/no)	Confidential		Confidential		
Application of the STP sludge on agricultural soil (yes/no)	Confidential		Confidential		
Discharge rate of STP (m ³ .d ⁻¹)	Confidential		Confidential		
River flow rate (m ³ .d ⁻¹)	Confidential		Confidential		

INDUSTRIAL USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics				
	ES14			
Annual tonnage (t. year ⁻¹)	Confidential			
Release times per year (days. year ⁻¹)	Confidential			
Release fraction to water before wastewater (%)	Confidential			
Release fraction to soil (%)	Confidential			
Release fraction to air (%)	Confidential			
Fraction tonnage to region (%)	Confidential			
Fraction used at main source (%)	Confidential			
Connected to STP municipal (yes/no)	Confidential			
Application of the STP sludge on agricultural soil (yes/no)	Confidential			
Discharge rate of STP (m ³ .d ⁻¹)	Confidential			
River flow rate (m ³ .d ⁻¹)	Confidential			
WIDE DISPERSIVE USE: additives resulting in inclusion into a matrix, including application in coatings, adhesives and plastics				
	ES15	ES16	ES17	ES18
Annual tonnage (t. year ⁻¹)	Confidential	Confidential	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential	Confidential	Confidential
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential

WIDE DISPERSIVE USE: cosmetic end-products					
	ES19	ES20	ES21	ES22	ES23
Annual tonnage (t. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential	Confidential	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential	Confidential	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential	Confidential	Confidential	Confidential
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	Confidential
SERVICE LIFE: plastic articles					
	ES24	ES25	ES26	ES27	
Annual tonnage (t. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	
Release times per year (days. year ⁻¹)	Confidential	Confidential	Confidential	Confidential	
Release fraction to water before wastewater (%)	Confidential	Confidential	Confidential	Confidential	
Release fraction to soil (%)	Confidential	Confidential	Confidential	Confidential	
Release fraction to air (%)	Confidential	Confidential	Confidential	Confidential	
Fraction tonnage to region (%)	Confidential	Confidential	Confidential	Confidential	
Fraction used at main source (%)	Confidential	Confidential	Confidential	Confidential	
Connected to STP municipal (yes/no)	Confidential	Confidential	Confidential	Confidential	
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential	Confidential	Confidential	
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential	
SERVICE LIFE: articles including plastic articles					

	ES28	ES29	ES30	ES31
Annual tonnage (t. year ⁻¹)	Confidential	Confidential	Confidential	Confidential
Release times per year (days. year ⁻¹)	Confidential	Confidential	Confidential	Confidential
Release fraction to water before wastewater (%)	Confidential	Confidential	Confidential	Confidential
Release fraction to soil (%)	Confidential	Confidential	Confidential	Confidential
Release fraction to air (%)	Confidential	Confidential	Confidential	Confidential
Fraction tonnage to region (%)	Confidential	Confidential	Confidential	Confidential
Fraction used at main source (%)	Confidential	Confidential	Confidential	Confidential
Connected to STP municipal (yes/no)	Confidential	Confidential	Confidential	Confidential
Application of the STP sludge on agricultural soil (yes/no)	Confidential	Confidential	Confidential	Confidential
Discharge rate of STP (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential
River flow rate (m ³ .d ⁻¹)	Confidential	Confidential	Confidential	Confidential

NR: not relevant (no emission *via* STP). * Risk management measure (RMM) to ensure control of risks: site-specific information based on the very low solubility of octocrilene. ** RMM (air): filtration – RMM (water): incineration – RMM (soil): incineration

B.8.3.2. Tonnage approach - Regional exposure assessment

Octocrilene is distributed between the compartments according to its properties and the environmental behaviour. Consequently, several types of environmental fate processes are considered in the regional assessment:

- ✓ releases in STP, air, water, industrial soil, and agricultural soil;
- ✓ biotic and abiotic degradation in all compartments;
- ✓ transport, e.g. gas absorption, volatilisation, deposition, run-off, and erosion.

Due to its high potential of adsorption and its low solubility, the dissipation of octocrilene from aqueous to sediment phase is very quick, leading to no possible determination of octocrilene (< LOD) in the water column at the end of the simulation studies (as seen in B.4.1.2.1.2). Consequently, a differentiated 2 steps approach is carried out for the aquatic and sediment compartments in order to take into account the dissipation/degradation of octocrilene in these two respective compartments when it is relevant. Two separate modeling runs have been conducted: one for the water phase, another for the sediment. This approach is based on the results of the two simulations tests in freshwater and estuarine water (see B.4.1.2.1.2). Finally, no rate constant for hydrolysis in surface water is used considering that hydrolysis process is included in the dissipation/degradation of the simulation studies.

Step 1: To calculate the regional PEC in freshwater and in estuarine water, the DT50_{aqueous} dissipation values estimated in the simulation studies have been used to take into account the dissipation mechanism. No degradation in sediment is used to calculate the concentrations in the water phase:

- ✓ **DT₅₀ "aqueous dissipation" = 1.7 days (at 12°C) for freshwater;**
- ✓ **DT₅₀ "aqueous dissipation" = 4.1 days (at 12°C) for estuarine water.**

Step 2: To calculate the regional PEC in sediment (there is no possible distinction between freshwater sediment and marine water sediment in EUSES model), the DT₅₀ for the whole system have been preferred to the DT₅₀ degradation in sediment, in order not to underestimate the potential degradation of octocrilene in water before dissipation to sediment phase. However, no dissipation in water phase is used, as dissipation is probably linked to adsorption of the substance to sediment:

- ✓ **DT₅₀ "degradation, whole system" = 45.5 days (at 12°C) for sediment (provided in the freshwater simulation study).**

For the PEC_{regional} calculations, in contrast to PEC_{local}, an average percentage connection rate to STPs should be included in the calculation. This leads to a more realistic estimation of the likely background concentration at a regional scale. For the purposes of the generic regional model, a STP connection rate of 80% (the EU average according to data available before the implementation of the Urban Waste Water Treatment Directive and currently used in EUSES) will be assumed as proposed in the ECHA guidance (although this rate only reflects the situation for domestic water; industrial water is all collected), leading to direct releases to aquatic compartment for 20% of the emissions.

The regional releases associated with the different identified uses are based on the tonnage at regional level for each use and the same release factors used at local scale.

It was assumed that all direct emissions to soil occurred *via* industrial soil (no direct emission to agricultural soil).

The concentration at the regional scale is used as background concentration for the local scale. For this purpose, the concentration in unpolluted soil needs to be applied ("natural soil", only input through deposition); the industrial or agricultural soil are not presented as they are not used in the risk assessment as background concentrations. Finally, industrial soil is not considered a protection target for direct releases under Reach.

All regional releases (E_{regional}) associated with the different identified uses, both industrial and wide disperse sources, are cumulated to estimate the total regional release (kg/day) to surface water, wastewater, air and soil.

Table 31: total regional releases.

Release route	Total regional releases per day
Wastewater	247 kg/day
Surface water	61.7 kg/day
Air	0.104 kg/day
Industrial Soil	5.619 kg/day
Agricultural soil	0 kg/day

All regional releases associated with the different identified uses of cosmetic products, for both industrial and wide disperse sources, are cumulated to estimate the total regional release (kg/day) to surface water, wastewater, air and soil.

Table 32: regional releases for uses of cosmetic products (industrial and wide disperse sources)

Release route	Total regional releases per day Uses of cosmetic products
Wastewater	205 kg/day
Surface water	51.1 kg/day
Air	1.71E-03 kg/day
Industrial Soil	0 kg/day
Agricultural soil	0 kg/day

B.8.3.3. Tonnage approach – Environmental risk assessment

B.8.3.3.1. Local scale assesment

See Annex XV report.

B.8.3.3.2. Regional scale assesment

See Annex XV report.

B.8.4. Refined approach for the dispersive use of cosmetics end-products - Environmental exposure assessment and associated risk

B.8.4.1. Refined tonnage approach

See Figure 7: Conceptual release pathway of octocrilene to the environment through the use of cosmetic products (releases in tonnes per year determined through a tonnage approach).below describes the release pathway of octocrilene to the environment through the use of cosmetic products (with both tonnage values for the different steps).

Table 33: Parameters used to estimate local PEC values after release to a municipal WWTP.

FORMULATION: Wide dispersive use of cosmetic products			
	Low value	High value	Reference
Annual tonnage (t. year ⁻¹)	1 473	1 665	
Release times per year (days. year ⁻¹)	365		SPERC Cosmetics Europe 8a.1a.v2*: dispersive use - continuous release
Release fraction to water before wastewater (%)	100		SPERC Cosmetics Europe 8a.1a.v2*: Rinse-off products such as shampoo and soap products that are disposed of with the waste water such that 100% of the product ingredients enter the waste water system. Product residues
Release fraction to non agricultural soil (%)	0		
Release fraction to air (%)	0		

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		remaining on the substrate are likely to be washed off in the next washing event.
Fraction tonnage to region (%)	5.3	SPERC Cosmetics Europe 8a.1a.v2*: <i>Price, et al., 2010</i> : value given for hair care products.
Fraction used at main source (%)	0.075	SPERC Cosmetics Europe 8a.1a.v2*: value correspond a homogeneous distribution of octocrilene emissions over 1333 STP in a region instead of the default value of 500 STP.
Connected to STP municipal (yes/no)	yes	Default settings of the REACH Guidance.
Application of the STP sludge on agricultural soil (yes/no)	yes	Default settings of the REACH Guidance.
Discharge rate of STP (m ³ .d ⁻¹)	2000	Default settings of the REACH Guidance.
River flow rate (m ³ .d ⁻¹)	18000	Default settings of the REACH Guidance.

* SPERC Cosmetics Europe 8a.1a.v2: Wide Dispersive Use of Cosmetic Products - 'Down the drain' products - hair and skin care products (Consumers and Professionals)

** Price, O. R., Hughes, G. O., Roche, N. L., & Mason, P. J. (2010). Improving emissions estimates of home and personal care products ingredients for use in EU risk assessments. *Integrated Environmental Assessment And Management*, 6(4), 677-684. <https://doi.org/10.1002/ieam.88>.

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

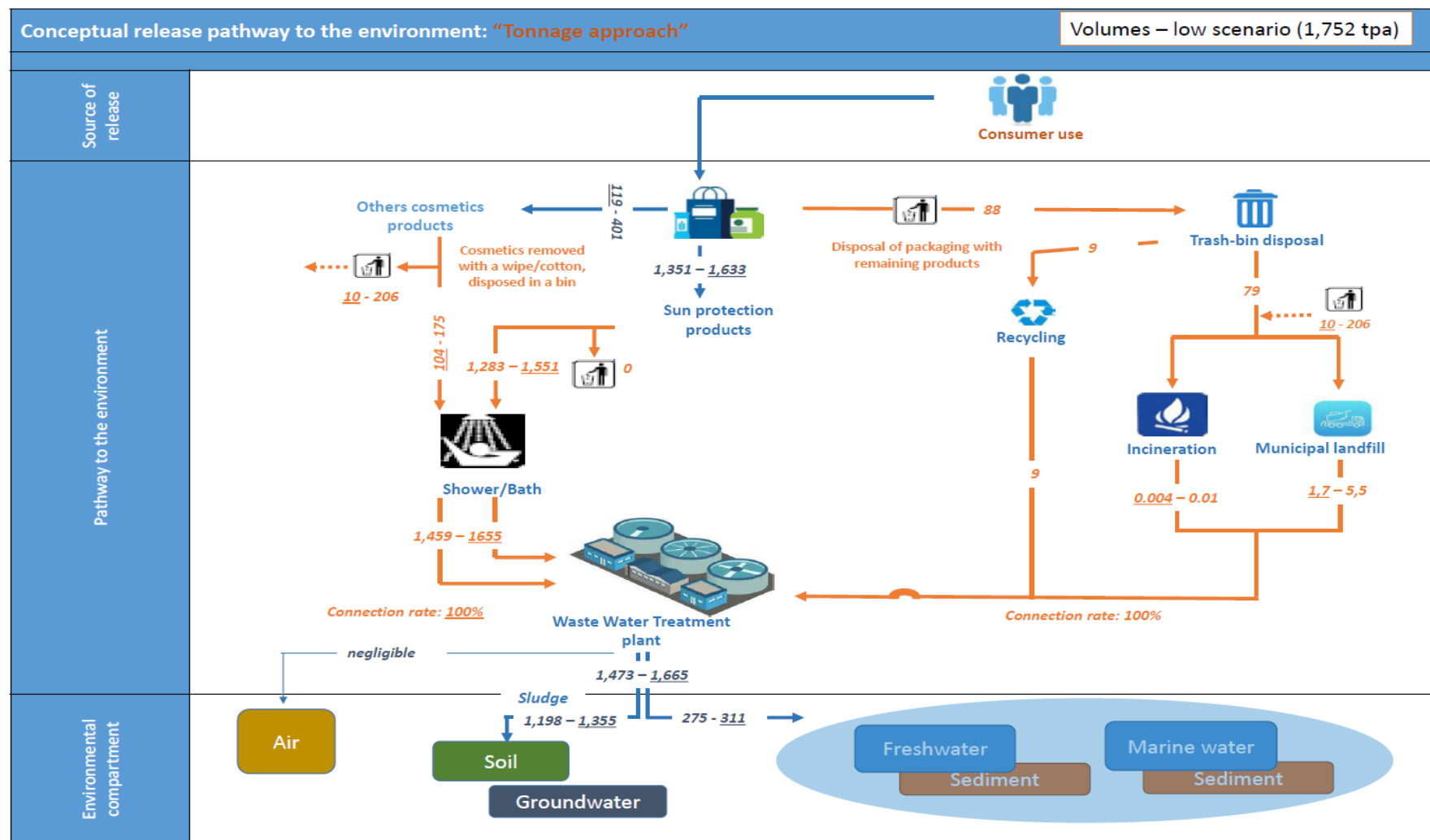


Figure 7: Conceptual release pathway of octocrilene to the environment through the use of cosmetic products (releases in tonnes per year determined through a tonnage approach).

B.8.4.1. Consumer approach

An analysis of the input parameters is carried in the different tables below:

Table 34: Quantity of sunscreen lotion applied per day by a consumer ($Q_{\text{sunscreen_appl}}$) – applicable to swimming and shower scenarios

Value	Unit	Reference and description
18	[g.d ⁻¹]	This single value was proposed for adult consumers who used sunscreen product on the whole body and comes from the SPERC Factsheet Guidance Document "Direct release scenario for UV-filters used in sunscreen products during swimming in freshwater lakes and marine coastal waters", October 2018. SCCS (EC, 2023).
40-80		Recommendation of use by the American Academy of Dermatology: 28 g of sunscreen for the entire adult body and frequent reapplication. "Considering an application of 2 mg.cm ⁻² and a variation of the body area from 1 to 2 m ² of skin area for children to adult men, a single application would use 20 to 40 g per person per day, and two applications would use 40 to 80 g per person per day". National Academies Press eBooks. (2022)
Spray <u>Age 3-13</u> : 6.49 ♀ ; 7.41 ♂ <u>Age ≥ 29</u> : 9.44 ♀ ; 6.96 ♂		"Consumption study based on 95 French families (337 participants) during summer holidays. The first inclusion criterion was the regular use of a high protection sunscreen product in a pump spray form. The others inclusion criteria were as follow: - Between 1 and 3 children with an age ranging from 3 to 13 per family; - Vacation in the sun. A SPF 50 face and body sunscreen pump spray was provided to each member of the 95 families. Participants were invited to use the product in the closest possible way to their personal usage patterns during summer holidays 2014. The amount of cosmetic product used per day was calculated by dividing the total amount of product used during holidays by the period of use (g/day). The values reported in the table correspond to the 95th percentile." Gomez-Berrada et al., 2018

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Spray <u>Age 3-9</u>: 16.65 ♀♂ <u>Age 10-14</u>: 16.56 ♀♂ <u>Age ≥ 15</u>: 16.69 ♀ <u>Age ≥ 15</u>: 16.24 ♂ Sunscreen milk <u>Age 3-9</u>: 11.32 ♀♂ <u>Age 10-14</u>: 10.77 ♀♂ <u>Age ≥ 15</u>: 8.49 ♀♂		<i>"Consumption study from 75 clinical safety studies conducted between 2006 and 2016 conducted in Mauritius (57), Spain (9), France (8), Italy (1). Studies were carried out over a 3-4 week-period. One product was provided for each participant. For sunscreen products, people were intended to have regular sun exposures during the study. The frequencies of use, close to real life conditions of use, were defined by the protocol as follows: at least twice a day for sunscreens. The amount of product used per application depended on the usage patterns specific to each participant. For young children, the cosmetic product was applied by their parents. Each tested cosmetic product was weighed at the beginning and at the end of the study. The total amount of product used per subject was obtained by subtracting the final weight to the initial weight. The values reported in the table correspond to the 95th percentile."</i> Gomez-Berrada et al., 2017
Adult: 9 (16g after 2 applications)		<i>"This study aimed to investigate how much two consecutive sunscreen applications increased the quantity of sunscreen applied and decreased the skin area left without sunscreen (missed area) compared to a single application. Participants were enrolled from a Danish website. Thirty-one healthy volunteers wearing swimwear were included and applied sunscreen two consecutive times in a laboratory environment. The overall quantities of sunscreen were calculated from the weighed amount of sunscreen used divided by the skin area. The values reported in the table correspond to the median."</i> Heerfordt et al., 2018
Age ≥ 15 ♀ <u>Spray</u>: 11.1 <u>Cream/Milk</u>: 33.6 <u>Oil</u>: 7.9 Age ≥ 15 ♂ <u>Spray</u>: 8.9 <u>Cream/Milk</u>: 47.2 <u>Oil</u>: 11.8 Age 0-3 <u>Spray</u>: 29 <u>Cream/Milk</u>: 37	[g.application⁻¹]	<i>"This study was conducted on 1078 French people (Brest, Paris, Nancy, Toulouse,). The study subjects entered a lab room specially arranged for cosmetic tests. Products usually used by volunteers were provided by the laboratory. Participants were invited to use these products in the closest possible way to their personal usage patterns. The amount of each cosmetic used was determined by differential weighing before and after use. The values reported in the table correspond to the 95th percentile."</i> Ficheux et al., 2016

Conclusion:

Sunscreen use varies among individuals. How much sunscreen is applied to the skin and how often will vary based on: habits, climate, time spent outside vs indoors (due to occupation or recreation), cost of sunscreens, socioeconomic factors, organisational, perception of skin cancer risk, susceptibility to sunburn and whether sunscreen is available when needed. This makes it difficult to estimate an exposure amount that covers all different situations for the European population that can be used for risk assessments. Therefore, the value of 18 g.d⁻¹ will be used in the exposure scenario in accordance with the SPERC Factsheet Guidance Document as median value.

Table 35: Number of swimmers (N_{swimmer}) in function of the water body dimension ($V_{\text{waterbody}}$) – applicable to swimming scenario only

Value	Unit	Reference and description
Daily number of swimmers: 1 500	[swim.d ⁻¹]	<i>"Basis is the daily average number of swimmers during summer and the average of the volume of surface water from one study conducted in the Netherlands and one study carried out in the German Federal Land Hesse. The 90th percentiles of the number of swimmers were considered a realistic worst case scenario."</i> Emission scenario document for Product Type 19 – Repellents and attractants, 2015. ECHA. Appendix 6.3.
Volume of water body : 435 000	[m ³]	Value proposed in the SPERC Factsheet Guidance Document – October 2018 - Direct release scenario for UV-filters used in sunscreen products during swimming in freshwater lakes and marine coastal waters
Daily number of people on the beach: 5 514	[people.d ⁻¹]	<i>"Located in Marseille, the second largest town in France (nearly 900,000 inhabitants), the study area was Prophète beach. The beach area is approximately 7,000 m². On July 18, 2018, the number of users at the beach was determined using an observation protocol implemented between 8 a.m. and 8 p.m., i.e., 12 h of continuous observation. The total number of persons counted at the two entrances of the beach between 8 a.m. and 8 p.m. was 5 514. Between 2 p.m. and 7 p.m., more than 150 bathers could be counted every hour, reaching a maximum of 322 persons at 4 p.m., with more than two thirds in the supervised bathing area."</i> Toubiana et al., 2021

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Daily number of people on the beach: 3 000		<i>"Located in Marseille, three urban beaches (Lave Beach, Pointe Rouge Beach, the Prophète Beach) selected as study sites were chosen based on geographical and socio-demographic criteria. They are located in various stretches of Marseille's seafront. Spite their small size (less than 1 ha), each of the three beaches is intensely frequented during the summer season, which requires specific management by local authorities from June to September (beach patrol, user services, additional cleaning and maintenance, etc.). A daily beach attendance of $N = 3000$ people was used for this estimation as it is typical for the Prophète Beach studied in this work. Bathing is a very common practice, with >75% of respondents having a bath every time they go to the beach and <3% never bathing. When going to the beach, attendees (40–60%) bathe 2.6 times on average and 91.5% of them practice whole body immersion. On average, 68.4% of the interviewees use a suncare product and apply it to their skin 2.6 times per visit to the beach."</i>
Volume of water body: 4 530	[m ³]	<i>"85 m length x 64 m width and bathymetric data ranging from 0 to 3 m depth, giving 4530 cubic meters of water. In a coastal system such as this, constant water renewal should be considered as a source of dilution, but here water renewal is limited by the dike. To compensate for this effect, we used an average water residence time of 24 h, giving a renewal of 29% of this volume after 7 h of recreation time cumulated at the 4:00 pm sampling time. This gave a total water column volume of $V_{col} = 5850 \text{ m}^3$."</i> Labille et al., 2020

Conclusion:

Two scenarios are taking into account the environmental exposure assessment distinguishing two approaches in function of the pair number of swimmer/volume of water body:

- **1 500 swim.d⁻¹ / 435 000 m³: realistic worst case scenario only for inland water bodies (ponds, shallow lakes, lakes)**
- **3 000 swim.d⁻¹ / 4 530 m³: representative scenario only for marine coastal waters**

Table 36: Fraction of swimmers using a sunscreen product ($F_{\text{swim sunscreen}}$) – applicable to swimming scenario only

Value	Unit	Reference and description
0.7	[-]	More than 75% of the 471 interviewees reported bathing every time they go to the beach, with 68% using a suncare product 2.6 times on average. Labille et al., 2020

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

<u>Average air temperature conditions (<29°C)</u> Men: 0.64 Women: 0.67 Children: 0.97 <u>Average air temperature conditions (>29°C)</u> Men: 0.69 Women: 0.72 Children: 1		<i>"This study was conducted on the Atlantic beach of Arcachon, a popular seaside resort (11,557 inhabitants). The survey was conducted on (...) a representative beach population of 2,076 people with 55% women, 32% men and 13% children (under 12 years old). Regarding the sunscreen use habits for average air temperature conditions (i.e. below 29 °C), the results show that a high proportion of children both bath and use sunscreen (96.6%). Other categories, i.e. men and women, were less represented (63.9% and 66.7% respectively)."</i> Milinkovitch et al., 2024.
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Conclusion:

The value of 0.7 for the fraction of swimmers using a sunscreen product will be used in the exposure scenario in accordance with the values found in the scientific literature as realistic worst case.

Table 37: Fraction of sunscreen products containing octocilene (F_{sunscreen_OC}) – applicable to swimming scenario only

Value	Unit	Reference and description
0.7	[-]	Registrant data provided in the CSR dossier
0.56		<i>"For the 471 people interviewed at the three beaches. Data recovered from photographs of 124 products."</i> Labille et al., 2020
0.36		Datasets obtained from the YUKA (YUKA, 2024) providing a detailed overview of cosmetic products containing Octocilene.

Conclusion:

The value of 0.36 will be used in the exposure scenario in accordance with the market data. However, this value cannot be considered a realistic worst-case value. This value considers a homogeneous distribution over Europe without taking into account the spatial variability of the market.

Table 38: Fraction of octocrilene released after swimming ($F_{\text{after_swimming}}$) – applicable to swimming scenario only

Value	Unit	Reference and description
0.5	[-]	<i>"In the EU, sunscreen product is considered as water proof, if at least 50 % of the product (including the UV-filters) remains on the skin after two times of 20 min swimming in the water. As most of the sunscreens fulfill the requirement "water resistant", the release rate of 50% into the water due to swimming activities is used as worst case assumption."</i> Registrant data
0.2		SETAC Séville (2024) Poster
0.06		<i>"Test samples included a total of eight samples consisting of four formulations (two lotions, one stick, and one spray) with and without UVFs. The method for simulating sunscreen rinse-off in seawater follows part (a) of US FDA test method 352.76 with modification to obtain data for more time intervals. However quantitative data are not presented for the spray formula, recovered less than 20% (tendency to stick to the walls of the experimental apparatus). Porcine abdominal skin was selected as a model for human skin. A pre-weighed portion of sunscreen product was applied to the skin and allowed to dry at room temperature for 20 min. After drying, the skin patch was dipped into and immediately removed from a glass jar containing 250 mL artificial seawater at 28 °C. The skin patch was then placed into a second, identical seawater jar, agitated at 100 rpm for 10 min, and removed (sample SW t = 10). This was repeated once more in a third jar (sample SW t = 20). The skin patch was then allowed to dry at room temperature for 20 min followed by agitation in a fourth and final, identical jar for 20 min (sample SW t = 60). Seawater rinsates from each jar were processed and analyzed separately."</i> Saxe et al.,2021 <u>Relevance analysis of this study:</u> Accurate rates of rinse-off from human skin during aquatic recreational activities vary by the chemical properties of the UV filter, the formulation of the product such as its water-resistance, and the application rate. The rates of rinse-off may also depend on the environmental context (e.g., salinity, water temperature, presence of dissolved or particulate organic matter). ⇒ The proposed value is not representative of : - spray use ; - all lotions and sticks formulations available on the market ; - environmental conditions ; - people in movment in bathing activities

Conclusion:

The fraction of octocrilene released after swimming of 0.2 will be used in the exposure scenario as intermediate value but cannot be considered as a realistic worst case approach due to uncertainties about the release of octocrilene in real conditions.

Table 39: Number of emission days (T_{emission}) – applicable to swimming scenario only

Value	Unit	Reference and description
1	[d]	Emission scenario document for Product Type 19 – Repellents and attractants, 2015. ECHA. Appendix 6.3.
91		In the framework of the revised Bathing Water Directive, European Member States have to report the bathing seasons to the Commission. The following table contains bathing seasons of selected European countries, as published by the European Environment Agency. The maximum duration of the bathing season is lowest in the northern European countries Denmark (92 d), Finland (77 d), and Sweden (60 d), whereas all other countries have maximum bathing seasons of more than 100 days. Germany has the longest maximum bathing season with more than 190 days. According to HLUG (2013a), noteworthy swimming in Hesse is to be expected from the beginning to the middle of May until the end of August/middle of September. This time period would comprise less than 140 maximum swimming days. For determining the bathing season relevant for assessment purposes, the following considerations have been made. Swimming in surface water bodies is highest during July and August due to suitable water and air temperatures and the fact that there is holiday season. Extensive swimming in surface water bodies during April and May is not to be expected since water temperatures and air temperatures (under moderate climate conditions) are in most cases not high enough. Swimming outside the holiday season is limited to weekends due to employment of adults and children going to school. Therefore, it is considered a realistic worst case to take 91 swimming days (3 months) for emission calculations.

Conclusion:

These values (1 d and 91 d) will be used in the exposure scenario in a realistic worst-case approach

Table 40: All remaining parameters

Parameter	Symbol	Value	Unit	Reference and description	Scenario
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APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

C_{product}	Percentage of octocrilene in sunscreen product	10	[%]	Octocrilene can be considered as safe when used as a UV filter in cosmetic products at a concentration up to 10%. Berardesca et al., 2019	Swimming and Bathing/showering scenarios
N_{local}	Number of inhabitants feeding one sewage treatment plant	10 000	[-]	Guidance on information requirements and chemical safety assessment. Chapter R.16: Environmental Exposure Estimation. Draft version 2.0 25/01/2010. ECHA	Bathing/showering scenario
T_{tidal}	Tidal period	12.41	[h]	Emission scenario document for Product Type 21: Antifouling products, 2004, Table 0.3 According to ESD for PT 21 (p. 118), the average tidal height in the OECD regions is 1.5 m. Taking into account that the majority of swimmers will be in the coastal area of maximum 2 m water depth, it can be concluded that at least 50 % of the water is replaced by fresh water during each tidal period. Therefore, 50 % water replacement is considered as a worst case.	Swimming scenario - Refinement
DIL_{tide}	DILUTION	0.5	[-]		Swimming scenario - Refinement
$k_{\text{dilution tide}}$	Dilution constant for tide	1.34	[d ⁻¹]	$=(\ln(T_{\text{period}})-\ln(T_{\text{period}}/2))/(T_{\text{period}}/24)$	Swimming scenario - Refinement
K_{degwater}	First order rate constant for biodegradation in surface water	1.54E-02 1.47E-02	[d ⁻¹]	Degradation in the whole system (water/sediment). Value for the freshwater system and the estuarine water system respectively.	Swimming scenario

B.8.5. Environmental monitoring data

B.8.5.1. Wastewater

Table 41: results of wastewater monitoring studies in Europe

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2007	Jan.	WWTP Galicia (7 500 equiv of inhab) – Spain Biological with P and N removal	Influent	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	NA	18	ng.l ⁻¹	Rodil et al., 2008
		March			Mean	36						
		July			Mean	<LOD						
		Jan.		Effluent	Mean	<LOD						
		March			Mean	20						
2	2008	Jan.to July	5 WWTPs (urban, agricultural, and industrial activities) – Spain Biological TTT and clarification	Effluent	Mean	42	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Gomez et al., 2009
					Min	15	ng.l ⁻¹					
					Max	70	ng.l ⁻¹					
					Frequency	40	%					
3	2010	July	WWTP (urban) – Spain Biological TTT	Influent	Mean	831	ng.l ⁻¹	GC-MS/MS	11	NA	ng.l ⁻¹	Negreira et al., 2011
		July		Effluent	Mean	106	ng.l ⁻¹					
		Sep.		Influent	Mean	437	ng.l ⁻¹					
		Sep.		Effluent	Mean	111	ng.l ⁻¹					
		Oct.		Influent	Mean	209	ng.l ⁻¹					
		Oct.		Effluent	Mean	65	ng.l ⁻¹					
4	2014	July	WWTP (urban and industrial activities) – Spain Biological with N removal	Influent – dissolved fraction	Value	1 050	ng.l ⁻¹	GC-MS/MS	<0.01	<0.01	ng.l ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	200	ng.l ⁻¹					
		Oct.			Value	125	ng.l ⁻¹					
		Nov.			Value	<LOD	ng.l ⁻¹					
		Dec.			Value	<LOD	ng.l ⁻¹					
	2015	Jan.			Value	<LOD	ng.l ⁻¹					
		Feb.			Value	<LOD	ng.l ⁻¹					
		Apr.			Value	9 060	ng.l ⁻¹					
		May			Value	7 790	ng.l ⁻¹					
		Jun.			Value	13 000	ng.l ⁻¹					
4	2014	July	WWTP (urban and industrial activities) – Spain Biological with N removal	Influent – Particulate fraction	Value	149 000	ng.g ⁻¹	GC-MS/MS	0.02	0.01	ng.g ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	65 000	ng.g ⁻¹					
		Oct.			Value	53 100	ng.g ⁻¹					
		Nov.			Value	27 500	ng.g ⁻¹					
		Dec.			Value	28 400	ng.g ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
	2015	Jan.			Value	32 100	ng.g ⁻¹					
		Feb.			Value	37 800	ng.g ⁻¹					
		Apr.			Value	31 200	ng.g ⁻¹					
		May			Value	76 000	ng.g ⁻¹					
		Jun.			Value	240 000	ng.g ⁻¹					
4	2014	July	WWTP (urban and industrial activities) – Spain Biological with N removal	Effluent – dissolved fraction	Value	<LOD	ng.l ⁻¹	GC-MS/MS	<0.01	<0.01	ng.l ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	<LOQ	ng.l ⁻¹					
		Oct.			Value	<LOQ	ng.l ⁻¹					
		Nov.			Value	<LOD	ng.l ⁻¹					
		Dec.			Value	<LOD	ng.l ⁻¹					
	2015	Jan.			Value	<LOD	ng.l ⁻¹					
		Feb.			Value	<LOD	ng.l ⁻¹					
		Apr.			Value	3 930	ng.l ⁻¹					
		May			Value	885	ng.l ⁻¹					
		Jun.			Value	56	ng.l ⁻¹					
		July			Value	13 100	ng.g ⁻¹	GC-MS/MS	0.02	0.01	ng.g ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	33 700	ng.g ⁻¹					
		Oct.			Value	14 300	ng.g ⁻¹					
		Nov.			Value	16 800	ng.g ⁻¹					
		Dec.			Value	5 690	ng.g ⁻¹					
4	2014	Jan.	WWTP (urban and industrial activities) – Spain Biological with N removal	Effluent – Particulate fraction	Value	13 100	ng.g ⁻¹					
		Feb.			Value	19 000	ng.g ⁻¹					
		Apr.			Value	20 400	ng.g ⁻¹					
		May			Value	21 400	ng.g ⁻¹					
		Jun.			Value	95 000	ng.g ⁻¹					
	2015	Jan.			Value	13 100	ng.g ⁻¹	GC-MS/MS	0.02	0.01	ng.g ⁻¹	Biel-maeso et al., 2019
		Feb.			Value	19 000	ng.g ⁻¹					
		Apr.			Value	20 400	ng.g ⁻¹					
		May			Value	21 400	ng.g ⁻¹					
		Jun.			Value	95 000	ng.g ⁻¹					
		July			Value	13 100	ng.g ⁻¹					
		Sep.			Value	33 700	ng.g ⁻¹					
5	2019	May	WWTP (urban) 1– Spain Biological TTT	Influent	Mean	1 940	ng.l ⁻¹	LC-MS/MS	152	45.6	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
		May		Effluent	Mean	<LOD	ng.l ⁻¹					
		Jun.		Influent	Mean	10 600	ng.l ⁻¹					
		Jun.		Effluent	Mean	1 060	ng.l ⁻¹					
		July		Influent	Mean	8 270	ng.l ⁻¹					
		July		Effluent	Mean	1 280	ng.l ⁻¹					
		Aug.		Influent	Mean	8 150	ng.l ⁻¹					
		Aug.		Effluent	Mean	790	ng.l ⁻¹					
		Sep.		Influent	Mean	6 960	ng.l ⁻¹					
		Sep.		Effluent	Mean	890	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
		Oct.		Influent	Mean	9 990	ng.l ⁻¹					
		Oct.		Effluent	Mean	640	ng.l ⁻¹					
5	2019	May	WWTP (urban) 2–Spain Biological TTT and microfiltration	Influent	Mean	21 700	ng.l ⁻¹	LC-MS/MS	152	45.6	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
		May		Effluent	Mean	<LOD	ng.l ⁻¹					
		Jun.		Influent	Mean	5 740	ng.l ⁻¹					
		Jun.		Effluent	Mean	1 310	ng.l ⁻¹					
		July		Influent	Mean	27 000	ng.l ⁻¹					
		July		Effluent	Mean	320	ng.l ⁻¹					
		Aug.		Influent	Mean	3 950	ng.l ⁻¹					
		Aug.		Effluent	Mean	1 820	ng.l ⁻¹					
		Sep.		Influent	Mean	2 920	ng.l ⁻¹					
		Sep.		Effluent	Mean	<LOD	ng.l ⁻¹					
		Oct.		Influent	Mean	1 670	ng.l ⁻¹					
		Oct.		Effluent	Mean	630	ng.l ⁻¹					
5	2019	May	WWTP (urban) 3–Spain Biological TTT	Influent	Mean	208 000	ng.l ⁻¹	LC-MS/MS	152	45.6	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
		May		Effluent	Mean	1 430	ng.l ⁻¹					
		Jun.		Influent	Mean	157 000	ng.l ⁻¹					
		Jun.		Effluent	Mean	1 220	ng.l ⁻¹					
		July		Influent	Mean	150 000	ng.l ⁻¹					
		July		Effluent	Mean	350	ng.l ⁻¹					
		Aug.		Influent	Mean	127 000	ng.l ⁻¹					
		Aug.		Effluent	Mean	390	ng.l ⁻¹					
		Sep.		Influent	Mean	9 360	ng.l ⁻¹					
		Sep.		Effluent	Mean	<LOD	ng.l ⁻¹					
		Oct.		Influent	Mean	21 600	ng.l ⁻¹					
		Oct.		Effluent	Mean	720	ng.l ⁻¹					
6	NA	NA	WWTP Tábor (municipal and	Influent	Mean	4 100	ng.l ⁻¹	LC-MS/MS	NA	NA	ng.l ⁻¹	Amankwah et al., 2024

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
7	2015		industrial activities) - Czech Republic	Effluent	Mean	78	ng.l ⁻¹					
			WWTP Tábor-Klokoty (municipal and hospital activities) - Czech Republic	Influent	Mean	150	ng.l ⁻¹					
				Effluent	Mean	<32	ng.l ⁻¹					
		Autumn	WWTP Skravelsjö (30-40 inhab.) – Sweden Soil bed Treatment	Influent	Mean	8 600	ng.l ⁻¹	GC-MS/MS	29	7.1	ng.l ⁻¹	Blum et al., 2016
		Autumn		Effluent	Mean	<LOQ	ng.l ⁻¹					
		Autumn	WWTP Hjöggbjölle (60 inhab.) – Sweden Soil bed treatment	Influent	Mean	4 200	ng.l ⁻¹					
		Autumn		Effluent	Mean	160	ng.l ⁻¹					
		Autumn	WWTP Bjännsjö (14 inhab.) – Sweden Soil bed treatment	Influent	Mean	6 800	ng.l ⁻¹					
		Autumn		Effluent	Mean	1 200	ng.l ⁻¹					
		Autumn	WWTP Mörkö (80-90 inhab.) – Sweden Soil bed treatment	Influent	Mean	1 200	ng.l ⁻¹					
		Autumn		Effluent	Mean	25	ng.l ⁻¹					
		Autumn	WWTP Drotting (39 inhab.) – Sweden Soil bed treatment	Influent	Mean	6 400	ng.l ⁻¹					
		Autumn		Effluent	Mean	76	ng.l ⁻¹					
		Autumn	WWTP Holm (2930 inhab.) – Sweden Chemical, Biological, Sedimentation, Disinfection treatment	Influent	Mean	1 400	ng.l ⁻¹					
		Autumn		Effluent	Mean	53	ng.l ⁻¹					
		Autumn	WWTP Sävar (80000 inhab.) – Sweden	Influent	Mean	4 000	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
		Autumn	Chemical, Biological, Sedimentation, Disinfection, bio bed, treatment	Effluent	Mean	26	ng.l ⁻¹					
		Autumn	WWTP Gässlösa (94978 inhab.) – Sweden	Influent	Mean	1 600	ng.l ⁻¹					
		Autumn	Chemical, Biological, Sedimentation, Disinfection, treatment	Effluent	Mean	32	ng.l ⁻¹					
		Autumn	WWTP Umeå (171400 inhab.) – Sweden	Influent	Mean	5 300	ng.l ⁻¹					
		Autumn	Chemical, Biological, Sedimentation, Disinfection, treatment	Effluent	Mean	<LOQ	ng.l ⁻¹					
		Autumn	WWTP Kungsängsverket (784100 inhab.) – Sweden	Influent	Mean	6 700	ng.l ⁻¹					
		Autumn	Chemical, Biological, Sedimentation, treatment	Effluent	Mean	<LOQ	ng.l ⁻¹					
8	2013	Sep. Oct.	WWTP Slemmestad – municipal, 580 000 inhab) – Norway Chemical, Biological treatment	Effluent	Min	181	ng.l ⁻¹	GC-HRMS LC-HRMS	5	NA	ng.l ⁻¹	Langford et al., 2015
	2013	Sep. Oct.		Effluent	Median	258	ng.l ⁻¹					
	2013	Sep. Oct.		Effluent	Max	538	ng.l ⁻¹					
	2013	Jun. July	WWTP Ottestad – municipal, 52 000 inhab) – Norway Chemical, Biological treatment	Effluent	Min	7	ng.l ⁻¹					
	2013	Jun. July		Effluent	Median	158	ng.l ⁻¹					
	2013	Jun. July		Effluent	Max	227	ng.l ⁻¹					
	2013	July	WWTP Tromsø – municipal, 38 400 inhab) – Norway mechanical treatment only	Effluent	Min	1 701	ng.l ⁻¹					
	2013	July		Effluent	Median	2 167	ng.l ⁻¹					
	2013	July		Effluent	Max	6 969	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Nº	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
9	2014	Oct.	WWTP Slemmestad – municipal, 580 000 inhab) – Norway Chemical, Biological treatment	Effluent	Min	310	ng.l ⁻¹	GC-MS	NA	NA	ng.l ⁻¹	Krzeminski et al., 2017
	2014	Oct.			Median	537	ng.l ⁻¹					
	2014	Oct.			Max	763	ng.l ⁻¹					
10	2013	May	WWTP Bragança (41955 hab.eq./year), Domestic, industrial (residual) activities – Portugal Biological treatment (activated sludge)	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP Cachãoa (10000 hab.eq./year), Domestic, industrial (mainly) activities – Portugal Biological treatment (activated sludge)	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP Chaves (57748 hab.eq./year), Domestic, industrial (residual) activities – Portugal Biological treatment (activated sludge) and UV disinfection	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	March	WWTP Viana do Castelo Cidade (45257 hab.eq./year), Domestic, hospital, industrial activities – Portugal Biological treatment (activated sludge) and UV disinfection Discharge in Ocean	Influent	Mean	659.8	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
		May			Mean	<LOQ	ng.l ⁻¹					
		Jun.			Mean	689.6	ng.l ⁻¹					
		Jun.		Effluent	Mean	154.3	ng.l ⁻¹					
10	2013	May	WWTP Ave Cidade (25557 hab.eq./year), Domestic, industrial	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Nº	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
			activities – Portugal Biological treatment (activated sludge) and UV disinfection									
10	2013	March	WWTP Gaia Litoral (300000 hab.eq./year), Domestic activities – Portugal Biological treatment (activated sludge) and UV disinfection Discharge in Ocean	Influent	Mean	159.5	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
		May		Influent	Mean	88.1	ng.l ⁻¹					
		Jun.		Influent	Mean	247.6	ng.l ⁻¹					
10	2013	May	WWTP Choupal (213000 hab.eq./year), Domestic, industrial activities – Portugal Trickling Filters	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP S. Pedro (6850 hab.eq./year), Domestic, hospital, industrial activities – Portugal Biological treatment (activated sludge)	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP Frielas, Domestic, industrial activities – Portugal Biological treatment (activated sludge) and biofiltration	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP Beirolas, Domestic, industrial activities – Portugal Biological treatment (activated sludge) and disinfection	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	March	WWTP Alcântara, Domestic activities – Portugal Biofiltration	Influent	Mean	491.6	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
		March		Effluent	Mean	353.5	ng.l ⁻¹					
		May		Influent	Mean	49.1	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Nº	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
		June		Influent	Mean	687.8	ng.l ⁻¹					
		June		Effluent	Mean	3357.4	ng.l ⁻¹					
10	2013	May	WWTP Évora (60000 hab.eq./year), Domestic, hospital, industrial activities – Portugal Biological treatment (activated sludge) and disinfection	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May	WWTP Reguengos de Monsaraz (8700 hab.eq./year), Domestic activities – Portugal Biological treatment (activated sludge) and disinfection	Influent	Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	March	WWTP Vale Faro (49547 hab.eq./year), Domestic activities – Portugal Biological treatment (activated sludge) and disinfection Discharge in Ocean	Influent	Mean	235.4	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May			Mean	<LOQ	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	Jun.			Mean	785.5	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	Jun.		Effluent	Mean	124.6	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	March	WWTP Vila Real de Santo António (30766 hab.eq./year), Domestic activities – Portugal Secondary treatment and disinfection	Influent	Mean	19.6	ng.l ⁻¹	GC-MS	10	3	ng.l ⁻¹	Cunha et al., 2015
10	2013	May			Mean	<LOQ	ng.l ⁻¹					
10	2013	Jun.			Mean	546	ng.l ⁻¹					
11	2017	Jun.	WWTP Porto Secondary treatment	Influent	Mean	554.3	ng.l ⁻¹	GC-MS/MS	0.4	0.1	ng.l ⁻¹	Ramos et al., 2019b
				Effluent	Mean	101.3	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
12				Influent	Mean	671	ng.l ⁻¹	GC-MS/SIM	NA	10	ng.l ⁻¹	Balmer et al., 2005
				Effluent	Mean	<LOD	ng.l ⁻¹					
		Apr.	WWTP Horgen (20560 inhab.) – Switzerland Mechanical, biological, and chemical treatment	Influent	Mean	200	ng.l ⁻¹					
				Effluent	Mean	20	ng.l ⁻¹					
		Apr.	WWTP Meilen (22000 inhab.) – Switzerland Mechanical, biological, and chemical treatment	Influent	Mean	100	ng.l ⁻¹					
				Effluent	Mean	<LOD	ng.l ⁻¹					
		Jun.	WWTP Meilen (22000 inhab.) – Switzerland Mechanical, biological, and chemical treatment	Influent	Mean	12 000	ng.l ⁻¹					
				Effluent	Mean	60	ng.l ⁻¹					
		Apr.	WWTP Küsnacht (16475 inhab.) – Switzerland Mechanical, biological, and chemical treatment	Influent	Mean	NA	ng.l ⁻¹					
				Effluent	Mean	<LOD	ng.l ⁻¹					
		Apr.	WWTP Männedorf (9044 inhab.) 4340 m ³ .d ⁻¹ - Switzerland	Influent	Mean	NA	ng.l ⁻¹					
				Effluent	Mean	50	ng.l ⁻¹					
				Influent	Mean	NA	ng.l ⁻¹					
				Effluent	Mean	300	ng.l ⁻¹					
		2003 Sep.		Influent	Mean	6 100	ng.l ⁻¹					
				Effluent	Mean	90	ng.l ⁻¹					
		2002 Jun..	WWTP Thalwil (19548 inhab.) - Switzerland Mechanical, biological, and chemical treatment	Influent	Mean	3 200	ng.l ⁻¹					
				Effluent	Mean	10	ng.l ⁻¹					
		2003 Sep.		Influent	Mean	2 100	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
13	2002	Apr.	WWTP Wädenswil (19000 inhab.) – Switzerland Mechanical, biological, and chemical treatment	Effluent	Mean	30	ng.l ⁻¹	GC-MS/SIM	17	5	ng.l ⁻¹	Kupper et al., 2006
				Influent	Mean	1 100	ng.l ⁻¹					
				Effluent	Mean	100	ng.l ⁻¹					
				Influent	Mean	2 700	ng.l ⁻¹					
				Effluent	Mean	10	ng.l ⁻¹					
				Influent	Mean	2 100	ng.l ⁻¹					
				Effluent	Mean	10	ng.l ⁻¹					
				Effluent	Mean	10	ng.l ⁻¹					
13	2002	Apr.	WWTP Mittleres Emmental, Hasle (23 000 inhab), domestic, industrial activities – Switzerland Biological treatment (activated sludge)	Influent	Value	1 970	ng.l ⁻¹	GC-MS/SIM	17	5	ng.l ⁻¹	Kupper et al., 2006
					Value	3 060	ng.l ⁻¹					
					Value	2 410	ng.l ⁻¹					
					Value	1 150	ng.l ⁻¹					
					Value	950	ng.l ⁻¹					
					Value	950	ng.l ⁻¹					
					Value	1 300	ng.l ⁻¹					
					Mean	1 680	ng.l ⁻¹					
					Median	1 300	ng.l ⁻¹					
				Effluent	Value	<LOQ	ng.l ⁻¹	GC-MS/SIM	17	5	ng.l ⁻¹	Kupper et al., 2006
					Value	<LOQ	ng.l ⁻¹					
					Value	<LOQ	ng.l ⁻¹					
					Value	<LOQ	ng.l ⁻¹					
					Value	<LOQ	ng.l ⁻¹					
					Value	20	ng.l ⁻¹					
					Value	<LOQ	ng.l ⁻¹					
					Value	<LOQ	ng.l ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.				
					Median	<LOQ	ng.l ⁻¹									
14	2011	Jun.	WWTP La Spezia (200 000 inhabitants) - Italy	Influent	Value	71	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a				
					Value	28	ng.l ⁻¹									
					Value	29	ng.l ⁻¹									
					Value	36	ng.l ⁻¹									
					Value	463	ng.l ⁻¹									
				Effluent	Value	<LOQ	ng.l ⁻¹									
					Value	<LOQ	ng.l ⁻¹									
					Value	<LOQ	ng.l ⁻¹									
					Value	13	ng.l ⁻¹									
					Value	<LOQ	ng.l ⁻¹									
15	2011	Apr.	WWTP Genoa (631,000 inhabitants) - Italy	Influent	Value	12	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012b				
					Value	50	ng.l ⁻¹									
		May			Value	27	ng.l ⁻¹									
					Value	69	ng.l ⁻¹									
		Jun.			Value	178	ng.l ⁻¹									
					Value	119	ng.l ⁻¹									
					Value	16	ng.l ⁻¹									
					Value	118	ng.l ⁻¹									
		July			Value	53	ng.l ⁻¹									
					Value	73	ng.l ⁻¹									
					Value	390	ng.l ⁻¹									
					Value	195	ng.l ⁻¹									
		Aug.			Value	62	ng.l ⁻¹									
					Value	339	ng.l ⁻¹									
					Value	209	ng.l ⁻¹									
		Sep.			Value	31	ng.l ⁻¹									

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

N°	Year	Period	Location	Media	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
					Value	123	ng.l ⁻¹					
					Value	76	ng.l ⁻¹					
15	2011	Apr.	WWTP Genoa (631,000 inhabitants) - Italy	Effluent	Value	6	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/b
					Value	7	ng.l ⁻¹					
					Value	7	ng.l ⁻¹					
		May			Value	6	ng.l ⁻¹					
					Value	65	ng.l ⁻¹					
					Value	26	ng.l ⁻¹					
		Jun.			Value	4	ng.l ⁻¹					
					Value	9	ng.l ⁻¹					
					Value	7	ng.l ⁻¹					
		July			Value	5	ng.l ⁻¹					
					Value	31	ng.l ⁻¹					
					Value	17	ng.l ⁻¹					
		Aug.			Value	4	ng.l ⁻¹					
					Value	126	ng.l ⁻¹					
					Value	65	ng.l ⁻¹					
		Sep.			Value	6	ng.l ⁻¹					
					Value	13	ng.l ⁻¹					
					Value	9	ng.l ⁻¹					

B.8.5.1.1. Sludge**Table 42: results of sludge monitoring studies in Europe**

Nº	Year	Period	Location	Treatment	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2010	March	WWTP 1 (urban) - Spain	Primary sludge	Mean	2 242	ng.g ⁻¹ _{wwt}	GC-MS	NA	33	ng.g ⁻¹ _{wwt}	Negreira et al., 2011
		March	WWTP 2 (urban) - Spain	Biological sludge	Mean	3 263						
		Feb.-May	WWTP 3 (urban) - Spain	Primary sludge	Mean	1 039						
		Feb.-May	WWTP 4 (urban) - Spain	Biological sludge	Mean	377						
		Feb.-May	WWTP 5 (urban) - Spain	Mixture sludge: Prim. / Bio.	Mean	1 766						
		Feb.-May	WWTP 6 (urban) - Spain	Mixture sludge: Prim. / Bio.	Mean	1 038						
		Feb.-May	WWTP 7 (urban) - Spain	Mixture sludge: Prim. / Bio.	Mean	1 934						
		Feb.-May	WWTP 8 (urban) - Spain	Mixture sludge: Prim. / Bio.	Mean	523						
		Feb.-May	WWTP 9 (urban) - Spain	Mixture sludge: Prim. / Bio.	Mean	2 240						
2	2009	Jun.	WWTP Girona - Spain	Biological (activated sludge) with P and N removal + Tertiary	Value	2 600	ng.g ⁻¹ _{dwt}	UPLC-MS/MS	18	60	ng.g ⁻¹ _{dwt}	Gago-Ferrero et al., 2011b
		Jun.	WWTP St Feliu de Llobretat - Spain	Biological (activated sludge) + Tertiary	Value	2 870						
		Jun.	WWTP Rubi - Spain	Biological (activated sludge) with N removal	Value	9 170						
		Jun.	WWTP Granollers - Spain	Biological (activated sludge) with P and N removal	Value	2 610						
		Jun.	WWTP Reus - Spain	Biological (activated sludge)	Value	5 390						
		Jun.	WWTP Montcada - Spain	Biological (activated sludge)	Value	4 150						
		Jun.	WWTP Llagosta - Spain	Biological (activated sludge) with P removal	Value	4 490						

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		Jun.	WWTP Lleida - Spain	Biological (activated sludge)	Value	2 860						
		Jun.	WWTP Vilafranca - Spain	Biological (activated sludge) with P and N removal	Value	2 160						
		Jun.	WWTP Mataro - Spain	Biological (activated sludge)	Value	3 630						
		Jun.	WWTP Prat de Llobregat - Spain	Biological (activated sludge) with P and N removal + Tertiary	Value	6 600						
		Jun.	WWTP Terrassa - Spain	Biological(activated sludge)	Value	2 250						
		Jun.	WWTP Gava - Spain	Biological (activated sludge)	Value	3 860						
		Jun.	WWTP Teia - Spain	Biological (activated sludge)	Value	3 000						
		Jun.	WWTP Vic - Spain	Biological (activated sludge) with P and N removal	Value	1 060						
3	2014	July	WWTP Jerez de la Frontera (urban and industrial activities) - Spain	Biological (activated sludge) with N removal Influent of anaerobic digester	Value	9 030	ng.g ⁻¹	GC-MS/MS	0.0 2	0.0 1	ng.g ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	7 780	ng.g ⁻¹					
		Oct.			Value	742	ng.g ⁻¹					
		Nov.			Value	1 220	ng.g ⁻¹					
		Dec.			Value	1 850	ng.g ⁻¹					
	2015	Jan.			Value	2 130	ng.g ⁻¹					
		Feb.			Value	3 310	ng.g ⁻¹					
		Apr.			Value	1 710	ng.g ⁻¹					
		May			Value	3 340	ng.g ⁻¹					
		Jun.			Value	7 640	ng.g ⁻¹					
	2014	July	WWTP Jerez de la Frontera (urban and industrial activities) - Spain	Biological (activated sludge) with N removal Effluent of anaerobic digester	Value	5 280	ng.g ⁻¹	GC-MS/MS	0.0 2	0.0 1	ng.g ⁻¹	Biel-maeso et al., 2019
		Sep.			Value	7 200	ng.g ⁻¹					
		Oct.			Value	654	ng.g ⁻¹					
		Nov.			Value	1 500	ng.g ⁻¹					
		Dec.			Value	1 520	ng.g ⁻¹					
	2015	Jan.			Value	2 110	ng.g ⁻¹					
		Feb.			Value	2 360	ng.g ⁻¹					
		Apr.			Value	1 690	ng.g ⁻¹					
		May			Value	2 630	ng.g ⁻¹					
		Jun.			Value	7 060	ng.g ⁻¹					
4	2013	Sep. Oct.	WWTP Slemmestad	Chemical and Biological	Min	3 448	ng.g ⁻¹ _{dwt}	GC-	5	NA	ng.g ⁻¹ _{dwt}	Langford et

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		Sep. Oct.	– municipal, 580 000 inhab) – Norway	(activated sludge)	Median	6 257	ng.g ⁻¹ _{dwt}	HRMS LC-HRMS	5	NA	ng.g ⁻¹ _{dwt}	al., 2015
		Sep. Oct.		Max	12 661	ng.g ⁻¹ _{dwt}						
		Jun. July	WWTP Ottestad – municipal, 52 000 inhab) – Norway	Chemical and Biological (activated sludge)	Min	26 823	ng.g ⁻¹ _{dwt}	GC-HRMS LC-HRMS				
		Jun. July			Median	35 973	ng.g ⁻¹ _{dwt}					
		Jun. July			Max	41 610	ng.g ⁻¹ _{dwt}					
5	2013-2017	Winter	WWTP Oporto (municipal, 80 000 inhab) - Portugal	Biological (activated sludge)	Mean	59 557	ng.g ⁻¹ _{dw}	GC-MS/MS	19	6	ng.g ⁻¹ _{dw}	Ramos et al., 2018
		Summer		Mean	87 999	ng.g ⁻¹ _{dw}	GC-MS/MS	19	6	ng.g ⁻¹ _{dw}	Ramos et al., 2018	
6	2002	Apr.	WWTP Mittleres Emmental, Hasle (23 000 inhab), urban, industrial activities – Switzerland	Biological (activated sludge) Influent of anaerobic digester	Value	4 150	ng.g ⁻¹	GC-MS/SIM	20	6	ng.g ⁻¹	Kupper et al., 2006
		Apr.			Value	1 420	ng.g ⁻¹					
		Apr.			Value	1 200	ng.g ⁻¹					
		Apr.			Value	3 240	ng.g ⁻¹					
		Apr.			Value	4 680	ng.g ⁻¹					
		Apr.			Value	1 990	ng.g ⁻¹					
		Apr.			Value	2 370	ng.g ⁻¹					
		Apr.			Value	2 720	ng.g ⁻¹					
		Apr.			Mean	2 370	ng.g ⁻¹					
		Apr.			Median	4 150	ng.g ⁻¹					
		Apr.	Biological treatment (activated sludge)	Biological (activated sludge) Effluent of anaerobic digester	Value	3 040	ng.g ⁻¹	GC-MS/SIM	20	6	ng.g ⁻¹	Kupper et al., 2006
		Nov.		Value	4 950	ng.g ⁻¹						
		Apr.		Biological (activated sludge) Storage tank	Value	1 980	ng.g ⁻¹					
		Nov.		Value	9 520	ng.g ⁻¹						
7	2001	Jan.	WWTP Chevilly (210 inhab.), separate sewer system, rural activities - Switzerland	NA	Value	850	ng.g ⁻¹ _{dwt}	GC-MS/SIM	NA	6	ng.g ⁻¹ _{dwt}	Plagelat et al., 2006
		July		NA	Value	900	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Cronay (284 inhab.), separate sewer system, rural activities - Switzerland	NA	Value	1 410	ng.g ⁻¹ _{dwt}					
		July		NA	Value	1 590	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Thierrens (514 inhab.),	NA	Value	1 190	ng.g ⁻¹ _{dwt}					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		July	separate sewer system, rural activities - Switzerland	NA	Value	2 480	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Prahins (214 inhab.), separate sewer system, rural activities - Switzerland	NA	Value	320	ng.g ⁻¹ _{dwt}					
		July	WWTP Prahins (214 inhab.), separate sewer system, rural activities - Switzerland	NA	Value	530	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Echallens (5700 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	6 440	ng.g ⁻¹ _{dwt}					
		July	WWTP Echallens (5700 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	7 860	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Wohlen (8460 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	6 300	ng.g ⁻¹ _{dwt}					
		July	WWTP Wohlen (8460 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	6 930	ng.g ⁻¹ _{dwt}					
		Feb.	WWTP Wenslingen (674 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	4 400	ng.g ⁻¹ _{dwt}					
		July	WWTP Wenslingen (674 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	2 580	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Seuzach (6500 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	7 760	ng.g ⁻¹ _{dwt}					
		July	WWTP Seuzach (6500 inhab.), separate sewer system, rural activities with a combined sewer system - Switzerland	NA	Value	4 280	ng.g ⁻¹ _{dwt}					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		Jan.	WWTP Konolfingen (7860 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	3 740	ng.g ⁻¹ _{dwt}					
		July	WWTP Konolfingen (7860 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	4 560	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Muri (7800 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	1 830	ng.g ⁻¹ _{dwt}					
		July	WWTP Muri (7800 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	1 800	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Reinach (17140 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	1 800	ng.g ⁻¹ _{dwt}					
		July	WWTP Reinach (17140 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	1 600	ng.g ⁻¹ _{dwt}					
		Nov.	WWTP Affoltern (14610 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	10 030	ng.g ⁻¹ _{dwt}					
		Aug.	WWTP Affoltern (14610 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	18 740	ng.g ⁻¹ _{dwt}					
		Jan.	WWTP Gossau (11300 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	2 800	ng.g ⁻¹ _{dwt}					
		July	WWTP Gossau (11300 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	6 930	ng.g ⁻¹ _{dwt}					
	2003	Nov.	WWTP Zürich-Werdhölzli (10950 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	10 950	ng.g ⁻¹ _{dwt}					
		Aug.	WWTP Zürich-Werdhölzli (10950 inhab.), combined sewer system, urban, industrial activities - Switzerland	NA	Value	14 950	ng.g ⁻¹ _{dwt}					

B.8.5.1. Aquatic ecosystems

B.8.5.1.1. Freshwater environment

B.8.5.1.1.1. Lake

(In red, value > $PNEC_{aquatic}$ (freshwater): 266 ng.l⁻¹)

Table 43: results of lake monitoring studies in Europe

N°	Year	Period	Country	Location	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	1998	April	Switzerland	Lake Zurich	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	July	Switzerland	Lake Zurich	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	Aug.	Switzerland	Lake Zurich	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	Oct.	Switzerland	Lake Zurich	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	April	Switzerland	Lake Hüttnersee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	July	Switzerland	Lake Hüttnersee	Value	27	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
1	1998	Sep.	Switzerland	Lake Hüttnersee	Value	4	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Poiger et al., 2004
2	2002	July	Switzerland	Lake Jörisee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Sep.	Switzerland	Lake Jörisee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Aug.	Switzerland	Lake Zürichsee	Value	3	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Aug.	Switzerland	Lake Zürichsee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Sep.	Switzerland	Lake Zürichsee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Sep.	Switzerland	Lake Zürichsee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Sep.	Switzerland	Lake Zürichsee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	Aug.	Switzerland	Lake Greifensee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	July	Switzerland	Lake Greifensee	Value	5	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	July	Switzerland	Lake Hüttnersee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
2	2002	July	Switzerland	Lake Hüttnersee	Value	<LOD	ng.l ⁻¹	GC-MS	NA	2	ng.l ⁻¹	Balmer et al., 2005
3	2007	May	Switzerland	Lake Cospuden	Mean	10	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
3	2007	Jun.	Switzerland	Lake Cospuden	Mean	250	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
3	2007	May	Switzerland	Lake Bagger	Mean	73	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
3	2007	Jun.	Switzerland	Lake Bagger	Mean	126	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
4	2009	Summer	Switzerland	Lake Cospuden	Mean	4 381	ng.l ⁻¹	LC-MS/MS	NA	8.5	ng.l ⁻¹	Rodil et al., 2009
5	2021	Aug.	Serbia	Lake Sava	Mean	23	ng.l ⁻¹	LC-MS/MS	NA	1.7	ng.l ⁻¹	Lukić et al., 2022
5	2021	Aug.	Serbia	Lake Sava	Max	58	ng.l ⁻¹	LC-MS/MS	NA	1.7	ng.l ⁻¹	Lukić et al., 2022
5	2021	Aug.	Serbia	Lake Sava	Min	4.20	ng.l ⁻¹	LC-MS/MS	NA	1.7	ng.l ⁻¹	Lukić et al., 2022

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

B.8.5.1.1.1. River

Table 44: results of river monitoring studies in Europe

Nº	Year	Period	Country	Location	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2007	Jan.	Spain	Mero river (Galicia)	Value	<LOD	ng.l ⁻¹	LC-MS/MS	NA	18	ng.l ⁻¹	Rodil et al., 2008
1	2007	March	Spain	Mero river (Galicia)	Value	<LOD	ng.l ⁻¹	LC-MS/MS	NA	18	ng.l ⁻¹	Rodil et al., 2008
1	2007	July	Spain	Mero river (Galicia)	Value	<LOD	ng.l ⁻¹	LC-MS/MS	NA	18	ng.l ⁻¹	Rodil et al., 2008
2	2008	Jan-July	Spain	River centre of Spain	Min	13	ng.l ⁻¹	GC-MS/SIM	NA	2	ng.l ⁻¹	Gomez et al., 2009
2	2008	Jan-July	Spain	River centre of Spain	Mean	59	ng.l ⁻¹	GC-MS/SIM	NA	2	ng.l ⁻¹	Gomez et al., 2009
2	2008	Jan-July	Spain	River centre of Spain	Max	283	ng.l ⁻¹	GC-MS/SIM	NA	2	ng.l ⁻¹	Gomez et al., 2009
3	2018	July	Spain	Ebro Delta river	Mean	1 436	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta river	Median	1 451	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta river	Max	>2 000	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta river	Min	677	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta river	Mean	391	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta river	Median	326	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta river	Max	788	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta river	Min	159	ng.l ⁻¹	SBSE/GS/MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
4	2016	Sep.	Spain	Ave river	Max	9.5	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Ave river	Min	7.6	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Ave river	Median	7.7	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Leça river	Max	118.7	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Leça river	Min	60.7	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Leça river	Median	57.4	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Antua river	Max	39.3	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Antua river	Min	20.1	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Antua river	Median	15.8	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Certima river	Max	128.6	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Certima river	Min	53.9	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2016	Sep.	Spain	Certima river	Median	26.8	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Ave river	Max	9.8	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Ave river	Min	10.9	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Ave river	Median	10.9	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Leça river	Max	14.1	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Leça river	Min	9.6	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Leça river	Median	9.2	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Antua river	Max	31.6	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Antua river	Min	22.4	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Antua river	Median	23.5	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Certima river	Max	44.5	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

4	2017	Feb.	Spain	Certima river	Min	29	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
4	2017	Feb.	Spain	Certima river	Median	29	ng.l ⁻¹	GC-MS/MS	4.3	1.3	ng.l ⁻¹	Homem et al., 2022
5	2007	May	Germany	Elsterbecken river	Mean	<LOD	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
5	2007	May	Germany	Parthe river	Mean	16	ng.l ⁻¹	GC-MS	11	7	ng.l ⁻¹	Rodil et al., 2007
6	2011	April	Italy	Sturla river (Genoa)	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/a
6	2011	May	Italy	Sturla river (Genoa)	Mean	112	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/a
6	2011	Jun.	Italy	Sturla river (Genoa)	Mean	95	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/a
6	2011	July	Italy	Sturla river (Genoa)	Mean	19	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/a
6	2011	Aug.	Italy	Sturla river (Genoa)	Mean	11	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012/a

B.8.5.1.2. Marine environment

(In red, value > PNEC_{aquatic} (marine water): 27 ng.l⁻¹)

Details by study is given below:**Spain****Galicia**

In *Rodil et al., 2008*, all samples were collected in Galicia (North-Western Spain) in 2007. Seawater samples were collected in March and July at Santa Cristina beach, near the mouth of the river Mero. All concentrations are below 18 ng.l⁻¹ (LOD).

Ebro Delta

The Ebro Delta area is located in the north-eastern Spain; ecosystem of crucial importance for the environment with also a high economic activity (agriculture (rice) farming, shellfish culture, and fisheries), chemical industries, a nuclear powerplant and several urban wastewater treatment plants (WWTPs)).

In *Aminot et al., 2019*, samples were collected between May and June 2016 along the mussel lines, at two points (150m and 500m away to a treated wastewater outfall from a 16 046-population equivalent WWTP). A maximum value of 303.5 ng.l⁻¹ was reported at a sample point situated within 500 m of the outfall of the WWTP. **All sample values were above the PNEC for aquatic organisms.**

In *García-Pimentel et al., 2023*, samples were carried out in summer (July 2018) and in winter (March 2019) at a depth of 0–15 cm. Octocrilene was detected in all samples, the median and maximum concentration in summer being significantly higher than in winter (mean >2 000 and 157.1 ng.l⁻¹, maximum >2 000 ng.l⁻¹ and = 394.6 ng.l⁻¹ respectively). The highest concentrations in winter were found at the points closest to the inner area of the bay, where a lower hydrodynamics take place. **Environmental risks for aquatic organisms were observed throughout the year and at all sampling points.**

Mar Menor Lagoon

Mar Menor is a hypersaline restricted coastal lagoon located in the South East of Spain. It is isolated from the Mediterranean Sea by a 24 km long sand bar that is only crossed by five channels, causing large seasonal fluctuations of temperature and salinity. It is located close to the Campo de Cartagena area, relevant in terms of agricultural, tourism, industrial and recreational activities. Moreover, it also receives wastewater effluents continuously.

In *García-Pimentel et al., 2023*, Samples were carried out in summer (July 2018) and in winter (February 2019) at a depth of 0–15 cm. Octocrilene was detected in all samples, the median and maximum concentration in summer being lower than in winter (median= 54.6 ng.l⁻¹ and 104.2 ng.l⁻¹, maximum = 479.1 ng.l⁻¹ and 540.6 ng.l⁻¹ respectively). The authors indicate that the high concentrations of octocrilene are likely a consequence of its **common use as a stabiliser and sunscreen in many products commonly used in urban areas, including not only cosmetic products, but also plastic packaging. In this study, environmental risks for aquatic organisms were clearly observed throughout the year.**

In *Mozas-Blanco et al. 2023*, measures were carried out during August 2021, collected from the upper part of the water column (approximately 20 cm). A median concentration of 296 ng.l⁻¹ and a maximum of 307 ng.l⁻¹ were found. These results are higher than in the publication by *García-Pimentel et al., 2023*. The fact that the samples were taken in August (and not in July), when the weather was warmer and tourist numbers were higher, may explain this difference.

These two studies showed environmental risks for aquatic organisms throughout the year with a risk which increases in closed and semi-enclosed bays with less water exchange with the open sea.

Canaries Island

Several studies have been carried out in the Canary Islands (Spanish archipelago located close to the northwest African coast, where tourism is the main economic activity). It is the only European region where the bathing season is extended throughout all the year due to its mild weather. Moreover, Gran Canaria is one of most visited and populated island of the archipelago. Urban and resort areas are located close to the coast which are exposed to high human pressures.

In *Rodríguez et al., 2015b*, six beaches located in Gran Canaria Island were selected for this study. Four of them (Maspalomas, MP; Puerto Rico, PR; Amadores, AM and Mogán, MG) are located in the southwest of the island, where most touristic activity is focused, and the remaining ones (Las Alcaravaneras, AL and Las Canteras, CA) in Las Palmas de Gran Canaria (northeast), the main city of the archipelago. Beach water samples were collected twice a month from May to October 2011 and monthly for the rest of 2011. One sample was collected on each beach on each sampling date and a total of 108 samples were collected. Samples were taken between 10 a.m. and 5 p.m at 1–1.5 m depth.

In *Cadena-Aizaga et al., 2022a*, three different beaches were selected for their particular geomorphological characteristics, their level of tourist pressure and type of users: Las Canteras, Arinaga and Playa del Inglés. Sampling lasted 6 months (May–October 2019) at low tide around noon; the seawater samples were collected at 2 m from the coast in the middle of the bathing zone and approximately at 50 cm below the water surface.

Environmental risks for aquatic organisms were observed for all selected sites (except to Arinaga beach). Seasonal variations were noted.

Moreover, the highest concentrations in *Rodríguez et al., 2015b* were determined for “**semi-enclosed**” profile beaches (**with minimal water exchange**): Puerto Rico, Mogán and Amadores.

By comparing results of these two studies for Canteras beach, Rodríguez et al., 2015b detected higher concentrations than Cadena-Aizaga, et al. 2022a. The authors suggest that these variations can be explained by a larger number of tourists in 2019 than in 2011 (25%). Nevertheless in all situations, risks for the aquatic organisms were clearly demonstrated.

In *Mozas-Blanco et al., 2023*, 14 beaches were sampled along the coast of Cádiz (Atlantic Ocean), in areas close to the Natural Park of the Bay of Cádiz, as well as near ports and other urban areas, in May 2021. Beaches were selected with a wide range of anthropogenic influence and pressures: areas close to natural or protected areas that are less frequented by humans, as well as areas of difficult access, and highly urbanized areas, which have a large tourist influx. A median concentration of 297 ng.l⁻¹ and a maximum of 308 ng.l⁻¹ was

recorded, **much higher than the PNEC value for the aquatic organisms.**

In *Agawin et al., 2024*, sampling was done in seven *Posidonia Oceanica* beds in Formentera (Balearic Islands) in July 2022 plus a control site (Cale Barbaria). The sites were selected based on their varying degrees of health status of *P. Oceanica* and popularity among tourists. The presence of sewage outlet pipe on two sites (i.e. La Savina, Els Pujols) were considered as potential discharge of octocrilene to seawater from residential and hotel wastes. The sewage pipe in La Savina maintained constant functionality, while the Els Pujols sewage pipe exhibited operational activity solely during periods of intense precipitation. Seawater samples were taken from the shoreline (the shores were approximately 50 to 260 m from the *Posidonia Oceanica* beds).

The concentrations were clearly higher near the shore in the **bathing zone**; concentrations were **above the PNEC aquatic value** for all locations of sampling (except to espalmador beach). Octocrilene concentrations were also higher in the water of beaches where **sewage outlet pipes** are present (Els Pujols and La Savina), demonstrating the **influence of the indirect releases in the marine compartment contamination and the environmental risk.**

Portugal

Sixteen beaches were selected in *Mozas-Blanco et al., 2023*, along the Portuguese coast, mainly in the Lisbon and Algarve area. In Lisbon region: the surveyed beaches (with high tourist influx in summer even if difficult to access) were close to protected natural areas. Algarve's beaches were expected to be the most populated. Samples were collected in August 2021, from the upper part of the water column (approximately 20 cm). On each beach, samples were taken at 3 different points twice a day with a 12-h difference.

Octocrilene was detected in 44% of the samples with a mean concentration of **298 ng.l⁻¹** and a maximum of **321 ng.l⁻¹**. *Mozas-Blanco et al., 2023* showed **risks for the aquatic organisms in all beaches, related to high touristic activity**, and on sites **close to environmentally protected areas.**

In *Aminot et al., 2019*, samples were collected between May and June 2016 at the mouth of the Aveiro lagoon in Portugal, where aquaculture (oysters, mussels), tourism and a major harbour are the main activities of the area. The three points of sample were close to the coast, outside the discharge offshore of the treated sewage of the whole Aveiro region. The levels clearly showed **risks for aquatic organisms in the Aveiro lagoon in Portugal**, with ranging from **27 to 46 ng.l⁻¹**, related to the touristic activity.

Greece

In *Aminot et al., 2019*, samples were collected between May and June 2016 in a fish farm on the Adriatic Sea. The area has a very small population (nearest town of 600 inhabitants 10 km away), with no industrial or town sewage discharge. Samples were collected closest to the coast and about 200 m offshore. The levels clearly showed high levels of contamination with **observed risks at the highest concentration (10.4 to 46 ng.l⁻¹)**.

Albania

In *Aminot et al., 2019*, samples were collected between May and June 2016 in a in the Butrint lagoon, a semi enclosed water body in a rural area of low population. The lagoon is one of the 2 major mussel farming areas of the country. The levels clearly showed high

levels of contamination with **observed risks at the highest concentration (10.4 to 509 ng.l⁻¹)**.

Norway

In *Aminot et al., 2019*, samples were collected between May and June 2016 in a salmon farm and a bivalve farm situated in the Bergen fjord. The area is **not densely populated, with no major pollution sources identified**. Nevertheless; octocrilene was detected with ranging from **1.6 to 42.2 ng.l⁻¹**.

In *Langford and Thomas., 2008*, water samples were collected in the Norwegian coastal zone with direct (bathing areas and marinas) and indirect (sites receiving wastewater effluent) releases. Samples were collected at a depth of 50 cm in the peak of summer on a weekend in July 2007 when recreational activities were at a maximum with extensive boating and swimming activity. Sampling was also conducted in May 2007, before the summer, as the ice melted and before the small boats were back in the water and before any swimming activity. Sample locations are The Sandvika beach and marina samples, Kalvøya and Ostøya Channel are all enclosed areas with very limited exposure to currents, wind or freshwater inputs. The VEAS and fjord samples are in the middle of the fjord, resulting in more tidal and weather influence. The VEAS sample point is in close proximity to the discharge point of a wastewater treatment serving the west of Oslo city and surroundings. The beach at Huk faces the inner Vestfjord so has more exposure to water exchange than the other beach locations. Samples from Sandvika beach were collected off the beach amongst approximately 200 swimmers, the majority being children, with sample 1 being the most densely populated with children.

The octocrilene levels showed **a seasonal variation** with concentrations below or close to the LOD in spring compared to **summer. At this time very high levels were measured in the bathing areas (posing an aquatic risk) showing that washing of octocrilene directly from the skin is an important point source**. As expected, the two samples with the highest concentrations were taken where children were swimming (Sandvika Beach) and concentrations decreased further from the swimming area. The samples collected amongst the swimmers in this study demonstrated very high concentrations of **7 301 ng.l⁻¹**.

France

In *Picot-Groz et al., 2018*, three sites located in the Gulf of Lion along the French Mediterranean coast were selected. Site A was subject to the highest touristic pressure, followed by sites B and C. Seawater sampling were conducted during 1 day at the beginning of July, at a distance from the beach ranging from 50 to 80m and at 20- to 100-cm depth. Seawater was collected six times a day at 8:00 a.m., 10:00 a.m., 12:00 a.m., 3:00 p.m., 5:00 p.m., and 7:00 p.m. Concentrations in excess of the PNEC were observed at all 3 sampling sites. The highest concentrations were obtained at site A, which had the **highest recreational activity pressure. A diurnal variation** of UV filter concentrations in seawater was observed correlated with the bathing activity.

In *Labille et al., 2019d*, three urban beaches, located in Marseille's seafront in France were studied. Despite their small size (less than 1 ha), each of the three beaches was intensely frequented during the summer season. Water samples were collected at 4:00 pm on Saturday, July 15th, during peak recreational activity. The samples were taken in the bathing area at two depths: the top surface layer (~1 cm), and the water column (at 40 cm, depth). Octocrilene was **detected at concentration above the PNEC** in the top surface layer at the three beaches sites and clearly also **in the water column of the**

bathing zone. Results suggests that octocrilene can be found in the water column despite its hydrophobic character. In *Thallinger et al., 2023*, a 2-year sampling campaign (2020-2021) was carried out on a beach in south of France (with very **high anthropic pressure**) **in the bathing zone**; 3 sampling times per day were realized, at 12:00, 16:00 and 18:00, so as to cover the beginning, full peak, and end of the beach affluence. **All results were above the PNEC whatever the month or the time of the day.** The authors indicated that **the summer periods, characterized by warm temperatures in air and water, were associated to peak attendance on the beach together with peak levels of UV filter concentration in the bathing water.** These results evidence the direct link between the octocrilene pollution in coastal seawater and its **recreational origin** favoured during the **warm season**. Moreover, it suggests that such effect might become more and more pronounced in a close future with the expected multiplication of extreme meteorological events.

The evolution of octocrilene concentrations **in function of time (at the day scale)** in seawater was also observed, confirming that the maximum measured concentration generally coincides with the **peak attendance on the beach**, i.e. when sampled in the middle to the end of the afternoon.

Regarding the vertical distribution between the top surface layer of water and the bulk column, octocrilene tends to **concentrate in the top surface layer even if very high concentrations are observed in the water column.**

Italy

In *Magi et al., 2012a*, data are not enough sufficient to point out a seasonal trend. Nevertheless, value above the PNEC was observed in May in an unusual hot and dry weather day.

In *Aminot et al., 2019*, samples were collected between May and June 2016, in a fish farm site in the northern Tyrrhenian Sea. The levels clearly showed **risks for the aquatic organisms in north Italian fish-farming site**, with ranging from **19 to 77 ng.l⁻¹**.

Netherlands

In *Schaap et al. 2018*, very high concentrations were observed in lagoon of a Caribbean Island in the water surface (max= 1 950 ng.l⁻¹ at 0-1 cm depth) and in the water column (max= 810 ng.l⁻¹). The lagoon comprises important nursery habitats, such as a back reef, mangroves, and seagrass beds, which supports reef fish abundance in the reefs around the island. Although authors expected a built up of concentrations in November (high touristic activities), lower concentrations of octocrilene were detected (but above the PNEC in a location point, 100 ng.l⁻¹), which could be explained by the weather conditions (a rain shower passed by and affected the number of beach goers and thus the release, and the concentrations of octocrilene by dilution). Consequently, and as mentioned by authors, the fact that octocrilene (posing potentially a risk) were detected during rainy weather (November sampling) including subsequently low or no bathers, indicates that the **persistence of octocrilene, water movement and area characteristics all play a role in maintaining octocrilene at risk levels in fragile ecosystems.**

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Table 45: results of marine water in Europe

N°	Year	Period	Country	Location	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2007	March	Spain	Santa Christina beach (Galicia)	Value	<LOD	ng.l ⁻¹	LC-MS/MS	n.a.	18	ng.l ⁻¹	Rodil et al., 2008
1	2007	July	Spain	Santa Christina beach (Galicia)	Value	<LOD	ng.l ⁻¹	LC-MS/MS	n.a.	18	ng.l ⁻¹	Rodil et al., 2008
2	2016	May-Jun.	Portugal	Aveiro lagoon	Value	45.8	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Portugal	Aveiro lagoon	Value	29.1	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Portugal	Aveiro lagoon	Value	27.4	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	United Kingdom	Exe estuary (south coast)	Value	4.3	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	United Kingdom	Exe estuary (south coast)	Value	4.8	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	United Kingdom	Exe estuary (south coast)	Value	3.8	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	3.3	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	4.7	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	4.3	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	42.2	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	1.6	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Norway	Bergen fjord	Value	2.3	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Spain	Ebro River delta	Value	38.4	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Spain	Ebro River delta	Value	303.5	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Spain	Ebro River delta	Value	46	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	1.2	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	7.7	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	3.5	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	77.5	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	18.6	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Italy	Northern Tyrrhenian Sea	Value	53.7	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Greece	Adriatic Sea	Value	23.8	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Greece	Adriatic Sea	Value	86.6	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Greece	Adriatic Sea	Value	10.4	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Albania	Butrint lagoon	Value	509	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
2	2016	May-Jun.	Albania	Butrint lagoon	Value	18.5	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

2	2016	May-Jun.	Albania	Butrint lagoon	Value	107	ng.l ⁻¹	GC-MSD	0.7	0.2	ng.l ⁻¹	Aminot et al., 2019
3	2018	July	Spain	Mar Menor Lagoon: Total	Median	54.6	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: Total	Median	104.2	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: North	Median	402.3	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: North	Median	139.4	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: Center	Median	54.6	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: Center	Median	61.2	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: South	Median	34.1	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: South	Median	526.9	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta sea	Median	>2000	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta sea	Median	157.1	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: Total	Max	479.1	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: Total	Max	540.6	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: North	Max	479.1	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: North	Max	276.1	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: Center	Max	55.6	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: Center	Max	69.0	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Mar Menor Lagoon: South	Max	82.7	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	Feb.	Spain	Mar Menor Lagoon: South	Max	540.6	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta river	Max	>2000	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2018	July	Spain	Ebro Delta sea	Max	>2000	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023
3	2019	March	Spain	Ebro Delta sea	Max	394.5	ng.l ⁻¹	GS-MS	12.5	3.7	ng.l ⁻¹	García-Pimentel et al., 2023

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

												2023
4	2021	Spring	Spain	Cadiz-Malaga	Median	297	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Spring	Spain	Cadiz-Malaga	Mean	298	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Spring	Spain	Cadiz-Malaga	Max	308	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Cadiz-Malaga	Median	<LOD	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Cadiz-Malaga	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Cadiz-Malaga	Max	<LOD	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Mar Menor lagoon	Median	296	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Mar Menor lagoon	Mean	297	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Summer	Spain	Mar Menor lagoon	Max	307	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
5	2011	Annual	Spain	Canaries Island - Las Canteras	Min	<LOD	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Las Canteras	Median	52.7	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Las Canteras	P95	620.5	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Las Canteras	Max	768.5	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Maspalomas	Min	<LOQ	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Maspalomas	Median	23.8	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Maspalomas	P95	177.4	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island - Maspalomas	Max	359.1	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Puerto Rico	Min	61.2	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Puerto Rico	Median	374.5	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Puerto Rico	P95	891	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island –	Max	973.1	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Puerto Rico								
5	2011	Annual	Spain	Canaries Island – Amadores	Min	30.7	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Amadores	Median	160.4	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Amadores	P95	667.2	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Amadores	Max	766.7	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Mogán	Min	37.8	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Mogán	Median	497.8	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Mogán	P95	1 263.2	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Mogán	Max	1 324.9	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Las Alcaravaneras	Min	<LOD	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Las Alcaravaneras	Median	20.9	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Las Alcaravaneras	P95	160.2	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
5	2011	Annual	Spain	Canaries Island – Las Alcaravaneras	Max	183.2	ng.l ⁻¹	HPLC	9.3	2.8	ng.l ⁻¹	Rodríguez et al., 2015b
6	2019	May	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Jun.	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	July	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Aug.	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	172 000	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Sept.	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	5 640	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Oct.	Spain	Canaries Island - Las Canteras (50cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	May	Spain	Canaries Island – Arinaga (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Jun.	Spain	Canaries Island – Arinaga (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	July	Spain	Canaries Island –	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al.,

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Arinaga (50 cm depth)								2022a
6	2019	Aug.	Spain	Canaries Island – Arinaga (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Sept.	Spain	Canaries Island – Arinaga (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Oct.	Spain	Canaries Island – Arinaga (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	May	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Jun.	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	July	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Aug.	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	2 520	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Sept.	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
6	2019	Oct.	Spain	Canaries Island - Playa del Inglés (50 cm depth)	Mean	<LOD	ng.l ⁻¹	LC-MS/MS	42.4	12.7	ng.l ⁻¹	Cadena-Aizaga et al., 2022a
7	2022	July	Spain	Formentera – Espalmador (shore)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Playa Illetes (shore)	Value	33	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Playa Illetes (shore)	Value	53	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (shore)	Value	80	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (shore)	Value	84	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Els Pujols (shore)	Value	208	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Els Pujols (shore)	Value	246	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Migjorn (shore)	Value	0.5	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Migjorn	Value	31	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				(shore)								
7	2022	July	Spain	Formentera – La Savina (shore)	Value	118	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – La Savina (shore)	Value	282	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cap de Barbaria (shore)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cap de Barbaria (shore)	Value	105	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Espalmador (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Playa Illetes (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (surface)	Value	8.6	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Els Pujols (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Migjorn (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – La Savina (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cap de Barbaria (surface)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Espalmador (Canopy water)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Playa Illetes (Canopy water)	Value	19.2	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Playa Illetes (Canopy water)	Value	19.2	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (Canopy water)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cala Saona (Canopy water)	Value	8.4	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Els Pujols (Canopy water)	Value	12.4	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Els Pujols (Canopy water)	Value	13.2	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Migjorn (Canopy water)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

7	2022	July	Spain	Formentera – Migjorn (Canopy water)	Value	4	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – La Savina (Canopy water)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – La Savina (Canopy water)	Value	9.2	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
7	2022	July	Spain	Formentera – Cap de Barbaria (Canopy water)	Value	<LOD	ng.l ⁻¹	GC-MS/MS	ND	<0.2	ng.l ⁻¹	Agawin et al., 2024
4	2021	Aug.	Portugal	Portuguese coast (16 beaches)	Median	294	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Aug.	Portugal	Portuguese coast (16 beaches)	Mean	298	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
4	2021	Aug.	Portugal	Portuguese coast (16 beaches)	Max	321	ng.l ⁻¹	LC-MS/MS	22	7	ng.l ⁻¹	Mozas-Blanco et al., 2023
8	2007	May	Norway	Innerfjord 1	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Innerfjord 1	Mean	12.5	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Innerfjord 2	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Innerfjord 2	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Huk	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Huk	Mean	29.9	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Kalvøya	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Kalvøya	Mean	105.1	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Sandvika Beach 1	Mean	7 301	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Sandvika Beach 2	Mean	2 780.7	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Sandvika Beach 3	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Sandvika Beach 3	Mean	65.3	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Sandvika Marina	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Sandvika Marina	Mean	90.1	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Off Sandvika Marina	Mean	24.8	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Off Sandvika Marina	Mean	71.2	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Sandvika Mid channel	Mean	63.2	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	Ostøya Channel	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	Ostøya Channel	Mean	58.5	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	0 m off VEAS	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	0 m off VEAS	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	10 m off VEAS	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	10 m off VEAS	Mean	31.2	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	May	Norway	100 m off VEAS	Mean	<LOD	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
8	2007	July	Norway	100 m off VEAS	Mean	24.8	ng.l ⁻¹	GC-ToF-MS	ND	5	ng.l ⁻¹	Langford et al. 2008
9	2014	July	France	Gulf of Lion A: 8h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 10h am	Min	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

9	2014	July	France	Gulf of Lion A: 10h am	Max	24	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 12h am	Min	12	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 12h am	Max	46	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 3h pm	Min	134	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 3h pm	Max	369	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 5h pm	Min	120	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 5h pm	Max	313	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 7h pm	Min	67	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion A: 7h pm	Max	160	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 8h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 10h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 12h am	Min	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 12h am	Max	23	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 3h pm	Min	23	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 3h pm	Max	64	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 5h pm	Min	21	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 5h pm	Max	31	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 7h pm	Min	65	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion B: 7h pm	Max	102	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 8h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 10h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 12h am	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 5h pm	Min	11	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 5h pm	Max	22	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 7h pm	Min	50	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
9	2014	July	France	Gulf of Lion C: 7h pm	Max	59	ng.l ⁻¹	LC-MS/MS	2	1	ng.l ⁻¹	Picot-Groz et al., 2018
10	2017	July 15th	France	Lave Beach: Bathing zone (BZ)1 - 0-1 cm depth	Value	85	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Lave Beach: BZ2 - 0-1 cm depth	Value	419	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Lave Beach: BBZ - 0-1 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Lave Beach: BZ1 - 40 cm depth	Value	148	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Lave Beach: BZ2 - 40 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Lave Beach: BBZ - 40 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Pointe Rouge: BZ1 - 40 cm depth	Value	149.8	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

10	2017	July 15th	France	Pointe Rouge: BZ2 - 40 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Pointe Rouge: BBZ - 40 cm depth	Value	141.4	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BZ1 - 0-1 cm depth	Value	44.8	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BZ2 - 0-1 cm depth	Value	44.4	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BBZ - 0-1 cm depth	Value	141.6	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BZ1 - 0-1 cm depth	Value	31.4	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BZ2 - 0-1 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
10	2017	July 15th	France	Prophète: BBZ - 0-1 cm depth	Value	<LOD	ng.l ⁻¹	UPLC-MS/MS	15	5	ng.l ⁻¹	Labille et al., 2019d
11	2020	Jun. 6	France	Palavas-les-Flots – 1:30pm – 50 cm depth	Mean	170	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Jun. 6	France	Palavas-les-Flots – 4:30pm – 50 cm depth	Mean	370	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Jun. 6	France	Palavas-les-Flots – 7:30pm – 50 cm depth	Mean	480	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Aug. 8	France	Palavas-les-Flots 0:30pm – 50 cm depth	Mean	953	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Aug. 8	France	Palavas-les-Flots 3:30pm – 50 cm depth	Mean	3 483	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Aug. 8	France	Palavas-les-Flots 6:30pm – 50 cm depth	Mean	2 743	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Oct. 28	France	Palavas-les-Flots 0:30pm – 50 cm depth	Mean	40	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Oct. 28	France	Palavas-les-Flots 3:30pm – 50 cm depth	Mean	60	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2020	Oct. 28	France	Palavas-les-Flots 6:30pm – 50 cm depth	Mean	40	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	May 20	France	Palavas-les-Flots 1:30pm – 0-1 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	May 20	France	Palavas-les-Flots 1:30pm – 50 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	May 20	France	Palavas-les-Flots 4:30pm – 0-1 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

11	2021	May 20	France	Palavas-les-Flots 4:30pm – 50 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	May 20	France	Palavas-les-Flots 7:00pm – 0-1 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	May 20	France	Palavas-les-Flots 7:00pm – 50 cm depth	Mean	<LOD	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 1:30pm – 0-1 cm depth	Mean	1 440	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 1:30pm – 50 cm depth	Mean	1 093	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 4:30pm – 0-1 cm depth	Mean	1 593	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 4:30pm – 50 cm depth	Mean	1 077	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 7:00pm – 0-1 cm depth	Mean	1 043	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	July 7	France	Palavas-les-Flots 7:00pm – 50 cm depth	Mean	1 147	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	Aug. 15 (Heat waves)	France	Palavas-les-Flots 1:30pm – 0-1 cm depth	Mean	3 133	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	Aug. 15 (Heat waves)	France	Palavas-les-Flots 1:30pm – 50 cm depth	Mean	1 937	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	Aug. 15 (Heat waves)	France	Palavas-les-Flots 4:30pm – 0-1 cm depth	Mean	4 857	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
11	2021	Aug. 15 (Heat waves)	France	Palavas-les-Flots 4:30pm – 50 cm depth	Mean	5 013	ng.l ⁻¹	UPLC-MS/MS	610	200	ng.l ⁻¹	Thallinger et al., 2023
12	2011	April	Italy	Foce beach in Genoa	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a
12	2011	May	Italy	Foce beach in Genoa	Mean	32	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a
12	2011	Jun.	Italy	Foce beach in Genoa	Mean	20	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a
12	2011	July	Italy	Foce beach in Genoa	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a
12	2011	Aug.	Italy	Foce beach in Genoa	Mean	19	ng.l ⁻¹	LC-MS/MS	11.1	3.3	ng.l ⁻¹	Magi et al., 2012a
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Sorobon	Mean	1950	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Beach – 0-1 cm depth								
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Sorobon Beach – 30 cm depth	Mean	810	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Mangrove Forest – 0-1 cm depth	Mean	30	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Mangrove Forest – 30 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Reef channel Cai – 0-1 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Aug 23	Netherlands	Caribbean Island Bonaire - Reef channel Cai – 30 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Sorobon Beach – 0-1 cm depth	Mean	100	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Sorobon Beach – 30 cm depth	Mean	160	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Mangrove Forest – 0-1 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Mangrove Forest – 30 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Reef channel Cai – 0-1 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018
13	2016	Nov 18	Netherlands	Caribbean Island Bonaire - Reef channel Cai – 30 cm depth	Mean	<LOQ	ng.l ⁻¹	LC-MS/MS	20	3.2	ng.l ⁻¹	Schaap et al., 2018

B.8.5.1. Sediment**B.8.5.1.1. Freshwater environment****Table 46: results of Sediment in freshwater environment in Europe**

Nº	Year	Period	Country	Location	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2009	Dec.	Spain	Ebro River Basin - Reinososa (Cantabria)	Value	130	ng.g ⁻¹ dw	UPLC-MS/MS	2.2	7.3	ng.g ⁻¹ dw	Gago-Ferrero et al., 2011a
				Ebro River Basin - Miranda de Ebro (Burgos)	Value	300	ng.g ⁻¹ dw					
				Ebro River Basin - Audinaka (Alava)	Value	35	ng.g ⁻¹ dw					
				Ebro River Basin - Haro (La Rioja)	Value	130	ng.g ⁻¹ dw					
				Ebro River Basin - Najera (La Rioja)	Value	37	ng.g ⁻¹ dw					
				Ebro River Basin - Logrono (La Rioja)	Value	170	ng.g ⁻¹ dw					
				Ebro River Basin - Estella (Navarra)	Value	39	ng.g ⁻¹ dw					
				Ebro River Basin - Tudela (Navarra)	Value	540	ng.g ⁻¹ dw					
				Ebro River Basin - Alsasua (Navarra)	Value	70	ng.g ⁻¹ dw					
				Ebro River Basin - Puente de la Reina (Navarra)	Value	49	ng.g ⁻¹ dw					
				Ebro River Basin - Grisen (Zaragoza)	Value	18	ng.g ⁻¹ dw					
				Ebro River Basin - Zaragoza (Zaragoza)	Value	2 400	ng.g ⁻¹ dw					
				Ebro River Basin - Caldearenas (Huesca)	Value	34	ng.g ⁻¹ dw					
				Ebro River Basin - San Mateo de Gallego (Zaragoza)	Value	34	ng.g ⁻¹ dw					
				Ebro River Basin - Presa de Pina (Zaragoza)	Value	870	ng.g ⁻¹ dw					
				Ebro River Basin -	Value	60	ng.g ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Sastago (Zaragoza)			dw					
				Ebro River Basin - Alcolea de Cinca (Huesca)	Value	<LOD	ng.g ⁻¹ dw					
				Ebro River Basin - Flix (Tarragona)	Value	64	ng.g ⁻¹ dw					
				Ebro River Basin - Amposta (Tarragona)	Value	84	ng.g ⁻¹ dw					
				Ebro River Basin - Delta del Elbre (Tarragona)	Value	32	ng.g ⁻¹ dw					
2	2008	Spring-Fall	Germany	Lake Badesee Rodgau	Mean	43.45	ng.g ⁻¹	GC-MS	0.2	NA	ng.g ⁻¹	Kaiser et al., 2012
				Lake Weisssee	Mean	36.73	ng.g ⁻¹					
				Lake Bärensee	Mean	66.42	ng.g ⁻¹					
				Lake Kinzigsee	Mean	159.6	ng.g ⁻¹					
				Lake Badesee Mainflingen	Mean	41.58	ng.g ⁻¹					
				Lake Spessartblick	Mean	72.57	ng.g ⁻¹					
				Lake SchuttheiB-weiher	Mean	7.877	ng.g ⁻¹					
				Lake Ostsee	Mean	62.96	ng.g ⁻¹					
				Lake Badesee Walldorf	Mean	92.83	ng.g ⁻¹					
				River Bieber	Mean	24.1	ng.g ⁻¹					
				River Rodau	Mean	3.8	ng.g ⁻¹					
				River Geräthsbach	Mean	8.9	ng.g ⁻¹					
				River Hahnwiesen-bach	Mean	4.7	ng.g ⁻¹					
				River Bieber	Mean	5	ng.g ⁻¹					
				River Rodau	Mean	3.5	ng.g ⁻¹					
				River Geräthsbach	Mean	5.4	ng.g ⁻¹					
				River Schwarzbach	Mean	2.2	ng.g ⁻¹					
				River Schwarzbach	Mean	4.6	ng.g ⁻¹					
3	2013	Jun.	Norway	Lake Mjøsa	Max	<7	ng.g ⁻¹ dw	LC-HRMS & GC-HRMS	NA	NA	ng.g ⁻¹ dw	Langford et al., 2015
4	2015	Feb.	Italy	Adige River SITE1	Value	<LOD	ng.g ⁻¹	HPLC-MS	0.08	0.02	ng.g ⁻¹	Mandaric et al., 2017
					Value	<LOD	ng.g ⁻¹					
				Adige River SITE2	Value	3.92	ng.g ⁻¹					
					Value	7.6	ng.g ⁻¹					
					Value	<LOD	ng.g ⁻¹					
				Adige River SITE3	Value	633	ng.g ⁻¹					
					Value	0.68	ng.g ⁻¹					
					Value	<LOD	ng.g ⁻¹					
					Value	<LOD	ng.g ⁻¹					

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Adige River SITE4	Value	<LOD	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
				Adige River SITE5	Value	2.5	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
				Adige River SITE6	Value	3.16	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
				Adige River SITE7	Value	<LOD	ng.g ⁻¹						
					Value	4.44	ng.g ⁻¹						
					Value	4.76	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
					Value	87.7	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
					Value	<LOD	ng.g ⁻¹						
5	2016	May-Jun.	Spain	Ebro River delta SPS1	Value	12	ng.g ⁻¹ dw	GC-MS	2.7	9.2	ng.g ⁻¹ dw	Aminot et al., 2019	
				Ebro River delta SPS2	Value	127	ng.g ⁻¹ dw						
				Ebro River delta SPS3	Value	15.9	ng.g ⁻¹ dw						

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

B.8.5.1.2. Marine water environment

Table 47: results of Sediment in marine water environment in Europe

Nº	Year	Period	Country	Location	Parameter	Value	Unit	Method	LOQ	LOD	Unit	Ref.
1	2013	Aug.	Norway	Oslofjord	Median	<7	ng.g ⁻¹ dw	LC-HRMS & GC-HRMS	NA	NA	ng.g ⁻¹ dw	Langford et al., 2015
1	2013	Aug.	Norway	Oslofjord	Min	<7	ng.g ⁻¹ dw	LC-HRMS & GC-HRMS	NA	NA	ng.g ⁻¹ dw	Langford et al., 2015
1	2013	Aug.	Norway	Oslofjord	Max	82.1	ng.g ⁻¹ dw	LC-HRMS & GC-HRMS	NA	NA	ng.g ⁻¹ dw	Langford et al., 2015
2	2014	Oct.	Italy	Northern Adriatic Sea	Min	4	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Northern Adriatic Sea	Max	40.7	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Northern Adriatic Sea	Mean	16.3	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Central Adriatic Sea	Min	0.8	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Central Adriatic Sea	Max	33.7	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Central Adriatic Sea	Mean	7.6	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Southern Adriatic Sea	Min	0.9	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Southern Adriatic Sea	Max	19	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Southern Adriatic Sea	Mean	6.8	ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Northern Adriatic Sea	Min		ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
2	2014	Oct.	Italy	Northern Adriatic Sea	Max		ng.g ⁻¹	GC-MS/MS	0.1	NA	ng.g ⁻¹	Combi et al., 2016
3	2011	NA	Spain	Cadiz Bay	Min	4.6	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
3	2011	NA	Spain	Cadiz Bay	Max	43.6	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
3	2011	NA	Spain	Cadiz Bay	Mean	20.8	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
3	2011	NA	Spain	Huelva Estuary	Min	1.7	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
3	2011	NA	Spain	Huelva Estuary	Max	38	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
3	2011	NA	Spain	Huelva Estuary	Mean	15.1	ng.g ⁻¹ dw	GC-MS/MS	NA	NA	ng.g ⁻¹ dw	Pintado-Herrera et al., 2017
4	2017	Jun. - Sep.	Germany	German bight	Mean	1.3	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	German bight	Min	<(LOD)	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	German bight	Max	9.7	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2016	Jan.	Germany	Skagerrak and Kattegat sweater	Mean	<LOQ	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2016	Jan.	Germany	Skagerrak and Kattegat sweater	Min	<LOD	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2016	Jan.	Germany	Skagerrak and Kattegat sweater	Max	0.1	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	Baltic Sea	Mean	0.25	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

4	2017	Jun. - Sep.	Germany	Baltic Sea	Min	<LOD	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	Baltic Sea	Max	1.7	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Oct. - Sep.	Germany	Rhine-Meuse-delta	Min	0.27	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Oct. - Sep.	Germany	Rhine-Meuse-delta	Max	2.5	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	Elbe Estuary	Min	0.54	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2017	Jun. - Sep.	Germany	Elbe Estuary	Max	2.7	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2015-2017	Oct. - Sep.	Germany	Oder Lagoon	Min	<LOQ	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
4	2015-2017	Oct. - Sep.	Germany	Oder Lagoon	Max	0.76	ng.g ⁻¹ dw	LC-MS/MS	0.1	0.03	ng.g ⁻¹ dw	Apel et al., 2018
5	2017	Jun.- July	France	Capbreton Canyon (0-10 cm (200 m depth))	Mean	3.6	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
5	2017	Jun.- July	France	Capbreton Canyon (0-10 cm (200 m depth))	Min	<LOQ	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
5	2017	Jun.- July	France	Capbreton Canyon (0-10 cm (200 m depth))	Max	29.2	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
5	2017	Jun.- July	France	Capbreton Continental shelf (0-10 cm ((40-136 m depth))	Mean	1.6	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
5	2017	Jun.- July	France	Capbreton Continental shelf (0-10 cm ((40-136 m depth))	Min	<LOQ	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
5	2017	Jun.- July	France	Capbreton Continental shelf (0-10 cm ((40-136 m depth))	Max	12.7	ng.g ⁻¹ dw	GC-MS	1.9	NR	ng.g ⁻¹ dw	Azaroff et al., 2019
6	2016	May-Jun.	NR	Adriatic Sea GRF1	Value	23.7	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	NR	Adriatic Sea GRF2	Value	<LOQ	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	NR	Adriatic Sea GRF3	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Norway	Bergen fjord: salmon farm NOF2	Value	<LOQ	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Norway	Bergen fjord: Salmon farm NOF3, scallops farm NOS3	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	UK	Exe estuary: UKS1	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	UK	Exe estuary: UKS2	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	UK	Exe estuary: UKS3	Value	12.8	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

6	2016	May-Jun.	Portugal	Aveiro lagoon: PTL1	Value	12.5	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Portugal	Aveiro lagoon: PTL2	Value	23.5	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Portugal	Aveiro lagoon: PTL3	Value	34.4	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Albania	Butrint lagoon: ALS1	Value	39.3	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Albania	Butrint lagoon: ALS2	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
6	2016	May-Jun.	Albania	Butrint lagoon: ALS3	Value	<LOD	ng.g ⁻¹ dw	GC-MS	9.2	2.7	ng.g ⁻¹ dw	Aminot et al., 2019
7	2022	July	Spain	Formentera (Balearic Islands), Migjorn	Range (shore)	35.5	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Migjorn	Range (shore)	217	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Migjorn	Range (meadow)	26.7	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Migjorn	Range (meadow)	182.1	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Playa Illetes	Range (shore)	1 535	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Playa Illetes	Range (shore)	2 196	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Playa Illetes	Range (meadow)	13.7	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Playa Illetes	Range (meadow)	118.3	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), La Savina	Value (meadow)	1 342	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), La Savina sewage	Value (meadow)	1 591	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), La Savina sewage	Value (meadow)	2 575	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Cala Saona	Range (shore)	312.3	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Cala Saona	Range (shore)	834.7	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Cala Saona	Range (meadow)	139.8	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Cala Saona	Range (meadow)	500.4	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols	Range (shore)	635.4	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols	Range (shore)	2 116	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols	Range (meadow)	46.7	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols	Range (meadow)	125.1	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols sewage	Range (meadow)	50.8	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Els Pujols sewage	Range (meadow)	215.7	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Espalmador	Range (shore)	384.4	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Espalmador	Range (shore)	507.1	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
7	2022	July	Spain	Formentera (Balearic Islands), Cap de Barbaria	Value (meadow)	115.8	ng.g ⁻¹ dw	GC-MS/MS	nd	<0.95	ng.g ⁻¹ dw	Agawin et al., 2024
8	2021	March – Aug.	Spain	La Herradura Bay	Min	0.4	ng.g ⁻¹	GC-MS/MS	NA	0.187	ng.g ⁻¹	Tovar-Salvador et al., 2025
8	2021	March – Aug.	Spain	La Herradura Bay	Max	36	ng.g ⁻¹	GC-MS/MS	NA	0.187	ng.g ⁻¹	Tovar-Salvador et al., 2025
8	2021	March – Aug.	Spain	La Herradura Bay	Mean	6	ng.g ⁻¹	GC-MS/MS	NA	0.187	ng.g ⁻¹	Tovar-Salvador et al., 2025

B.8.5.2. Terrestrial compartment

To date, only few data are available on the occurrence of octocrilene in terrestrial compartment in European area.

Among entry routes of hydrophobic compounds as octocrilene into soils, there is releases from conventional WWTPs, *via* application of sewage sludge as fertilizer and/or re-use of reclaimed water for irrigation of agricultural soils. *Biel-Maeso et al., 2018* studied soil samples from 20 cm to 150 cm depth collected at Jerez de la Frontera WWTP gardens (Spain). Those soils are irrigated roughly five times per week during summer months with wastewater effluents whereas during the rest of the year the irrigation frequency decreases to approximately twice per week. In this study, octocrilene was measured in a range of 20.9-69.7 ng.g⁻¹dw in winter 2015) and a range of 11.9-35.1 ng.g⁻¹dw in summer 2015. No trend was observed in the vertical distribution profile of octocrilene in soil and concentrations of octocrilene remained relatively constant between the seasons despite the higher inputs from reclaimed wastewater during the warmer months.

Ramos et al., 2021a collected different types of soil in Portugal from the top layer (0–30 cm). All the sampled soil contained octocrilene (near a beach: 6 ng.g⁻¹dw; on agricultural lands: 5.2 ng.g⁻¹dw, in gardens: 3.3 ng.g⁻¹dw, on industrial areas: 3 ng.g⁻¹dw, and in schoolyards: 137 ng.g⁻¹dw). The highest octocrilene concentration was measured in the schoolyards loamy sand soil and in agricultural land silty loam soil with respectively the highest percentage of organic carbon (14.7%) and clay (4.8%), giving them high adsorption capacity compared to the others soil type. Higher octocrilene concentration was detected in the amended agricultural soil (84 ng.g⁻¹dw) than in the initial soil (5.1 ng.g⁻¹dw) after an application of a sewage-sludge-based fertilizer which contains 1519 ng.g⁻¹dw of octocrilene. Higher concentration of octocrilene in tomato (147 ng.g⁻¹dw in fruit) that grew up in this amended agricultural soil than in the initial soil (33 ng.g⁻¹dw) was shown in Ramos et al. (2021b) and underline the capacity of octocrilene to be transferred into edible plants.

B.8.5.3. Biota

Data on the occurrence of octocrilene in biota sampled in European countries are available mainly in aquatic organisms and throughout the food chain (Table 48).

Table 48: Occurrence of Octocrilene in the aquatic biota in European countries

Location	Biota		Concentration (ng.g ⁻¹ _{dw}) – max values	References
Macrophytes				
Spain	Seagrass	<i>Cymodocea nodosa</i>	19369	Cadena-Aizaga et al., 2022b
Spain		<i>Posidonia oceanica</i>	2112	Agawin et al., 2024
Molluscs				
Spain	Cockle	<i>Cerestoderma edule</i>	141	Rodil et al., 2019
	Mussel	<i>Mytilus galloprovincialis</i>	19	Lindo-Atichati et al., 2019
Portugal		<i>Mytilus galloprovincialis</i>	3800	Picot-Groz et al., 2014
France		<i>Mytilus galloprovincialis</i>	7112	
France		<i>Mytilus edulis</i>	23	
France		<i>Mytilus galloprovincialis</i>	967	Picot-Groz et al., 2018
EU		<i>M. sp.</i>	56	Cunha et al., 2017
Spain	Clam	<i>Ruditapes philippinarum</i>	159	Pintado-Herrera et al., 2020
Spain	Snail	<i>Phorcus atratus</i>	602	Cadena-Aizaga et al.,

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

		(Herbivorus)		2022c	
		<i>Stramonita haemastoma</i> (Carnivorous)	24.8		
	Sea hare	<i>Aplysia dactylomela</i>	1735		
Portugal	Oyster	<i>Crassostrea gigas</i>	21	Gadelha et al., 2019	
Echinoderms					
Spain	Sea cucumber	<i>Holothuria sanctori</i>	166	Cadena-Aizaga et al., 2022c	
Spain	Coral	<i>Astroides calycularis</i> <i>Dendrophyllia ramea</i>	102 ng.g ⁻¹	Tovar-Salvador et al., 2025	
Crustacean					
Norway	Shrimps	<i>Pandalus borealis</i>	23.1 ng.g ⁻¹ _{ww}	Langford et al., 2015	
Norway	Crab	<i>Carcinus meanas</i>	<10		
Fish					
Spain	Freshwater fish	<i>Luciobarbus sclateri</i>	30.4	Gago-Ferrero et al., 2015	
		<i>Cyprinus carpio</i>	<LOD		
Spain		<i>Silurus glanis</i>	25.7		
Spain		<i>Anguila anguila</i>	30		
Spain		<i>Cyprinus carpio</i> , <i>Leponis gibbosus</i> , <i>Barbus graellsii</i>	50.7	Picot et al., 2019	
Spain	Tuna (muscle, liver)	<i>Thunnus thynnus</i>	2698, 145	Pintado-Herrera et al., 2024	
Spain		<i>Sarda sarda</i>	61.6, 31.3		
Spain		<i>Katsuwonus pelamis</i>	79.5, 471		
EU	Seafood	Tuna (canned)	57.6	Cunha et al., 2017	
EU		Tuna	5		
EU		Mackerel (canned)	18.5		
EU		Mackerel	43.2		
EU		Salmon (aquaculture, muscle)	5		
EU		Seabream (aquaculture, muscle)	103.3		
EU		Cod	39.1		
EU		Monkfish	19.3		
Norway	Cod	<i>Gadus morhua</i>	193 ng.g ⁻¹ _{ww}	Magnuson et al., 2024	
Norway	Cod	<i>Gadus morhua</i> (liver)	86 ng.g ⁻¹ _{ww}	Langford et al., 2015	
Norway	Freshwater fish	<i>Lota lota</i> , <i>Perca fluviatilis</i> , <i>Coregonus lavaretrus</i> (filet)	<2–2.1 ng.g ⁻¹ _{ww}		
Switzerland		<i>Perch</i>	25 ng.g ⁻¹ _{lipid}		Balmer et al., 2005
Switzerland		<i>Salmo trutta</i>	2400 ng.g ⁻¹ _{lipid}		Buser et al., 2006
Marine mammals					
Spain	Dolphin (blubber)	<i>Tursiops truncatus</i>	108	Gonzalez-Bareiro et al., 2023	

Octocrilene (OC) was found in primary marine consumers (sea snails, sea cucumber) sampled in Gran Canary islands touristic sites with high affluence of bathers. Octocrilene was the second most frequently detected compound (40%) out of eight UV-filters and was measured at the highest concentration of 1735 ng.g⁻¹_{dw} in the **algal feeder sea hare** (Cadena-Aizaga et al., 2022c). This concentration corresponds to the largest studied organism, which can be attributed to its greater feeding behaviour *versus* the other analysed organisms (herbivorous and carnivorous sea snails). Octocrilene had also the highest concentrations among the eight monitored UV-filters in **macrophytes** (19369 ng.g⁻¹_{dw}; Cadena-Aizaga et al., 2022b) and in seawater (172000 ng.l⁻¹; Cadena-Aizaga et al., 2022a) sampled at the same sites and time as sea snails. It is worth noting that if octocrilene was among the most commonly found compound in primary marine consumers

and macrophytes, it was not in seawater (more easily distributed in solid matrices since its high Koc value 48898 L.kg⁻¹).

The level measured in the sea snail is comparable to those found in **mussels** on the French Mediterranean coast at 967 ng.g⁻¹_{dw} (Picot-Groz *et al.*, 2018). Octocrilene was also measured in others european countries in cockles (141 ng.g⁻¹_{dw}; Rodil *et al.*, 2019), clams 159 ng.g⁻¹_{dw}; Pintado-Herrera *et al.*, 2020) and in mussels from 19 to 3800 ng.g⁻¹_{dw} (Lindo Atichati *et al.*, 2019; Negreira *et al.*, 2013; Cunha *et al.*, 2017; Picot Groz *et al.*, 2014, 2018). The highest octocrilene concentrations in mussels was recorded at 7112 ng.g⁻¹_{dw} along the mediterranean coast in France (Bachelot *et al.*, 2012). Those concentrations were correlated with the air temperature (21.8 to 24 °C), corresponding to the highest tourist activity and thus recreational pressure in july and august and those octocrilene maxima were confirmed for the ten sites studied. Such correlations were also observed in oysters: octocrilene concentrations (one of the two UVs filters detected among the five analysed) were lower in winter (9 ng.g⁻¹_{dw}) than in summer (20 ng.g⁻¹_{dw}; Gadelha *et al.*, 2019). Diurnal variations of OC concentrations in mussels was also observed, the lowest concentrations recorded in the morning and increasing throughout the day but with no decrease at the day end, as it was measured in the water (Picot Groz *et al.*, 2018). The authors underlined that if mussels could rapidly uptake waterborne octocrilene, they also could rapidly excrete ingested octocrilene as shown in laboratory by Pintado-Herrera *et al.*, (2024b) then, a decrease in octocrilene concentrations in organisms at the end of the day was thus expected within a few hours.

Octocrilene was found in all the samples of two species of **cold-water corals** collected in a natural protected area (Spain). It was detected at the highest concncentration among the 49 monitored substances (101.7 ng.g⁻¹; Tovar-Salvador *et al.*, 2025). In this study, no significant differences in octocrilene concentrations between winter and summer were found in coral samples.

Agawin *et al.*, 2024 found octocrilene in epiphytes (organisms that grows on the plant surface) and several tissues of **Posidonia** sampled in balearic island in sites contaminated where octocrilene was detected in water and sediment. The maximum concentrations was measured in leaf epiphytes in the most exposed site at 2112.1 ng.g⁻¹_{dw} that is 20× higher than in the control site. The maximum concentrations in leaves, roots and rhizomes were 596.7 ng.g⁻¹_{dw}, 203.8 ng.g⁻¹_{dw} and 38.3 ng.g⁻¹_{dw}, respectively. The octocrilene concentrations in the leaves, epiphytes and roots were positively correlated with the concentrations in the sediment underlying the role of the root system in the octocrilene absorption, accumulation and translocation to seagrass tissues.

Octocrilene was found in several **freshwater fish** sampled in four Spanish river basin (range <LOQ to 30.4 ng.g⁻¹_{dw}; Gago-Ferrero *et al.*, 2015; Pico *et al.*, 2019), in a Norwegian lake (range <2 to 2.1 ng.g⁻¹_{ww}; Langford *et al.*, 2015) and small Swiss rivers (range 40 to 2400 ng.g⁻¹ lipid; Buser *et al.* 2006) with WWTPs and/or recreational activities. Balmer *et al.*, (2005) also sampled fish from different lakes in Switzerland selected to cover different direct and indirect input of UV-filters. Octocrilene concentration were <LOD in fish from all sites, except at Huttnensee lake measured at 25 ng.g⁻¹_{lipid} in perch. In spite of the absence of WWTP located in the watershed of this lake and therefore no indirect inputs of UV filters expected, there is a public access and during summer recreational activities are present. The number of bathers relative to the lake volume and water through flow is high, so that this lake represent a worst-case situation with respect to possible contamination with UV filters from direct input. In general, lower concentrations were observed in lake fish compared to river fish. According to Balmer *et al.*, 2005, octocrilene was observed in lake fish in only one case whereas it was observed in all river fish. Possible explanations for such an observation include different transport, sorption,

and degradation processes as well as biological factors such as different uptake rates and metabolic processes between fish species (lake fish are on a lower trophic level compared to river fish). Elimination from surface water in lakes and/or different bioaccumulation or metabolic processes in different fish species are responsible for the generally higher concentrations and the changed UV filter composition in river and lake fish.

In **marine fish**, the highest concentration of octocrilene was reported by Langford et al. (2015) at 11875 ng.g⁻¹ wet weight in cod livers from Norway. However, according to the authors, this high concentration is a potential outlier that if removed reduce the median concentration of octocrilene in cod liver to 86 ng.g⁻¹. In this study, octocrilene was detected in 80% of the cod liver samples. Similar concentrations were found in muscle of cod caged during 28d throughout the Stavanger fjord to WWTP outflow (range 53-193 ng.g⁻¹) (Magnuson et al., 2024). In this study, the lowest octocrilene concentration (77.4 ng.g⁻¹) was found in fish in the site closest to the WWTP discharge point compared to the site addressed as reference (154.5 ng.g⁻¹) and the intermediate site not directly in the discharge plume (169.8 ng.g⁻¹). This suggests that additional inputs, such as those associated with recreational and swimming activities might be important sources of octocrilene to consider in marine systems, aside from WWTP effluent.

Octocrilene was one of the two most concentrated UV filter compounds detected in three species of **tuna** sampled in gulf of Cadiz, an highly touristic area that receive river inputs carrying agricultural waste and sewage from upstream croplands and cities (Pintado-Herrera et al., 2024). The mean values measured in muscle and liver tissues were, respectively, 367.7 and 49.8 ng. g⁻¹ dw for *Thunnus thynnus*, 22.2 and 24.6 ng.g⁻¹dw for *Sarda sarda*, 26.9 and 197 ng.g⁻¹ dw for *Katsuwonus pelamis*. A differential pattern of UV filter compounds was observed among species. Octocrilene is more concentrated in liver except for *Thunnus thynnus* where octocrilene predominated in muscle up to 2698 ng.g⁻¹ dw. Tuna are large marine predators that require considerable prey consumer rate. The diet disparity among those tuna species explains inter-specific differences in octocrilene bioaccumulation patterns. Their high trophic position in marine food webs makes them highly susceptible to accumulate persistent organic pollutant via biomagnification.

Octocrilene was also studied in **marine mammals**, in blubber samples of bottlenose dolphin specimens (*Tursiops truncatus*) stranded on the coast of the Canary Islands (Gonzalez-Bareiros et al., 2023). Octocrilene was detected in all the samples and also presented the highest concentration of the analyzed compounds within the range from 52.61 to 108 ng g⁻¹ dw. The results seem to demonstrate that dolphins can accumulate relatively high levels of organic pollutants in their bodies because they occupy a high trophic level in the marine food chain and their metabolic activity is relatively low.

Field bioaccumulation factors was estimated in marine **clam** *Ruditapes philippinarum* caged in cadiz bay (Spain) to be Log BAF= 2.9 (mean) and 4.89 (max) (Pintado-Herrera et al., 2020). A BCF value of 2204 L.kg⁻¹ was observed in laboratory study on marine clams (Pintado-Herrera et al., 2024b) and confirmed that octocrilene may be considered bioaccumulative (ECHA 2023). Based on measured values of octocrilene in **sea snails**, in their food (macrophytes) and in seawater sampled in the same site and time, bioconcentration was shown in the red, brown and green **macrophytes** with the highest BCF observed in the most touristic site. Bioaccumulation in the primary marine consumers sea snails was also calculated and the LogBAF range from 5.4 to 8.4, whereas the BMF are lower than 0.3 (Cadena-Aizaga et al., 2022c). The Log BAF values were also calculated at 3.8 and 4.3 in the two species of cold-water **corals** sampled in Spain (Tovar-Salvador et al., 2025). Those data suggest that octocrilene could be considered as a 'bioaccumulative' compound.

It has to be underlined the tendency of octocrilene to form octocrilene -fatty acids conjugated in some organisms as demonstrated in marine Coral (*Stien et al., 2019*) and marine sediment worm species (*Clergeaud et al., 2022*). The authors showed that octocrilene was transformed into fatty acid conjugates via oxidation of the ethylhexyl chain, yielding very lipophilic octocrilene analogues that accumulate in tissues. The formation of octocrilene analogues suggests that octocrilene concentrations measured in the environment, and organisms may have been largely underestimated.

Appendix C: Justification for action on a Union-wide basis

See Annex XV report.

Appendix D: Baseline

See Annex XV report.

Appendix E: Impact Assessment

E.1. Risk Management Options

E.1.1. Discarded restriction options

In addition to the proposed restriction options, the Dossier Submitter initially considered the following measures:

- Restriction on placing octocrilene on the market in consumer and professional cosmetic end-products consisting in the definition of an updated and lower concentration limit (higher than the LoD);
- Elimination of octocrilene by treatment of sewage plant effluents (degradation by micro-organisms, membrane filtration, chlorination).

The following sections detail the assessment of the above-mentioned options and the reasons why they were discarded by the Dossier Submitter.

E.1.1.1. Definition of an updated and lower concentration limit for octocrilene in cosmetic products

The Dossier submitter initially considered a restriction option consisting in lowering the maximum concentration limit for octocrilene in cosmetic products (higher than the LoD). Such a measure is considered by the Dossier Submitter as not allowing a proper risk management of the risks arising from the use of octocrilene and cosmetic products containing octocrilene, based on the environmental risk assessment, and the suspected ED properties of octocrilene. Indeed, exposure to the environment is driven by three main factors:

- Octocrilene concentration in cosmetic products;
- Applied quantity;
- Size of the concerned population

Acting solely on concentration (moderate decrease in maximum concentration) does not

guarantee effective risk management for the environment, since two of the three factors mentioned above are not controlled. Octocrilene is authorised to be used in cosmetic products as UV-filter to a maximum of 10 %.

E.1.1.2. Elimination of octocrilene by treatment of sewage plant effluents

Octocrilene is present in WWTPs due to the use of this substance in many cosmetic products and their subsequent release into the wastewater system from showering/bathing.

Indeed, once in the WWTP system, the UV filters can either:

- i) stay sorbed to the sludge and thus be removed with the solid phase;
- ii) stay solubilized and be present in the effluents;
- iii) be degraded by microorganisms in the WWTP (or the anaerobic sludge digester).

Due to its low solubility and its strong adsorption potential, the primary mechanism for eliminating octocrilene is sorption onto sludge. Octocrilene might also undergo other processes, such as degradation by microorganisms. But scientific studies show that octocrilene can be degraded only by a specific set of microorganisms (Suleiman et al., 2019, Fagervold and Lebaron, 2022). Consequently, considering the complex microbial communities process in WWTP and given the complexity of the chemical cocktails found in WWTP influents, it is unrealistic to consider, today, the elimination of octocrilene from municipal WWTP by adapted or specialised microorganisms.

Even if the sorption on sludge is the primary mechanisms in WWTP for eliminating octocrilene before release to river, high concentrations of octocrilene are observed in effluents leading to unacceptable risks for aquatic organisms and sediment-dwelling organisms. Several membrane filtration processes for the post-treatment of municipal WWTP effluent (ultrafiltration, nanofiltration, reverse osmosis,...) showed some efficiency in removing octocrilene. These treatment generally correspond to quaternary treatment. Nevertheless, it is not feasible to extend this type of treatment to all municipal WWTP. The cost (mainly associated with energy demand and membrane replacement) of treatment would be far too high for the taxpayer (Tsui et al., 2014). As an example, Germany estimates at €1.3 billion the total annual cost of adding a quaternary stage to its treatment plants processing more than 5,000 PE (Population Equivalent)⁷. According to an impact assessment performed by the European Commission on the update of the Urban Wastewater Treatment Directive (UWWTD) in relation to the European Green Deal and Zero Pollution Action Plan, the annual cost to treat micropollutants (in which octocrilene is belonging) in WWTP for PE > 10 000 is of 2 651 millions €/y by 2040 for the most ambitious future policy (option 3). The recommendation from the impact assessment was to choose option 2, corresponding to an annual cost of 1185.51 millions €/y (all plants > 100 kPE and plants 10 k to 100 k PE in areas at risks)⁸.

Moreover, some processes used to clean water, such as chlorination, can be used to degrade octocrilene but this mechanism will generate potential adverse metabolite, and is not implementable as such (Medici et al., 2022).

⁷ <https://learnandconnect.pollutec.com/en/micropollutants-in-water-entering-the-age-of-quaternary-treatment/>

⁸ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52022SC0541>

E.1.2. Other Union-wide risk management options than restriction

See Annex XV report.

E.2. Alternatives

E.2.1. Description of the use and function of the restricted substance(s)

The light emitted by the sun, called solar light, correspond to electromagnetic radiation. The radiation that can be perceived by the human eye correspond to the visible spectrum and consists of wavelengths ranging from approximately 380 nm (purple or violet) to 700 nm (red). Ultraviolet (UV) radiation is subdivided into UV-C (100–290 nm), UV-B (290–320 nm), UV-A1 (320–340 nm) and UV-A2 (340–400 nm). UV-A radiation are able to penetrate the skin deeply to induce adverse effect leading mainly to early photoaging. UV-B radiation can also penetrate into the skin, to a lesser depth than UV-A, but it can induce DNA damage and sunburn through interactions with the upper skin layers (*Gholap et al., 2024*). The UV-A radiation energy bring to the skin can led to photosensitized oxidation through radical formation or production of singlet molecular oxygen. DNA damage from these UV-A results from generating reactive oxygen species through these processes (*Gholap et al., 2024*, illustrated in Figure 8). Both UV-A and UV-B can potentially adversely affect skin cells, harming the skin and eyes, leading to erythema (even sunburn), thereby increasing the risk of skin cancer (*Addor et al., 2022*, illustrated in Figure 9) at the opposite of what is occurring with UV-C which is fully absorbed by Earth's atmosphere and does not reach the surface (*Rajasekar et al., 2024*). Chronic UV exposure can cause various adverse effects for human health such as degenerative changes in cells, blood vessels, and fibrous tissues. It can also cause freckling, hyperpigmentation, itching, rashes, pain, and blister formation. Macular degeneration and cataracts may also occurred after chronic exposure to UV radiation. Photoaging and immune system dysregulation are also reported as arising after chronic exposure to UV radiation (*Tang et al., 2024*).

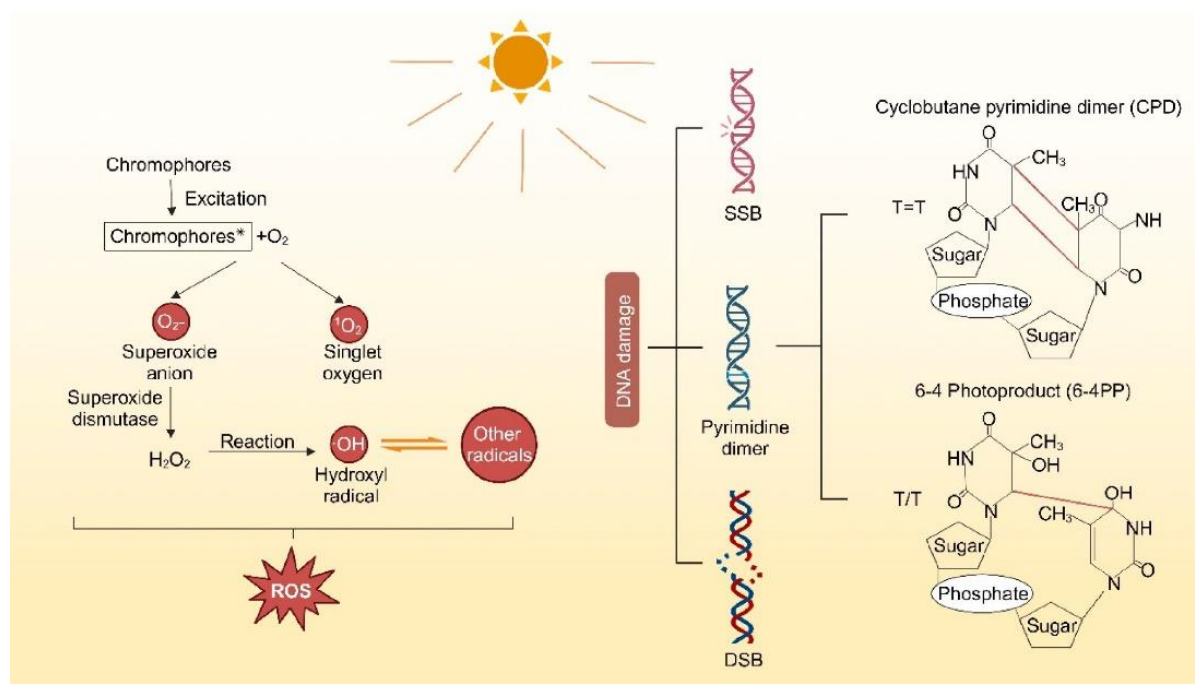


Figure 8: Illustration of potential adverse effect on DNA from UV radiation (from Tang et

al., 2024).

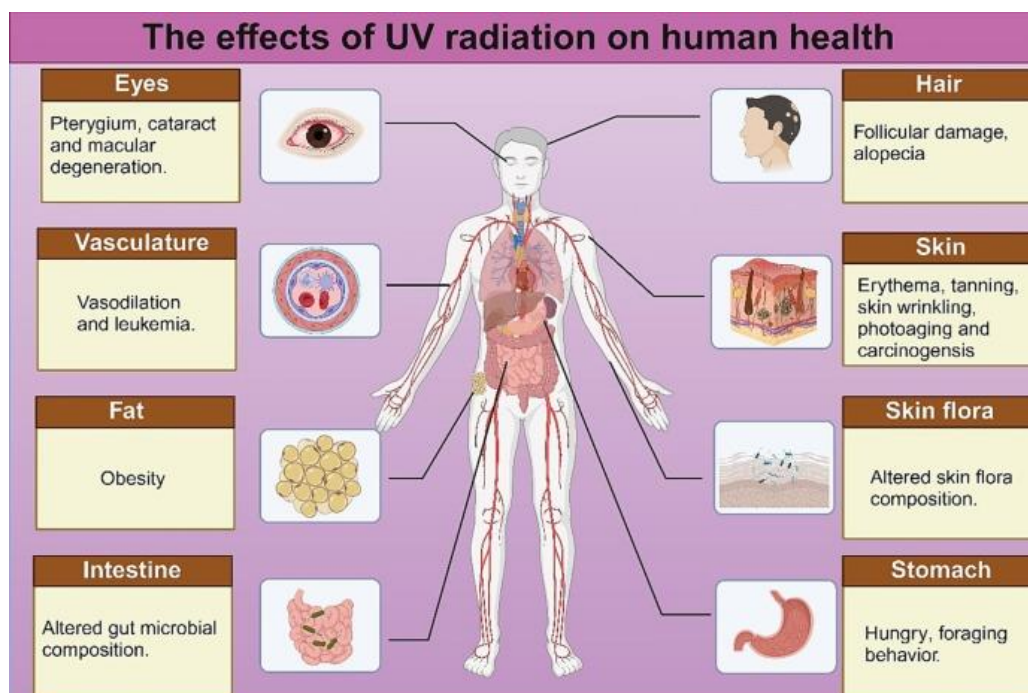


Figure 9: Illustration of potential adverse effect on Human Health from UV radiation (from Tang et al., 2024).

Nevertheless, it is important to note that UV radiation also has beneficial effects for health; it interacts with 7-dehydrocholesterol (provitamin D) and converts it into vitamin D. which plays a crucial role in facilitating the absorption of calcium, necessary for the formation and maintenance of healthy bones (Nitulescu et al., 2023). There are several possibilities to protect the skin from the UV radiation, and as clearly stated by the Center for Disease and Prevention (CDC), sun safety needs to be practice⁹. These practices advices are derivated from the International Agency for Research on Cancer (IARC) that clearly state that "Protection of the skin from solar damage ideally involves a number of actions which include wearing tightly woven protective clothing that adequately covers the arms, trunk and legs, a hat that provides adequate shade to the whole of the head, seeking shade whenever possible, avoiding outdoor activities during periods of peak insolation and use of sunscreens." Another solution, as evocated by the IARC, is to use sunscreens, which are cosmetic products.

Cosmetic regulation define a cosmetic products as 'any substance or mixture intended to be placed in contact with the external parts of the human body (epidermis, hair system, nails, lips and external genital organs) or with the teeth and the mucous membranes of the oral cavity with a view exclusively or mainly to cleaning them, perfuming them, changing their appearance, protecting them, keeping them in good condition or correcting body odours'. Specifically, a sunscreen is a skin product that is define in this regulation as a "cosmetic product which is intended to be applied on the skin"; and A 'UV-filters' means substances which are exclusively or mainly intended to protect the skin against certain UV radiation by absorbing, reflecting or scattering UV radiation.

A sunscreen is a multicomponent product that contains at least one UV filter, but frequently

⁹ [https://www.cdc.gov/skin-cancer/prevention/index.html#:~:text=Wear%20clothing%20that%20covers%20your,SPF\)%20of%2015%20or%20higher](https://www.cdc.gov/skin-cancer/prevention/index.html#:~:text=Wear%20clothing%20that%20covers%20your,SPF)%20of%2015%20or%20higher)

a mixture of them, and additional substances like surfactants, preservatives, and stabilizers. Sunscreens are categorised into two main types - physical or mineral and chemical - based on the type of filters present in their formulations. The term “physical filter” can be misleading because mineral filters are chemicals. Mineral filters in sunscreens are zinc oxide and/or titanium dioxide as active protective components. Chemical sunscreens typically include a blend of UV filters to covert to the maximum extent the full spectra of UV radiation. Upon application, these chemical UV filters are absorbed into the skin.

UV filters are characterized primarily by the value of their absorption peak, lambda max (λ_{max}), where the substances absorbs the most of wavelength of light and the value of the molar absorptivity coefficient (ξ) at the peak’s wavelength. These parameters are directly influenced by the chemical structure of the UV filter and determine the range of UV protection. Chemical structure of the UV filter influence their optical properties and UV absorption, in which the most important element is the structure of the chromophore. Generally UV filters contain an aromatic ring and carbonyl group coupled with various structural elements essential for their specific physico-chemical properties such as branched alkyl or cycloalkyl fragment or ionizable groups (*Nitulescu et al., 2023*).

Octocrilene is a colourless, odourless and oily fluid that performs three main technical functions in cosmetic products. Octocrilene absorbs UV radiation efficiently from 290 to 340 nm with a peak absorption at 307 nm (*Rai et al., 2012*). Based on these specific properties against UV radiation, Octocrilene is therefore used as a UV (mainly UVB) filter in cosmetic products offering sun protection, i.e. in primary sunscreens as well as in secondary photo-protection products¹⁰.

When the skin is subsequently exposed to UV radiation, these chemical active components absorb UV radiation and transform it into thermal energy (heat). However, this process may generate free radicals as a by-product, which can be detrimental to the skin (*Ambruso et al., 2023*). To avoid and encounters the potential adverse effects of these by-products, sunscreens formulations are often complemented with antioxidants such as vitamins C and E.

E.2.2. Identification of potential alternative substances and techniques fulfilling the function

E.2.2.1. Alternatives authorised in cosmetic regulation

The sole UV filters that can be used in cosmetics products are listed in the Annex VI of Regulation (EC) N° 1223/2009 on cosmetic products, also known as the positive list. These UV filters are also listed in section 3.2.4 of the main report. Specific information on potential endocrine disruptors¹¹. On 7 November 2018, the Commission adopted the review of Regulation (EC) No 1223/2009 on cosmetic products containing endocrine-disrupting substances. In the report, the Commission committed to creating a priority list of potential endocrine disruptors that were not covered by the regulation bans yet. After

¹⁰ ‘Secondary photo-protection products’ refer to products with a primary purpose other than sun protection that also contain sun screening agents such as daily face care cream (whose primary function is care, anti-aging, etc.), make up with a Sun Protection Factor (SPF), etc. In this case, the protection against UV radiation is an additional benefit.

¹¹ https://single-market-economy.ec.europa.eu/sectors/cosmetics/cosmetic-products-specific-topics/endocrine-disruptors_en

input from EU countries, industry, consumer organisations and the SCCS, the Commission consolidated a list of 28 substances in early 2019.

The Commission launched a first public call for data from 16 May -15 October on 14 substances that merit a higher priority for assessment. These substances include Benzophenone-3, kojic acid, 4-methylbenzylidene camphor, propylparaben, triclosan, resorcinol, octocrilene, triclocarban, butylated hydroxytoluene (BHT), benzophenone, homosalate, benzyl salicylate, genistein and daidzein. The Commission analyses the received data, mandating the SCCS to evaluate these substances as soon as possible. Then, the Commission takes action to prohibit or restrict the use of these substances in cosmetics if needed.

A second call for data related to the remaining 14 substances took place in the future, taking into consideration any relevant developments. These substances are Butylparaben, tert-butylhydroxyanisole/Butylated hydroxyanisole (BHA), ethylhexyl methoxycinnamate (EHMC), octylmethoxycinnamate (OMC), octinoxate, benzophenone-1 (BP-1), benzophenone-2 (BP-2), benzophenone-4 (BP-4), benzophenone-5 (BP-5), methylparaben, cyclopentasiloxane/decamethylcyclopentasiloxane (D5), Cyclomethicone, salicylic acid, butylphenyl methyl propanol (BMHCA), triphenyl phosphate and deltamethrin.

In regards of new available data, the SCCS was requested to provide, when necessary new input concerning authorised substances used in cosmetics and the evolution were transcript into this regulation. These modification and update occurs regularly based on SCCS works after specific request from the EU Commission for reassessing data in light of new scientific evidence.

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Table 49: Authorised UV filters with/without UV absorption and light stabilisation properties.

Reference number	Substances information						
	Chemical name/INN/XAN	Name of Common Ingredients Glossary	CAS number	EC number	UV Filter	UV absorber	Light stabilizer
2	<i>N,N,N</i> -Trimethyl-4-(2-oxoborn-3-ylidenemethyl) anilinium methyl sulfate	Camphor Benzalkonium Methosulfate	52793-97-2	258-190-8	✓	✓	
3	Benzoic acid, 2- hydroxy-, 3,3,5- trimethylcyclohexyl ester/Homosalate (8)	Homosalate	118-56-9	204-260-8	✓	✓	
4	2-Hydroxy-4-methoxy-benzophenone/Oxybenzone (6)	Benzophenone-3	131-57-7	205-031-5	✓	✓	✓
6	2-Phenylbenzimidazole-5-sulfonic acid and its potassium, sodium and triethanolamine salts/Ensulizole	Phenylbenzimidazole Sulfonic Acid	27503-81-7	248-502-0	✓	✓	✓
7	3,3'-(1,4-Phenylenedimethylene) bis(7, 7-dimethyl-2-oxobicyclo-[2.2.1]hept-1-yl-methanesulfonic acid) and its salts/Ecamsule	Terephthalylidene Dicamphor Sulfonic Acid	92761-26-7, 90457-82-2	410-960-6	✓	✓	✓
8	1-(4-tert-Butylphenyl)-3-(4-methoxyphenyl)propane-1,3-dione/Avobenzene	Butyl Methoxydibenzoylmethane	70356-09-1	274-581-6	✓	✓	✓
9	alpha-(2-Oxoborn-3-ylidene)-toluene-4-sulphonic acid and its salts	Benzylidene Camphor Sulfonic Acid	56039-58-8		✓	✓	✓
10	2-Cyano-3,3-diphenyl acrylic acid, 2- ethylhexyl ester/Octocrilene (6) (?)	Octocrilene	6197-30-4	228-250-8	✓	✓	✓
11	Polymer of N-[(2 and 4)-[(2-oxoborn-3-ylidene)methyl]benzyl]acrylamide	Polyacrylamidomethyl Benzylidene Camphor	113783-61-2		✓	✓	✓

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

12	2-Ethylhexyl 4-methoxycinnamate/Octinoxate	Ethylhexyl Methoxycinnamate	5466-77-3	226-775-7	✓	✓	✓
13	Ethoxylated ethyl-4-aminobenzoate	PEG-25 PABA	116242-27-4		✓	✓	✓
14	Isopentyl-4-methoxycinnamate/Amiloxate	Isoamyl p-Methoxycinnamate	71617-10-2	275-702-5	✓	✓	✓
15	2,4,6-Trianiino-(p-carbo-2'-ethylhexyl-1'-oxy)-1,3,5-triazine	Ethylhexyl Triazone	88122-99-0	402-070-1	✓	✓	✓
16	Phenol,2-(2H-benzotriazol-2-yl)-4-methyl-6-(2-methyl-3-(1,3,3,3-tetramethyl-1-(trimethylsilyl)oxy)-disiloxanyl)propyl)	Drometrizole Trisiloxane	155633-54-8	423-660-5	✓	✓	✓
17	Benzoic acid, 4,4-((6-((4-(((1,1-dimethylethyl)amino)carbonyl)phenyl)amino)-1,3,5-triazine-2,4-diyl)diimino)bis-, bis (2-ethylhexyl) ester/Iscotrizinol (USAN)	Diethylhexyl Butamido Triazone	154702-15-5	421-450-8	✓	✓	✓
18	3-(4-Methylbenzylidene)-d1 camphor/Enzacamene	4-Methylbenzylidene Camphor	38102-62-4/36861-47-9	- / 253-242-6	✓	✓	✓
20	2-Ethylhexyl salicylate/Octisalate	Ethylhexyl Salicylate	118-60-5	204-263-4	✓	✓	✓
21	2-Ethylhexyl 4-(dimethylamino)benzoate/Padimate O (USAN:BAN)	Ethylhexyl Dimethyl PABA	21245-02-3	244-289-3	✓	✓	✓
22	2-Hydroxy-4-methoxybenzophenone-5-sulfonic acid and its sodium salt/Sulisobenzene	Benzophenone-4, Benzophenone-5	4065-45-6 6628-37-1	223-772-2 -	✓	✓ ✓	✓ ✓
23	2,2'-Methylene-bis(6-(2H-benzotriazol-2-yl)-4-(1,1,3,3-tetramethylbutyl)phenol)/Bisoctrizole	Methylene Bis-Benzotriazolyl Tetramethylbutylphenol	103597-45-1	403-800-1	✓		✓

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

23a	2,2'-Methylene- bis(6-(2H-benzotriazol-2-yl)-4-(1,1,3,3-tetramethylbutyl)phenol)/Bisotrizole	Methylene Bis-Benzotriazolyl Tetramethylbutylphenol (nano)	103597-45-1	403-800-1	✓		✓
24	Sodium salt of 2,2'-bis(1,4-phenylene)-1H-benzimidazole-4,6-disulfonic acid) / Bisdisulizole disodium (USAN)	Disodium Phenyl Dibenzimidazole Tetrasulfonate	180898-37-7	429-750-0	✓	✓	✓
25	2,2'-(6-(4-Methoxyphenyl)-1,3,5-triazine-2,4-diyl)bis(5-((2-ethylhexyl)oxy)phenol) / Bemotrizinol	Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine	187393-00-6	425-950-7	✓	✓	✓
26	Dimethicodiethylbenzmalonate	Polysilicone-15	207574-74-1	426-000-4	✓	✓	✓
27	Titanium dioxide (2)	Titanium Dioxide	13463-67-7/1317-70-0/1317-80-2	236-675-5/215-280-1/215-282-2	✓	✓	
27a	Titanium dioxide (2)	Titanium Dioxide (nano)	13463-67-7/1317-70-0/1317-80-2	236-675-5/215-280-1/215-282-2	✓	✓	
28	Benzoic acid, 2-[-4-(diethylamino)-2-hydroxybenzoyl]-. Hexylester	Diethylamino Hydroxy benzoyl Hexyl Benzoate	302776-68-7	443-860-6	✓	✓	✓
29	1,3,5-Triazine, 2,4,6-tris[1,1'-biphenyl]-4-yl-, including as nanomaterial	Tris-biphenyl triazine Tris-biphenyl triazine (nano)	31274-51-8	479-950-7	✓	✓	✓
30	Zinc oxide	Zinc Oxide	1314-13-2	215-222-5	✓	✓	✓
30a	Zinc oxide	Zinc Oxide (nano)	1314-13-2	215-222-5	✓	✓	✓
31	3,3'-(1,4-Phenylene)bis(5,6-diphenyl-1,2,4-triazine)	Phenylene Bis-Diphenyltriazine	55514-22-2	700-823-1	✓		✓

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

32	2-ethoxyethyl (2Z)-2-cyano-2-[3-(3-methoxypropylamino) cyclohex-2-en-1-ylidene]acetate	Methoxypropylamino Cyclohexenylidene Ethoxyethylcyanoacetate	1419401-88-9	700-860-3	✓		✓
33	1,1'-(1,4-piperazinediyl)bis[1-[2-[4-(diethylamino)-2-hydroxybenzoyl]phenyl]-methanone	Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine	919803-06-8	485-100-6	✓		
34	1,1'-(1,4-piperazinediyl)bis[1-[2-[4-(diethylamino)-2-hydroxybenzoyl]phenyl]-methanone	Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine (nano)	919803-06-8	485-100-6	✓		

Despite being authorised as UV filters in cosmetic regulation annex VI, some UV filters are under assessment for potential health hazards, environmental hazards or both. These ongoing process and call for data generation to industry in order to remove regulatory uncertainties in regards to these potential hazards are detailed in Table 50.

Table 50 (summary of table 23 of the main report): Authorised cosmetic UV filters under ongoing REACH process.

Table 36 (Summary of table 25 of the main report): Authorised cosmetic UV filters under ongoing REACH process.						
	Substances information					
Reference number	Name of Common Ingredients Glossary	CAS number	EC number	Information requirements	Suspected Hazard	
2	Camphor Benzalkonium Methosulfate	52793-97-2	258-190-8	No relevant data✓		
3	Homosalate	118-56-9	204-260-8	Data generation under Compliance Check requesting: 1. Sub-chronic toxicity study (90-day), oral route (OECD TG 408) in rats; 2. Pre-natal developmental toxicity study (OECD TG 414) in a first species (rat or rabbit), oral route; 3. Extended one-generation reproductive toxicity study (OECD TG 443) in rats, oral route as follows: - Ten weeks premating exposure duration for the parental (PO) generation; - Dose level setting shall aim to induce some toxicity at the highest dose level; - Cohort 1A (Reproductive toxicity); - Cohort 1B (Reproductive toxicity) without extension to mate the Cohort 1B animals to produce the F2 generation; - Cohorts 2A and 2B (Developmental neurotoxicity); and - Cohort 3 (Developmental immunotoxicity), 4. Identification of degradation products using an appropriate test method.	Suspected ED, Suspected PBT, Reproductive toxicity Exposure of the Environment	

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Deadline: 25 February 2024 Data generation for point 1, 2 and 3 were suspended due to an appeal from industry in the EU Court of Justice of ECHA. This appeal was dismissed (22 November 2023) ¹² and data generation should be ongoing.	
4	Benzophenone-3	131-57-7	205-031-5	Substance under evaluation by Denmark.	Suspected ED
6	Phenylbenzimidazole Sulfonic Acid	27503-81-7	248-502-0	A Testing Proposal Evaluation and a Compliance Check in case of Dossier Evaluation are ongoing. No details available on the subject covered by this TPE and CCH.	
7	Terephthalylidene Dicamphor Sulfonic Acid	92761-26-7, 90457-82-2	410-960-6	No relevant data	
8	Butyl Methoxydibenzoylmethane	70356-09-1	274-581-6	A substance evaluation decision number SEV-D-2114357832-44-01/F ongoing requesting: i) Aerobic mineralisation in surface water – simulation biodegradation test, (OECD TG 309), ii) Aerobic and anaerobic transformation in aquatic sediment systems, (OECD TG 308), iii) Long-term toxicity testing on aquatic invertebrates (OECD TG 211), iv) Long-term toxicity testing on fish (OECD TG 210). Based on the available information, the evaluating MSCA considers that the Substance is potentially PBT as the substance is considered to fulfil the P and B criteria; information on T is currently considered inconclusive despite the availability of standard long-term studies on aquatic organisms. An amphibian metamorphosis assay (OECD TG 231) Deadline: 16 February 2026	Suspected ED and PBT, Exposure of the environment

¹²

<https://curia.europa.eu/juris/document/document.jsf?jsessionid=5EEE21635DEEA5CBBAF06F32427304E5?text=&docid=244530&pageIndex=0&doclang=EN&mode=lst&dir=&occ=first&part=1&cid=1590412>

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

9	Benzylidene Camphor Sulfonic Acid	56039-58-8		No relevant data	
10	Octocrilene	6197-30-4	228-250-8	A substance evaluation decision number SEV-D-2114287467-34-01/F ongoing requesting: The Larval Amphibian Growth and Development Assay (LAGDA, OECD TG 241). Deadline: 04 June 2024 (expected 31 October 2024) A Testing Proposal was also ongoing: A. Information required from the Registrants subject to Annex IX of REACH 1. Sediment simulation testing (OECD TG 308), 2. Identification of degradation products (OECD TG 308), 3. Effects on soil micro-organisms (OECD TG 216), 4. Long-term toxicity to terrestrial plants (OECD TG 208 with at least six species). B. Information required from the Registrants subject to Annex X of REACH 1. Long-term toxicity testing to sediment organisms (OECD TG 225 and OECD TG 233 and OECD TG 239). Deadline: 03 December 2023, submitted in June 2024.	Potential PBT, Potential ED for human health and the environment
11	Polyacrylamidomethyl Benzylidene Camphor	113783-61-2		No relevant data	
12	Ethylhexyl Methoxycinnamate	5466-77-3	226-775-7	No data requested	Uncertainties related to genotoxicity, Potential ED
13	PEG-25 PABA	116242-27-4		No relevant data	
14	Isoamyl p-Methoxycinnamate	71617-10-2	275-702-5	1. Either an Amphibian Metamorphosis Assay, (OECD 231), or a Larval Amphibian Growth and Development Assay, (OECD 241); 2. Fish Sexual Development Test, test method (OECD 234); 3. Provide information and justification for parameters in the environmental exposure assessment within the Chemical Safety Report. Combined assessment of OMC and IPMC. ECHA considers the results obtained in the FSDT and AMA with OMC as unsuitable to conclude on the ED properties for IPMC as well.	Suspected ED, Exposure of the environment.

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

				Combination of Fish short term reproduction assay (OECD TG 229) and Fish Sexual Development Test (OECD TG 234) Deadline: 19 December 2025	
15	Ethylhexyl Triazone	88122-99-0	402-070-1	No relevant data	
16	Drometrizole Trisiloxane	155633-54-8	423-660-5	No relevant data	
17	Diethylhexyl Butamido Triazone	154702-15-5	421-450-8	Initial ground for potential PBT. Substance is P, vP Inconclusive results for B vB. Classification: Not PBT, vPvB	
18	4-Methylbenzylidene Camphor	38102-62-4/36861-47-9	- / 253-242-6	SVHC dossier according to REACH Art. 57 (f) owing to its endocrine disrupting properties was prepared by Germany and later withdrawn due to data generation awaited. A follow up action for SVHC dossier by Denmark was done. Identified as a SVHC according to art. 57f in the 76 th meeting of the MSC. Listed for eventual inclusion in Annex XIV of REACH. Indication of potential PBT properties.	SVHC ED art. 57f Potential PBT.
20	Ethylhexyl Salicylate	118-60-5	204-263-4	1. Pre-natal developmental toxicity study (OECD TG 414) in a first species (rat or rabbit), oral route; 2. Extended one-generation reproductive toxicity study (OECD TG 443) in rats, oral route; 3. Long-term toxicity testing on aquatic invertebrates (OECD TG 211); 4. Long-term toxicity testing on fish (OECD TG 234)) Deadline: 25 February 2024 Data generation for point 1, 2 and 4 were suspended due to an appeal from industry in the EU Court of Justice of ECHA. This appeal was dismissed (22 November 2023) ¹³ and data generation should be ongoing.	Suspected ED Aquatic toxicity Reproductive toxicity

¹³

<https://curia.europa.eu/juris/document/document.jsf?jsessionid=81FF4CB4A599CB25D8A22F9F8F19271D?text=&docid=279983&pageIndex=0&doclang=en&mode=req&dir=&occ=first&part=1&cid=1802221>

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

21	Ethylhexyl Dimethyl PABA	21245-02-3	244-289-3	No relevant data	Indication of potential classification as Repro 1B from data
22	Benzophenone-4, Benzophenone-5	4065-45-6 6628-37-1	223-772-2 -	Benzophenone-4 <u>CCH 1:</u> 1. In vitro gene mutation study in bacteria (OECD TG 471); 2. Short-term toxicity testing on aquatic invertebrates (OECD TG 202); 3. Growth inhibition study aquatic plants (OECD TG 201). 4. In vitro cytogenicity study in mammalian cells (OECD TG 473) or In vitro micronucleus study (OECD TG 487); 5. If negative results are obtained in tests performed, then: In vitro gene mutation study in mammalian cells (OECD TG 476 or TG 490). Deadline: 20 August 2024¹⁴. <u>CCH 2:</u> 1. Short-term toxicity testing on aquatic invertebrates (OECD TG 202); 2. Growth inhibition study aquatic plants (OECD TG 201 or OECD TG 221). 3. In vitro gene mutation study in mammalian cells (OECD TG 476 or TG 490). Deadline: 20 August 2024¹⁵. Benzophenone-5 No relevant data	Suspected ED Considered safe by the SCCS (April 2024)¹⁶
23	Methylene Bis-Benzotriazolyl Tetramethylbutylphenol	103597-45-1	403-800-1	No relevant data	
23a	Methylene Bis-Benzotriazolyl Tetramethylbutylphenol (nano)	103597-45-1	403-800-1	No relevant data	
24	Disodium Phenyl Dibenzimidazole Tetrasulfonate	180898-37-7	429-750-0	Ongoing Dossier Evaluation for completeness and comprehension	

¹⁴ <https://echa.europa.eu/documents/10162/36364f70-c144-8623-7023-bfcb7250447d>

¹⁵ <https://echa.europa.eu/documents/10162/c6ac4ed0-2f4e-b974-5b02-84aacb7b4a8d>

¹⁶ https://health.ec.europa.eu/publications/benzophenone-4-cas-no-4065-45-6-ec-no-223-772-2_en

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

25	Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine	187393-00-6	425-950-7	No relevant data	P, vP
26	Polysilicone-15	207574-74-1	426-000-4	No active registration under REACH	
28	Diethylamino Hydroxy benzoyl Hexyl Benzoate	302776-68-7	443-860-6	No relevant data	
29	Tris-biphenyl triazine Tris-biphenyl triazine (nano)	31274-51-8	479-950-7	No relevant data	
31	Phenylene Bis-Diphenyltriazine	55514-22-2	700-823-1	No relevant data	
32	Methoxypropylamino Cyclohexenylidene Ethoxyethylcyanoacetate	1419401-88-9	700-860-3	No relevant data	
33	Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine	919803-06-8	485-100-6	No relevant data	
34	Bis-(Diethylaminohydroxybenzoyl Benzoyl) Piperazine (nano)	919803-06-8	485-100-6	No relevant data	

Table 51: Physico-chemical properties of best suitable alternatives to octocrilene according to stakeholders consultation.

Input	Ethylhexyl Triazone	Diethylhexyl Butamido Triazone	Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine	Diethylamino Hydroxy benzoyl Hexyl Benzoate
Physical form	Solid	White powder, Solid, odourless	Pale yellow Solid	White powder, Solid
Melting/freezing point at 101.3 kPa	114-130°C	88.3 – 91.4°C	80.4 °C	54 °C
Boiling point at 100.3 kPa	874.4°C	>= 343.5 - <=351.6°C	>400 °C	Substance decompose at 314° C, before boiling
Relative density	1.1 g/cm3 at 20 °C	1.15 g/cm3 at 20 °C	1.17 g/cm3 at 20 °C	1.16 g/cm3 at 20 °C
Vapour Pressure	0.000006 Pa at 20°C 0.0005 Pa at 50°C	< 0.00019 Pa at 25°C	5.9e-20 Pa at 25 °C.	0.000000029 hPa at 20°C
Water solubility	0.007 mg/L	< 0.00075 mg/L at 20.0±0.5°C, pH= 5.4 – 7.1	< 0.014 mg/L at 20°C	16 µg/l at 20° C
Oil solubility	High	High	High	High
Partition coefficient n-octanol/water (Log Kow)	7 (at 25°C)	5.925 ± 0.100 (at 25°C) 4.12 ± 0.20 (at 20°C, pH 6.2)	> 5.7 at 20 °C	6.2 at 24° C
Log Koc	> 5.63	8.934	5.7 - 6.8	5.1
Photostability	High	High	High	High
λ max	314 nm	310 nm	310 and 343 nm	354 nm
Light Spectrum	UVB, UVA2	UVB, UVA 1 and UVA2	UVB, UVA 1 and UVA2	UVA1, UVA2

E.2.2.2. Specific information on substances not used anymore as UV filters despite being authorised in annex VI of the cosmetic regulation

Benzyldienecamphor Derivatives

These substances presents α,β-unsaturated carbonyl structure with a double bond system acting as a chromophore showing great photostability. The UV absorption process for these substances occurs through photo-induced reversible trans–cis isomerization (Nitulescu et al., 2023).

Camphor Benzalkonium Methosulfate

This substance is a UVB filter with a $\lambda_{\text{max}} = 284 \text{ nm}$, chemically photostable authorised in Australia and the EU up to 6% in sunscreen products. No specific data were available indicating their potential hazards for human health and the environment.

Benzylidene Camphor Sulfonic Acid

This substance is capable of forming salts due to its sulfonic acid moiety. These salts are also authorised as UV filters by cosmetic regulation. This substance is a UVB filter ($\lambda_{\text{max}} = 294 \text{ nm}$), authorised in EU, Australia and Japan up to 6% (as an acid).

Polyacrylamidomethyl Benzylidene Camphor

This polymeric UVB filter ($\lambda_{\text{max}} = 297 \text{ nm}$) is not registered under REACH and is approved in China and in the EU to be used in sunscreens up to 6%.

Ethoxylated ethyl-4-aminobenzoate

This substance is an UVB filter ($\lambda_{\text{max}} = 308 \text{ nm}$) and a partial UVA-2 filter ($\lambda = 320\text{-}340 \text{ nm}$), with the advantage of being hydrosoluble. It is approved by the EU, China and Australia, but not by the FDA (solely used as a UV absorber). It can be used in all types of cosmetic products at concentrations up to 10%.

E.2.2.3. Other potential alternatives to be used as UV filters

E.2.2.3.1.1. Advanced nanotechnology-based sunscreen formulations

The preparation of these nanocarriers required specific technologies and can possibly be tricky. They all presents advantages and disadvantages depending on the required function in the sunscreen formulation or more generally, in any of the cosmetics. They are detailed in Figure 11.

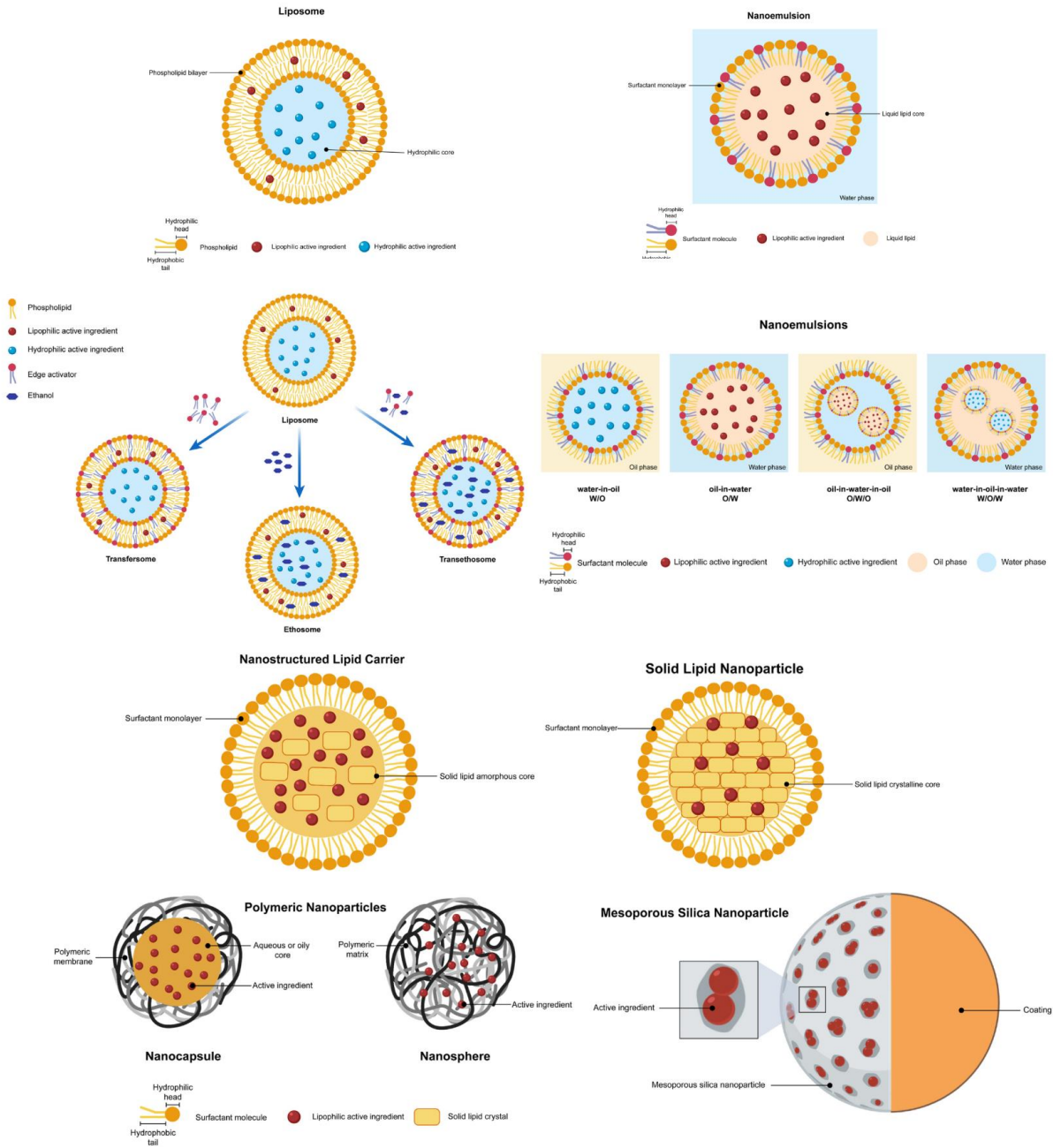


Figure 10: Illustration of nanocarriers used in sunscreen formulations (from Santos et al., 2022).

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Nanoparticles	Advantages	Disadvantages
Polymeric nanoparticles	• Easily can be utilized in tissue engineering • Higher effectiveness over conventional oral delivery • Very useful and effective in the case of volatile substances delivery.	• Presence of toxicity due to the non-biodegradable nature of the polymers.
Nanoemulsion	• Problems like coalescence and flocculation can be avoided with nanoemulsions • Non-irritant as well as non-toxic nature of these molecules enhances their topical use • Due to oil soluble nature of the molecules, it can be useful in the toxicity study of oil-soluble substances • With the small size of these molecules, a higher penetration rate across the skin can be achieved.	• pH, as well as temperature, can affect the stability • A higher amount of surfactant is required to maintain stability.
Nanocrystals	• Most likely for hydrophobic substances • Enhances drug solubility and permeation	• Not suitable for substances that are hydrophilic in nature.
Mesoporous silica particles	• Major advantages related to their excellent biodegradable nature as well nontoxic nature.	• Leads to melanoma.
Solid lipid nanoparticles	• Excellent biodegradable property • Easy to produce • Higher bioavailability of drug due to better skin permeation property.	• Low drug encapsulation property • Solution state stability issues.
Nanostructure lipid carriers	• Higher physical stability • Easy to formulate • Higher drug loading capacity • Higher penetration • Excellent value of benefit/risk ratio • Controlled drug delivery property	• Surfactant-induced irritancy • Cytotoxic effect of lipid portion.
Liposomes	• Act as a nanocarrier for hydrophilic as well as hydrophobic drugs • Excellent biodegradable property • Can produce longer action through sustained release	• It may be allergic in some cases • Lesser stability
Niosomes	• Higher bioavailability due to a higher rate of penetration • Higher drug loading capacity • Higher physical stability.	• More complex as well as time-consuming preparation • Highly costly in terms of production

Figure 11: Advantages and disadvantages of using nanocarriers in cosmetics formulation (adapted from Chavda et al., 2023).

E.2.3. Risk reduction, technical and economic feasibility, and availability of alternatives¹⁷

Octocrilene was demonstrated to be hazardous and leading to risk for the environment due to its large and dispersives uses as a UV filter in sunscreen products but also as a UV absorber and a light stabilizer for other cosmetic products. Regarding UV filter properties for sunscreen products, potential alternatives must be substances authorised, or possibly being authorised in the coming years, under cosmetic regulation. When considering this regulatory imperative, **30 potential substances are available and already authorised under cosmetic regulation**. Titanium dioxide and zinc oxide (non-nano and nano) were not considered further as potential alternative because, according to Cosmetics Europe, the substitution of octocrilene as a UV filter will likely be based on other organic filters (Cosmetics Europe, personal communication).

Table 52: Potential alternatives to octocrilene and current market price.

INCI Name	Chemical name	CAS Number	EC Number	Absorpti on spectrum	Market price ¹⁸ (€/kg)
Camphor Benzalkonium methosulfate	N,N,N-Triméthyl-4-(2-oxoborn-3-ylidene-methyl) anilinium methyl sulfate	52793-97-2	258-190-8	UVB	NA : No registrant in the EEA
Homosalate	Homomenthyl salicylate	118-56-9	204-260-8	UVB	20
Benzophenone-3	Oxybenzone 2-Hydroxy-4-methoxybenzophenone	131-57-7	205-031-5	UVA + UVB	No data available
Phenylbenzimidazole sulfonic acid	2-phenylbenzimidazole-5-sulphonic acid and its potassium, sodium and triethanolamine salts Ensulizole	27503-81-7	248-502-0	UVB	56
Terephthalylidene dicamphor sulfonic acid	3,3'-(1,4Phenylene-dimethylene) bis (7,7-dimethyl-2-oxo-bicyclo[2.2.1]hept-1-ylmethane sulphonic acid) and its salts Ecamsule	92761-26-7	410-960-6	UVA	No data available
Butylmethoxydibenzo yl methane	1-(4-Tert-butylphenyl)-3-(4-methoxyphenyl) propane-1,3-dione Avobenzene 4-Tert-butyl-4'-methoxydibenzoylmethane	70356-09-1	274-581-6	UVA	18
Benzylidene camphor sulfonic acid and salts	alpha-(2-oxoborn-3-ylidene) toluene-4sulphonic acid and its salts	56039-58-8		UVA + UVB	NA : No registrant in the EEA
Octocrilene	2-cyano-3,3-diphenylacrylic acid, 2-ethyl-hexyl ester	6197-30-4	228-250-8	UVB	19
Polyacrylamidomethyl benzylidene camphor	Polymer of N-(2 ans 4)-[(2-oxoborn-3-ylidene) methyl] benzyl} acrylamide	113783-61-2		UVB	NA : No registrant in the EEA
Ethylhexyl Methoxycinnamate / Octyl Methoxycinnamate	Octyl methoxycinnamate 2-ethylhexyl-4-methoxy-cinnamate Octinoxate	5466-77-3	226-775-7	UVB	NA : No registrant in the EEA

¹⁷ Assess each alternative using the headings set out below adding additional headings as necessary. If it is considered more feasible to describe all alternatives in parallel, use headings: Assessment of alternatives; Availability of alternatives; Human health risks related to alternatives; Environmental risks related to alternatives; Technical and economical feasibility of alternatives and other information of alternatives.

¹⁸ The market price indicated here corresponds to the average of the market prices communicated by the industrial stakeholders.

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

PEG-25-PABA	Ethoxylated ethyl-4-amino-benzoate	116242-27-4		UVB	NA : No registrant in the EEA
Isoamyl p-methoxycinnamate	Isopentyl-4-methoxy-cinnamate	71617-10-2	275-702-5	UVB	No data available
Ethylhexyl Triazone	Octyl triazone 2,4,6-Trianiino-(p-carbo-2'-ethylhexyl-1'-oxy)-1,3,5 triazine 2,4,6-tris(4-(2-ethylhexyloxy-carbonyl) aniline)-1,3,5 triazine	88122-99-0	402-070-1	UVB	41
Drometrizole trisiloxane	Phenol,2-(2H-benzo-triazol-2-yl)-4 methyl-6- (2-methyl-3-(1,3,3,3 tetramethyl-1-(trimethyl-silyl)oxy)-disiloxanyl)propyl)	155633-54-8		UVA + UVB	No data available
Diethylhexyl butamido triazone	Benzoic acid, 4,4-(((6-((1,1-dimethylethyl) amino) carbonyl) phenyl) amino)-1,3,5-triazine-2,4-diyl) diimino) bis-,bis(2-ethylhexyl)ester)	154702-15-5		UVB	73
4-methylbenzylidene camphor	3-(4'-methylbenzyl-idene)-d-1 camphre Enzacamene	36861-47-9 38102-62-4	253-242-6	UVB	NA : No registrant in the EEA
Ethylhexyl Salicylate / Octyl Salicylate	Octyl Salicylate 2-Ethylhexyl salicylate Octisalate	118-60-5	204-263-4	UVB	15
Ethylhexyl Dimethyl PABA	Octyl dimethyl PABA 4-Dimethyl-amino-benzoate of ethyl-2-hexyl Padimate O 2-Ethylhexyl p-dimethylamino benzoate	21245-02-3	244-289-3	UVA + UVB	No data available
Benzophenone-4	2-hydroxy-4-methoxybenzone-5-sulfonic acid and its trihydrate Sulisobenzone	4065-45-6	223-772-2		No data available
Benzophenone-5	Sodium hydroxymethoxybenzophenone sulfonate Sulisobenzone Sodium	6628-37-1		UVA + UVB	
Methylene bis-benzotriazolyl tetramethylbutylphenol	2,2'-methylene-bis-6-(2H-benzotriazol-2yl)-4-(tetramethyl-butyl)-1,1,3,3-phenol	103597-45-1	403-800-1	UVA + UVB	48
Methylene bis-benzotriazolyl tetramethylbutylphenol [nano]	2,2'-methylene-bis-6-(2H-benzotriazol-2yl)-4-(tetramethyl-butyl)-1,1,3,3-phenol [nano]			UVA	
Disodium Phenyl Dibenimidazole Tetrasulfonate	Monosodium salt of 2-2'-bis-(1,4-penylene)1H-benzimidazole-4,6-disulphonic acid) Bisymidazylate	180898-37-7	429-750-0	UVA + UVB	18-27
Bis-Ethylhexyloxyphenol Methoxyphenyl Triazine	(1,3,5)-triazine-2,4-bis((4-2-ethylhexyloxy)-2-hydroxy)-phenyl)-6-(4-methoxyphenyl) Anisotriazine Bemotrizinol	187393-00-6			79
Polysilicone-15	Dimethicodiethylbenzal malonate	207574-74-1	426-000-4	UVA	No data available
Diethylamino Hydroxybenzoyl Hexyl Benzoate	Benzoic acid, 2-[-4-(diethylamino)-2-hydroxybenzoyl]-,hexylester	302776-68-7	443-860-6	UVA + UVB	49

Tris-biphenyl triazine Tris-biphenyl triazine (nano)	1,3,5-Triazine, 2,4,6-tris [1,1'-biphenyl]-4-yl-, including as nanomaterial	31274-51-8	/	UVA + UVB	No data available
Phenylene bis-diphenyltriazine	3,3'-(1,4-phénylène) bis(5,6-diphényl-1,2,4-triazine)	55514-22-2	700-823-1 (L)	UVA	No data available
Methoxypropylamino Cyclohexenylidene Ethoxyethylcyanoacetate	2-ethoxyethyl (2Z)-2-cyano-2-[3-(3-methoxypropylamino) cyclohex-2-en-1-ylidene]acetate	1419401-88-9	700-860-3		No data available
Bis-(Diethylaminohydroxy benzoyl Benzoyl) Piperazine	1,1'-(1,4-pipérazinediyl)bis[1-[2-[4-(diéthylamino)-2-hydroxybenzoyl]phényl]-méthanone	919803-06-8	485-100-6	UVA	No data available
Bis-(Diethylaminohydroxy benzoyl Benzoyl) Piperazine (nano)	1,1'-(1,4-pipérazinediyl)bis[1-[2-[4-(diéthylamino)-2-hydroxybenzoyl]phényl]-méthanone	919803-06-8	485-100-6	UVB	

E.3. Restriction scenario(s)

See Annex XV report.

E.4. Economic impacts

E.4.1. Description of the cosmetic industry

E.4.1.1. Cosmetic Value Chain

Upstream operators of the cosmetic value chain, showed in Figure 12, are **ingredients producers**. They produce cosmetic raw materials and supply cosmetics manufacturers that produce cosmetic products by relying on formulators and laboratories.

Laboratories are essential in this value chain, not only to create new ingredients or new products, but also to ensure their efficacy and compliance with claims and regulations (such as that relative to safety for consumers). These laboratories may be part of the manufacturers or independent.

Manufacturers of cosmetic products reach consumers through exporters, wholesalers, retailers and beauty providers. However, there are two types of cosmetic manufacturers: manufacturers of national brand cosmetics and manufacturers of private labels. The first type can outsource some or all of its production to the second type. Private label producers can provide products to national brand manufacturer or to some retailers that want their own brand of cosmetics. These products can be bespoke (original equipment manufacturing) or not (original development and design manufacturing). A private label cosmetics manufacturer is a company specialised in the custom-development and manufacturing of custom-made cosmetic products. They formulate and supply cosmetic goods according to the client's specific needs. The client sells the product provided by private label manufacturers under its own name.

Wholesalers buy large quantities of cosmetics from manufacturers. They then sell these products in smaller quantities to **retailers**, which in turn resell them to consumers.

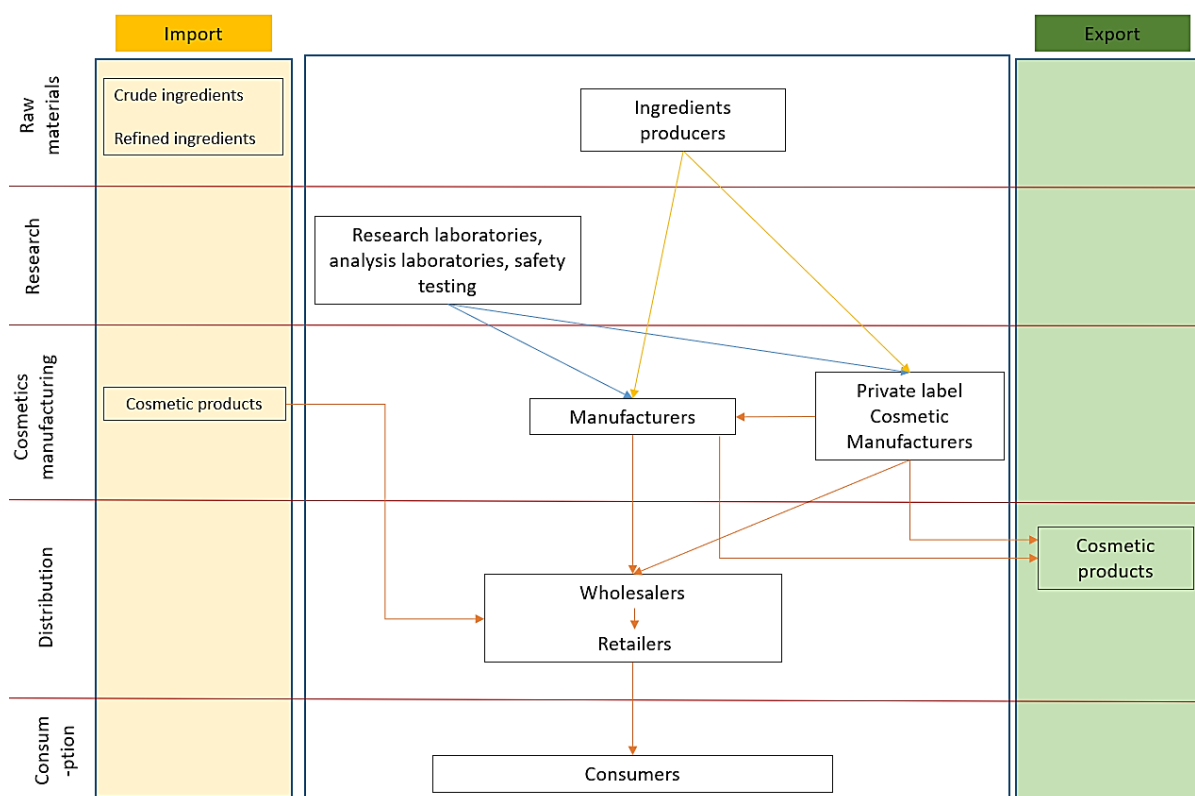


Figure 12: Cosmetic Value Chain.

(yellow arrow: supply of ingredients; blue arrow: laboratory services; orange arrow: supply of cosmetic products).

E.4.1.2. Cosmetic market

E.4.1.2.1. Market size

Market for cosmetic products in Europe (European Union + UK) is in constant growth. It was worth 80 billion euros at Retail Sales Price (RSP) in 2021 and 88 billion euros in 2022 and 96 billion euros in 2023 (Cosmetics Europe).

E.4.1.2.2. Heterogeneity of cosmetic products manufacturers

The manufacturing market of cosmetic products is asymmetric with a high number of small and medium-sized enterprises (SMEs) and some large manufacturers.

According to stakeholders consultation, SMEs are the one which use the most octocrilene. In 2022, there were 8,540 SMEs on this market in Europe (EU27, Norway, Switzerland and UK), 6,841 of them in EU-27 alone. The number of SMEs in Europe has risen by 23 % in one year and they account for 80% of cosmetic producers in Europe (stakeholders consultation). Not all manufacturers have their own brand, some are subcontractors.

In 2023, according to Euromonitor, the cosmetics market included 70 cosmetics

manufacturers in western Europe¹⁹ with a market share greater than 0.1%²⁰. These 70 manufacturers together account for 79% of total market share. More than one quarter of the market were owned by three firms²¹ and this share has decreased by 3.9 percentage points between 2014 (30.6%) and 2023 (26.5 %).

E.4.1.2.3. Distribution

According to Cosmetics Europe, in 2015, there were approximately 23,000 firms involved in the wholesale of cosmetics in Europe, the majority of which were located in Italy (17%), Spain (15%), and France (10%) (Eurostat, 2015). Retailers include salons, department stores, online stores and pharmacies. In 2015, there were approximately 46,400 specialist stores. There is a large heterogeneity among retailers in terms of size and among countries in terms of type of distribution channels.

E.4.1.3. Sun care market

E.4.1.3.1. Market size

This cosmetic market²² can be breakdown into seven main categories: sun care, oral care, skin care, hair care, body care, decorative cosmetics and perfume. The data provided herein come from Euromonitor.

The sun care market is valued at 3.5 billion euros in Western Europe, with 46 million of litres consumed in 2023. The market has been steadily growing in volume terms (litres) since 2009 (**Figure 14**) and has remained constant in terms of value (RSP²³ and MSP²⁴) until 2021, the date from which a slight growth begins to emerge. According to Euromonitor estimation, all else equal, the market should reach 54 million of litres in 2028.

E.4.1.3.2. Heterogeneity of sun care manufacturers

More than three quarters of the 70 firms involved in cosmetics with more than 0.1 % of the market share manufacture sun care products. This market is asymmetric. Almost half of the market are owned by three firms and this share has increased by 2.3 percentage points between 2014 (45.4 %) and 2023 (47.7).

E.4.1.3.3. Distribution

The following information relates to sun care products, but can be transposed to adult sun protection and child-specific products given their weight (80 %) in sun care products.

In value terms (RSP), 86.6 % of sales take place offline through non-grocery retailers mainly (61.7 % of offline sales), but also through grocery retailers (36.4 %) and in a lesser extent through direct selling (1.9 %). Grocery retailers includes convenience retail, warehouse clubs, small local grocers, discounters, supermarkets and hypermarkets. Large retailers account for 93% of sales in grocery retailers. As for, non-grocery retailers, they are composed essentially of health and beauty specialists that account for 89 % of sales in this type of retailers. The remaining 11% is sold through general merchandise stores,

¹⁹ Cf. section 3.4.1 information sources

²⁰ Market shares are analysed in value.

²¹ It should be noted that the firms in the top three may have changed between 2014 and 2023.

²² According to the sources consulted, different terms are used to describe the cosmetic market. Euromonitor uses the term "beauty and personal care market" and Cosmetics Europe uses the term "Cosmetics and personal care market".

apparel and footwear specialists, and others non-grocery retailers.

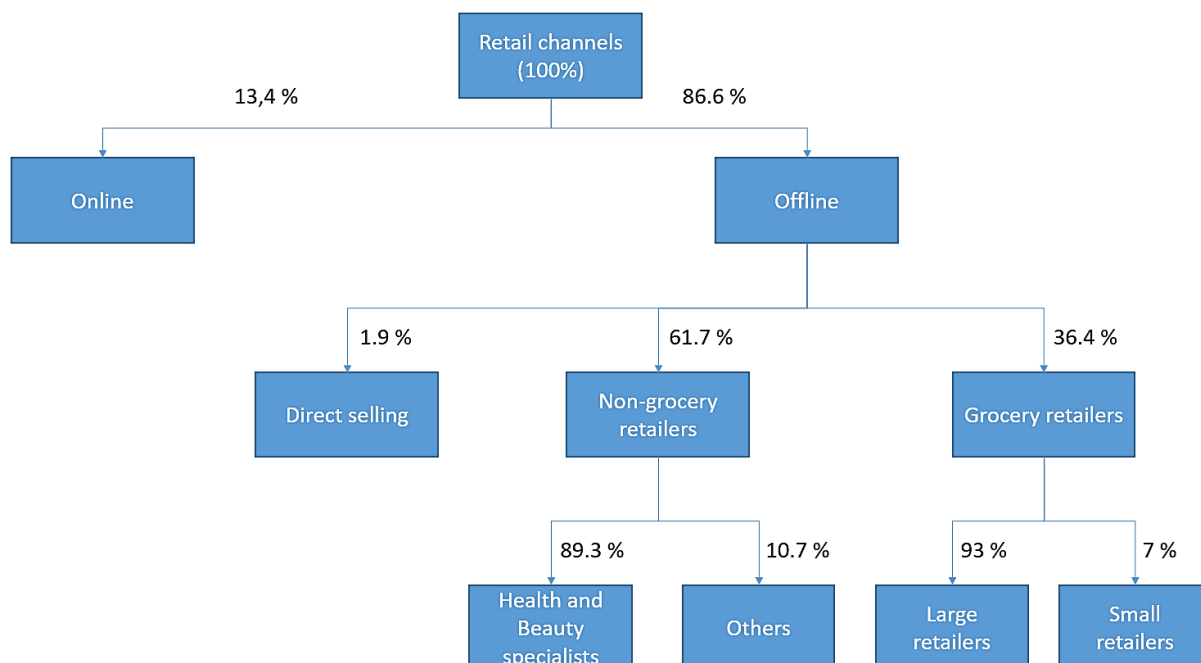


Figure 13: Share of retail channels in the distribution of sun care products in Western Europe.

(Source: Euromonitor)

E.4.1.4. Market for adult sun protection products and market for baby and child specific sun care products

The data provided herein come also from Euromonitor.

E.4.1.4.1. Markets' size

This sun care market can be breakdown into two main categories: baby and child specific sun care and the market for adult sun care. The latter is composed of sun protection, self-tanning and after-sun.

Baby and child-specific sun care products and adult sun protection are primary sunscreen products and are considered accounting for more than 80 % of the sun care market (the remaining 20% being self-tanning and after-sun)²⁵. This is why hereafter, only these markets are studied. **Figure 14** shows the evolution of the size of the markets for baby and child sun care products, adult sun protection and for sun care products in volume terms.

²⁵ In 2023, the baby and child specific sun care market represents more than 10 % of the sun care market in value and volume terms. This share remains steady since 2009. The adult sun protection market represents more than 72 % of the sun care market in value terms (RSP and MSP) and almost 70 % of it in volume (litres) in 2023.

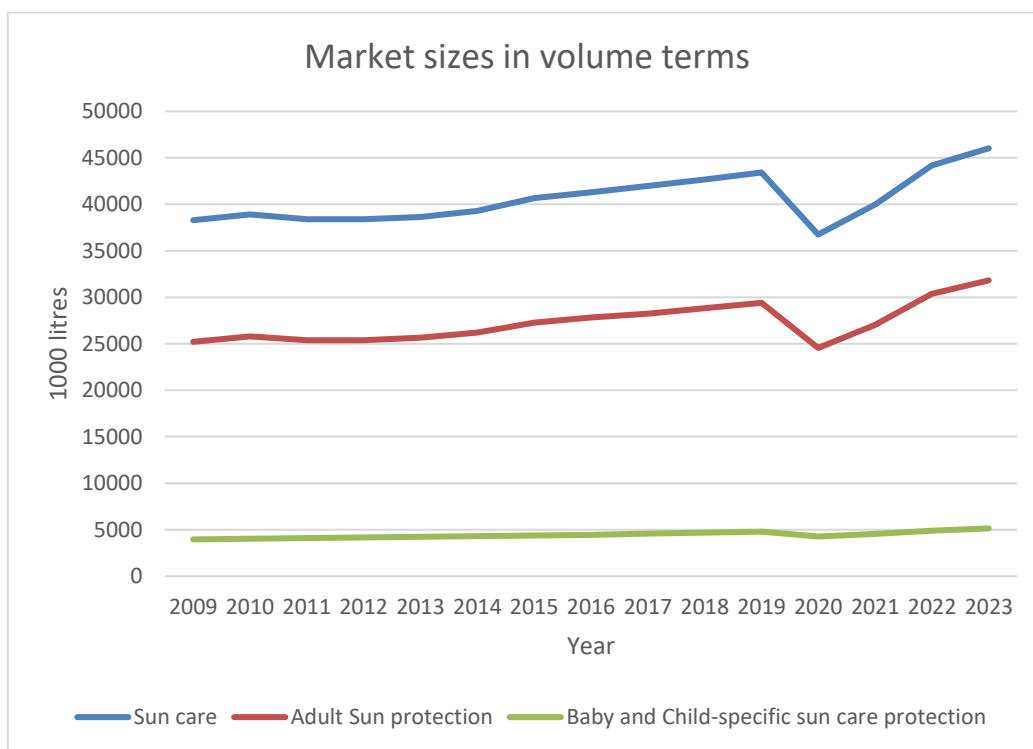


Figure 14: Evolution of markets' size over time (in litres) over 2009-2023.

(Source: Euromonitor)

Although the size of the markets has been increasing in volume terms, it has remained constant over the period in value terms. That is, the average price of cosmetics is decreasing over the period.

E.4.1.4.2. Heterogeneity of manufacturers

Almost all firms (91 %) that manufacture sun care products and own more than 0.1 % (among with 20 others not producing sun care products) of the market share manufacture adult sun protection products, and more than half (53 %) manufacture baby and child specific sun care products respectively. Nearly 1 manufacturer over 3 (29 %) produce both types of cosmetics. These figures remain constant between 2014 and 2023.

Given that the Herfindahl-Hirschmann Index²⁶ for those suncare markets ranges from 1000 and 2000, these markets can be considered as moderately concentrated (cf. **Table 53**)²⁷. Market for baby and child specific sun care products is more concentrated than adult sun protection market.

²⁶ The HHI is computed by summing the squares of the individual market shares of all the firms in the market. It ranges from close to zero (in an atomistic market) to 10 000 (in the case of a pure monopoly).

²⁷ The Official Journal of the European Union (2004/C 31/03) provides these thresholds. Although they are considered in the case of a merger, thus in the case of a change in the HHI, it appears that 1000 may be a threshold that can be interpreted in absolute terms: if the HHI is less than 1000, then the risk of impeding competition is low. If the HHI is more than 1000, then a monitoring may be necessary

Table 53: Herfindahl-Hirschmann Index of the relevant suncare markets under review.

Market	Adult sun protection	Baby and child specific sun care
HHI	1245	1858

In addition, these markets are asymmetric:

- In 2023, the baby and child sun care market is dominated by two firms (among 29), which together hold more than 50 % of the market in terms of value of sales (RSP). As for the adult sun protection market, more than half the market belongs to four manufacturers (out of 50). L'Oréal and Beiersdorf are the firms with the highest market shares (the top 2 henceforth) in markets for adult sun protection products and for baby and child-specific sun care products. Together, they own 42 % and 53 % of these two markets respectively.
- For each market, the top 3 is identical since 2014. Market for adult sun protection is dominated by L'Oréal, Beiersdorf and Pierre Fabre and the market for baby and child-specific sun care products is dominated by L'Oréal, Beiersdorf and ISDIN. The position of the companies on these top 3 has been consolidated over time. Regarding the baby and child specific sun care market, the market share of the top 3 increased from 53 % to 60 %. As for the adult sun protection market, the top 3 increased from 46 % to 49 % of market share. These firms operate, or even play a leading role, in related markets, and almost all of them provide several product ranges.

E.4.1.4.3. Distribution

The adult sun protection market and the baby and child specific sun care market provide a high variety of brands. In 2023, products from 75 brands of adult sun protection and 43 sun care for baby and children were purchased in Western Europe. Beiersdorf owns 4 brands of adult sun protection products (Eucerin, Eucerin sun, Nivea, Nivea sun) and L'Oréal Groupe owns 12 brands (Biotherm, CeraVe, Garnier, Garnier Ambre Solaire, La Roche-Posay, La Roche-Posay Anthelios, Lancôme Soleil, L'Oréal Paris, L'Oréal Paris Solar, Expertise, L'Oréal Sublime Sun and Vichy). As for baby and child-specific sun care products, Beiersdorf adds two brands to its portfolio (Coppertone and Nivea Baby) and L'Oréal adds one more (Garnier Ambre Solaire kids). The supply of product variety (product proliferation) enables firms to serve heterogeneous consumer preferences better. However, this can also increase barrier to entry and lead to an increase in price (that is not cost-related) (*Schmalensee., 1978; Connor., 1981; Lutz., 1997 ; Piazzai & Wijnberg., 2019*).

Finally, it has to reminded that octocrilene-free suncare products are already on the market. For instance, in France, the Dossier submitter could infer that there are 167 octocrilene-free products available among 475 suncare products. According to stakeholders, the market price of octocrilene-free products is higher than the one of products containing it. However, the database at the Dossier Submitter disposal did not allow to confirm or refute these statements. A more in-depth price study would be necessary to compare the prices of products without and with octocrilene.

E.4.2. Number of expected reformulations

The table below, based on the YUKA database, lists for each category of cosmetic products the number and proportion of octocrilene-containing and octocrilene-free product

references. The Dossier Summitter uses this initial information and the assumptions set out in section 3.4.2.1. of the main report to determine the reformulation rate - i.e., the proportion of formulations that would actually be reformulated within each category in the event of a restriction. For example, as the proportion of self-tanner containing octocrilene is less than 30%, the reformulation rate for this category is 5%.

This reformulation rate is applied to the number of formulations available within the EEA for each product category to determine the number of reformulations expected for this same product category. Thus, of the 33 self-tanning formulations containing octocrilene in the EEA, only 5% will be reformulated, i.e. around 2 formulations.

It should be noted that the number of formulations existing within the EEA for each product category is determined by extrapolation from the overall figure of 520,000 formulations, then distributed pro rata according to the weight of each category in the YUKA database. For instance, primary sunscreens represent 3.43% of products references listed in the YUKA database. On that basis, the Dossier Submitter considers that 3.43% of the 520,000 formulations available in the EEA are primary sunscreens.

Table 54: Number of expected reformulations per product category based on YUKA data

	Number of product references without octocrilene	Number of product references containing octocrilene	Proportion of product references containing octocrilene	Number of formulations at EEA scale containing octocrilene	Reformulation rate	Number of expected reformulations
Tanning	7 140	3 066	30.04%	6 776	50.00%	3 388
Sunscreen	5 108	2 966	36.74%	6 555	50.00%	3 278
Self-tanner	886	15	1.66%	33	5.00%	2
After sun	968	8	0.82%	18	5.00%	1
Lip balm with sunscreen	178	77	30.20%	170	50.00%	85
Facial care	63 193	1 401	2.17%	3 096	5.00%	155
Moisturisers/Face cream	14 588	594	3.91%	1 313	5.00%	66
Anti-age cream	6 027	398	6.19%	880	5.00%	44
Lip balm	3 773	130	3.33%	287	5.00%	14
Serums and treatments	14 074	230	1.61%	508	5.00%	25
Toners and mists	1 502	2	0.13%	4	5.00%	0
Eye moisturiser	4 045	20	0.49%	44	5.00%	2
Masks	13 350	19	0.14%	42	5.00%	2
Cleansers	5 834	8	0.14%	18	5.00%	1
Baby and Children's Products	780	4	0.51%	9	5.00%	0
Lotion	404	3	0.74%	7	5.00%	0
Other baby products	376	1	0.27%	2	5.00%	0
Hands and Nails	15 330	267	1.71%	590	5.00%	30

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Nail polish	6 572	183	2.71%	404	5.00%	20
Creams and lotions	8 171	77	0.93%	170	5.00%	9
Nail polish remover	587	7	1.18%	15	5.00%	1
Make up	27 820	279	0.99%	617	5.00%	31
Foundation/BB Cream	8 412	226	2.62%	500	5.00%	25
Tinted lip balm	652	11	1.66%	24	5.00%	1
Pressed powder	7 552	14	0.19%	31	5.00%	2
Lipstick	5 613	17	0.30%	38	5.00%	2
Concealer	2 575	5	0.19%	11	5.00%	1
Lip gloss	3 016	6	0.20%	13	5.00%	1
Bath and Body Products	41 304	117	0.28%	259	5.00%	13
Body lotion/Balm/Cream/Gel	19 488	96	0.49%	212	5.00%	11
Body oil	2 523	9	0.36%	20	5.00%	1
Other	6 098	5	0.08%	11	5.00%	1
Shower gel	13 195	7	0.05%	15	5.00%	1
Hair care	40 166	96	0.24%	212	5.00%	11
Serum/oil	4 073	37	0.90%	82	5.00%	4
Treatments	1 499	2	0.13%	4	5.00%	0
Masks	3 863	17	0.44%	38	5.00%	2
Styling cream	4 507	8	0.18%	18	5.00%	1
Hair spray	3 968	8	0.20%	18	5.00%	1
Hair color	2 009	2	0.10%	4	5.00%	0
Shampoo	16 146	14	0.09%	31	5.00%	2
Conditionner	4 101	8	0.19%	18	5.00%	1
Male grooming	1 646	4	0.24%	9	5.00%	0
Facial care	676	2	0.29%	4	5.00%	0
After shave	970	2	0.21%	4	5.00%	0
Fragrances	13 505	177	1.29%	391	5.00%	20
Perfume/Parfum/Eau de Parfum	13 505	177	1.29%	391	5.00%	20
Autres	18 957	20	0.11%	44	5.00%	2
Grand Total	229 841	5 431	2.31%	12 001	30.23%	3 629

E.4.3. Sensitivity analyses

E.4.3.1. Impact of alternative assumptions for reformulation cost per formulation and useful life of a formulation on reformulation costs

The Dossier Submitter assessed the cost of reformulation induced by the present restriction based on a main scenario. In this scenario, it is assumed that the useful life of a cosmetic formulation is 10 years and that the cost of one reformulation is equal to the mid-point values provided by the industrial stakeholders for sun protection products and other cosmetic products (see Table 55 and Table 56 below).

Table 55: Reformulation costs for major reformulation of sun protection products (cost per formulation in €2024).

	Minimum ^a	Mid-point	Maximum ^a
SMEs	€50,000-€70 000	€85,000	€100,000-€120,000
Larger companies	€90,000-€110 000	€175,000	€240,000-€260,000

a: As the cost data provided by stakeholders are confidential, the latter are expressed in the form of intervals.
Source: stakeholder consultations

Table 56: Reformulation costs for reformulation of 'other' cosmetic products (cost per formulation in €2024).

	Minimum	Mid-point	Maximum
SMEs	€10,000-€14 000	€17,000	€20,000-€24,000
Larger companies	€18,000-€22,000	€35,000	€48,000-€52,000

However, given the uncertainty over the value of these parameters, this section aims to identify the impact of alternative assumptions on the cost of a reformulation and the useful life of a formulation on the cost of reformulation induced by the proposed restriction. The results of this sensitivity analysis are shown in Table 57 below. It can be seen that the choice of the cost scenario has a significant impact on the overall reformulation cost: 77% and 44% increases can be observed when moving from a minimum scenario (lowest cost value) to a mid-point scenario and then from the mid-point scenario to the maximum scenario (highest cost value). However, the reformulation cost induced by the proposed restriction is more sensitive to the assumption concerning the useful life of a cosmetic formulation: moving from 10 to 20 years for the useful life of a formulation leads to a 93% increase in reformulation cost. An increase in the useful life of a cosmetic formulation leads to the implementation of a lower number of 'routine reformulations' each year all else equal and thus reduces the number of routine reformulations that can be coordinated with the reformulation imposed by the proposed restriction. The impact of the useful life of a cosmetic formulation on formulation cost increases with the length of the transition period, since the proportion of routine reformulations that can be coordinated with the reformulations implemented as part of the restriction falls drastically.

Table 57: Reformulation costs based on YUKA data for the three cost scenarios and two scenarios regarding the useful life of a formulation (2024 PV, 3% discount rate, 2 years transitional period).

	Minimum	Mid-point	Maximum
Partial reformulation (3 629 reformulations) - Reformulation cost in million euros			
10-year PV (main assessment)	62.72 - 84.46	111.22	137.98 - 159.73
20-year PV (sensitivity analysis)	121.27 - 163.31	215.05	266.79 - 308.83

E.4.3.2. Impact of alternative assumptions on the reformulation rate

The Dossier Submitter estimates the cost of reformulation based on the assumption that only a proportion of formulations containing octocrilene will actually be reformulated in the event of a restriction. The reformulation rates applied for each category of cosmetic products are shown in Table 54 above and the associated assumptions are explained in the main report (see section 3.4.2.1).

Nevertheless, the Dossier Submitter has conducted a sensitivity analysis to identify the impact of relaxing the partial reformulation hypothesis on the cost of reformulation.

Table 58 below compares the cost of reformulation in the two scenarios. It can be seen that the cost of reformulation is extremely sensitive to a variation in the rate of reformulation and therefore in the number of formulations reformulated: relaxing the partial reformulation hypothesis leads to a more than three-fold increase in the number of expected reformulations and almost a three-fold increase in reformulation cost. This is to be expected since the cost of reformulation is proportional to the number of formulations reformulated.

Table 58: Reformulation cost in million euros based on YUKA data for two scenarios regarding the reformulation rate (2024 PV, 3% discount rate, mid-point cost).

	Partial reformulation (3 629 reformulations)	Full reformulation (12 001 reformulations)
10-year PV	111.22 (Main scenario)	315.43
20-year PV	215.05	609.89

E.4.3.3. Impact of the length of the transitional period on reformulation costs

Table 59 below shows the results of the sensitivity analysis on the impact of the length of the transitional period on the cost of reformulation. It can be seen that an increase in the length of the transition period induces a linear decrease in the reformulation cost. This can be explained by the fact that an increase in the length of this period leads to an increase in the number of routine reformulations that would have been carried out during the transitional period, as well as those that could be coordinated with the restriction schedule. As a reminder, the Dossier Submitter considers the following assumption in its analysis: For products where the baseline major reformulation would have taken place during the 5 years after the end of the transitional period, that reformulation would be coordinated with

removal of octocrilene and done earlier, during the transition period. So there would be only the costs of bringing those reformulations forward.

Table 59: Impact of the length of the transitional period on reformulation cost.

Length transitional period (years)	Number of expected reformulations	Total number of reformulations	Number reformulations brought forward	Number of formulations reformulated twice during the study period	Reformulation cost ^a (€ million)
1	3 629	5 080.6	1 814.5 (50%)	1 451.6 (40%)	143.04
2 (proposed restriction/main assessment)	3 629	4 717.7	1 814.5 (50%)	1 088.7 (30%)	111.22
3	3 629	4 354.8	1 814.5 (50%)	725.8 (20%)	80.29
4	3 629	3 991.9	1 814.5 (50%)	362.9 (10%)	50.22
5 (main assessment)	3 629	3 629	1 814.5 (50%)	0	20.97

a : (PV 2024 at 3%; mid-point reformulation cost; lifetime of a formulation = 10 years)

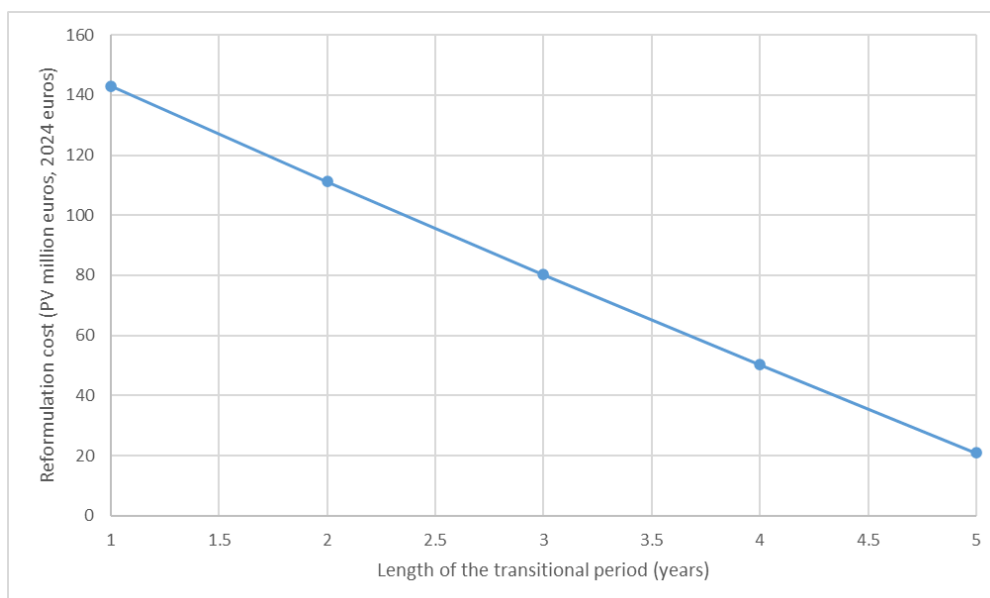


Figure 15: Impact of the length of the transitional period on reformulation cost.

E.4.3.4. Impact of voluntary reformulations on reformulation costs

As explained in section 3.2.1 of the Annex XV report, there are reasons to believe that there could be some voluntary move away from octocrilene in cosmetics even in the absence of a restriction. However, there is limited data that would allow the Dossier Submitter to estimate for what proportion of formulations this could happen voluntarily, or what the timeline would be for those voluntary substitutions.

To understand how assuming a baseline with voluntary substitution could affect the costs of the restriction and its cost-effectiveness, the Dossier Submitter has performed the following sensitivity analysis.

The total reformulation cost when assuming voluntary reformulations is compared with the total reformulation cost calculated for the main scenario for which a 2-year transitional period, partial reformulation, mid-point values for the reformulation cost and YUKA data are considered. These assumptions are maintained in the scenario considered for this section and **the only change is in the baseline** (which was assumed to be 'business as usual').

In this new baseline, the Dossier Submitter considered two levels of voluntary reformulation: 25% and 50%. Thus, it is considered that 25% (50%) of formulations containing octocrilene would be voluntarily withdrawn from the market (reformulated or discontinued).

- This voluntary withdrawal would take place throughout 5 years, starting from the date of Entry into Force of the restriction (assumed to be 2027);
- There is a 2-year transitional period;
- Mid-point values for the reformulation cost are used and
- The proportions of products that would be reformulated and withdrawn from the market are the same as assumed in section 3.4.2.1 of the Annex XV report.

Table 60 compares the number of reformulations per year between the 'business as usual' baseline scenario and the alternative baseline scenario for which it is assumed that voluntary reformulations are implemented. In particular, this table considers two time periods during the study period:

- 2027-2031: period during which voluntary reformulations AND 'routine reformulations' are implemented
- 2032-2036: period during which only 'routine reformulations' are implemented as voluntary reformulations are completed

Table 60: Number of reformulations per year under the three alternative baseline scenarios.

	Number of reformulations per year (2027-2031)			Number of reformulations per year (2032-2036)		
	Voluntary reformulations	Routine reformulations	Total	Voluntary reformulations	Routine reformulations	Total
Baseline 'business as usual'	0	362.9	362.9	0	362.9	362.9
Baseline '25% voluntary reformulations'	181.45	272.18	453.63	0	272.18	272.18
Baseline '50% voluntary reformulations'	362.9	181.45	544.35	0	181.45	181.45

Assuming the implementation of voluntary reformulations as part of the baseline, the proposed restriction is less costly: 24% less costly when assuming 25% voluntary reformulations and 48% when assuming 50% voluntary reformulations (see Table 61). In fact, in this case, the number of reformulations actually resulting from the proposed restriction is lower (i.e., routine reformulations). In addition, since voluntary reformulations are carried out over a short period of time (5 years at the beginning of the study period), they can be coordinated with the proposed restriction (see hypotheses in

section 3.4.2.1 of the Annex XV report), which has the only effect of accelerating the schedule of voluntary reformulations.

The impact of these voluntary reformulations on the raw material costs is also taken into account: the production volumes of formulations that are reformulated under the Baseline scenario are no longer the result of the proposed restriction.

Table 61: Total cost of the proposed restriction for the three alternative baseline scenarios in million euros (2024 PV, 3% discount rate).

	Baseline 'business as usual'	Baseline '25% voluntary reformulations'	Baseline '50% voluntary reformulations'
Reformulation costs	111.22	86.82	62.41
Raw material costs	242.47	181.86	121.24
Total costs	353.69	268.68	183.65

NB: Total cost of the proposed restriction has been calculated here based on YUKA data, considering partial reformulation, a 10-year useful life of a formulation, a 2-year transitional period as well as mid-point reformulation cost.

E.4.3.5. Costs of the restriction based on CosmEthics data

E.4.3.5.1. Reformulation costs

The Table 62 below provides the results of the quantification of reformulation costs when considering the CosmEthics database to determine the number of formulations containing octocrilene as well as the number of expected reformulations for each product category in the event of a restriction.

In comparison to the YUKA data, this database shows a higher proportion of products containing Octocrilene across the EEA (3.87% compared with 2.31% for YUKA data). Consequently, basing the quantification of reformulation costs on this database leads to a higher number of expected reformulations and higher reformulation costs all else equal.

As this database results in a more 'pessimistic' assessment of reformulation costs, CosmEthics data are used for the worst-case scenario.

Table 62: Reformulation costs based on CosmEthics data for three cost scenarios and two scenarios regarding the useful life of a formulation (2024 PV, 3% discount rate, 2 years transitional period).

	Minimum	Mid-point	Maximum
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Partial reformulation (6 270 reformulations) - Reformulation cost in million euros			
10-year PV	110.95 – 149.41	196.75	244.09 - 282.56
20-year PV	214.52 – 288.89	380.42	471.95 – 546.32
Full reformulation (20 148 reformulations) - Reformulation cost in million euros			
10-year PV	311.19 – 419.07	551.84	684.61 - 792.49
20-year PV	601.68 – 810.26	1,066.98	1,323.69 - 1,532.28

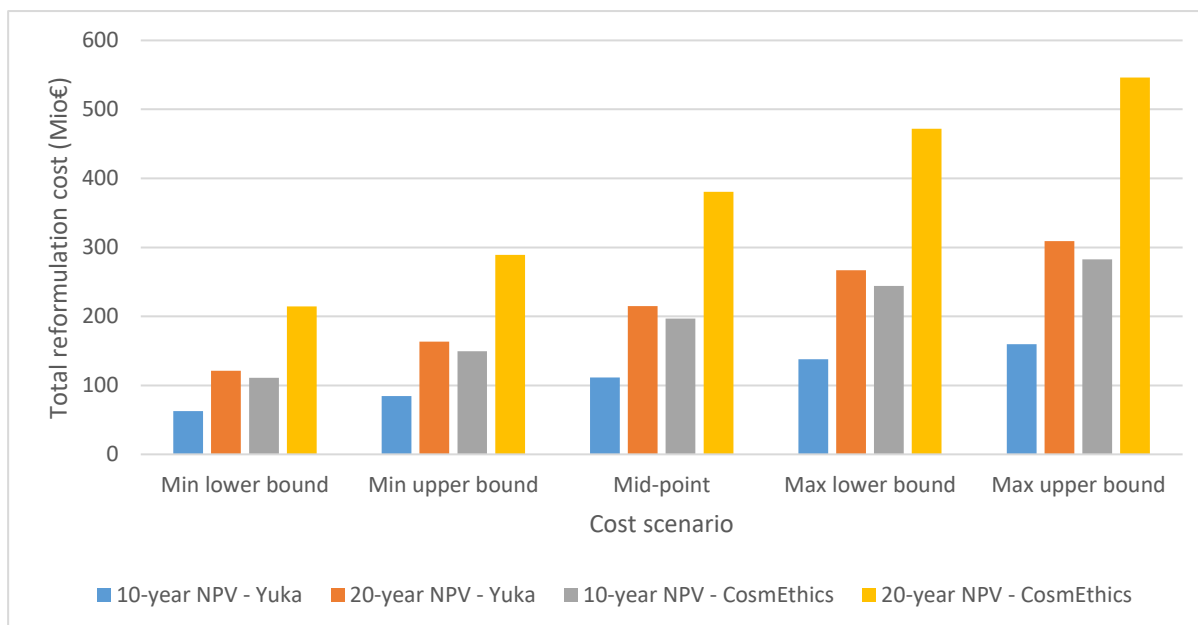


Figure 16: Comparison of reformulation costs calculated using YUKA vs CosmEthics data for three cost scenarios (2024 NPV, 3% discount rate, 2 years transitional periods and partial reformulation).

Figure 16 compares the reformulation costs calculated using YUKA data on the one hand (main assessment) and CosmEthics data on the other (sensitivity analysis). It can be seen that the reformulation cost calculated based on the CosmEthics database is 77% higher than the cost calculated based on the YUKA data.

E.4.3.5.2. Raw material costs

The Table 64 below provides the results of the quantification of raw material costs (yearly extra-cost) when considering the CosmEthics database to determine the share of sales volumes that would be affected by a reformulation and associated higher production costs due to the use of alternative raw materials. Table 63 shows the values considered regarding the formulation cost and the increase in formulation cost in the minimum, mid-point and maximum cost scenarios.

Table 63: Values considered regarding the formulation cost and the increase in formulation cost in the minimum, mid-point and maximum cost scenarios.

	Minimum	Mid-point	Maximum
Formulation cost	10€/kg	15€/kg	20€/kg

Increase in formulation cost	+10%	+30%	+50%
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Table 64: Raw material costs based on CosmEthics data for three cost scenarios and yearly extra cost (in million euros).

	Minimum	Mid-point	Maximum
Yearly extra cost – raw material costs	12.94	58.23	129.41

Raw material cost calculated based on the CosmEthics database is 50% higher than the cost calculated based on the YUKA data.

E.4.3.6. Total costs of the restriction under the worst-case scenario

In this section, the Dossier Submitter calculated the total costs induced by the proposed restriction based on all the most pessimistic assumptions (likely to lead to an overestimation of costs):

- Share of cosmetic products formulations containing octocrilene determined based on CosmEthics data (see section A.2.1.1)
- Reformulation rate: it is assumed that all formulations containing octocrilene are reformulated (20,148 reformulations expected).
- Cost of reformulation per formulation: the reformulation costs used in the Annex XV restriction proposal on the use of D4, D5 and D6 (ECHA, 2020) for larger companies are considered (€466,130 to reformulate a sun protection product and €93,226 to reformulate an 'other' cosmetic product).
- Useful life of a formulation: a useful life of 20 years is assumed.

The results of the cost calculation for this scenario are intended solely to discuss the proportionality of the proposed restriction. Thus, the cost induced by the restriction under the worst case scenario amounts to 5,108.55 million euros (5,109.23 including enforcement costs) for the entire study period (annualised cost of 314.24 million euros).

Table 65: Total costs of the restriction in million euros under the worst-case scenario (20 years PV 2024 at 3%)

	Reformulation costs^a	Raw material costs^b	Total
Sun protection products	1,860.59	3,162.86	5,023.4Mio€
Other cosmetic products	85.1	NA	85.1
Total	1,945.69	3,162.86	5,108.55 (5,109.23 including enforcement costs)

a: PV 2024 at 3%; worst-case cost of a reformulation; useful life of a formulation = 20 years; 2 years transitional period)

b: Quantified for primary sunscreens only.

E.4.4. Economic valuation of environmental impacts of the restriction options based on a stated preference

The following sections present the methodology and results of the stated preference study that underpins the evaluation of the benefits of the restriction conducted by the Dossier Submitter.

In this study, carried out by a team of consultants (Sotis et al., unpublished), people's willingness to pay (WTP) for the environmental benefits resulting from the restriction of the use of octocrilene in cosmetic products was estimated based on a discrete choice experiment (DCE) and a contingent valuation (CV) study. The changes in the environment were described in terms of the condition of invertebrate and fish species populations in aquatic ecosystems. The final survey was implemented online in September 2024 by a survey company (Ipsos).

E.4.4.1. Relevance of discrete choice experiment and contingent valuation for the benefit assessment of the proposed restriction

Discrete choice experiments (DCE) and contingent valuation (CV) are survey-based stated preference methods for eliciting preferences and estimating WTP for goods or services, particularly those not traded in conventional markets. In a DCE, respondents evaluate a series of choice tasks, with each task presenting multiple alternatives characterized by various attributes at different levels (Louviere and Hensher, 1982; Louviere and Woodworth, 1983). By including a cost attribute, the method enables estimation of willingness to pay. In CV, respondents are asked for their willingness to pay for a specific good or service, such as an improvement in environmental quality (Mitchell and Carson, 1989). The willingness to pay estimates from the DCE and CV are measures of the economic benefits from the environmental improvements resulting from the restriction. Stated preferences were chosen as the appropriate valuation method, as the changes in aquatic ecosystems are largely associated with non-use (passive use) values.²⁸

The DCE approach offers several distinct advantages when aiming at assessing the benefits of the proposed restriction. The DCE methodology aligns well with changes involving multiple dimensions. The restriction on the use of octocrilene impacting both freshwater and marine ecosystems and biodiversity, the implementation of a DCE allows capturing these dimensions and estimating both the value of individual attributes and the total benefits of the restriction. Besides, by asking respondents to make choices rather than directly state their willingness to pay, DCEs may reduce strategic behavior where respondents deliberately over- or understate their preferences.

However, while debate remains about empirical extent, if it is thought that DCE is prone to packaging effects this might caution emphasising only WTP values from such applications. To be specific the concern is that simply summing the parts (i.e., implicit attribute values from a DCE) may lead to overly high estimates of the total value of the project. By 'overly high' what is meant is that respondents may not be willing to pay those implied total amounts. One natural way of testing the extent of this is the addition of a contingent valuation (CV) question which asks people for their maximum WTP for the total change brought about by the policy. To the extent that packaging effects are thought to be

²⁸ Non-use values include values related to the existence of environmental goods without their use or consumption, while use values are related to the use of environmental goods, such as water recreation. Existence and good health of species, populations and ecosystems fall under non-use values. As the proposed restriction improves the condition of invertebrates and fish species, the values are mainly of non-use type.

present then reference to WTP metrics from this corresponding CV application might also frame thinking about the monetary value of policy benefits (CV producing more conservative estimates).

The CV measures the willingness to pay for the total improvement in water ecosystems rather than its component. Packaging effects may arise in the present study due to two main factors: the geographic scope of the proposed EU-level policy and the environmental attributes under consideration. The broad geographic scope means multiple areas would be affected simultaneously, potentially leading participants to consider overall rather than marginal benefits. Similarly, respondents might bundle freshwater and coastal water improvements in their valuation rather than considering them separately. In addition to its relevance to deal with packaging effects, CV is a well-established and well-studied methodology used for policy purposes in chemical impact valuation (e.g., OECD SWACHE projects).

E.4.4.2. Survey development

The development of the survey instrument required careful consideration to ensure both validity and reliability of the results. The process encompassed several key decisions and procedural steps to create a robust research tool. This section outlines these key decisions and the processes involved in creating the survey.

E.4.4.2.1. Scope

The initial methodological decision centered on determining the appropriate scope of the policy change to be valued (i.e., impacts on the environment). While octocrilene appears in numerous cosmetic products, most of its volume is primarily used in sun protection products (especially sunscreens). The substance poses risks to aquatic environments through two distinct exposure pathways: direct emissions through bathing and swimming at coastal sites primarily affect marine waters, while other cosmetics products lead to exposure through wastewater treatment plant discharges, predominantly affecting freshwater systems.

As the scope of the proposed restriction covers all cosmetic products containing octocrilene, the decision was taken to assess the benefits of this restriction for all aquatic ecosystems: freshwater and coastal ecosystems. This was reflected in the choice of the attributes for the DCE and the improvement scenarios presented in the CV.

E.4.4.2.2. Attributes and description of environmental improvements

The selection and definition of the environmental improvements was made based on a review of the scientific literature on octocrilene's environmental impacts and extensive consultations between the team of consultants and the Dossier Submitter and in particular its experts in eco-toxicology. Two environmental attributes were defined: improvements in coastal (marine) water quality and improvements in inland/freshwater quality. For each attribute, three levels of improvement were established (low, medium, and high). Qualitative descriptions were developed for each level, explaining the status of invertebrate and fish species populations in aquatic ecosystems. The improvement levels were defined in collaboration between the Dossier Submitter and the consultants on the basis of the results of regulatory tests studying the effect of octocrilene on indicator species. The decision to focus on these two broad categories of aquatic ecosystems stemmed from several considerations. First, scientific evidence indicates that octocrilene primarily affects aquatic ecosystems, with potential impacts on both marine and freshwater species. However, current evidence cannot establish that improvements in one ecosystem would

correspond directly to improvements in the other. This heterogeneity stems from two sources:

1. Marine and freshwater ecosystems experience different types and quantities of emissions. Marine ecosystems typically face direct emissions from bathing and swimming, while freshwater ecosystems primarily encounter indirect emissions through water treatment plants. Consequently, releases of octocrilene, resulting concentration — and thus the benefits of restriction—may differ between these ecosystem types.
2. These ecosystem types face different levels of overall pressure, particularly from chemicals, affecting the marginal impact of an octocrilene restriction.

Additionally, a participant's geographic location, typical holiday destinations, and engagement with aquatic ecosystems may influence their preference for either ecosystem type. The concepts of coastal and inland waters also prove easily comprehensible to the public, enhancing survey accessibility. While this approach is somewhat rudimentary, as it does not consider a specific dose response, it ensures that respondents understand the dimensions to be valued. Moreover, this design allows for separate valuations of coastal and freshwaters, allowing people to have distinct preferences for changes in each ecosystem.

In addition to these environmental attributes, a cost attribute was defined (see following section). The Table 66 below summarises the three attributes included in the DCE and the CV as well as the attributes levels.

Table 66: Attributes and attribute levels included in the survey

Attributes	Attribute levels
Improvement in aquatic species – freshwater ecosystems	• No: Many invertebrate species disappear from water bodies exposed to Octo. Fish species also have fewer individuals than if Octo were not present in these waterbodies. • Low: Return of some invertebrate species • Medium: Return of invertebrate species most sensitive to octocrilene, as well as recovery of other invertebrates • High: Significant improvement for all invertebrate species as well as recovery of notably some fish species which feed these invertebrates.
Improvement in aquatic species – coastal water ecosystems	• No: same as above • Low: same as above • Medium: same as above • High: same as above
Yearly increase in household expenditure on cosmetic products for the next 5 years	€10, €21, €52, €103, €155, €215 ²⁹

Finally, best practices in stated preference studies on chemicals in the natural environment were also followed by using visual aids that incorporate experts' inputs. Accordingly, figures and graphics – which were used in the survey – are widely used to communicate changes

²⁹ Bids for Germany provided as an illustration, the bid vector was adapted for other surveyed countries via Purchasing power parity (PPP) conversion.

in species or ecosystem health. Evidence from the literature has shown that visual aids can help convey scientific uncertainties and concepts more accurately and effectively, improving respondents' comprehension of the valuation scenarios (von Stackelberg and Hammitt, 2005; Hoffmann et al., 2017; Alberini and Scasny, 2018). In doing so, visual aids were developed in collaboration between the Dossier Submitter and the team of consultants in order to increase the credibility and accuracy of valuation scenarios (Bond et al., 2011; Van Houtven et al., 2014). This is particularly important to increase credibility and accuracy of the valuation scenarios. Furthermore, estimates obtained from these valuations would be scientifically relevant, which can then be linked to the proposed regulations and used for subsequent cost-benefit analyses (Van Houtven et al., 2014).

E.4.4.2.3. Provision mechanism and payment vehicle

The provision mechanism in stated preference studies refers to how the change in the environment will be implemented. In deciding the provision mechanism for the survey, established studies within the relevant literature were mirrored. Some studies have opted for either public good (e.g., government interventions) or private good (e.g., purchase of alternative products) provision mechanisms. Public good mechanisms - which can be posited as either specific or general government interventions - are predominantly used for non-health outcomes studies. For instance, King et al. (2021) considered sewage treatment plants upgrades to reduce microplastics in aquatic ecosystems. Public good provision mechanisms can also be deliberately abstract and general, as seen in Van Houtven et al. (2014) and Moore et al. (2018) where the authors framed the intervention programs as a bundle of policy measures.

Payment vehicle defines how the payments will be made. Regarding the payment vehicle, in the stated preference literature, common payment vehicles include service bills (Logar et al., 2014) and taxes (Moore et al., 2018; Choi & Ready, 2021). For example, Logar et al. (2014) used service bills as a payment vehicle for reducing micropollutants in water, while Moore et al. (2018) and Choi & Ready (2021) used taxes to address excess nutrients in water bodies. These payment vehicles are typically chosen for their realism, credibility, and the likelihood of being perceived as binding by respondents (Johnston et al., 2017). It is important that the payment vehicle is consistent with the provision mechanism.

In the context of the proposed restriction, reducing the risk associated with the use of octocrilene in cosmetic products cannot be achieved through public intervention such as a tax or the implementation of specific treatments in wastewater treatment plants. Risk reduction necessarily involves restricting the use of the substance in these products and using alternative substances. Consequently, the decision was made to consider a private good provision mechanism. This type of mechanism is often framed around individuals' willingness to pay (WTP) for safer, chemical-free or alternative products and has been commonly applied in studies examining the health impacts of chemical regulations (e.g., Alberini and Ščasný, 2014; Appéré et al., 2024). Specifically, respondents were asked about their willingness to pay for products that are octocrilene-free.

The payment vehicle was defined as an increase in household expenditure incurred by respondents due to higher prices of octocrilene-free products. While there are advantages and drawbacks to any payment vehicle, higher product prices were selected in part to align with emerging best practice in the context of valuing impacts of chemical use in regulatory proposals (e.g., OECD SWACHE) but also because they have the virtue of realism given that any restriction primarily impacts the (cosmetic) products that respondents buy that currently contain octocrilene. This expenditure-based payment vehicle was described to respondents as an annual payment for 5 years (as 5-years was specified to respondents as the maximum duration of these higher prices). The bid vector was agreed between the

Dossier Submitter and the team of consultants, following pre-testing with interviews and a pilot survey.

E.4.4.2.4. Survey structure

The survey was structured to provide respondents with the necessary context and information before presenting the choice tasks. The main sections of the survey included (see section E.4.4.9.1 for the full survey):

1. Introduction and consent: Explaining the purpose of the survey, ensuring respondents are 18 years or older, and obtaining informed consent from participants.
2. Screening questions: screening questions to ensure a representative sample. These questions include the year and month of birth, the respondent's age, their gender, their geographic location, their employment status (and in case they are employed their profession), the size of their household, their marital status and whether they have any children (if so how old).
3. Past experiences and familiarity:
 - a. Questions about the use of sunscreen and other cosmetic products containing UV filters
 - b. Frequency of holidays in areas with beaches, lakes, or rivers
 - c. Swimming habits and sunscreen reapplication practices
4. Background information on octocrilene:
 - a. Introduction to octocrilene (referred to as "Octo" for simplicity)
 - b. Description of products containing Octo
 - c. Quiz question to check understanding of Octo-containing products
5. Environmental impacts of Octo:
 - a. Explanation of how Octo enters surface waters
 - b. Description of environmental risks associated with Octo
 - c. Information on potential government action (ban on Octo)
 - d. Questions about familiarity with this information and helpfulness of visual aids
6. Payment mechanism explanation:
 - a. Introduction to the concept of "Octo-free" products
 - b. Explanation of potential price increases for these products
 - c. Question about willingness to pay for Octo-free products in principle
7. Choice experiment:
 - a. Explanation of the choice task format
 - b. Detailed description of attributes (coastal marine/estuarine species impact, freshwater species impact, extra monthly expenditure)
 - c. Example choice card with explanation
 - d. Series of choice tasks (6 per respondent)
 - e. Debriefing questions on task difficulty and certainty of responses
8. Contingent valuation question:
 - a. Willingness to pay question for a specific scenario
 - b. Follow-up certainty question

- c. Opportunity to revise stated willingness to pay
- 9. Environmental attitudes:
 - a. Self-rated knowledge and concern about environmental issues
 - b. Agreement with statements about survey response and information provided
- 10. Attitudes toward chemicals:
 - a. Perceived exposure to chemicals in various settings
 - b. Agreement with statements about chemical regulation and personal behaviours
- 11. Socio-economic and demographic questions:
 - a. Household income
 - b. Financial well-being questions
 - c. Education level
- 12. Final comments: Open-ended question for additional feedback

This structure was designed in line with best practices to gradually build respondents' understanding of the issue before presenting the valuation tasks. It also includes various debriefing and attitudinal questions to help interpret the valuation responses and assess the validity of the survey. The mix of closed-ended and open-ended questions allows for both quantitative analysis and qualitative insights into respondents' thought processes.

Besides, as mentioned previously, a two-stage approach was implemented: first eliciting WTP through a DCE, then following with a CV assessment for a specific environmental change. While CV typically imposes less cognitive burden (Colombo et al., 2022; Vossler et al., 2012) DCE provides superior flexibility in accounting for heterogeneous environmental improvements resulting from chemical regulations.

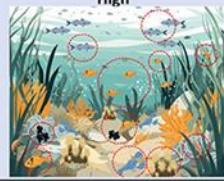
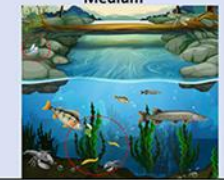
E.4.4.2.5. DCE experimental design

The allocation of attribute levels to the alternatives and alternatives to the choice tasks is called experimental design. The experimental design was a Bayesian efficient design, implemented through NGENE software to minimize standard errors of parameter estimates. Following Ferrini and Scarpa (2007), the design used priors from the final pilot study (see below), achieving a D-error of 0.03. Using 500 Halton draws, choice tasks were created that excluded dominated alternatives while placing no other constraints on the attribute levels.

In total 48 choice cards were generated, divided into 8 blocks of 6 cards each, with each respondent completing one block, i.e. 6 choice tasks. Each card presented three alternatives: a status quo scenario (no improvement, no cost) and two improvement alternatives affecting coastal water and freshwater quality, with associated yearly costs presented in local currencies. An illustration of a choice card is included in Figure 17 below.³⁰

³⁰ The illustrations in this report are taken from the English version of the questionnaire. This version was initially developed for methodological reasons. It was then adapted to the other case study countries (in particular PPP-conversion of the cost attribute) before the survey was implemented.

Please choose the option you would prefer overall

	Current	Option B	Option C
Positive impact on coastal water species	None 	Low 	High 
Positive impact on freshwater species	None 	High 	Medium 
Extra expenditure/year	0 £/year	60 £/year	180 £/year
Total expenditure in 5 years	0 £ for 5 years in total	300 £ for 5 years in total	900 £ for 5 years in total

☐ Current
☐ Option B
☐ Option C

[Previous](#)

Figure 17: Example of a choice card

E.4.4.3. Pre-testing of the survey

To ensure the quality and effectiveness of the survey instrument, several rounds of qualitative and quantitative testing and refinement were conducted. This included one-on-one interviews and two pilots. The following steps were

1. Expert consultations: experts in eco-toxicology, chemical regulation, and survey design were consulted to refine attribute definitions and ensure the scientific accuracy of the information provided to respondents.
2. Cognitive interviews: ten one-on-one interviews were conducted with members of the public to gauge their understanding of the issue, test the clarity of the explanations, and explore their decision-making processes when considering trade-offs between environmental improvements and costs. These interviews were used to test respondents' comprehension of the survey questions and identify any areas of confusion or misinterpretation.
3. Initial pilot: An initial pilot survey was conducted with a sample of 600 respondents spread across all six countries (see below). This allowed to test the full survey implementation, gather preliminary data to inform final design, and identify any remaining issues with question wording or survey flow. Specifically, following the initial pilot, the survey was updated with priors from the results obtained, while allowing for more uncertainty. In addition, apart from updating the design and choice cards, additional attribute level balance and block choice cards were introduced following this initial pilot.
4. Final pilot: The final pilot was conducted with a similar sample of 600 respondents

spread across all six countries. The key findings informed on the encouraging preliminary results using updated choice cards and allowed to perform minor modifications on survey flow and adjustment of inconsistencies for Bulgarian sample before the roll out of the survey.

E.4.4.4. Geographical scope and sampling

In this survey, respondents were asked to consider changes in the abovementioned attributes between options that were said to occur across the entirety of the EU and EEA (including the overseas territories of its Member States). This broad scope reflects the reality of the restriction if it were to be implemented. To aid participants in valuing the change proposed a map of the water bodies that would be affected in the survey was included. The approach followed in this study does not assume that respondents value changes across all EU/ EEA water bodies equally. Preferences and priority towards local water bodies (or otherwise) will be implicitly reflected in the choices that a respondent makes when presented with options.

The stated preference study was conducted on representative samples of the general national adult population of six EEA countries: Bulgaria, France, Germany, Hungary, Spain and Sweden (sample size = 7,200 with 1,200 respondents per country). The countries were selected to capture heterogeneity especially regarding preferences for the environment, economic conditions (average income/GDP per capita) and geographic location in the EEA. To assess the economic conditions and preferences for the environment, data from the Eurostat data on Volume indices of GDP and AIC per capita (2022) and the 2019 “Attitudes of Europeans towards the Environment” Eurobarometer survey were considered.

Representativeness of the general national adult population was aimed in terms of gender, age, occupation and geographical location. Occupation was used as a proxy for income representativeness.

Besides, samples were not stratified in terms of education, as previous surveys show that people are prone to misunderstand questions about their education and/or lie about their education level. A final dataset comprising 1,200 complete responses per country was collected (7,200 overall) after excluding low-quality responses and speeders. The sample size provides robust statistical power while remaining manageable within resource constraints. It allows for meaningful cross-country comparisons.

E.4.4.5. Model estimation

E.4.4.5.1. DCE estimation

DCEs are grounded in random utility theory, which posits that individuals select alternatives providing them with the highest utility. This utility comprises two components: a deterministic component based on observable attributes and a random component accounting for unobservable factors affecting choice. Analysis of respondents' trade-offs between attributes allows researchers to infer the relative importance of each attribute and derive monetary values for attribute-level changes.

In estimating WTP values from the DCE, the following steps were followed:

1. The model estimation was conducted applying a mixed logit model. This is an extension of the standard conditional logit model used in DCE by allowing one or

more of the parameters in the model to be randomly distributed. In doing so, the independence of irrelevant alternatives (IIA) assumption was relaxed and any heterogeneity in preferences across respondents could be accounted for (Scarpa et al., 2008). Model estimation was conducted in WTP space rather than in preference space. This means estimating directly the monetary valuations rather than the preference coefficients associated with the attributes and, only at a second stage, the monetary valuations³¹.

2. Whilst the mixed logit models provide coefficients for the mean and standard deviation of the distribution associated to each attribute, it is advisable to also unveil individual WTPs. These provide insights into how preferences vary across individuals or groups within a population. This helps identify diverse valuation patterns that may be hidden when only aggregate WTP is analyzed. Individual WTPs per attribute were generated at the post-estimation level. Specifically, the expected values of β conditional on a given response pattern y_n and a set of alternatives characterised by x_n is given by (Hess, 2007):

$$E[\beta|y_n, x_n] = \frac{\int \beta \prod_{t=1}^T \prod_{j=1}^J \left[\frac{\exp(x'_{njt}\beta)}{\sum_{j=1}^J \exp(x'_{njt}\beta)} \right]^{y_{njt}} f(\beta|\Theta) d\beta}{\int \prod_{t=1}^T \prod_{j=1}^J \left[\frac{\exp(x'_{njt}\beta)}{\sum_{j=1}^J \exp(x'_{njt}\beta)} \right]^{y_{njt}} f(\beta|\Theta) d\beta}$$

The above can be thought of as the conditional mean of the coefficient distribution for the sub-group of individuals who face the same alternatives and make the same choices.

3. Following the model estimations and derivation of individual WTP, a few corrections were applied to ensure consistency of results between CE and CV, avoid overestimation of WTP and enhancing robustness of the results. Specifically, individual WTPs were set to zero for respondents who stated that they were not willing to pay anything for the environmental improvements. In addition, individual WTPs were subsequently set to zero for respondents who provided a zero WTP in the CV question, even though they responded to the DCE and their inferred WTP from the DCE was greater than 0.

Mixed logit models offer several key advantages over simpler models such as the conditional/multinomial logit model. First and as mentioned above, mixed logit relaxes the Independence of Irrelevant Alternatives (IIA) assumption, which unrealistically assumes that the relative probabilities of choosing between any two alternatives remains constant regardless of what other alternatives are available. Second, mixed logit allows for random taste variation across respondents, capturing unobserved heterogeneity in preferences. Third, it allows for correlation in unobserved factors over time, which is important given that each respondent makes multiple choices in our study.

³¹ In preference space estimation, parameters correspond to the utility coefficient for each attribute. To determine the monetary value of a specific attribute, the ratio of the non-monetary attribute's coefficient to the monetary attribute's coefficient is computed. When using a conditional logit model, which assumes homogeneous preferences, either preference space or WTP space estimation can be applied without issues. However, with mixed logit models, where both non-monetary and monetary coefficients follow specified distributions, their ratio may lack defined statistical moments. Some practitioners have previously attempted to address this by fixing the monetary attribute's coefficient while allowing only the non-monetary coefficients to follow distributions. Whilst computationally simpler, this approach is flawed, as it fails to account for potential heterogeneity in preferences toward the monetary attribute. WTP space estimation addresses this challenge by directly estimating the monetary valuation as a coefficient. In this framework, models are parametrised in WTP distributions and the parameters are the (marginal) WTP for each attribute (see for instance Mariel et al., 2021, p 83-84).

The willingness to pay estimates were derived from a model where the dependent variable is the choice made in each choice situation. The models were estimated separately for each country, with standard errors clustered at the individual level to account for multiple responses per individual. Choice scenarios defined the choice sets, and individual respondent IDs were used for clustering the standard errors. All monetary values are expressed in Euro-PPP per year per household.

Models specifying the environmental improvements as linear and non-linear (according to the level of improvement in aquatic ecosystems) were estimated. To estimate the marginal effects for coastal waters (freshwaters) for the non-linear specification model, binary variables were included for different levels of coastal water (freshwater) and freshwater (coastal water) quality improvements.

The deterministic component of the utility function in the linear specification is set as:

$$V = \alpha SQ + \beta_1 Coastal + \beta_2 Freshwater + \beta_3 Cost + \varepsilon$$

Where α represents the coefficient associated with the Status quo (SQ) effect. The variables "Coastal" and "Freshwater" contain four levels each (no improvement, low, medium, and high).

The deterministic component of the utility function, with non-linear specification, is set as:

$$V = \alpha SQ + \beta_1 Coastal_{low} + \beta_2 Coastal_{medium} + \beta_3 Freshwater_{low} + \beta_4 Freshwater_{medium} + \beta_5 Cost + \varepsilon$$

Where α represents the coefficient associated with the status quo (SQ) effect and the non-monetary attributes enter the utility function identifying specific levels. It should be noted that in the case of the non-linear specification, the preferences (and hence WTP) associated with a particular attribute's level are to be interpreted with respect to the level omitted from the function. That is, coastal with low or medium improvement against high improvement. Same applies to freshwater improvements.

Detailed estimation results are available in section 212E.4.4.9.4.

Individual coefficients in mixed logit models are derived by estimating the expected value of the random coefficients conditional on an individual's observed choices and the set of specific alternatives they faced. This process integrates over the distribution of the random coefficients using simulation techniques. Specifically, the expected value is calculated as the conditional mean of the coefficient distribution for those who make the same choices, accounting for heterogeneity in preferences. This is achieved through iterative simulations, where the model estimates the likelihood of observed choices for various draws of the random coefficients, then averages these draws weighted by the likelihood of the observed choices. This approach ensures that the coefficients reflect individual-specific preferences while considering the variability across the sample.

The performance of the linear and non-linear specifications were compared based on log likelihood, AIC and BIC criteria. As Table 67 shows, the goodness of fit appears overall better for the linear model: AIC and BIC³² values are always lower when considering the

³² AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are metrics used to evaluate and compare statistical models by balancing model fit and complexity. AIC focuses on minimizing information loss by weighing the trade-off between goodness of fit and the number of parameters, with a lower AIC indicating a model that better explains the data without overfitting.

linear specification with the only exception for Hungary, indicating a better fit and hence not justifying the inclusion of additional parameters.³³ The willingness-to-pay values used to assess the proposed restriction are therefore derived from the linear model estimate.

The estimation of non-linear specifications, however, allows to better understand how people value different levels of environmental improvements in coastal water and freshwater ecosystems and to determine whether people place proportionally different values on small versus large improvements. Addressing this question is important for policy decisions, given the uncertainty over the impacts of releasing octocrilene in the environment and consequently over the benefits triggered by the proposed restriction. If additional scientific evidence becomes available values can be adjusted to reflect more precisely the level of environmental improvement.

Table 67: Comparison of goodness of fit of linear and non-linear models.

Country	Linear Specification			Non-Linear specification*		
	Log Likelihood	AIC	BIC	Log Likelihood	AIC	BIC
Bulgaria	-5720.64	11457.28	11520.39	-5756.042	11536.08	11630.75
Germany	-5501.46	11018.92	11081.98	-5585.644	11195.29	11289.87
Spain	-5467.67	10951.35	11014.46	-5552.126	11128.25	11222.92
France	-5575.09	11166.18	11229.23	-5628.36	11280.72	11375.3
Hungary	-5908.64	11833.28	11896.38	-5805.664	11635.33	11729.97
Sweden	-5812.25	11640.5	11703.77	-5903.15	11830.31	11925.21

*For CSW and FSW, estimation of 2 effects each: Low and Medium (reference high). Status quo includes no improvement.

In addition to estimating willingness to pay, robustness checks were carried out to study the determinants of WTP. The results, available in section E.4.4.9.4, show that WTP evolves in a consistent manner, particularly as a function of income level, level of education and level of preference for the environment. The results of the estimations of the non-linear models also constitute useful robustness tests, since they show that WTP evolves as a function of the level of environmental improvement.³⁴

E.4.4.5.2. Contingent valuation

To complement the DCE approach and address potential packaging effects, a contingent

BIC, on the other hand, applies a stronger penalty for model complexity, particularly in larger datasets, making it more conservative than AIC. Models with lower AIC and BIC values are preferable because they achieve a good fit while avoiding excessive complexity, ensuring more reliable and parsimonious representations of the data.

³³ It should be noted that average WTP derived from the non-linear model for Hungary do match the values derived from the linear model.

³⁴ However, the results obtained for Bulgaria are not in line with the expected effects. This could be explained by a problem with the quality of the sample, which was noted on several occasions during the analysis.

valuation (CV) question was also used to elicit respondents' willingness to pay for the proposed restriction on octocrilene.

The CV component consisted of a two-stage elicitation process. In the first stage, respondents were randomly assigned to one of three scenarios describing either low, medium, or high levels of environmental improvement (corresponding to the attribute levels used in the DCE for both coastal and freshwater environments). They were then asked to state their maximum willingness to pay for achieving these improvements using an open-ended elicitation format.

This random assignment of improvement scenarios enables to investigate how WTP varies with the magnitude of environmental benefits. This CV approach serves as a valuable cross-check against the DCE results, particularly in light of potential packaging effects, as it prompts respondents to consider the total value of the proposed restriction as a complete policy package rather than as a sum of individual attribute changes.

To analyze the CV responses, a series of regression models was employed to understand the determinants of willingness to pay. A base model examined the impact of the the scope of the environmental change (low, medium and high, with low as the reference category) on WTP. Additional controls including country fixed effects, demographic characteristics, and environmental attitudes and awareness were progressively integrated. Respondents' personal use of sunscreen and octocrilene-containing products was also controlled for. To account for potential heterogeneity across countries, both pooled models and country-specific regressions were estimated. This comprehensive analytical approach allows to identify not only the core relationship between improvement scenarios and WTP, but also how this relationship might be moderated by individual characteristics and cross-country differences. More specifically, OLS regressions were run with robust standard errors on the pooled samples as follows:

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \gamma X + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \delta Z + \gamma X + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \gamma X + \beta_3 \text{NoSunscreen} + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \gamma X + \beta_3 \text{NoOcto} + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \gamma X + \delta Z + \beta_3 \text{NoOcto} + \varepsilon$$

$$Y = \beta_1 \text{Medium} + \beta_2 \text{High} + \mu C + \theta_1 \text{Medium} \times C + \theta_2 \text{High} \times C + \gamma X + \delta Z + \beta_3 \text{NoOcto} + \varepsilon$$

Where Y (the dependent variable) is either the WTP in the first response, Medium is a binary variable capturing whether the participant was assigned to the medium level of improvement, High is a binary variable capturing whether the participant was assigned to the high level of improvement, C is a vector of binary variables, one per country, to evaluate the country specific WTP, X is a vector of demographic variables specific to the participant (age, gender, employment status, income, education, and presence of children), Z is a vector with environmental attitudes (including knowledge of chemical

risks, environmental concerns, and views on chemical regulation), *NoSunscreen* is a binary that takes value 1 if the participant does not use sunscreen personally, *NoOcto* is a binary that takes value 1 if the participant does not use any product containing octocrilene (we note that for participants who do not personally use products with octocrilene there is always at least a family member that uses products that contain it) and finally is the error term. Similarly, for each country we run the following regressions: ε

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \gamma X + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \delta Z + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \gamma X + \delta Z + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \gamma X + \beta_3 \text{NoSunscreen} + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \gamma X + \beta_3 \text{NoOcto} + \varepsilon$$

$$Y = \alpha + \beta_1 \text{Medium} + \beta_2 \text{High} + \gamma X + \delta Z + \beta_3 \text{NoOcto} + \varepsilon$$

All the tables referring to this analysis can be found in the section E.4.4.9.4. As the CV results in one total value estimate for the entire change in coastal and freshwater ecosystems, this means that it does not allow determining the WTP for each attribute. On the other hand, it imposes lesser cognitive strain on respondents and focuses on a value that is more holistic.

E.4.4.6. Descriptive statistics – Results and discussion

In the following sections, main findings are summarized with a particular focus on WTP results. Discussions make both reference to main results presented in several tables that are included in the main text but also to additional tables presented in section E.4.4.9.

E.4.4.6.1. Summary statistics

A total of 7,200 complete responses were collected, with 1,200 respondents evenly distributed across six countries: France, Bulgaria, Germany, Hungary, Spain, and Sweden. The average time taken to complete the survey was 15 minutes. The 7,200 responses do not include the 189 and 182 respondents respectively either refused to answer the survey or were automatically screened out. The final sample also excludes the 1,325 participants who did not complete the survey in full. It should be noted that the vast majority of these participants (760) left the survey in the sections collecting demographic data and previous experience.

E.4.4.6.2. Socio-demographics

Responses are distributed equally between female and male, with the former making up 51% of the responses collected. In terms of age, collected responses are well distributed and represented across all bands, with no observable over-representation in specific age bands with respect to the demographic composition of each country. That said, fewer respondents are from ages between 18 to 24, and 73 to 99, while most respondents fall

within the age band of 35 to 44. The overall sample is also observed to be relatively educated, with most of the sample having a mid to high level of education. Detailed demographic characteristics are provided in section E.4.4.9.

E.4.4.6.3. Past experience and familiarity

An overwhelming majority of the sample indicated having past experiences and familiarity with products containing UV filters. Specifically, over 85% of the sample indicated usage of sunscreen and cosmetic products containing UV filters when spending time outdoors, of which more than half opted for SPF of at least 30 or more as the preferred SPF level when it comes to sunscreen application. This level of familiarity is also extended to members of their households, with over 65% indicating family members' usage of sunscreens. Similarly, a high level of experience and familiarity is also echoed in cosmetic product usages. It was made sure that respondents were directly or indirectly responsible for some octocrilene emissions through their personal use and that of their household. It should be noted that participants that do not use any product that could contain octocrilene (sun protection products and/or cosmetic products) are always part of a household in which other people use products that could be responsible for releases of octocrilene. For this reason, no participant were excluded from the analysis on that basis.

In addition, over 85% of the sample has been on holiday at least once to destinations with beaches, lakes or rivers. Unsurprisingly, most of these respondents have indicated (1) a high likelihood of engaging in water activities and/or using outdoor showers during such holidays; and (2) spending at least 1 hour or more in the water on a typical beach/swimming day. When it comes to sunscreen reapplication during the time spent by the water, a quarter of respondents indicated they only apply sunscreen once. As octocrilene is also released in the environment through showering/bathing after using of a great diversity of other cosmetics apart from sunscreens the choice was made not to exclude the participants who do not regularly go to destinations with aquatic systems, at they or their household use products containing octocrilene.

E.4.4.6.4. Environmental attitudes and knowledge level

Overall, the sample has a high level of environmental attitudes and concern. 82% of the sample indicated at least a fair level of environmental knowledge and over 88% indicated at least average level of concern for the environment in general. Specific to octocrilene, after being presented with the initial information, approximately 95% of the sample correctly chose consumer products that are known to contain octocrilene, a reassuring result that the respondents, at least in principle, found the information presented in the survey highly comprehensible.

E.4.4.6.5. Survey perceptions

The responses collected to assess survey perceptions were categorized into four key metrics (see Table 72 in section E.4.4.9.3), all of which provided strong support for the reliability of the survey's estimates. Notably, 84% of respondents indicated that they answered the survey as they would in real-life situations. Additionally, 73% agreed that the information provided was sufficient, and 69% found it accurate. Furthermore, 63% of respondents expressed confidence in the information, reinforcing the credibility of the estimates derived from the survey.

E.4.4.7. Willingness to pay estimates

E.4.4.7.1. Results of the discrete choice experiment

The WTP values obtained through a mixed logit model are displayed in Table 68 below. From the DCE, the mean yearly WTP for coastal water quality improvement is €221.98 per household after PPP adjustments in per capita terms. In terms of freshwater water quality improvement, the mean value is €82.97 per household per year.

Table 68: Mean WTP by country from individual WTP (Values are expressed in Euro PPP per household and per year for one level of improvement)

	Mean willingness to pay – improvement for coastal water ecosystems	Mean willingness to pay – improvement for freshwater ecosystems	Total
Bulgaria	166.75 ^{M,SD***}	14.95	181.7
Germany	301 ^{M,SD***}	106.2 ^{M***}	407.2
Spain	241 ^{M,SD***}	74.75 ^{M***}	315.75
France	219.12 ^{M,SD***}	128.26 ^{M,SD***}	347.38
Hungary	38.41 ^{SD***}	35.88 ^{SD***}	74.29
Sweden	365.6 ^{M,SD***}	137.8 ^{M,SD***}	503.4
Simple average (NS included)	221.98	82.97	304.95

Note: Values are derived from estimation of mixed logit models in WTP space, with coastal water and freshwater improvements set as normally distributed, whilst the cost attribute was log-normally distributed. The dependent variable is the choice of the alternative presented in each choice task. The valuations represent the average effect across the environmental improvements (i.e. linear specification in attributes). The mean and medians were calculated on the individual (respondent level) distributions of WTP computed at the postestimation stage. Mixed logit models were run with 500 draws. The time frame of the payments was 5 years. When estimated mean *and* standard deviation coefficients were significant for alpha less than 1%: ^{M,SD***}. When estimated mean coefficients were significant for alpha less than 1%: ^{M***}. When estimated standard deviation coefficients were significant for alpha less than 1%: ^{SD***}.

The results show significant variation in willingness to pay (WTP) for coastal water improvements across the six countries. Sweden has the highest mean WTP (€365.6), reflecting strong preference for coastal water protection. Germany and Spain also display relatively high mean WTPs of €301 and €241, respectively. On the other hand, it should be noted that Hungary shows lower WTP. However, the mean coefficient is not significantly different from zero (see detailed estimation results available in section E.4.4.9.4). Besides, the estimation results highlight significant heterogeneity in willingness to pay within the Hungarian sample, suggesting that a subset of respondents expressed preferences for coastal water protection.

For freshwater improvements, Sweden has the highest mean WTP of €137.8. France follows closely, with comparable mean (€128.26), highlighting consistent willingness to pay for freshwater resources. Germany and Spain report moderate WTP figures, while Bulgaria and Hungary rank lowest. Hungary's mean WTP is €35.88, suggesting a lack of widespread willingness to pay for freshwater improvements. Bulgaria also shows low figures, with a mean of €14.95, underscoring limited concern for freshwater enhancements.

in these regions. However, it should be noted that the mean coefficient is not significantly different from zero for these two countries (see detailed estimation results available in section E.4.4.9.4).

Overall, WTP for coastal waters is higher than for freshwaters across the countries, with an average mean WTP of €221.98 for coastal water compared to €82.97 for freshwater.

E.4.4.7.2. Results of the contingent valuation

In addition to the DCE estimates, the contingent valuation (CV) estimates from this study also shed light on how people value a ban on products containing octocrilene. Table 69 displays the mean WTP in Euro PPP per household per year for each of the case-study country as well as for the overall sample. More detailed results are provided in section E.4.4.9.4. The mean PPP-adjusted WTP is €57.15, €71.23 and €74.99 for a low, medium and high level of improvement respectively, with some variation across countries. When compared to DCE, the CV-WTP elicited represents a more conservative value. Regression results from OLS regressions are presented in section E.4.4.9.4.5 with the WTP as the dependent variable and three sets of controls: 1) demographic characteristics, 2) environmental attitudes, and 3) stated use of octocrilene-containing products. In this analysis zero WTP are considered at face value. The results of this analysis show that WTP evolves in a consistent manner, particularly as a function of income level, level of education, level of preference for the environment as well as of the level of environmental improvement.

Table 69: Mean WTP by Country (Values are presented in Euro PPP per household and per year)

Country	Low improvement	Medium improvement	High improvement
Bulgaria	72.70	86.88	94.76
Germany	65.13	79.29	85.30
Spain	49.56	63.67	64.17
France	51.69	67.34	72.65
Hungary	41.17	58.04	61.50
Sweden	62.22	72.97	71.56
Overall	57.15	71.23	74.99

While the CV provides a direct stated value, the DCE estimate reflects respondents' preferences through a choice-based trade-off framework. The difference in these values may be attributed to the varying methodologies, with the CV potentially being more conservative due to its direct questioning format.

E.4.4.8. Policy implications

In summary, this study provides estimates of the mean level of WTP from both DCE and CV reflecting the environmental benefits associated with a restriction on the use of octocrilene in cosmetic products. This study followed, as closely as possible, best practices established by OECD in their ongoing work programme on valuing the impacts of chemicals on human health and the environment using stated preference methods.

Throughout the study, importance and care have been emphasised to ensure that findings are practical and can be used to support policy recommendations and evidence needs. Thus, in this final section, the main findings are discussed to provide comments on their use in subsequent cost-benefit analysis or CBA (OECD, 2018). In doing so, two observations are drawn.

First, while debate remains about empirical extent, if it is thought that DCE is prone to so-called packaging effects this might caution emphasising only WTP values from such applications. To be specific the concern is that simply summing the parts (i.e., implicit attribute values from a DCE) may lead to overly high estimates of the total value of the project. By 'overly high' what is meant is that respondents may not be willing to pay those implied total amounts. One natural way of testing the extent of this is the addition of a contingent valuation (CV) question which asks people for their maximum WTP for the total change brought about by the policy. To the extent that packaging effects are thought to be present then reference to WTP metrics from this corresponding CV application might also frame thinking about the monetary value of policy benefits.

Second, the results from the DCE itself indicate heterogeneity of preferences. In this study, this heterogeneity is of at least two types: variation across countries (given cross-country application in six EU/ EEA states) and variation within countries. The results also show how relative valuation of improvements in aquatic ecosystems in coastal waters and freshwaters differs across countries and across respondents.

To recap WTP findings, in terms of per annum values, from the DCE, the mean yearly WTP for coastal water quality improvement is €221.98 after PPP adjustments per household. In terms of freshwater water quality improvement, the mean value is €82.97. More detailed analysis including the WTP per improvement can be derived from the results presented in this report and should serve policymakers when more scientific evidence on the impact of octocrilene in the environment is established. The aforementioned values are simple averages across the mean WTP values for the six study countries.

It is also important to note once more that the results from the contingent valuation question, which followed the DCE application, indicates a mean WTP (overall) ranging from €57.15 (low improvement) to €74.99 (high improvement) per household, serving as a good robustness check for the DCE estimates.

E.4.4.9. Additional information

E.4.4.9.1. Full survey

The illustrations in this report are taken from the English version of the questionnaire. **This version was initially developed for methodological reasons. It was then adapted to the other case study countries** (in particular PPP-conversion of the cost attribute) **before the survey was implemented.**

What is your date of birth?

Year:

Month:

[Previous](#)

[Next](#)

Are you...?

☐ Female

☐ Male

Previous

Next

Where do you live?

Postcode: -

☐ Prefer not to answer

Please note:

This question may be considered personal. We would like to remind you that your participation is strictly voluntary and that your responses are used for research purposes only. The answers that you provide will be presented in aggregate form and none of them will be linked back to you in any way. All data will be collected and processed in adherence to the Market Research Society's Code of Conduct and the General Data Protection Regulation (GDPR).

If "prefer not to answer" selected

Where do you live?

☐ North East

☐ North West

☐ Yorkshire and Humberside

☐ West Midlands

☐ East Midlands

☐ East of England

☐ South West

☐ South East

☐ Greater London

☐ Wales

☐ Scotland

☐ Northern Ireland

Previous

Next

What is your current employment status?

Select only one

- ☐ Employed full-time
- ☐ Employed part-time
- ☐ Self employed
- ☐ Unemployed but looking for a job
- ☐ Unemployed and not looking for a job/Long-term sick or disabled
- ☐ Full-time parent, homemaker
- ☐ Retired
- ☐ Student/Pupil

Previous

Next

Which profession best corresponds to your own?

Single answer

- ☐ Senior executive
- ☐ Intellectual or scientific professional: engineer; scientific, technical, IT or communications expert; health professional; administrative, financial or sales manager; legal, social sciences or cultural professional; teacher
- ☐ Intermediate occupation: intermediate occupation in the field of science, technology, IT, communications, health, finance, sales or administration; legal services; social services; non-commissioned officer
- ☐ Small-business owner (not an employee): farmer, store owner, tradesperson
- ☐ Skilled employee: office worker, receptionist, caregiver, childcare worker, teachers' aide, law enforcement or civil service professional, member of the Armed Forces
- ☐ Skilled worker
- ☐ Service staff or store clerk, low-skilled worker: factory worker, maintenance worker, farm worker

Previous

Next

How many people are living or staying at your current address?

(Include yourself and any other adults or children who are currently living or staying at this address for at least two months)

Previous

Next

How many children under the age of 18 are living in your household?

Please reference only the children for which you are the parent or guardian.

Previous

Next

How many children do you have in each of the following age groups?*Enter the number of children for each range, enter 0 if you do not have any in the range*

3 years old or younger:	0
Between 4 and 14 years old :	0
Between 15 and 18 years old:	0
Older than 18 years old:	0

[Previous](#)[Next](#)**Welcome!**

This survey will ask your views on some proposals to lower the exposure to a chemical substance that can cause problems in the natural environment. Please read all the information and answer the questions carefully.

There are no right or wrong answers to the questions asked in this survey. It is your honest opinion that matters to us. The survey can be completed on a mobile device, but we recommend doing it on a larger device, such as a tablet, laptop or desktop.

We will ask some questions related to your familiarity with this chemical substance, your use of products which contain this substance and attitudes. Rest assured that a "Prefer not to answer" option will be available for you to select, at your discretion. Your answers throughout this survey will be kept confidential. Participation in this study is voluntary and you have the right to withdraw your consent at any time by writing to support or simply stopping the survey. This study respects the ICC/ESOMAR codes of conduct. Before agreeing, please also read this information sheet.

By agreeing to take part in this survey, you confirm that:

- (a) you are 18 years or older.
- (b) you are competent to provide this consent.
- (c) you have read the information about the survey.
- (d) you are taking part in this survey of your own free will.

Thank you in advance for your interest!

Do you agree to participate given the above conditions?

- ☐ Yes, I have read the information above and agree to take part in the survey
- ☐ No, I do not agree to take part in this survey

[Previous](#)[Next](#)

In this section, we will ask you a few questions related to your previous experience (if any) with the chemical substance which is the focus of this survey and the everyday products in which it is used.

[Previous](#)[Next](#)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

How often do you use sunscreen or other cosmetic products containing ultraviolet (UV) filters, that protect the skin from the harmful effects of sunrays, when spending time outdoors? These products typically specify their Sun Protection Factor (SPF).

Single answer

- ☐ Never
- ☐ Sometimes
- ☐ About half the time
- ☐ Most of the time
- ☐ Always

Previous

Next

Do you regularly use any of the following cosmetic products?

Select all that apply

- ☐ Body care products
- ☐ Facial care products
- ☐ Lip Balms
- ☐ Make-up
- ☐ Hair Products
- ☐ None of the above

Previous

Next

Do other members of your household use sunscreen or other cosmetic products containing UV filters, on a regular basis, when spending time outdoors?

Single answer

- ☐ Yes
- ☐ No
- ☐ Don't know

Previous

Next

How often do you go on holiday or spend time in areas with beaches, lakes or rivers in a typical year?

Single answer

- ☐ Never
- ☐ Once
- ☐ 2-3 times
- ☐ 4-6 times
- ☐ More than 6 times

Previous

Next

How many days does your typical holiday last?

Single answer

- ☐ 1-2 days
- ☐ Around a week
- ☐ Around two weeks
- ☒ Over two weeks
- ☐ I don't know/ it depends

Previous

Next

On a typical beach/swimming day, how many hours do you spend in the water? Please also count the time in which you are in the water but not swimming, e.g. playing with a ball, walking by the river bank/seashore.

Single answer

- ☐ Less than 1 hour
- ☐ 1-2 hours
- ☐ 2-4 hours
- ☐ More than 4 hours

Previous

Next

How often do you re-apply sunscreen when you spend time by the water?

Single answer

- ☐ More often than every hour (over 6 times in half a day by the water)
- ☐ Every hour (about 6 times in half a day by the water)
- ☐ Every two hours (about 3 times in half a day by the water)
- ☐ Every four hours (About twice in half a day by the water)
- ☐ Never (I only apply sunscreen once)

Previous

Next

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Octocrylene is a chemical widely used as an ingredient in sunscreen and other cosmetic products. It is a complicated word, so **in this survey we'll refer to it as "Octo"**. All the information related to Octo that you will see in this survey has been compiled and checked by researchers at ANSES (the French Agency for Food, Environmental and Occupational Health & Safety).

Octo is a ultraviolet (UV) filter that helps protect the skin from some of the UV rays from the sun and so protects against sunburn and reduces the chances of people developing longer-term conditions such as skin cancer.

The range of products in which Octo is found is described in the following image. This includes sunscreens as well as many other cosmetic products like lotions, creams, perfumes, sprays, and lip balms.

[Previous](#)[Next](#)

Which of the following consumer products are known to contain Octo?

Select all that apply.

- ☐ Makeup
- ☐ Sunscreen
- ☐ Facial Care
- ☒ Tanning
- ☐ Food
- ☐ Clothes

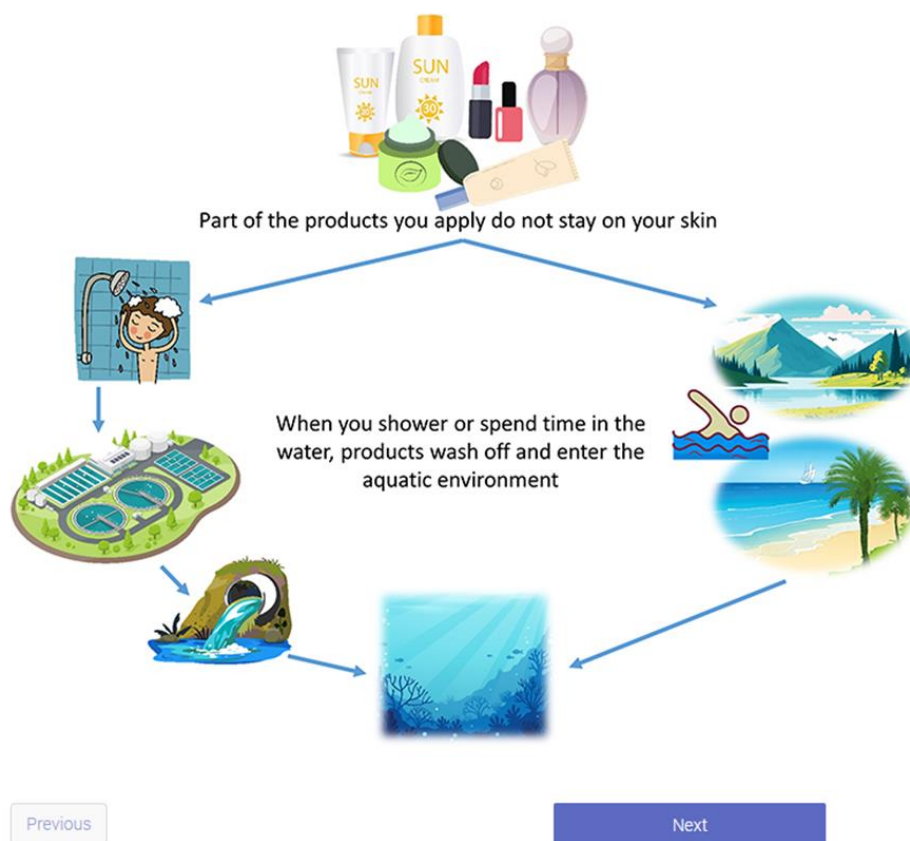
[Previous](#)[Next](#)

While Octo is considered safe for people, **there are concerns about its potential impact on the environment**, especially in coastal seas and inland freshwater bodies such as rivers and lakes.

Through swimming or other water-based recreation, Octo in cosmetic products can end up in these water-bodies. In addition, after bathing or showering, via wastewater treatment systems Octo can also end up in these water-bodies.

[Previous](#)[Next](#)

The image below illustrates how Octo can move from consumer products into water-bodies.



The evidence is that Octo does not remain in these water bodies for very long. **The problem is the quantity, and continuous release, of Octo, especially in recreational areas** during the peak of tourist seasons.

We will consider the **impacts of Octo on aquatic species**. Octo does not affect the recreational activities that people use water-bodies for and does not present a risk to humans for drinking water quality.

[Previous](#)

[Next](#)

Risks

There is scientific evidence that Octo is toxic to some aquatic species, and particularly it is **highly toxic to invertebrates** such as starfish, sea anemones, lobsters, molluscs, etc.

Octo concentrations in water can lead to death among individuals in local populations of these species. **Other aquatic species such as fish can be affected** through the food chain (as there are fewer invertebrates to eat) but the scientific evidence is more uncertain currently.

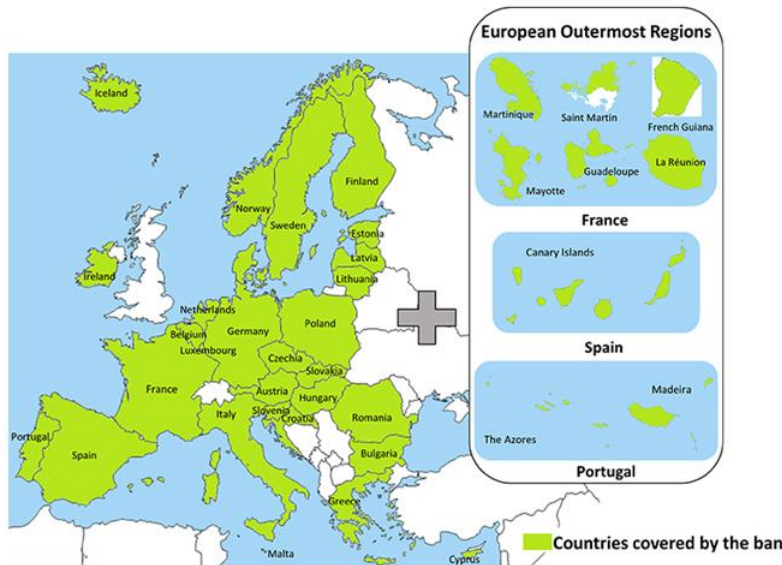
[Previous](#)

[Next](#)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Government Action

As a result of its proven and possible impacts, **there is some discussion in Europe about banning the use of Octo in sunscreens and other cosmetic products**. If this ban happens, it will apply to all countries which are members of the European Economic Area. This includes all member states of the European Union as well as Iceland, Liechtenstein and Norway and part of their overseas territories.



But if **no action is taken**, then Octo in sunscreens and cosmetic products will continue to be released into the environment over the coming years.

[Previous](#)[Next](#)

How familiar were you with this information on the impacts of Octo on the natural environment?

Single answer

- ☐ Not familiar at all
- ☐ Slightly familiar
- ☐ Moderately familiar
- ☐ Very familiar
- ☐ Extremely familiar

[Previous](#)[Next](#)

If a ban on using Octo in sunscreens and other cosmetics takes place, **producers will be required to certify that these products are “Octo-free”**. Of course: Octo provides people with important protections from sunrays and product formulations with Octo already exist. As a result, producers will replace Octo with an alternative substance, or substances. **Octo alternatives are safe for humans** (they undergo testing under the Cosmetics regulation) **and do not harm the environment**. **Octo alternatives provide the same protection against UV rays**, but producers will need to research the best formulations.

These certified ‘Octo Free’ products **would be identical to the products your household would normally purchase** (e.g., same quality, smell, effectiveness, etc.), except for the guaranteed absence of Octo in them. Octo-free products would always be easily available for purchase.

[Previous](#)[Next](#)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Imagine that buying these Octo-free products would lead to an increase in your yearly household spending, as these products are more expensive to produce than those that contain Octo (e.g., businesses would need to change production processes). Suppose that this leads to higher prices for these products for the next 5 years, after which the prices would come down again due to research and innovation.

Taking into account the information presented previously and what having an Octo-free water-bodies environment in Europe would mean to you and your household, would you be willing, in principle, to increase your yearly household spending to switch to cosmetic products certified as 'Octo Free'?

Single answer

- ☐ Yes, definitely
- ☐ Yes, but it depends on how much
- ☐ No
- ☐ I don't know

Previous

Next

In a moment, you will be shown a series of cards. Each card presents a choice between three different options and contains a series of attributes that are described in the next section.

Previous

Next

A ban, in Europe, on Octo in sunscreen and other cosmetic products will result in Octo-free water-bodies across the region. This would lead to an **improvement in the state of aquatic species** in coastal water- and freshwater-bodies.

There is some uncertainty, however, of the scale of this improvement. It could be a relatively "low" (i.e. small) improvement. It could be a "medium" (i.e. moderate) improvement. It could be a relatively "high" (i.e. large) improvement. By contrast, if there is no Octo-ban (i.e. the current situation), then there is "no improvement" at all.

This is described below in the next series of images, starting with "no improvement" and then low, medium and high levels of improvement.

Previous

Next

NO octo ban: NO improvement in the state of aquatic species (current situation)

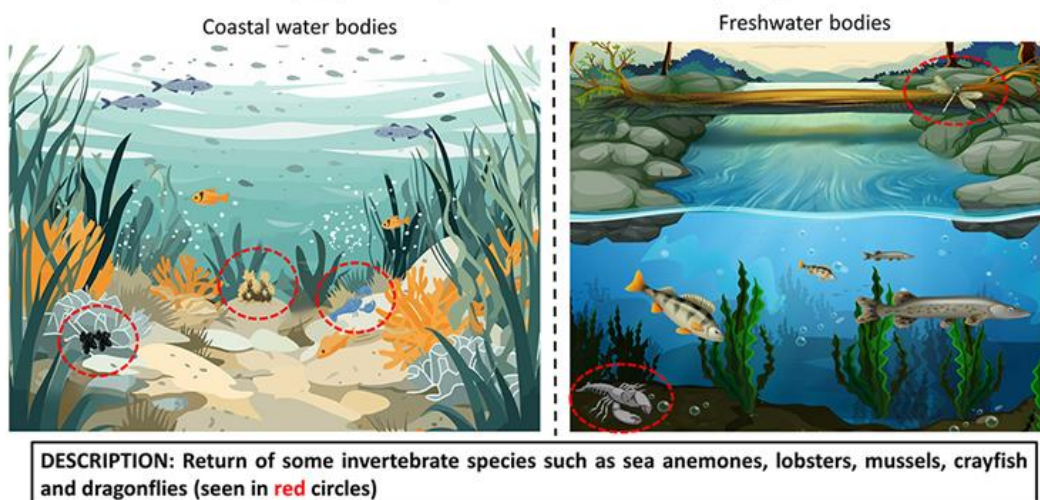


DESCRIPTION: Many invertebrate species (such as shrimps, lobster, sea anemones and mussels / snails, worms, crayfish and dragonflies) disappear from waterbodies exposed to Octo. Fish species also have fewer individuals than if octo were not present in these waterbodies.

Previous

Next

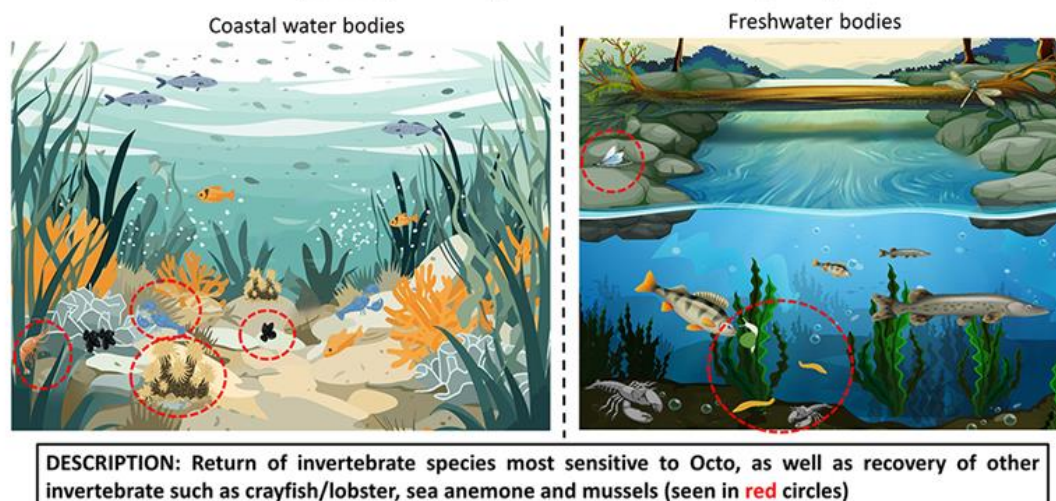
Octo ban : **LOW** level of improvement in the state of aquatic species



Previous

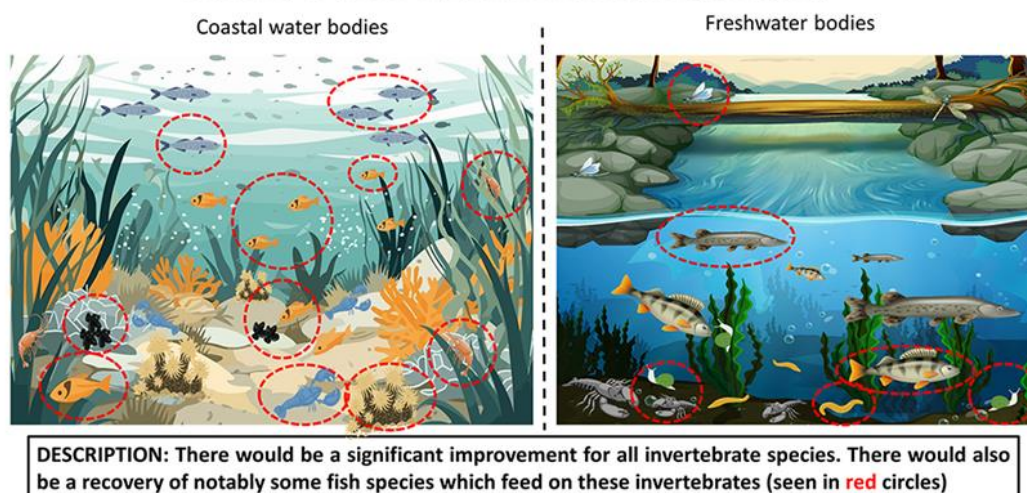
Next

Octo ban : **MEDIUM** level of improvement in the state of aquatic species



Previous

Next

Octo ban : HIGH level of improvement in the state of aquatic species

Previous

Next

	Current	Option B	Option C
Positive impact on coastal water species	None 	High 	Medium
Positive impact on freshwater species	None 	Medium 	High
Extra expenditure/year	0 £/year	180 £/year	60 £/year
Total expenditure in 5 years	0 £ for 5 years in total	900 £ for 5 years in total	300 £ for 5 years in total

Above is an example of the cards that you will be shown. Each card presents a choice between three different options. Assume the first option represents the **Current Situation**, and then there are two alternative options (B and C) that can be achieved by purchasing products free of Octo. Each option is described by a series of attributes and extra yearly household expenditure on products which are Octo-free.

This **extra yearly expenditure** is because of higher prices for Octo-free products. Assume that these higher prices would remain for the next 5 years, after which those prices would come down again due to research and innovation.

Suppose that in the Current Situation, there would be no positive impact on coastal and freshwater species. Please take a moment to imagine what this would mean to you.

In **Option B**, if you pay an extra £180 per year for products free of Octo, imagine that you would have a high positive impact on coastal species and a medium positive impact on freshwater species. Please take a moment to imagine what this would mean to you.

In **Option C**, if you pay an extra £60 per year for products free of Octo, you would have a medium positive impact on coastal species and a high positive impact on freshwater species. Please take a moment to imagine what this would mean to you.

Considering that the payments need to be made **each year for the next 5 years**, which of the three options would you prefer? Think about whether you would prefer keeping the current situation with no extra payments, or paying extra in your yearly spending for Option B or Option C.

Previous

Next

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Now let's get started with selecting your most preferred options (choice card will appear 6 times, each with different combination of options)!

Please don't agree to pay an amount if you cannot afford to pay it, if you think the proposed change is not worth it, or if you feel that there are more important things to spend your money on. Try to think about where any additional money might come from and if your budget allows the expense.

Previous

Next

Please choose the option you would prefer overall

	Current	Option B	Option C
Positive impact on coastal water species	None 	Low 	High 
Positive impact on freshwater species	None 	High 	Medium 
Extra expenditure/year	0 £/year	60 £/year	180 £/year
Total expenditure in 5 years	0 £ for 5 years in total	300 £ for 5 years in total	900 £ for 5 years in total

- ☐ Current
☐ Option B
☐ Option C

Previous

Please give us some feedback on how you found completing the previous questions. How difficult was it to answer the previous questions?

Single answer

- ☐ Very difficult
☐ Fairly difficult
☐ Not very difficult
☐ Not at all difficult

Previous

Next

How certain are you that you would actually be prepared to pay the extra amounts in the choice cards that you were presented with (on a scale from 1 to 10, where 1 is 'not at all certain', and 10 is 'extremely certain')



Previous

Next

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

You said it was difficult to respond to the questions in the previous section. Please explain:

[Previous](#)

[Next](#)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Please treat the following choice as a new situation when answering the following question.

Now suppose that, if Octo was banned in all cosmetic products, including sunscreens, there would be some positive impact on aquatic species living in: (a) coastal waters and (b) freshwater bodies across Europe. This improvement is LOW, and the ban would guarantee that water-bodies are Octo-free. These improvements are described in the images below.

The following question will ask you how much extra your household would be willing to pay, to secure these improvements.

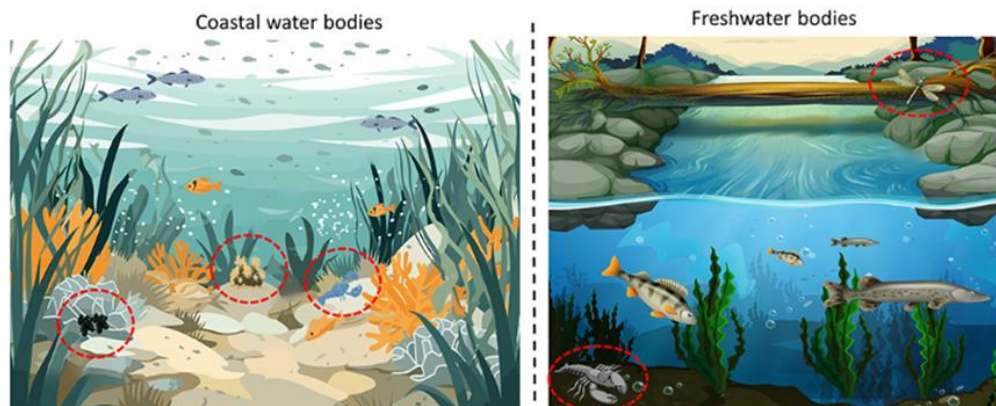
Please note: If you choose to pay 'Nothing' then these improvements will not occur and Octo will continue to be present in water-bodies across Europe (e.g. coasts, rivers and lakes). But please don't agree to pay an amount if you cannot afford to pay it, if you think the proposed change is not worth it, or if you feel that there are more important things to spend your money on.

NO octo ban: NO improvement in the state of aquatic species (current situation)



DESCRIPTION: Many invertebrate species (such as shrimps, lobster, sea anemones and mussels / snails, worms, crayfish and dragonflies) disappear from waterbodies exposed to Octo. Fish species also have fewer individuals than if octo were not present in these waterbodies.

Octo ban : LOW level of improvement in the state of aquatic species



DESCRIPTION: Return of some invertebrate species such as sea anemones, lobsters, mussels, crayfish and dragonflies (seen in **red** circles)

[Previous](#)

[Next](#)

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

What would be the most you would be willing to pay more per year for products which do not contain Octo? Please indicate the amount below.

Remember that the extra expenditure will be incurred every year for the next 5 years. Please keep in mind your household budget as you answer.

GBP

Previous

Next

How certain are you that you would actually be prepared pay the extra amounts you indicated?

On a scale from 1 to 10, where 1 is 'not at all certain', and 10 is 'extremely certain'

< 1 2 3 4 5 6 7 8 9 10 >

Previous

Next

A ban on Octo is only one possible proposal to tackle chemicals that are used in the everyday products that we buy. Governments and public bodies in Europe are also considering restrictions on other chemical substances in the months and the years ahead. **This could also have the additional effect of making other products that you buy more expensive.**

Given these possible extra expenses you might face, and the impact it might have on your budget, **would you like to revise the amount** you just said was the maximum you would pay for Octo-free sunscreen and cosmetic products?

Single answer

☐ Yes

☐ No

Previous

Next

What is the revised maximum amount that you would be willing to pay extra per year, for sunscreen and cosmetic products, which contain alternatives to Octo over and above your normal household yearly shopping bill?

GBP

Previous

Next

This is the final part of the survey! Please keep in mind that all your responses will remain confidential and will only be used for the purposes of the research.

Previous

Next

How would you rate your knowledge and understanding of environmental issues?

Single answer

- ☐ Very poor
- ☐ Poor
- ☐ Fair
- ☐ Good
- ☐ Excellent
- ☐ I don't know
- ☐ Prefer not to say

Previous

Next

How would you describe your concern for the environment in general?

Single answer

- ☐ Well below average
- ☐ Below average
- ☐ Average
- ☐ Above average
- ☒ Well above average
- ☐ I don't know
- ☐ Prefer not to say

Previous

Next

How much do you agree or disagree with the following statements?

Single answer per item

▼ The survey provided me with enough information to make informed choices

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

► I responded to the survey as I would have done in real life

Previous

Next

Do you agree with the description of the impacts of Octo provided in this survey?

Single answer

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☒ Strongly agree

Previous

Next

How confident are you that the information on the impacts of Octo that has been provided in this survey is correct?

Single answer

- ☐ Very doubtful
- ☐ Somewhat doubtful
- ☐ Neither doubtful nor confident
- ☐ Somewhat confident
- ☐ Very confident

Previous

Next

How would you describe your knowledge of Octo before having taken this survey?

Single answer

- ☐ Very poor
- ☐ Poor
- ☐ Fair
- ☐ Good
- ☐ Excellent

Previous

Next

How would you describe your knowledge of Octo after having taken this survey?

Single answer

- ☐ Very poor
- ☐ Poor
- ☐ Fair
- ☐ Good
- ☐ Excellent

Previous

Next

How much do you agree or disagree with the following statements

Single answer per item

▼ I am aware of the environmental risks associated with harmful chemicals

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

> I am aware of the impacts of Octo

> The use of harmful chemicals is sufficiently regulated in my country

> We have an obligation to future generations to do whatever we can to reduce exposure to harmful chemicals

> I try to reduce my use of harmful chemicals and chemical products in my daily life

> Business and industry should take stronger action to reduce the presence of harmful substances in products of daily use

> Business and industry should take stronger action to reduce emissions to the environment of harmful substances

> Governments should take stronger action to reduce the presence of harmful substances in products of daily use

> Governments should take stronger action to reduce emissions to the environment of harmful substances

Previous

Next

Can you make ends meet with the actual income of your household?

Single answer

- ☐ With great difficulty
- ☐ With difficulty
- ☐ Rather easily
- ☐ Easily
- ☐ Very easily
- ☐ Prefer not to say

Previous

Next

Do you think your household is financially better off or worse off now than it was a year ago?

Single answer

- ☐ Much worse off
- ☐ Somewhat worse off
- ☒ About the same
- ☐ Somewhat better off
- ☐ Much better off
- ☐ I don't know
- ☐ Prefer not to say

Previous

Next

And looking ahead, do you think your household will be financially better off or worse off a year from now?

Single answer

- ☐ Much worse off
- ☐ Somewhat worse off
- ☐ About the same
- ☐ Somewhat better off
- ☐ Much better off
- ☐ I don't know
- ☐ Prefer not to say

Previous

Next

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

Thank you for having completed the survey. If you have any additional comment, feel free to write below:

[Previous](#)

[Next](#)

E.4.4.9.2. Additional information on socio-demographics**Table 70: characteristics of the surveyed samples.**

Characteristics	France	Bulgaria	Germany	Hungary	Spain	Sweden
Gender						
Female	52.33	52.25	51.08	52.75	51.58	50.17
Male	47.67	47.75	48.92	47.25	48.42	49.83
Age Range						
18-24	10.25	8.33	8.75	9.42	8.75	11.00
25-34	14.17	15.00	15.17	16.75	13.33	16.17
35-44	16.17	18.50	16.58	18.33	17.42	17.58
45-54	17.75	19.58	16.83	19.92	19.58	17.25
55-64	16.42	16.17	19.33	14.50	16.75	15.67
65+	25.25	22.42	23.33	21.08	24.17	22.33
Education						
Low	4.75	3.17	38.88	10.30	29.89	38.48
Mid	42.25	31.58	58.68	23.75	41.21	33.98
High	53.00	65.25	2.44	65.95	28.89	27.55
Marital Status						
Married	41.08	40.33	46.58	39.75	51.58	34.17
Partner (not married)	22.50	23.33	10.25	26.50	13.50	23.58
Not in a relationship	25.50	19.50	29.42	20.00	24.67	27.17
Other	10.92	16.83	13.75	13.75	10.25	15.08
Area						
Urban	42.83	59.58	41.50	33.17	59.67	31.92
Suburban	21.25	24.00	37.42	34.17	26.92	35.50
Rural	35.92	16.42	21.08	32.67	13.42	32.58
Children						
No	67.67	67.58	75.50	68.58	69.33	69.00
Yes	32.33	32.42	24.50	31.42	30.67	31.00
Employment						
Unemployed	41.00	37.58	46.42	41.75	51.58	37.92
Employed	59.00	62.42	53.58	58.25	48.42	62.08
Income						
Low	50.22	11.84	38.88	10.30	29.89	38.48
Mid	30.08	33.48	58.68	23.75	41.21	33.98
High	19.70	54.68	2.44	65.95	28.89	27.55

Note: Percentages may not sum to 100 due to rounding.

Table 71: Characteristics of national populations.

Characteristics	France	Bulgaria	Germany	Hungary	Spain	Sweden
Gender						
Female	51.62	51.75	50.70	52.75	51.04	49.66
Male	48.38	48.25	49.30	48.03	48.96	49.65
Age Range						
20-24	7.61	5.36	6.64	6.71	6.40	7.22
25-34	14.98	9.10	15.44	15.61	13.66	18.01
35-44	16.30	11.65	15.62	17.48	17.54	16.49
45-54	16.85	12.61	16.08	19.72	20.12	16.40
55-64	16.62	11.24	19.00	14.82	17.29	15.43
65+	27.63	27.64	27.22	25.65	24.99	26.48
Education						
Low	16.7	15.6	16.8	13.00	36.60	11.70
Mid	41.70	54.50	51.00	57.60	23.00	39.70
High	41.60	29.90	32.30	29.40	40.70	29.90
Income						
Low	11.30	9.10	11.60	11.50	9.90	11.00
Mid	44.30	40.60	44.00	44.90	44.80	45.50
High	44.30	50.30	44.40	43.60	45.30	43.50

1: Age range 20-24 is used due to OECD's data availability.

2: Education levels are classified where Low (Lower secondary and below); High (Tertiary and above) and Mid is everything between these bands.

3: Income levels are classified where Low (lower than 25th percentile); Mid (25th to 75th percentile) and High (above 75th percentile)

E.4.4.9.3. Additional information on survey perceptions**Table 72: Survey perception by country (in %).**

Agreement Level	France	Bulgaria	Germany	Hungary	Spain	Sweden	Total
Behaved as in Real Life							
Strongly disagree	1.75	1.25	1.00	1.58	2.25	1.00	1.47
Somewhat disagree	5.08	2.92	3.58	5.25	2.92	3.67	3.90
Neither agree nor disagree	8.83	8.67	9.17	17.42	12.33	10.25	11.11
Somewhat agree	38.92	21.75	22.75	27.17	31.83	22.92	27.56
Strongly agree	45.42	65.42	63.50	48.58	50.67	62.17	55.96
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Enough Information Provided							
Strongly disagree	2.58	0.92	2.00	1.83	3.17	2.25	2.12
Somewhat disagree	7.17	4.92	7.67	8.08	5.75	6.58	6.69
Neither agree nor disagree	15.25	12.58	18.58	22.50	19.17	19.33	17.90
Somewhat agree	46.08	34.17	37.58	39.67	40.17	38.92	39.43
Strongly agree	28.92	47.42	34.17	27.92	31.75	32.92	33.85
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Survey Description Accuracy							
Strongly disagree	0.92	1.17	1.17	1.58	2.50	0.50	1.31
Somewhat disagree	3.92	4.42	3.00	3.00	3.67	2.25	3.38
Neither agree nor disagree	23.75	18.17	27.58	28.00	27.25	33.00	26.29
Somewhat agree	46.33	37.17	38.33	43.67	44.17	34.42	40.68
Strongly agree	25.08	39.08	29.92	23.75	22.42	29.83	28.35
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Confidence in Survey Info							
Very doubtful	0.92	1.92	2.17	2.00	1.67	3.00	1.94
Somewhat doubtful	3.75	8.50	8.17	6.75	5.67	7.25	6.68
Neither doubtful nor confident	30.58	26.33	33.50	38.67	38.67	39.50	34.54
Somewhat confident	48.75	42.67	41.92	37.75	42.33	39.67	42.18
Very confident	16.00	20.58	14.25	14.83	11.67	10.58	14.65
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00

E.4.4.9.4. Detailed estimation results**E.4.4.9.4.1. Mixed logit models-before derivation of Individual WTPs with correction and with linear specification****Table 73: Estimation results - Bulgaria.**

Variable	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
status_quo	-5610.12	515.50	-10.88	0.0	-6620.496	-4599.73
csw	192.85	54.44	3.54	0.0	86.14318	299.56
fsw	18.05	35.71	0.51	0.613	-51.94	88.04
mprice	-7.51	0.080	-93.57	0.0	-7.66	-7.35

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

SD(status_quo)	-7938.02	720.01	11.02	0.0	6526.823	9349.22
SD(csw)	1019.32	80.61	12.64	0.0	861.322	1177.31
SD(fsw)	-40.83	47.15	-0.87	0.387	-133.244	51.5837
SD(mprice)	1.04	0.1054	9.83	0.0	0.829	1.24313
Log Likelihood	-5720.64					
AIC	11457.28					
BIC	11520.39					

Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

Table 74: Estimation results – Germany.

Variable	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
status_quo	-1904.61	157.00	-12.13	0.0	-2212.33	-1596.89
csw	382.23	29.89	12.79	0.0	323.64	440.81
fsw	140.09	19.51	7.18	0.0	101.84	178.34
mprice	-6.61	0.057	-114.3	0.0	-6.721	-6.49
SD(status_quo)	-2383.36	164.10	-14.52	0.0	-2705.00	-2061.72
SD(csw)	704.03	36.14	19.48	0.0	633.2	774.8
SD(fsw)	104.28	66.65	1.56	0.118	-26.36	234.92
SD(mprice)	0.977	0.081	12.02	0.0	0.818	1.137
Log Likelihood	-5501.46					
AIC	11018.92					
BIC	11081.98					

Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

Table 75: Estimation results – Spain.

Variable	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
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APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

status_quo	-2171.08	165.63	-13.11	0.0	-2495.71	-1846.44
csw	297.61	32.25	9.23	0.0	234.38	360.84
fsw	105.55	20.77	5.08	0.0	64.81	146.27
mprice	-6.72	0.062	-107.48	0.0	-6.83	-6.59
SD(status_quo)	3508.65	236.57	14.83	0.0	3044.98	3972.32
SD(csw)	754.79	43.39	17.39	0.0	669.73	838.85
SD(fsw)	-49.64	33.85	-1.47	0.143	-116.003	16.70
SD(mprice)	1.059	0.090	11.71	0.0	0.882	1.23
Log Likelihood	-5467.67					
AIC	10951.35					
BIC	11014.46					

Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

Table 76: Estimation results – France.

Variable	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
status_quo	-2173.93	172.69	-12.59	0.0	-2512.41	-1835.44
csw	277.79	32.22	8.62	0.0	214.62	340.94
fsw	172.71	23.45	7.36	0.0	126.73	218.67
mprice	-6.76	0.055	-122.57	0.0	-6.86	-6.65
SD(status_quo)	-2824.72	201.84	-13.99	0.0	-3220.33	-2429.10
SD(csw)	641.23	43.75	14.66	0.0	555.47	726.98
SD(fsw)	171.80	91.51	1.88	0.06	-7.57	351.1
SD(mprice)	0.813	0.081	9.99	0.0	0.654	0.973
Log Likelihood	-5575.09					
AIC	11166.18					
BIC	11229.23					

Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

Table 77: Estimation results – Hungary.

Variable	Coefficient	Std. Err.	Z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
status_quo	-1772.53	112.57	-15.75	0.0	-1993.16	-1551.89
csw	59.17	37.84	1.56	0.118	-15.01	133.35
fsw	49.83	32.09	1.55	0.121	-13.08	112.74
mprice	-6.82	0.056	-121.56	0.0	-6.926	-6.70
SD(csw)	-858.15	51.33	-16.72	0.0	-958.77	-757.53
SD(fsw)	-674.91	46.85	-14.40	0.0	-766.74	-583.07
SD(mprice)	0.949	0.083	11.33	0.0	0.784	1.11
Log Likelihood	-5908.64					
AIC	11833.28					
BIC	11896.38					

Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

Table 78: Estimation results – Sweden.

Variable	Coefficient	Std. Err.	z	P> z	[95% Conf. Interval] Lower	[95% Conf. Interval] Upper
status_quo	-1836.55	155.47	-11.81	0.0	-2141.285	-1531.82
csw	400.7	33.43	11.99	0.0	335.1793	466.22
fsw	148.23	27.088	5.47	0.0	95.13524	201.31
mprice	-6.84	0.061	-111.39	0.0	-6.96107	-6.72
SD(status_quo)	-2713.84	198.76	-13.65	0.0	-3103.40	-2324.27
SD(csw)	857.90	43.17	19.87	0.0	773.27	942.53
SD(fsw)	327.77	38.20	8.58	0.0	252.89	402.64
SD(mprice)	0.938	0.085	10.93	0.0	0.770	1.10
Log Likelihood	-5812.25					
AIC	11640.5					

BIC	11703.77					
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Note: CSW stands for coastal water improvement; FSW stands for freshwater improvement; mprice refers to the monetary attribute; SD indicates the estimates relative to the Standard Deviation.

E.4.4.9.4.2. Non-linear model specifications

Non-linear WTP are to be interpreted with respect to the reference level omitted from the estimation, i.e.: high improvement. That is, the WTP associated with low improvement represents how much less (if negative sign) or how much more (if positive sign) individuals are willing to pay compared to a high improvement scenario. In line with expectations, across all countries and attributes considered, we find that low and medium improvements are associated with a lower valuation as opposed to high improvements. This is a reassuring finding in terms of internal consistency of the choice experiments results. To provide an example of interpretation, it can be seen that for the Bulgarian sample, a low level of improvement of coastal water is associated with a WTP lower of about 49.5 euros compared to a high improvement; or, in other words, individuals are WTP 49.5 euros more for the high improvement as opposed to a low improvement when it comes to coastal water. It should be noted that WTP values displayed in Table 79 to Table 84 below are expressed at the individual level (conversion from household to individual level based on household size data). This is a monotonic transformation (simple multiplication by a factor), so the same observations would be made on a WTP at household level : low and medium improvements associated with a lower valuation as opposed to high improvements.

Table 79: Mean and Median WTP for Bulgaria from non-linear individual WTP.

	Coastal water		Freshwater	
Bulgaria	Mean	Median	Mean	Median
	(C.I.)		(C.I.)	
Improvement: LOW	-49.5 ^{M,SD***} (-56.2 - 42.7)	-34.8	-4.75 ^{SD***} (1.5-7.9)	0
Improvement: MEDIUM	-60.9 (-69.9-51.8)	-30.5	28.2 ^{M*} (27.4-28.9)	31.8

Reference level: HIGH IMPROVEMENT

Table 80: Mean and Median WTP for Germany from non-linear individual WTP.

	Coastal water		Freshwater	
Germany	Mean	Median	Mean	Median
	(C.I.)		(C.I.)	
Improvement: LOW	- 100.9 ^{M,SD***} (-105.1-	-102.8	-46.8 ^{M,SD***} (-49.8-43.9)	-44.4

	96.7)			
Improvement: MEDIUM	-78.3 M,SD***	-65.5	-18.7 ^{M**} (-19.2-18.2)	-21.2
	(-84.5-72.1)			
Reference level: HIGH IMPROVEMENT				

Table 81: Mean and Median WTP for Hungary from individual WTP.

	Coastal water		Freshwater	
Hungary	Mean	Median	Mean	Median
	(C.I.)		(C.I.)	
Improvement: LOW	-26.2 ^{M,SD***} (-27.4)	-28.5	-24.4 ^{M***} (-25.8-23.1)	-23.8
Improvement: MEDIUM	-6.4 (-6.6-6.1)	-7.1	-8.1 ^{M*} (-8.3-7.8)	-9.7

Reference level: HIGH IMPROVEMENT

Table 82: Mean and Median WTP for Spain from non-linear individual WTP.

	Coastal water		Freshwater	
Spain	Mean	Median	Mean	Median
	(C.I.)		(C.I.)	
Improvement: LOW	-69.1 ^{M,SD***} (-73.3-64.9)	-66.1	-23.7 ^{M,SD***} (-26.2-21.1)	-18.5
Improvement: MEDIUM	-64.5 M,SD*** (-68.6-60.4)	-60	-6.6 (-7.2-6.1)	-5.7

Reference level: HIGH IMPROVEMENT

Table 83: Mean and Median WTP for France from non-linear individual WTP.

	Coastal water		Freshwater	
France	Mean	Median	Mean	Median
	(C.I.)		(C.I.)	
Improvement: LOW	-65.8 ^{M,SD***} (-68.9-62.8)	-69.2	-52.9 ^{M,SD***}	-53.4

			(-55.9-50)	
Improvement: MEDIUM	-70.1 M,SD***	-60.2	-27.2 ^{M***}	-31.5
	(-75.4-64.9)		(-28.9-25.4)	
Reference level: HIGH IMPROVEMENT				

Table 84: Mean and Median WTP for Sweden from non-linear individual WTP.

	Coastal water		Freshwater	
Sweden	Mean (C.I.)	Median	Mean (C.I.)	Median
Improvement: LOW	- 128.1 ^{M,SD***} (-133.6- 122.6)	-130.9	-57.1 (-59.5-54.7)	-57.2
Improvement: MEDIUM	-117.9 M,SD*** (-124.6)	-111.5	-9.3 ^{M,SD***} (-9.8-8.8)	-9.2
Reference level: HIGH IMPROVEMENT				

E.4.4.9.4.3. Robustness checks – Discrete choice experiment results

Two tables are presented: one for the regression where the dependent variable is the individual WTP of coastal water improvement (CSW), one where the dependent variable is the individual WTP for freshwater improvement (FSW). The results for **CSW** indicate that certainty, education, and income band significantly and positively influence willingness to pay (WTP), with certainty having the largest effect. Perceived difficulty reduces WTP significantly, while age shows a marginally significant positive effect. Other variables, such as gender and concerns about accuracy or insufficient information, do not have statistically significant impacts on WTP. For **FSW**, certainty, education, and income band also show significant positive effects on WTP, with certainty again being associated with the strongest effect. Factors such as lack of real-life relevance and absence of sunscreen are associated with significant negative impacts on WTP. Gender, age, and doubts about accuracy or sufficiency of information appear to have little to no significant effect on WTP in this context.

Table 85: Determinants of WTP for coastal water improvement.

Variable	Coefficient	Std. Err.	t	P> t	[95% Conf. Interval]
male	13.008	12.686	1.03	0.305	[-11.86, 37.87]
age	0.75116	0.387	1.94	0.053	[-0.00, 1.51]
nosunscreen	-7.2596	18.89	-0.38	0.701	[-44.30, 29.78]

APPENDIX TO THE ANNEX XV RESTRICTION REPORT – OCTOCRILENE

difficult	-46.557	13.71	-3.39	0.001	[-73.4, -19.6]
certainty	252.30	14.10	17.89	0.0	[224.6, 279.95]
noreallife	-52.971	30.83	-1.72	0.086	[-113.42, 7.4797]
notenoughinfo	14.289	24.88	0.57	0.566	[-34.48, 63.064]
notaccurate	-59.899	35.55	-1.68	0.092	[-129.6, 9.80]
doubtful	-1.3291	25.18	-0.05	0.958	[-50.7, 48.04]
edu	31.557	10.34	3.05	0.002	[11.27, 51.83]
incomeband	20.128	8.68	2.32	0.021	[3.10, 37.15]
co1 (Bulgaria)	-184.23	22.09	-8.34	0.0	[-227.55, -140.92]
co2 (Germany)	-22.124	21.74	-1.02	0.309	[-64.75, 20.50]
co3 (Spain)	-97.712	21.55	-4.53	0.0	[-139.96, -55.46]
co4 (France)	-104.20	21.49	-4.85	0.0	[-146.34, -62.05]
co5 (Hungary)	-304.93	22.44	-13.58	0.0	[-348.94, -260.93]
_cons	50.560	37.14	1.36	0.174	[-22.25, 123.37]

Table 86: Determinants of WTP for freshwater improvement.

Variable	Coefficient	Std. Err.	t	P> t	[95% Conf. Interval]
male	2.497	4.64	0.54	0.59	[-6.59, 11.59]
age	0.0054	0.141	0.04	0.969	[-0.27, 0.283]
nosunscreen	-15.96	6.911	-2.31	0.021	[-29.51, -2.416]
difficult	-5.974	5.017	-1.19	0.234	[-15.81, 3.86]
certainty	47.18	5.15	9.15	0.0	[37.07, 57.29]
noreallife	-38.5	11.27	-3.42	0.001	[-60.65, -16.43]
notenoughinfo	-9.479	9.100	-1.04	0.298	[-27.31, 8.359]
notaccurate	-9.70	13.00	-0.75	0.456	[-35.19, 15.79]
doubtful	-8.12	9.21	-0.88	0.378	[-26.18, 9.93]
edu	7.765	3.78	2.05	0.04	[0.34, 15.18]
incomeband	6.61	3.17	2.08	0.037	[0.383, 12.83]
co1 (Bulgaria)	-125.46	8.08	-15.53	0.0	[-141.30, -109.62]
co2 (Germany)	-14.54	7.95	-1.83	0.068	[-30.13, 1.050]
co3 (Spain)	-50.25	7.88	-6.38	0.0	[-65.710, -34.80]
co4 (France)	17.87	7.86	2.27	0.023	[2.46, 33.28]
co5 (Hungary)	-91.69	8.210	-11.17	0.0	[-107.78, -75.59]
_cons	81.34	13.58	5.99	0.0	[54.710, 107.97]

E.4.4.9.4.4. Detailed results of the contingent valuation**Table 87: WTP estimates for a low improvement in coastal and freshwater ecosystems (Values are presented in Euro PPP per household and per year).**

Country	France	Bulgaria	Germany	Hungary	Spain	Sweden	Total
Number of observations	476 (16.9%)	470 (16.7%)	476 (16.9%)	473 (16.8%)	471 (16.7%)	455 (16.1%)	2,821 (100%)
WTP ^a	51.690 (75.965)	72.697 (90.331)	65.129 (85.719)	41.168 (66.812)	49.561 (69.547)	62.221 (82.185)	57.149 (79.3)

^a: mean and standard deviation in brackets.**Table 88: WTP estimates for a medium improvement in coastal and freshwater ecosystems**

(Values are presented in Euro PPP per household and per year).

Country	France	Bulgaria	Germany	Hungary	Spain	Sweden	Total
Number of observations	362 (16.6%)	364 (16.6%)	361 (16.5%)	363 (16.6%)	365 (16.7%)	372 (17.0%)	2,187 (100%)
WTP ^a	67.336 (66.525)	86.876 (87.478)	79.289 (85.388)	58.041 (63.138)	63.669 (66.353)	72.966 (85.744)	71.231 (76.6)

^a: mean and standard deviation in brackets.

Table 89: WTP estimates for a high improvement in coastal and freshwater ecosystems
(Values are presented in Euro PPP per household and per year).

Country	France	Bulgaria	Germany	Hungary	Spain	Sweden	Total
Number of observations	362 (16.5%)	366 (16.7%)	363 (16.6%)	364 (16.6%)	364 (16.6%)	373 (17.0%)	2,192 (100.0%)
WTP ^a	72.648 (74.126)	94.763 (97.882)	85.296 (87.270)	61.503 (72.783)	64.168 (72.045)	71.564 (88.200)	74.992 (83.389)

^a: mean and standard deviation in brackets.

E.4.4.9.4.5. Robustness checks – contingent valuation results**Table 90: Determinants of WTP – Pooled model.**

	Model 1	Model 2	Model 3	Model 4
Medium improvement	14.08***	12.73***	9.253***	9.136***
	(0.000)	(0.000)	(0.000)	(0.000)
High improvement	17.82***	17.12***	13.33***	13.18***
	(0.000)	(0.000)	(0.000)	(0.000)
France	53.11***	41.26***	-27.68***	-26.09**
	(0.000)	(0.000)	(0.008)	(0.013)
Bulgaria	73.78***	64.21***	-7.052	-5.503
	(0.000)	(0.000)	(0.513)	(0.613)
Germany	65.87***	52.28***	-15.45	-14.20
	(0.000)	(0.000)	(0.132)	(0.168)
Hungary	43.02***	38.53***	-29.61***	-27.87***
	(0.000)	(0.000)	(0.003)	(0.006)
Spain	48.60***	37.49***	-30.17***	-28.80***
	(0.000)	(0.000)	(0.004)	(0.006)
Sweden	58.55***	45.04***	-19.88*	-18.40*
	(0.000)	(0.000)	(0.053)	(0.075)
Age		-0.401***	-0.470***	-0.464***
		(0.000)	(0.000)	(0.000)
Female		-0.304	-2.031	-2.635
		(0.878)	(0.312)	(0.200)
Has Children		0.403	0.221	0.170
		(0.767)	(0.874)	(0.903)
Employed		-2.547	-0.520	-0.734
		(0.234)	(0.809)	(0.733)
Income		0.00674***	0.00613***	0.00610***
		(0.000)	(0.000)	(0.000)
Education		6.497***	2.859*	2.746
		(0.000)	(0.090)	(0.104)
Aware of chemicals risk			-1.990*	-1.943*
			(0.072)	(0.079)
Aware of Octo impacts			7.109***	7.007***
			(0.000)	(0.000)
Sufficient chemicals regulation			-3.719***	-3.712***
			(0.000)	(0.000)
Obligation to future			5.759***	5.820***
			(0.000)	(0.000)
Reduces use of chemicals			1.100	1.102
			(0.299)	(0.298)
Companies should reduce chemicals			-1.285	-1.278
			(0.468)	(0.470)
Companies should reduce emissions			0.0653	0.0362
			(0.972)	(0.984)
Govts should reduce chemicals			1.864	1.843
			(0.294)	(0.300)
Govts should reduce emissions			-2.692	-2.686
			(0.121)	(0.121)
Perceived environmental knowledge			4.318***	4.262***
			(0.001)	(0.001)
Environmental concerns			9.385***	9.325***
			(0.000)	(0.000)
No personal use of Octo				-7.845*
				(0.054)
Observations	7200	6710	6373	6373
R²	0.425	0.449	0.487	0.487

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 91: Determinants of WTP – Bulgaria.

	Model 1	Model 2
Medium improvement	12.27*	9.302
	(0.055)	(0.149)
High improvement	19.36***	16.93**
	(0.003)	(0.011)
Age	-0.654***	-0.614***
	(0.001)	(0.003)
Female	-8.580	-9.379*
	(0.116)	(0.087)
Has Children	-1.844	-4.167
	(0.645)	(0.331)
Employed	6.854	8.964
	(0.274)	(0.156)
Income	0.00834***	0.00663***
	(0.000)	(0.003)
Education	6.132	0.0251
	(0.275)	(0.996)
Aware of chemicals risk		-7.546**
		(0.030)
Aware of Octo impacts		8.044***
		(0.008)
Sufficient chemicals regulation		-0.805
		(0.671)
Obligation to future		3.659
		(0.258)
Reduces use of chemicals		2.689
		(0.421)
Companies should reduce chemicals		-2.543
		(0.595)
Companies should reduce emissions		12.61**
		(0.016)
Govts should reduce chemicals		-1.855
		(0.686)
Govts should reduce emissions		-3.090
		(0.509)
Perceived environmental knowledge		1.598
		(0.660)
Environmental concerns		11.33***
		(0.001)
Constant	72.61***	-19.63
	(0.000)	(0.530)
Observations	1132	1102
R²	0.054	0.096

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 92: Determinants of WTP – France.

	Model 1	Model 2
Medium improvement	13.40***	11.89**
	(0.008)	(0.026)
High improvement	18.66***	15.39***
	(0.001)	(0.006)
Age	-0.429***	-0.352**
	(0.007)	(0.030)
Female	7.540*	8.174*
	(0.085)	(0.082)
Has Children	-0.0246	1.450
	(0.994)	(0.646)
Employed	-4.745	-6.308
	(0.357)	(0.237)
Income	0.00600***	0.00550***
	(0.000)	(0.001)
Education	10.56***	9.333**
	(0.005)	(0.016)
Aware of chemicals risk		-6.311**
		(0.015)
Aware of Octo impacts		6.379**
		(0.019)
Sufficient chemicals regulation		-5.437***
		(0.001)
Obligation to future		5.351*
		(0.069)
Reduces use of chemicals		1.886
		(0.414)
Companies should reduce chemicals		-6.711*
		(0.080)
Companies should reduce emissions		3.527
		(0.429)
Govts should reduce chemicals		-2.197
		(0.614)
Govts should reduce emissions		0.231
		(0.958)
Perceived environmental knowledge		6.726**
		(0.028)
Environmental concerns		4.105
		(0.167)
Constant	31.27**	8.660
	(0.037)	(0.724)
Observations	1137	1045
R²	0.055	0.097

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 93: Determinants of WTP – Germany.

	Model 1	Model 2
Medium improvement	11.07*	7.729
	(0.069)	(0.198)
High improvement	19.02***	13.15**
	(0.002)	(0.029)
Age	-0.194	-0.209
	(0.259)	(0.212)
Female	6.099	3.120
	(0.237)	(0.541)
Has Children	11.39**	12.47***
	(0.018)	(0.009)
Employed	-2.912	-0.460
	(0.586)	(0.930)
Income	0.00746***	0.00627***
	(0.000)	(0.000)
Education	6.036	4.192
	(0.143)	(0.310)
Aware of chemicals risk		-5.968*
		(0.053)
Aware of Octo impacts		13.96***
		(0.000)
Sufficient chemicals regulation		-4.906***
		(0.006)
Obligation to future		8.838***
		(0.002)
Reduces use of chemicals		0.803
		(0.740)
Companies should reduce chemicals		6.237
		(0.162)
Companies should reduce emissions		-8.774*
		(0.050)
Govts should reduce chemicals		-0.550
		(0.901)
Govts should reduce emissions		-3.703
		(0.390)
Perceived environmental knowledge		2.678
		(0.487)
Environmental concerns		12.13***
		(0.000)
Constant	33.73***	-38.20
	(0.010)	(0.114)
Observations	1132	1089
R²	0.079	0.166

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 94: Determinants of WTP – Hungary.

	Model 1	Model 2
Medium improvement	14.43***	11.77***
	(0.001)	(0.010)
High improvement	18.38***	14.14***
	(0.000)	(0.004)
Age	-0.613***	-0.636***
	(0.000)	(0.000)
Female	8.286**	5.580
	(0.034)	(0.150)
Has Children	-3.094	-2.310
	(0.198)	(0.362)
Employed	-8.043*	-5.502
	(0.071)	(0.207)
Income	0.0145***	0.0129***
	(0.000)	(0.000)
Education	8.937**	1.397
	(0.024)	(0.717)
Aware of chemicals risk		0.690
		(0.738)
Aware of Octo impacts		7.804***
		(0.000)
Sufficient chemicals regulation		-3.213**
		(0.039)
Obligation to future		6.771***
		(0.002)
Reduces use of chemicals		0.401
		(0.852)
Companies should reduce chemicals		4.699
		(0.109)
Companies should reduce emissions		-4.496
		(0.150)
Govts should reduce chemicals		-1.461
		(0.616)
Govts should reduce emissions		1.409
		(0.610)
Perceived environmental knowledge		0.304
		(0.882)
Environmental concerns		4.056
		(0.175)
Constant	29.62**	-33.26*
	(0.017)	(0.074)
Observations	1116	1052
R²	0.090	0.154

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 95: Determinants of WTP – Spain.

	Model 1	Model 2
Medium improvement	14.66***	10.29**
	(0.003)	(0.038)
High improvement	15.69***	12.68**
	(0.003)	(0.020)
Age	-0.133	-0.215
	(0.356)	(0.160)
Female	-3.288	-2.006
	(0.460)	(0.657)
Has Children	2.665	1.785
	(0.419)	(0.598)
Employed	-0.388	1.679
	(0.933)	(0.721)
Income	0.00469***	0.00475***
	(0.001)	(0.000)
Education	3.648	-0.344
	(0.214)	(0.909)
Aware of chemicals risk		1.975
		(0.374)
Aware of Octo impacts		2.085
		(0.381)
Sufficient chemicals regulation		-1.805
		(0.285)
Obligation to future		3.831
		(0.148)
Reduces use of chemicals		3.112
		(0.200)
Companies should reduce chemicals		-2.088
		(0.659)
Companies should reduce emissions		-3.208
		(0.478)
Govts should reduce chemicals		3.143
		(0.510)
Govts should reduce emissions		-0.920
		(0.837)
Perceived environmental knowledge		10.13***
		(0.000)
Environmental concerns		5.943**
		(0.030)
Constant	36.25***	-36.67*
	(0.001)	(0.080)
Observations	1104	1065
R²	0.037	0.094

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 96: Determinants of WTP – Sweden.

	Model 1	Model 2
Medium improvement	10.22	2.867
	(0.107)	(0.660)
High improvement	10.20	4.038
	(0.108)	(0.519)
Age	-0.269	-0.697***
	(0.143)	(0.000)
Female	-6.194	-13.37**
	(0.254)	(0.015)
Has Children	-2.486	-4.729
	(0.452)	(0.156)
Employed	-6.426	-4.816
	(0.289)	(0.440)
Income	0.00471***	0.00459***
	(0.003)	(0.004)
Education	7.891*	6.823
	(0.085)	(0.143)
Aware of chemicals risk		4.660*
		(0.080)
Aware of Octo impacts		5.336**
		(0.010)
Sufficient chemicals regulation		-7.308***
		(0.001)
Obligation to future		4.496
		(0.119)
Reduces use of chemicals		-2.647
		(0.339)
Companies should reduce chemicals		-4.965
		(0.285)
Companies should reduce emissions		5.394
		(0.238)
Govts should reduce chemicals		9.248**
		(0.041)
Govts should reduce emissions		-8.607**
		(0.046)
Perceived environmental knowledge		3.865
		(0.226)
Environmental concerns		15.78***
		(0.000)
Constant	51.84***	-20.31
	(0.001)	(0.441)
Observations	1089	1020
R²	0.018	0.121

p-values in parentheses

* *p* < 0.10, ** *p* < 0.05, *** *p* < 0.01

E.4.5. Benefits assessment of the proposed restriction based on DCE results

E.4.5.1. Main assessment

The Dossier Submitter also assessed the benefits of the proposed restriction based on the DCE results (see previous section). To do so, the Dossier Submitter first established an EEA-wide reference value, following the SEAC recommendations for the transfer of WTP values (ECHA, 2023): this EEA-wide reference value (in €/household/year) corresponds to a simple average over the surveyed countries (country-specific WTP values having already undergone a PPP-conversion). **When considering WTP values derived from the DCE** and as mentioned above, some of the estimated mean coefficients for willingness to pay are not statistically significant. For the main assessment, the Dossier Submitter chose to include non-significant coefficients obtained for some of the case-study countries in the calculation of the average WTP, while being aware of the associated limitations. Including these non-significant values reduces the level of precision with which it measures the population's willingness to pay to avoid the impacts of octocrilene on the aquatic environment. Thus, the average willingness to improve the state of coastal and freshwater ecosystems amount to **222 euros and 83 euros per year per household** respectively. In total, surveyed households are willing to pay on average **305 euros per year during five years** in order to reduce the risks associated with the use of octocrilene in cosmetic products by one level.

Table 97: Mean willingness to pay to reduce the risks associated with the use of octocrilene in cosmetic products for freshwater and coastal water ecosystems – Results of the discrete choice experiment.

	Mean willingness to pay – improvement for coastal water ecosystems ^a		Mean willingness to pay – improvement for freshwater ecosystems		Total
Bulgaria	166.75		14.95	NS	181.7
Germany	301		106.2		407.2
Spain	241		74.75		315.75
France	219.12		128.26		347.38
Hungary	38.41	NS	35.88	NS	74.29
Sweden	365.6		137.8		503.4
Simple average^b	221.98		82.97		304.95
Simple average – NS as zero^c	215.58		74.50		290.08
Weighted average Simple average – NS excluded	258.69		111.75		370.45

Note: for two of the countries surveyed (Hungary for both types of ecosystems and Bulgaria for freshwater ecosystems only), estimated mean coefficients for willingness to pay are not statistically significant (see Appendix E.4.4 for details of the statistical analysis). Thus, these coefficients reflect the population's willingness to pay with a low level of precision. "NS" means "not significant" in the table.

a: Values are expressed in Euro PPP per household and per year for one level of improvement.

b: Mean willingness to pay over the 6 surveyed countries. Willingness to pay values for the countries for which results were not statistically significant are included.

c: Mean willingness to pay over the 6 surveyed countries. Willingness to pay values for the countries for which results were not statistically significant are set to zero.

d: Willingness to pay values for the countries for which results were not statistically significant are excluded. Mean willingness to pay over 5 countries for coastal water and over 4 countries for freshwater.

As mentioned above, for methodological reasons households' willingness to pay has been elicited for the next five years. However, this 5-year timeframe does not correspond to the study period considered to assess the impacts of this restriction (i.e., 10 or 20 years). It is therefore necessary to 'convert' the aggregate willingness to pay established at the EEA level for this 5-year period to the study period under consideration. To do so, the Dossier Submitter determined the corresponding annuity while assuming that the benefits would only start to materialize after the transitional period (i.e. for a 10-year study period: after 2 years and for the 8 remaining years for RO1; after 5 years and for the 5 remaining years for RO2)³⁵. In addition, the Dossier Submitter wished to take into account two main sources of uncertainty in its assessment. As discussed in the main report, the water bodies affected by the proposed restriction are likely to be heterogeneous in terms of the level of exposure to octocrilene but also in terms of the overall environmental pressure faced. This heterogeneity, as well as the uncertainty about the initial state of the water bodies, results in uncertainty about the level of improvement triggered by the restriction. The Dossier Submitter therefore assessed the benefits of the restriction considering that water bodies would only benefit from a low level of improvement³⁶. Furthermore, given the persistent nature of the substance, there is uncertainty as to when the benefits will start to accrue. To account for this uncertainty, the Dossier Submitter considered the following three scenarios³⁷:

1. Immediate materialisation of benefits: the recovery of water ecosystems is complete at the end of the transition period;
2. Moderate lag in materialisation of benefits: the recovery of water ecosystems is gradual and complete four years after the end of the transition period;
3. Significant lag in the materialisation of benefits: the recovery of water ecosystems is gradual and complete eight years after the end of the transition period³⁸.

For scenarios #2 and #3, the annuity (benefits for the overall EEA) is prorated in relation to the level of recovery of the environment. Figure 18 shows the improvement levels (i.e., ratios) considered to do so for each of the abovementioned scenarios. For instance, when considering a 'moderate lag' scenario under RO1, only 50% and 75% of the annuity (yearly benefit) has been accounted for for the years 2030 and 2031 respectively when assessing the benefits over the whole study period (i.e., computing the 2024 PV).

Table 98 shows the benefits of the restriction calculated on that basis while considering WTP values derived from the DCE (PV 2024 at 3%).

Table 98: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE (simple

³⁵ WTP values for the future years were discounted to present values using a 3% discount rate.

³⁶ As the WTP values resulting from the DCE are derived from the estimation of a linear model, this implies considering one level of improvement. The use of a statistical model assuming linearity in the level of improvement induced by the proposed restriction for the environment does not require to define the initial state.

³⁷ In the absence of data to characterise more precisely the time required for beneficial environmental impacts to materialise, the Dossier Submitter has defined ad hoc scenarios to account for the impact of a lag on the benefits assessment of the restriction.

³⁸ This scenario was only considered for RO1 as for RO2 there are only five years between the end of the transitional period and the end of the study period (10 years).

average - 6 countries, NS included).

Scenario	RO1	RO2
Immediate full recovery of water ecosystems	259.58	237.55
Full recovery of water ecosystems after transitional period + 4 years	206.77	163.47
Full recovery of water ecosystems after transitional period + 8 years	140.98	Not considered
WORST CASE - Full recovery of water ecosystems after transitional period +20 years	125.58	Not considered

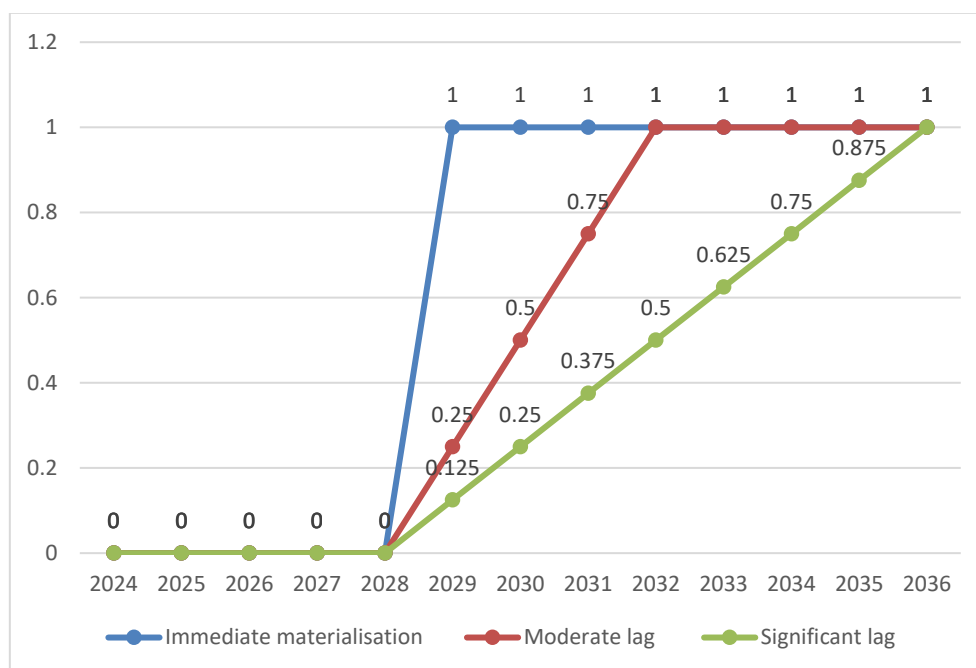


Figure 18: Share of improvement achieved under the three benefits assessment scenarios (under RO1).

In addition to these three scenarios, the Dossier Submitter assessed the benefits of the restriction for a worst-case scenario under RO1 for comparison with the worst-case scenario established for costs. Under this scenario, the Dossier Submitter considered a 20-year study period and assuming that the recovery of water ecosystems is gradual and complete only at the end of the study period (i.e., after 18 years). Under this scenario, the benefits of the restriction amount to 125.58 billion euros (PV 2024, 3%).

E.4.5.2. Sensitivity analysis – Processing of statistically insignificant WTP coefficients in the benefits assessment of the proposed restriction

As mentioned in the previous section, the mean WTP-values determined from the DCE are based, for some countries, on non-significant coefficients. This is the case for Hungary and Bulgaria for the mean WTP to reduce the risks associated with the use of octocrilene in cosmetic products for freshwater ecosystems and for Hungary for the mean WTP for coastal

water ecosystems. In this context, the Dossier Submitter considered the most appropriate way of taking these results into account when assessing the benefits of the restriction at the EEA level. For the main assessment, the Dossier Submitter chose to include non-significant coefficients obtained for some of the case-study countries in the calculation of the average WTP. The Dossier Submitter carried out a sensitivity analysis to identify the impact of this decision. Table 99 and Table 100 below show the results of the benefit assessment when processing insignificant WTP coefficients in alternative ways. The result of this sensitivity analysis shows that the way insignificant WTP coefficients are processed impacts significantly the results of the benefits assessment of the restriction. However, this does not impact the results of the CBA.

Table 99: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE (simple average - 6 countries – NS as zero).

Scenario	R01	R02
Immediate full recovery of water ecosystems	246.92	225.97
Full recovery of water ecosystems after transitional period + 4 years	196.68	155.5
Full recovery of water ecosystems after transitional period + 8 years	134.11	Not considered
WORST CASE - Full recovery of water ecosystems after transitional period +20 years	119.46	Not considered

Table 100: Benefits of the restriction in billion euros (2024 PV, 3%) - WTP from DCE Simple average – (NS excluded : average over 5 countries for coastal water and 4 for freshwater).

Scenario	R01	R02
Immediate full recovery of water ecosystems	315.33	288.57
Full recovery of water ecosystems after transitional period + 4 years	251.17	198.58
Full recovery of water ecosystems after transitional period + 8 years	171.26	Not considered
WORST CASE - Full recovery of water ecosystems after transitional period +20 years	152.55	Not considered

E.5. Practicality and monitorability

As an example of the possibility to detect, identify and quantify Octocrilene in Sunscreen Products you could report to the publication by ML Hudson-Davis, et al. by High Performance Liquid Chromatography-Diode Array Detection (HPLC-DAD)(CARTS No. IR01925).

Appendix F: Assumptions, uncertainties and sensitivities

See Annex XV report.

Appendix G: Stakeholder information

In the frame of the preparation of this restriction proposal, a call for evidence was open on the ECHA website from 15/11/2023 to 12/02/2024. This call for evidence (referred to as CfE in this report) was targeted to EC 228-250-8. This call for evidence aimed at gathering information on the manufacture, import, use and placing on the market of octocrilene, as well as on the possibility for its substitution and potential alternatives as part of the production articles/mixtures, in particular of the following:

- Plastisol,
- Cosmetic products (perfumes, fragrances, etc.) including sunscreens.

Additionally, information on life cycle, emissions/risks to the environment and socio-economic impacts of a potential restriction for the above-mentioned uses was also of interest.

The background note, and the list of questions asked during the call for evidence are available on the ECHA website through the following link:

- CfE: <https://echa.europa.eu/documents/10162/d2f94d14-3354-a5d8-bed0-428f5b30c4e6>

In this call for evidence, interested parties such as private companies (manufacturers, suppliers, recyclers, downstream users, distributors, importers, alternative providers etc.), trade associations, scientific organisations, NGOs and other stakeholders or Member State Authorities holding relevant information were particularly invited to participate. Moreover, both EU/EEA and non-EU/ EEA stakeholders were encouraged to participate.

The Dossier Submitter also performed hearings of several stakeholders in the course of the preparation of this Annex XV proposal. More specifically, the hearings were conducted with Cosmetics Europe as the European trade association for the cosmetics and personal care industry, COSMED as a French representative association of cosmetic professionals, as well as with five producers of cosmetic products/ R&D and formulation companies. The Dossier Submitter sought, as far as possible, to involve a wide range of stakeholders in terms of company size and location, product positioning but also in terms of their use of octocrilene and alternative substances. Some of these hearings gave rise to additional data collection via e-mail exchanges.

The following document was used to prepare and conduct the hearings.

Table 101: Guide for stakeholders hearing.

1) Describe briefly your organization and your position
Characterization of the cosmetics industry in the European Union
2) How many companies are active in the cosmetics market in the EEA?
3) What proportion is made up of SMEs and what proportion is made up of large companies?
4) How many companies are active in the sun care market in the EEA?
5) What proportion is made up of SMEs and what proportion is made up of large companies?
6) How many formulations (i.e., end products) are available to consumers on the cosmetics market in the EEA?
7) How many formulations (i.e., end products) are available to consumers on the sun care market in the EEA?
Use of Octocrilene in cosmetic and sunscreen products
8) In the EEA, what share of the total supply of cosmetic/sunscreen products contains octocrilene: - In number of formulations available on the market? - In sales volumes?
9) What are the overall volumes of octocrilene used by manufacturers (in tpa): - By the cosmetics industry? - More specifically for the production of sunscreen products?
10) Where are the production sites concerned located in the EU?
11) What is the average market price of Octocrilene (€/kg)?
12) At what concentration is octocrilene used in cosmetic products (including sunscreen products)? If octocrilene is used at different concentration levels, please describe these different cases (type of products and function of octocrilene).
13) What is the lowest concentration at which octocrilene is used in cosmetic products? In which type of end products can this concentration be found and what is the function of octocrilene at this concentration?
14) What is the average formulation cost (€/kg) of a sunscreen product containing octocrilene?
15) What UV filters can replace octocrilene (or combinations of UV filters as alternatives to octocrilene-containing combinations)? Can these alternatives be used in the same proportions as octocrilene in a formulation?
16) What are the technical properties of these alternative UV filter(s) (advantages/disadvantages) compared with octocrilene when considering the production process?
17) What is the average market price (€/kg) of each of these alternatives?
18) What is the average formulation cost (€/kg) of a product based on these alternative UV filters?
19) What characteristics (positive or negative) do these alternative UV filters give to the end products (color, smell, photo-stability, protection against UVA, etc.)?
20) Do you want to share any other information on these alternatives (supply issues, etc.)?
21) How often does a company reformulate its end products/formulations ("routine reformulations")? Are there different kind of reformulations? If so, can you describe the different reformulation scenarios (ex: minor versus major ingredient) and provide an estimate of associated costs? To what extent does the cost of a reformulation vary according to the size of a company?
22) Restricting the use of octocrilene would require the reformulation of (some) end products containing it. Can you describe all the steps involved in such a reformulation, its expected duration and the associated cost?
23) Does replacing octocrilene with alternative substances require adapting production processes (new processes, new equipment, etc.)?

24)	Are you considering abandoning the use of octocrilene (or have already done so)?
25)	If so, what alternative substance do you prefer and why?

Moreover, the Dossier Submitter, benefited from the support of the French association *UFC Que Choisir* and company YUKA that both provided data on cosmetic products containing octocrilene (see section 1.3.1 for details regarding the YUKA database).

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